



**Verified Carbon  
Standard**

## ARAGUAIA REDD+ BOCALON PROJECT



|                                  |                                                                    |
|----------------------------------|--------------------------------------------------------------------|
| <b>Project title</b>             | ARAGUAIA REDD+ BOCALON PROJECT                                     |
| <b>Project ID</b>                | 5044                                                               |
| <b>Monitoring period</b>         | 01-July-2023 to 30-Jun-2083                                        |
| <b>Crediting period</b>          | 01-July-2023 to 30-Jun-2043                                        |
| <b>Original date of issue</b>    | -                                                                  |
| <b>Most recent date of issue</b> | -                                                                  |
| <b>Version</b>                   | V1                                                                 |
| <b>VCS Standard Version</b>      | V4.5                                                               |
| <b>Prepared by</b>               | NOVAAGRI INFRA-ESTRUTURA DE ARMAZENAGEM E ESCOAMENTO AGRICOLA S.A. |

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# 1 PROJECT DETAILS

## 1.1 Summary Description of the Project

NOVAAGRI, the proponent and developer of the present project, is a company founded in 2008 as a visionary and innovative player in the Agribusiness sector. The company is specialized in providing independent storage services for agricultural commodities. Its activities involve from the origination of grains, through the logistical flow, to the exportation for destination or sale in domestic markets.

As it is known, Brazil has a territorial extension with great potential for agriculture, which is very advantageous for our business model. However, we are fully aware that this magnitude requires a stricter diligence, especially at environmental issues, since the country has sensitive areas and biomes that must be preserved, given the richness of biodiversity and carbon pools of its forests. For this reason, we drive a strict control under the process of farmers' prospection and selection, adopting internal criteria for analysis and approval of the growing/production areas, such as compliance with the national forestry legislation and other socio-environmental regulations related to combating deforestation, work conditions and conservation of sensitive areas and indigenous lands. This control is carried out through geo-monitoring systems, by using satellite images, applied to 100% of the commercial contracts carried out at NovaAgri. In accordance with its environmental commitment, NovaAgri has also been certified in some important and recognized programs. For instance: 2BSVS (Sustainable Certification to reach new markets); Soy Moratorium Audit (Signed zero deforestation pact in Amazon Rainforest, so we do not buy grain from farmers who have deforested); and RTRS (Round Table on Responsible Soy).

Although the development of Carbon Credits projects is not a main service of NovaAgri's portfolio, neither intend to make it a core source of revenue to the company, these projects are totally strategic and aligned to the Group's beliefs and philosophy, which counts on challenging targets of carbon reduction for the medium and long term – the carbon credit projects are vital to orientate the company into the carbon neutrality path, encouraging good actions among its value chain players. Illustrative of this environmental commitment is the Toyota Tsusho Group's global commitment to reduce its greenhouse gas emissions by 50% before 2030, ultimately striving for carbon neutrality by 2050.

A trustful partnership with farmers is central to our strategic focus. The Tapirapé Farm represents the property area of the present study, which is situated within the transition between Amazon and Cerrado biomes that stands as some of the planet's most ecologically diverse and carbon reservoir-richness ecosystems. It has historically encountered significant vulnerabilities attributable to deforestation, including the land-use transformation by agricultural activities. The convergence of our attention towards this biome underscores our commitment to not only capitalize on emergent carbon credit opportunities but also to contribute to the preservation of



this invaluable ecological domain. With this scenario, **we believe in a business strategy that will enable the farmers to choose a more sustainable way for their productive activities.**

In alignment with its principles, NovaAgri also endeavors to make a substantial impact to social prosperity. The Group's values manifest through the cultivation of an ethical business and the care of human resources assured to make meaningful contributions to local communities. For that reason, every carbon project developed and managed by NovaAgri intends to ensure a share of the project's financial returns to implement social actions within the local communities located in the surroundings of the project area. The objective is to generate real impacts for the society beyond those for tackling climate change.

The ARAGUAIA REDD+ BOCALON PROJECT is taken in Santa Terezinha, a municipality from the Mato Grosso State in Brazil. The property's is composed by an area of 24,100 hectares in total, and its project area composes an area of 13,926 hectares of native forest.

The project area is located in the Amazon biome, presenting an annual deforestation rate of 2,09% in its reference boundary. The activities planned to be implemented refer to AUD (Avoiding Unplanned Deforestation) mechanisms, based on the principles of REDD+ and following the methodology VM0015 from VERRA. The baseline assumption for this Project acknowledges the heightened deforestation pressure in threatened biomes in Mato Grosso, reflecting the substantial carbon pool and biodiversity importance, further compounded by the challenges posed by the non-compliance with environmental laws in Brazil. To accomplish this objective, NovaAgri is committed to investing in a trustable and modern technology of GIS (georeferenced image system) with deforestation near real-time alerts alongside a high-resolution satellite imagery resource, dedicating an exclusive corporate department to this end.

The project expects to avoid net emissions of 1,683,481.14 tCO<sub>2</sub>e, which means an average of 28,058 tCO<sub>2</sub>e per year and 10,012 hectares preserved during a horizon of 60 years.

## 1.2 Audit History

The development and monitoring of this VCS project is under validation and first verification simultaneously.

**Table 1. Audit History of Validation and Verification**

| Audit type   | Period                     | Program | Validation/verification body name | Number of years |
|--------------|----------------------------|---------|-----------------------------------|-----------------|
| Validation   | XX-XX-2024                 | VCS     | RINA Services S.p.A               | -               |
| Verification | 01-JUL-2023 to 30-JUN-2024 | VCS     | RINA Services S.p.A               | 1 year          |

### 1.3 Sectoral Scope and Project Type

|                                           |                                                    |
|-------------------------------------------|----------------------------------------------------|
| <b>Sectoral scope</b>                     | AFOLU '14. Agriculture, Forestry, Land Use'        |
| <b>AFOLU project category<sup>1</sup></b> | REDD+ Avoided Unplanned Deforestation (AUD)        |
| <b>Project activity type</b>              | Forest maintenance and protection / Single Project |

### 1.4 Project Eligibility

#### 1.4.1 General eligibility

The project activity is included under the scope of the VCS Program and not excluded under Table 2.1 of the VCS Standard.

**Table 2. Project Activities' inclusion under the VCS Program Scope**

| VCS Program Scope                                                                                                                                               | Applicable / included in the project activities                                                                                                                                                    | Not applicable                                                                                                                                                   |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>The seven Kyoto Protocol greenhouse gases.</i>                                                                                                               | <i>The methodologies, rationales, calculations, and other considerations of the activities of this project follow the principles of the Kyoto GHG Protocol.</i>                                    | -                                                                                                                                                                |
| <i>Ozone-depleting substances (ODS).</i>                                                                                                                        | -                                                                                                                                                                                                  | <i>The ODS gases, such as CFCs and HCFCs are not applicable for this specific project once the emissions' characteristics are solely related to REDD+ AFOLU.</i> |
| <i>Project activities supported by a methodology approved under the VCS Program through the methodology development and review process.</i>                     | <i>This project uses an eligible methodology under the VCS Program. In this case, the version 1.2 of the Methodology VM0015 has been fully applied, including any referenced tools or modules.</i> | -                                                                                                                                                                |
| <i>Project activities supported by a methodology approved under an approved GHG program, unless explicitly excluded (see the Verra website for exclusions).</i> | <i>The project activities are referenced by international consensus of relevant GHG programs and agreements.</i>                                                                                   | -                                                                                                                                                                |

<sup>1</sup> See Appendix 1 of the VCS Standard

|                                                                                                                                      |   |                                                                                                                                                                                                                                                                                                              |
|--------------------------------------------------------------------------------------------------------------------------------------|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                                      |   |                                                                                                                                                                                                                                                                                                              |
| <i>Jurisdictional REDD+ programs and nested REDD+ projects as set out in the Jurisdictional and Nested REDD+ (JNR) Requirements.</i> | - | <i>Although this project is not part of a Jurisdictional REDD+ program, this is aligned to the most recent criteria and trends for an integrated network of carbon initiatives, including the latest methodologies' updates from VERRA, and the recent evolution of an official Carbon Market in Brazil.</i> |

This project assumes to NOT have generated any GHG emissions primarily for the purpose of their subsequent reduction, removal, or destruction.

This project does NOT have any of the following project activities under the circumstances indicated below:

- Grid-connected electricity generation activities using hydroelectric power plants.
- Grid-connected electricity generation<sup>1</sup> activities using wind, geothermal, or solar photovoltaic (PV) power plants.
- Activities recovering waste heat for combined cycle electricity generation, or to heat/cool via cogeneration or trigeneration.
- Activities generating electricity and/or thermal energy for industrial use from the combustion of non-renewable biomass, agro-residue biomass, or forest residue biomass.
- Activities generating electricity and/or thermal energy using fossil fuels, and activities that involve switching from a higher to a lower carbon content fossil fuel.
- Activities replacing electric lighting with more energy-efficient electric lighting, such as the replacement of incandescent electrical bulbs with compact fluorescent lights (CFLs) or light emitting diodes (LEDs).
- Activities installing and/or replacing electricity transmission lines and/or energy-efficient transformers.
- Activities that reduce hydrofluorocarbon-23 (HFC-23) emissions.

The project meets requirements related to the pipeline listing deadline, the opening meeting with the validation/verification body, and the validation deadline.

- The proponent is aware of the requirements related on the document 'Registration and Issuance Process v4.4'

- The proponent has contacted and hired a qualified validation/verification body (VVB), which will perform a rigorous assessments of VCS methodologies and methodology elements to confirm that they conform with VCS Program requirements and AFOLU-specific best practices.
- The proponent has agreed alongside the VVB about the validation/verification deadlines.
- The project is eligible under the VCS Program
  - The project applies the latest version of the applicable methodology in all cases, ensuring compliance unless a grace period applies. The project updates to the latest version of the methodology when reassessing the baseline or renewing a crediting period. This project does not violate any applicable local or regional laws, regardless of whether or not the law is enforced.
  - The project and its implementation activities do not violate any applicable law, regardless of whether or not the law is enforced.
- The project is not a fragmented part of a larger project or activity, as demonstrated by the documents provided to the VVB, and also publicly on this PD.

#### 1.4.2 AFOLU project eligibility

The project is eligible under the AFOLU-specific matters:

- The project description identifies any implementation partners involved in collaboration with the project proponent. These partners delineate their roles and responsibilities, encompassing implementation, management, and monitoring of the project throughout the project crediting period.
- The project demonstrates the implementation of activities leading to the intended GHG benefit during each verification period in alignment with the project design.
- The project commits to reassessing the baseline every six years.
- The baseline reassessment will follow specific guidelines, including the application of the latest VCS Program rules and methodologies, validation during subsequent verifications, capturing changes in drivers, and updating relevant sections of the project description to reflect changes in baseline emissions quantifications.
- The project adheres to the stipulation that the latest version of the VCS Program rules and applied methodology, or its replacement, is utilized during baseline reassessment.
- The reassessment will capture changes in drivers and behavior affecting land use, hydrology, sediment supply, land or water management practices, and carbon stocks. These changes are incorporated into revised estimates of land-use change rates, patterns, and baseline emissions.

The project will reassess the validity of the original baseline scenario, evaluating the impact of new national and sectoral policies on its validity.

The project is eligible under the applicability conditions of the VM0015 v1.2:

- Baseline activities include unplanned agriculture and grazing activities, following the category: Avoiding Unplanned Deforestation according to the most recent VCS Standard.
- The project area includes one or a combination of forest types meeting the definition of “forest.”
- At the project start, the project area includes only land qualifying as “forest” for a minimum of 10 years prior to the project start date.
- The project area does NOT include forested wetlands growing on peat (e.g., peat swamp forests).
- This PD brings evidence that native ecosystems have not been converted, cleared, drained, or degraded to generate GHG credits. The evidence is represented by historical time-series in satellite images, data from PRODES<sup>2</sup>, georeferenced files, and other documents. Other complementary data or documents will be available for the VVB.

The project is eligible under the criteria of the Non-Permanence Risk Assessment

- The project prepares a non-permanence risk report using the AFOLU Non-Permanence Risk Tool v4.2 at both validation and verification.
- The project has a minimum project longevity of 40 years.
- Buffer credits will be deposited in the AFOLU pooled buffer account based on the non-permanence risk report assessed by the VVB.
- The project performs non-permanence risk analysis at every verification event and may be eligible for the release of buffer credits from the AFOLU pooled buffer account.
- Validation of non-permanence risk analyses is allowed to be conducted by the same VVB that validates or verifies the project.
- When an instance leaves a project before the end of project longevity, the Proponent is aware to assume a loss of all previously verified reductions and removals associated with the instance or continues monitoring for the remainder of project longevity.
  - Loss events are reported according to the requirements set out in the Registration and Issuance Process.
  - At a verification event after a loss event, the monitoring report restates the loss and calculates the net GHG benefit for the monitoring period.
- Reversals are managed based on whether they are unavoidable or avoidable, and the Proponent follows buffer account reconciliation requirements.
- Carbon stock permanence is monitored for a minimum of 40 years, with the option for Verra to agree to monitor projects with a shorter crediting period.

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<sup>2</sup> The PRODES project conducts satellite monitoring of clear-cut deforestation in the Legal Amazon and has been producing annual deforestation rates in the region since 1988, which are used by the Brazilian government to establish public policies. <http://www.obt.inpe.br/OBT/assuntos/programas/amazonia/prodes>

- The remaining balance of buffer credits is canceled at the end of the project longevity or at the end of monitoring, whichever is later.

### 1.4.3 Transfer project eligibility

This item does not apply to the project since it is not a transfer project or CPA seeking registration.

## 1.5 Project Design

This project has been designed as:

☒ Single location or installation

## 1.6 Project Proponent

|                          |                                                                                                                                                                                                                                                            |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Organization name</b> | NOVAAGRI INFRA-ESTRUTURA DE ARMAZENAGEM E ESCOAMENTO AGRICOLA S.A.                                                                                                                                                                                         |
| <b>Contact person</b>    | MAURICIO SCHIAVOLIN SILVA<br>VICTOR KENJI OSHIRO                                                                                                                                                                                                           |
| <b>Title</b>             | Carbon Credit Manager<br>Carbon Specialist                                                                                                                                                                                                                 |
| <b>Address</b>           | Alameda Vicente Pinzon, 51 - Vila Olímpia, São Paulo - SP, Brazil.<br>Postal Code 04547-130.<br><a href="http://www.novaagri.com.br">www.novaagri.com.br</a><br><a href="http://www.linkedin.com/company/novaagri/">www.linkedin.com/company/novaagri/</a> |
| <b>Telephone</b>         | +55 11 91736-3892<br>+55 11 3048 2888                                                                                                                                                                                                                      |
| <b>Email</b>             | mauricio.schiavolin@novaagri.com.br<br>victor.oshiro@novaagri.com.br<br>carbon.credit@novaagri.com.br                                                                                                                                                      |

## 1.7 Other Entities Involved in the Project

|                            |                                                           |
|----------------------------|-----------------------------------------------------------|
| <b>Organization name</b>   | BOCALON                                                   |
| <b>Role in the project</b> | Project area owner and protection activities´ responsible |

|                |                                                        |
|----------------|--------------------------------------------------------|
| Contact person | MAURO DONISETI SILVERIO RODRIGUES                      |
| Title          | Landowner                                              |
| Address        | RUA CAROLINA MARQUES 292 CENTRO, ARAGUARI, MG, Brazil. |
| Telephone      | -                                                      |
| Email          | bocalonm@uol.com.br                                    |

|                     |                                                                                                                                                                   |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Organization name   | AXELSPACE                                                                                                                                                         |
| Role in the project | Satellite Images solutions                                                                                                                                        |
| Contact person      | Tatsuhiko Fukasawa                                                                                                                                                |
| Title               | Head of AxelGlobe Business Division                                                                                                                               |
| Address             | Clip Nihonbashi Building 1-3F<br>3-3-3 Nihonbashi-Honcho<br>Chuo-ku, Tokyo, 103-0023 Japan<br><a href="https://www.axelspace.com/">https://www.axelspace.com/</a> |
| Telephone           | +81 3 6262 6105                                                                                                                                                   |
| Email               | fukasawa@axelspace.com                                                                                                                                            |

## 1.8 Ownership

The current Project ownership arises by virtue of a contractual right in the owners' land, which contemplates its vegetation, conservational and management process that generates GHG emission reductions and/or removals. The project area refers to a 13.926 ha of a vegetation land, contemplated in a private area, which belongs to the 'Fazenda Tapirapé'. The area is located in the municipality of Santa Terezinha – in the State of Mato Grosso (MT).

The location details are specified on the item 1.13 and the documentation evidence for land registration is presented on APPENDIX 2.

## 1.9 Project Start Date

|                    |            |
|--------------------|------------|
| Project start date | 01-07-2023 |
|--------------------|------------|

|               |                                                                                                                                                                                                                                                                                                                                      |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Justification | The project start date of this AFOLU project relates to the date on which activities that lead to the generation of reductions or removals were implemented. In addition, this project initiates the pipeline listing process after 1 year of the project start date, that means within three years, as defined by the VCS standard. |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## 1.10 Project Crediting Period

|                                                       |                              |
|-------------------------------------------------------|------------------------------|
| Crediting period                                      | 20 years                     |
| Start and end date of first or fixed crediting period | 01-July-2023 to 30-June-2043 |

## 1.11 Project Scale and Estimated GHG Emission Reductions or Removals

Indicate the estimated annual GHG emission reductions/removals (ERRs) of the project:

- ☒ < 300,000 tCO<sub>2</sub>e/year (project)
- ☐ ≥ 300,000 tCO<sub>2</sub>e/year (large project)

**Table 3. Estimated GHG emission reduction during the first crediting period (possible to extend due to renewals)**

| Calendar year of crediting period                                      | Estimated GHG emission reductions or removals (tCO <sub>2</sub> e) |
|------------------------------------------------------------------------|--------------------------------------------------------------------|
| 01-July-2023 to 30-June-2024                                           | 36.007                                                             |
| 01-July-2024 to 30-June-2025                                           | 37.107                                                             |
| 01-July-2025 to 30-June-2026                                           | 38.177                                                             |
| 01-July-2026 to 30-June-2027                                           | 39.219                                                             |
| 01-July-2027 to 30-June-2028                                           | 40.233                                                             |
| 01-July-2028 to 30-June-2029                                           | 85.257                                                             |
| 01-July-2029 to 30-June-2031                                           | 89.089                                                             |
| 01-July-2031 to 30-June-2033                                           | 92.418                                                             |
| 01-July-2033 to 30-June-2035                                           | 86.942                                                             |
| 01-July-2035 to 30-June-2037                                           | 83.580                                                             |
| 01-July-2037 to 30-June-2039                                           | 78.721                                                             |
| 01-July-2039 to 30-June-2041                                           | 75.731                                                             |
| 01-July-2041 to 30-June-2043                                           | 71.410                                                             |
| <b>Total estimated ERRs during the first or fixed crediting period</b> | <b>853.891</b>                                                     |



|                       |        |
|-----------------------|--------|
| Total number of years | 20     |
| Average annual ERRs   | 42.695 |

## 1.12 Description of the Project Activity

### Deforestation historical context

The reference area is located within the Arc of Deforestation in Brazil, which traces a concerning trajectory, representing the expansive frontier of deforestation that has trespassed upon the Amazon rainforest over the history. This arc refers to a crescent-shaped region that extends from the eastern to the southern Amazon. Diverse states within this arc include Maranhão, Tocantins, Piauí, Mato Grosso, and parts of Pará.

Historically, this arc has been a focal point for agricultural expansion, particularly driven by activities such as logging, cattle ranching, and soy cultivation. The allure of fertile soils and economic opportunities has spurred rapid land-use changes, transforming vast stretches of native rainforest into agricultural landscapes.



Image Source: *The Economist*, 2011 – Fiddling while the Amazon burns<sup>3</sup>

Even with the development of forestry laws in Brazil along the years, the deforestation in Brazil has continuing advancing. According to MAPBIOMAS (2022), in the whole Amazon Biome, around 3,267 ha have been deforested daily in 2022, which means 21 trees per second. About the

<sup>3</sup> The Economist. (2011, December 3). Fiddling while the Amazon burns. Retrieved from <https://www.economist.com/the-americas/2011/12/03/fiddling-while-the-amazon-burns>

impunity, the legal actions and embargo carried out by IBAMA and ICMBio - by May 2023 - reached only 10.2% of the deforested area identified from 2019 to 2022. That is a clear and worrying signal that the law against deforestation in Brazil is far away to be effective itself.

### Socioenvironmental Compliance

In the context of compliance, the monitoring system, integrated with other tools contracted by NovaAgri, facilitates the enforcement of socioenvironmental regulations. This is achieved by cross-referencing the public database in Brazil, which is built on a foundation of reliable sources and encompasses crucial socioenvironmental factors such as conservation units, delineated indigenous areas, legal reserve zones, historically mapped deforestation, among others, with the project's covered area<sup>4</sup>. These actions are part of the NovaAgri's rigorous Socioenvironmental Policy, initially designed for implementation across its credit/risk, sustainability, and commercial departments as a conventional process for selecting new customers.

### Monitoring System and Technology

The main objective of this Project is to avoid the areas to be converted into non-forest land, by conducting activities of engagement for preservation and reinforcing the partnership existing from the commercial relationship between NovaAgri and the farmer(s). To accomplish the main goal for a REDD+ AUD purpose, NovaAgri is committed to investing in a trustable and modern technology of GIS with deforestation near real-time alerts alongside a high-resolution satellite imagery resource, dedicating an exclusive corporate department to this end.

The project is going to integrate an advanced deforestation alert system provided by SCCON, a renowned expert in dashboards, data automation, and artificial intelligence. SCCON has international recognition and extensive track record of working with various public and private entities, exemplified by their successful deployment of a monitoring system for the 'Secretaria de Estado do Mato Grosso' (SEMA) – which can be accessed publicly on <https://alertas.scon.com.br/matogrosso/#/dashboard> –, inspire confidence in their capabilities. This project aims to implement a more refined version of such a system, incorporating micro-localized management for heightened alert precision and improved efficacy in deforestation prevention and remediation efforts. The integration of SCCON's expertise and technology aligns seamlessly with our commitment to modern practices and environmental stewardship in our carbon offset initiative.

By using satellite images of PlanetScope, which counts on the biggest satellite constellation (about 180 nanosatellites), which collect daily images from any spot on the planet – generating continuous imaging generated from PlanetScope orthorectified images with a 3-meter spatial

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<sup>4</sup> NovaAgri employs a tool called LYRA, developed by a company within the Imagem Group - ESRI, to facilitate internal verification activities in potential client and partner areas. LYRA is a technology that utilizes remote sensing and data intelligence to establish traceability, conduct socio-environmental diagnostics of individuals, companies, and territories, agroclimatic monitoring, and territorial intelligence. The system is built on layers of public and private socio-environmental and agricultural risk, adding value and mitigating reputational and operational risks. It features a GIS screen for navigation, providing a user-friendly interface for exploring new territories, especially in terms of socio-environmental monitoring, by checking daily compliance-related aspects for those involved and the monitored agroclimatic conditions. More information on: <https://vegamonitoramento.com.br/lyra/>

resolution, accessible with RGB viewing functionality in XYZ format, including levels 15 (4.77 meters spatial resolution) and 16 (2.38 meters spatial resolution) with an 8-bit radiometric resolution, and added up to an API (Application Programming Interface) integrated to a particular alerts' platform, the system might enable rapid detection of vegetation loss activities. This allows NovaAgri, alongside the owner and other entities, to 1) move internal guard and other specialized departments into the alert's focus to proceed with the emergency mitigation; or/and 2) activate authorities and other stakeholders to respond quickly to illegal logging or land clearing, preventing further damage to forests. The satellites used for this purpose are modern and cover vast areas of land, including remote and inaccessible regions. This extensive coverage allows for a comprehensive assessment of deforestation trends across different landscapes, including forests in difficult-to-reach areas.

The summarized characteristics and specifications of the monitoring system is explicit below:

- i) High frequency: Daily in-season revisit rates powered by 180 satellites.
- ii) Field-level details: 3-3-meter resolution satellite images.
- iii) Deforestation daily alerts: workflow integration with APIs and the NovaAgri's dashboard, created specifically for the ends of this project. The company SCCON, which has developed and provided this platform for NovaAgri, has a vast experience with monitoring systems, including robust government projects used by the Federal Policy and other State security institutions.

This system covers the total area of the Project, including the Reference Region, the Leakage Belt, the Project Area, and all surroundings within the Jurisdictional State (beyond the project boundaries). In addition, this provides an objective and transparent source of data, reducing the potential for bias in monitoring and reporting deforestation. This helps in building trust among stakeholders by supporting evidence-based reports for the MRV periodic verification. The alerts cover the following classes: - Clear-cutting; - Access; - Degradation; - Burn scar.

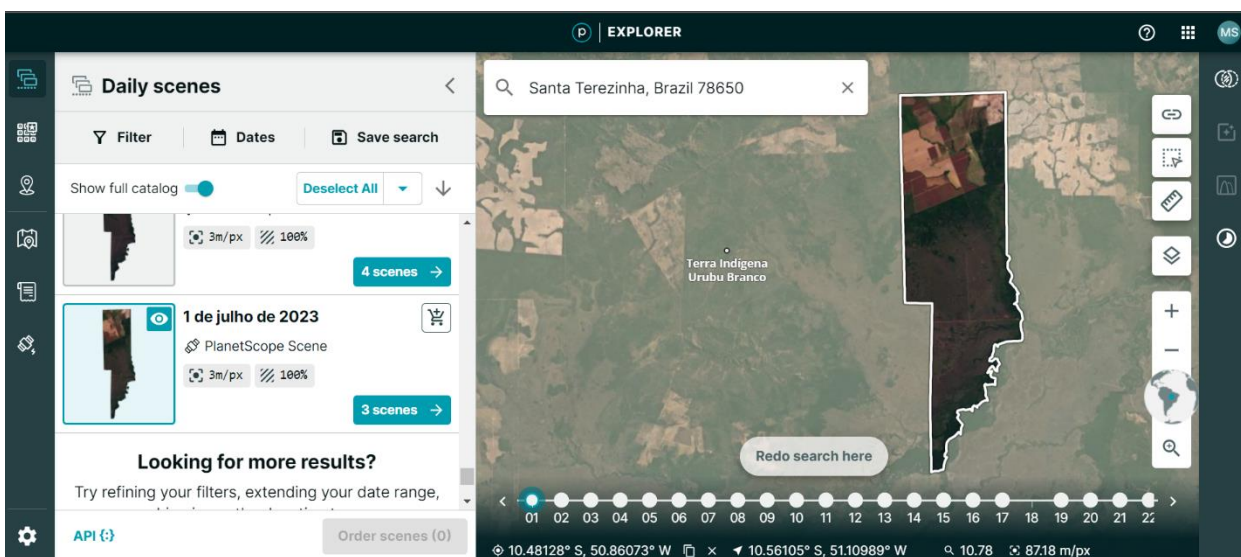


Image source: PLANET<sup>5</sup> – Start Date satellite monitoring

<sup>5</sup> <https://www.planet.com/>

By also using satellite images from the Japanese company Axelspace, which has a constellation of small satellites capable of collecting almost daily images of any point on the planet with a spatial resolution of 2.5 meters in the panchromatic band and 5 meters multispectral, with 8-bit radiometric resolution, integrated into the AxelGlobe platform for viewing image collections and data delivery via API (Application Programming Interface) it will allow NovaAgri to monitor emissions derived from deforestation in the project area and the leakage belts.

By empowering Japanese technology to measure VCUs (Verified Carbon Units) of the ARAGUAIA REDD+ BOCALON PROJECT, NovaAgri will increase confidence in the Measurement, Reporting and Verification (MRV) process by recording the maintenance of intact forests in the project area and the rate of deforestation in the project area of influence (leakage belts).

Looking at the long-term scenario, this system is going to facilitate extended monitoring of deforestation trends and changes in land use over time. By analyzing historical data, the proponent can identify patterns and understand possible drivers of deforestation in the surrounding areas. This might be especially important for the premises of the leakage areas.

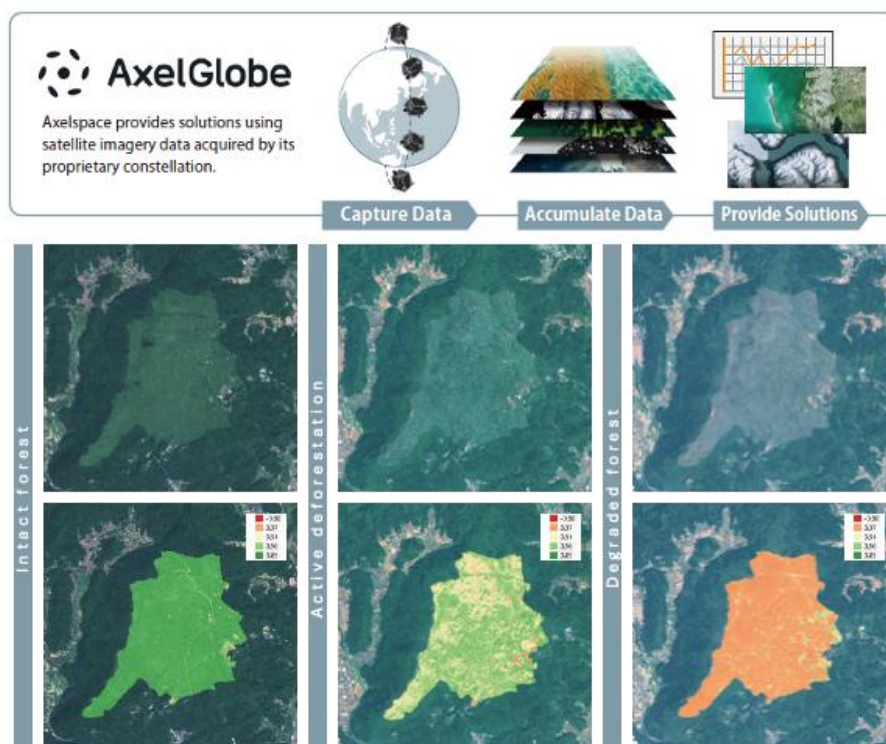


Image source: AXELSPACE – Macro vision of the solutions provided by Axelspace to support NovaAgri's Measurement, Reporting and Verification procedures.



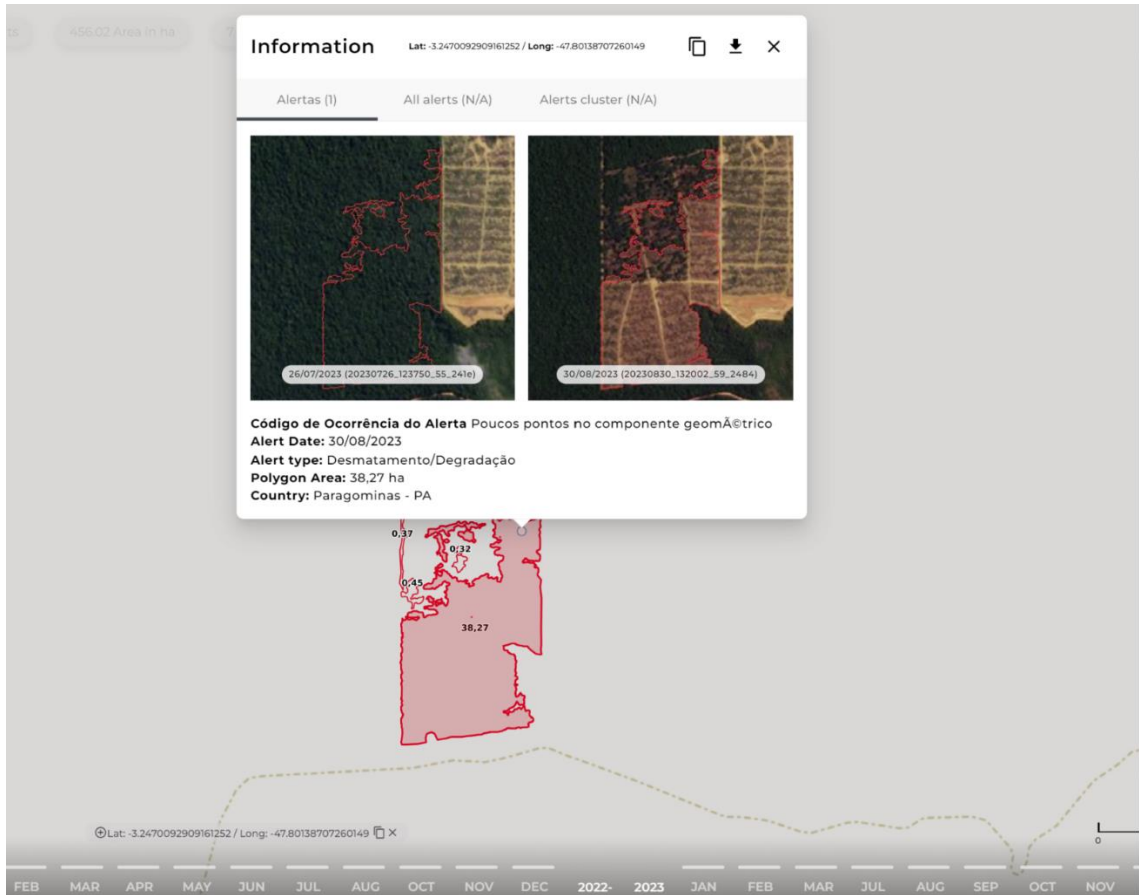


Image source: SCON – MRV dashboard deforestation alerts overview

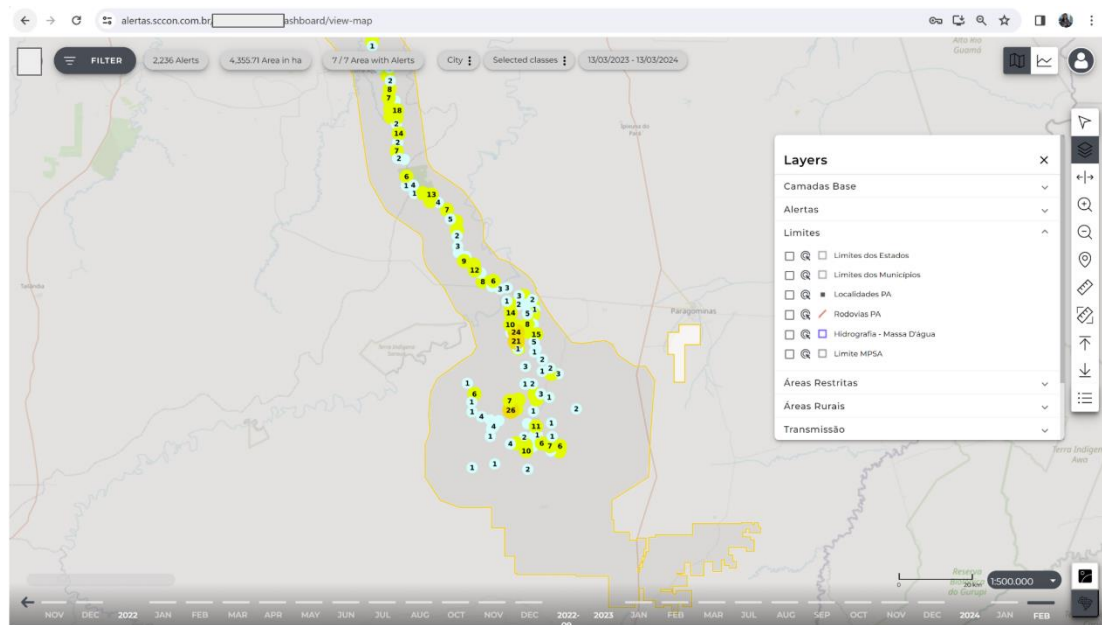
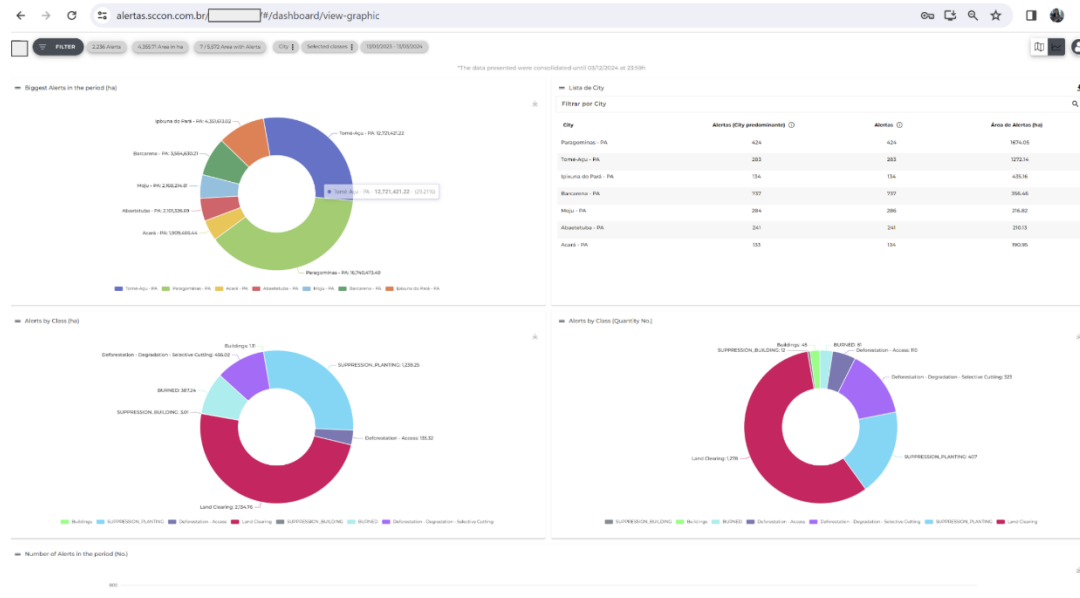


Image source: SCON – MRV dashboard georeferenced data layers overview



The proponent, responsible for technological implementation, will deploy **geospatial monitoring practices** using satellite imagery. By utilizing recent and regularly updated images, the system will identify wildfire outbreaks in nearby areas, allowing for proactive measures before the fire spreads into the forested project area. To facilitate and enhance the accuracy of this monitoring process, the establishment of a **fire near-real-time alert system<sup>6</sup>**, adapted to the web dashboard, is crucial for ensuring effective protection activities. This dashboard will generate alerts to our central system upon initial identification of fire signs. This technological integration aims to strengthen the project's ability to respond promptly to potential fire threats.



*Image source: SCON – MRV dashboard fire alerts overview*

### Engagement and Awareness actions

The project aims to fostering a comprehensive engagement among key stakeholders, including employees, local communities, and other participants. The strategy includes proactive initiatives that aim to raise awareness, build partnerships, and empower those involved.

A public audience was organized to provide a space for open dialogue. This was implemented on both VERRA's public platform and in a separate session, where members of the public,

<sup>6</sup> Utilization of continuous generation of fire spots' alerts throughout the project period. The method involves crossing the fire alert system by INPE with the daily Planet satellite images' system in order to confirm the fire spots more precisely. Files provided by IPNPE contains information on forest fire and wildfire incidents at various intervals, ranging from annual to near real-time (10 minutes). These data are available in CSV and KML formats. A perpetual and unrestricted license for the generated alerts will be provided for use, distribution, and modification.

environmental organizations, and other interested parties were invited to taking notice about the Project's details and sharing thoughts, concerns, and suggestions regarding it.

With the Project in course, comprehensive public awareness campaigns are expected to be launched to inform a wider audience about the objectives and benefits of the REDD+ project. Through media channels, educational materials, and community events, the Project aims to reach a broad spectrum of individuals and encourage their active involvement.

Other details, specifically about safeguards and engagement to avoiding negative externalities, and more information about the public consulting are presented on 'Section 2 – Safeguards' of this PD.

### Social Responsibility

In line to a holistic ESG purpose, NovaAgri integrates an essential social responsibility framework to its environmental actions. For that reason, the company is committed to **allocating 5% of its own returns** – generated from the sale of Verified Carbon Units (VCUs) – to fund and implement impactful social initiatives within the communities surrounding our project area. The social actions managed by NovaAgri will prioritize community development programs aimed at enhancing the quality of life for local residents. This may include, for instance, initiatives such as educational support, infrastructure development, access to natural resources etc.. The landowner is also committed and has been committed to allocate a portion of its revenue generated from its own credits to social initiatives in the region, as has been done on previous occasions.

As the AUD activities, the social activities will also follow high standards of transparency in as part of the company's commitment to accountability and openness, providing regular and detailed reports on the counterparties associated with these activities.

In response to the needs noted during the public consultation with stakeholders (as detailed in Section 2 - Safeguards), the project developer understands that the resources will be of great value in supporting the most vulnerable populations in the region, including low-income residents and traditional communities. For instance, there is an ongoing plan to establish an APAE (Association of Parents and Friends of Exceptional Children) in the municipality, which is a highly important and necessary demand in the area. APAE is a national non-profit network that has been dedicated to advocating for rights and providing services to people with disabilities in Brazil for over 60 years.

Projects related to education will also receive emphasis in investment, as it is a highly requested demand for the region, which has sectors requiring specialized workforce. Support for the development of technical schools, for instance, would also be welcome as part of the initiatives. The proposal may also include engaging in activities such as interactive sessions with young people in schools, fostering critical thinking through educational lectures, games, and awards for positive actions, all centered around the theme of sustainability and tackling climate changes. Recognizing that Mato Grosso is strong related to the agribusiness sector, the proponent



understands that activities promoting good agricultural practices connected to sustainable production methods are crucial. These activities not only contribute to the local families' income but also play a vital role in dispelling the misconception of unsustainable agribusiness. Instead, the goal is to shift towards an image of a sector that generates jobs and income without the need for deforestation or degradation of forested areas.

Such initiatives or other similar ones could be supported by the project's resources, as it is a commitment of the proponent to allocate a portion of its carbon credit-derived funds to priority social causes in the surroundings of the Project Area. It is important to mention that any social investment will reflect needs voiced by the local population itself. Another point is related to the transparency of these actions, which will strictly been reported to the stakeholders, which include the VVB by registering the proper accountability in further monitoring reports.

NovaAgri, as the proponent and developer of the project, adheres to a rigorous Compliance Policy expressed through its LATAM Code of Ethics from its Group. This project will be entirely governed by this instrument, and therefore, in all its activities, it will not tolerate any form of discrimination, coercion, inappropriate advertising, exposure of sensitive data, bribes, or other deviations. The company is committed to conducting a rigorous compliance assessment in all transactions, whether commercial or donation-related. To achieve this, it will rely on appropriate contractual instruments and internal verifications from the company's legal team.

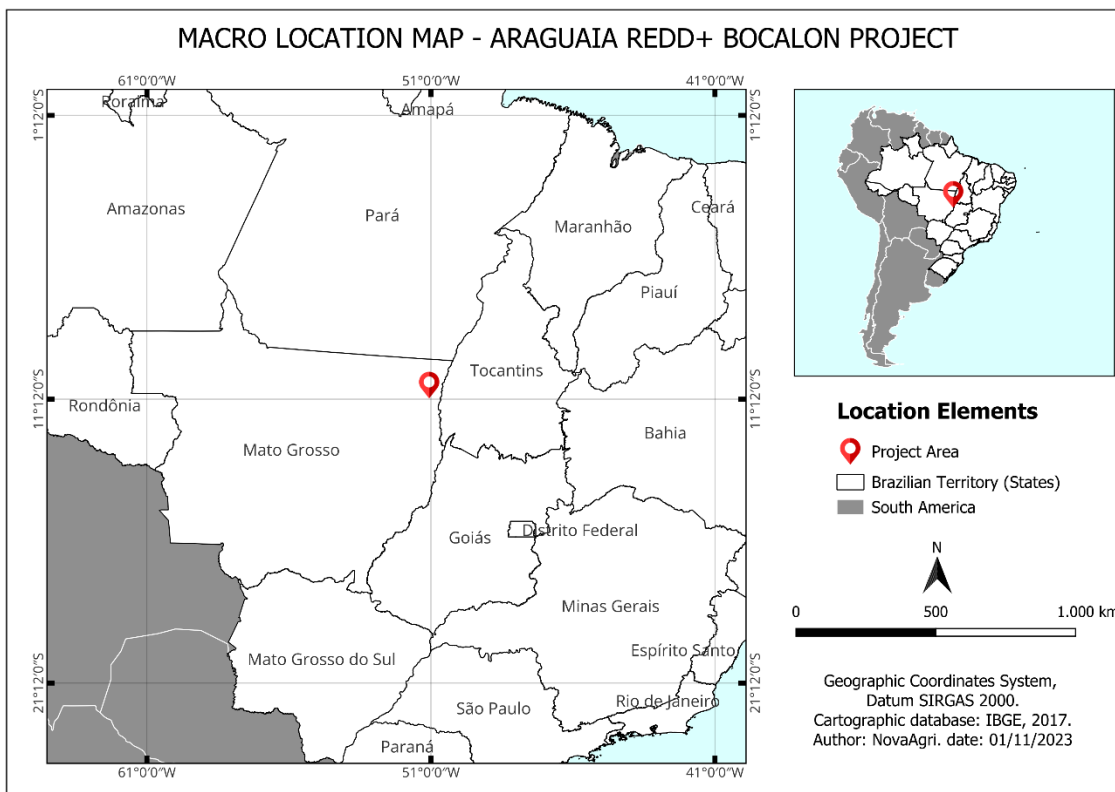
#### Definition of project activities' location

All the activities will be implemented in locations which are not located within a jurisdiction covered by a jurisdictional REDD+ Program. The geographic boundaries will be defined and better specified on Section 3 of this PD, including the coordinates separately as KML files. Further information is also detailed on the next topic 'Project Location'.

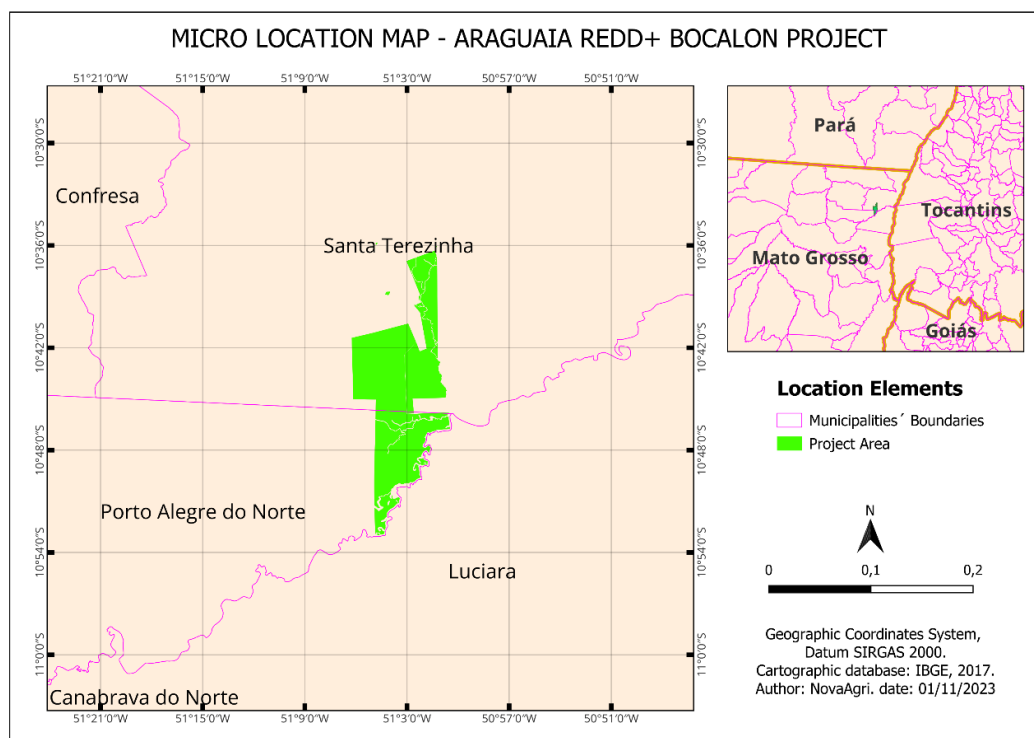
## 1.13 Project Location

### Macro-region description

The project area is situated in the municipality of Santa Terezinha, State of Mato Grosso / Brazil, near the border and frontier of the states of Pará (to the north) and Tocantins (to the east). The nearest airport is in Palmas – Tocantins State, which is 600km away from the location, by road.



Map Source: the author – macro vision (country level) of the region and Biome (boundary by IBGE)



Map Source: the author – localized vision (municipality level) of the area

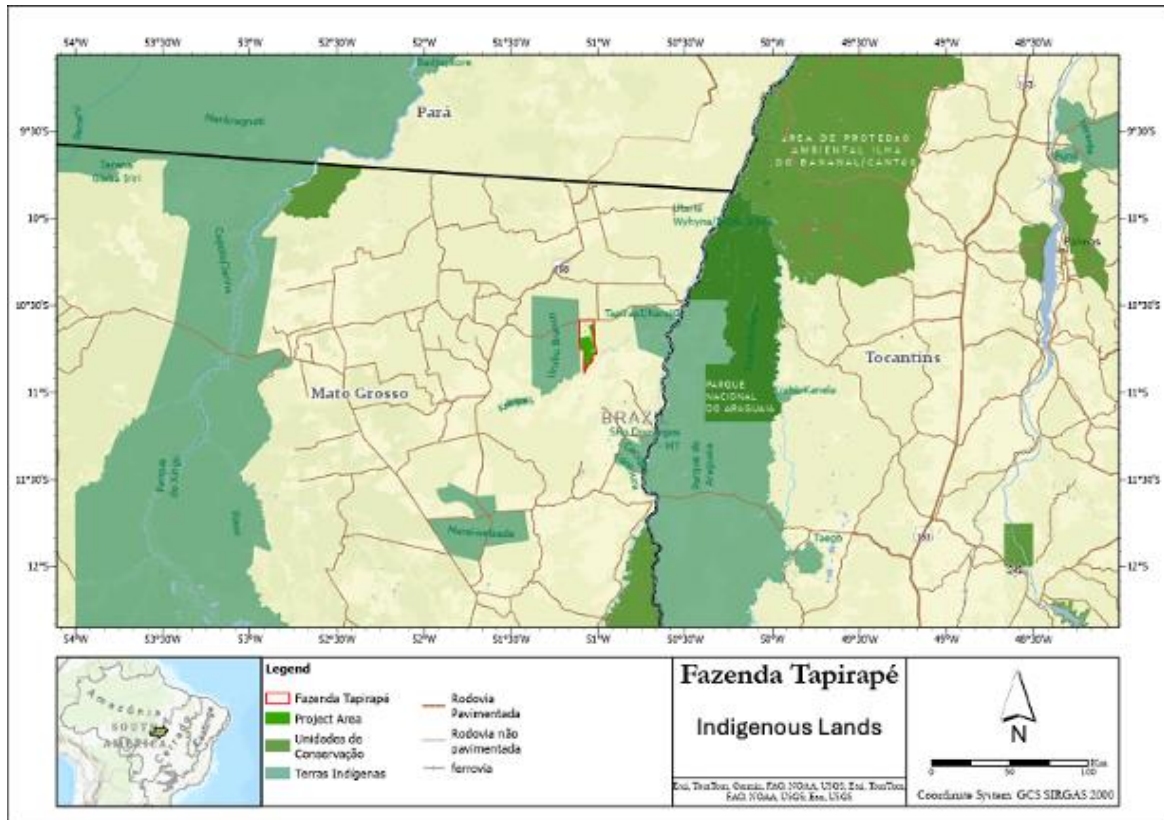


Image Source: the author – reference: Consolidated Indigenous areas public shapefile (update feb.,2023)

As depicted in the image above, the region is within the arc of deforestation in Brazil, situated amid one of the last portions of native forest in the area, and by side of a region safeguarded by a designated indigenous reserve (Urubu Branco). The significance of preserving this area is truly important by the fact that the adjacent zones of indigenous reserves fulfil their role in forest preservation through their traditional activities. However, the intermediary area, though part of the same ecosystem, remains vulnerable. Consequently, the carbon credit project assumes a pivotal role in conservation, employing the REDD+ methodology and associated activities, alongside a contractual agreement with the landowner (farmer proprietor). It is worth mentioning that even the indigenous reserves exhibit signs of deforestation, indicating that the threat persists due to various deforestation drivers in the region and the substantial pressure for land use change.

### Project Activities Instance Location

The project area is part of the ‘Tapirapé Farm’, which is established in the area since 2011. However, the deforested area is well established, and the area’s clearing for farming has probably started in the 70s period. The total area of the farm is 24,100 hectares, of which 13,926 hectares are native vegetation considered as the Project Area. The forested area crosses the municipalities of Santa Terezinha and Porto Alegre do Norte, both them in the same State – Mato Grosso (MT) in the eastern side. The area is officially registered in the CAR (Rural

Socioenvironmental Registry) and the official registry Brazilian entity, which can be consulted publicly in the national websites. Thus, the property has 1 CAR and 7 property registration documents (called 'matrículas' in Portuguese) that register all the exact coordinates, which will be presented further on the tables of this PD, and officialize the legal ownership in compliance with the environmental and land legislation in Brazil. To ensure this accuracy and fidelity of land ownership information, NovaAgri's Credit, Risk and Sustainability Department conducted a thorough internal audit. This meticulously scrutinized each point of the coordinates provided in the official registration documents, cross-referencing them with the visual data from the public shapefile outlining the property. As a result, the proponent can affirm that the delineations presented in the geospatial files align consistently with the coordinates documented in the official registry.

Furthermore, this audit served the additional purpose of confirming the absence of any overlap between the project area and public lands, conservation areas, territories of traditional communities, or any other areas conflicting with REDD+ methodology or VCS standards. The process strictly adheres to national and local legislation, ensuring compliance with all relevant regulations and guidelines.

Some of these documents are presented on the APPENDIX 2 of this PD, and the KML file is attached into the VERRA registry platform. The area does not overlap with the project area of any other VCS AFOLU projects.

**Table 4. PROPOERTY'S BOUNDARIES FROM REGISTRY OFFICIAL DOCUMENTS**

| Registry Document<br>(Matrícula) | Y Coordinate<br>(South)<br>UTM 22S | X Coordinate<br>(West)<br>UTM 22S |
|----------------------------------|------------------------------------|-----------------------------------|
| 2032                             | 494791,626                         | 8802474,644                       |
|                                  | 491022,044                         | 8802392,849                       |
|                                  | 491188,009                         | 8810517,168                       |
|                                  | 497201,630                         | 8810249,900                       |
|                                  | 497202,618                         | 8810162,570                       |
|                                  | 499126,776                         | 8810160,205                       |
|                                  | 499096,331                         | 8810134,917                       |
|                                  | 499079,448                         | 8810065,017                       |
|                                  | 499086,299                         | 8809984,810                       |
|                                  | 499140,827                         | 8809877,975                       |
|                                  | 499214,168                         | 8809588,833                       |
|                                  | 499229,451                         | 8809207,648                       |
|                                  | 499234,638                         | 8808934,452                       |
|                                  | 499172,098                         | 8808641,093                       |
|                                  | 498845,262                         | 8808487,824                       |

|  |            |             |
|--|------------|-------------|
|  | 498452,771 | 8808339,814 |
|  | 498176,486 | 8808296,584 |
|  | 497834,583 | 8808295,575 |
|  | 497782,207 | 8808271,743 |
|  | 497494,067 | 8808246,909 |
|  | 497183,126 | 8808246,146 |
|  | 497086,488 | 8808120,415 |
|  | 497052,438 | 8807868,187 |
|  | 497040,684 | 8807791,348 |
|  | 496976,330 | 8807818,184 |
|  | 496960,989 | 8807907,524 |
|  | 496906,575 | 8807963,779 |
|  | 496821,910 | 8807993,972 |
|  | 496792,988 | 8807981,588 |
|  | 496709,611 | 8807856,059 |
|  | 496671,247 | 8807634,068 |
|  | 496670,010 | 8807396,902 |
|  | 496552,055 | 8807044,181 |
|  | 496543,568 | 8806949,505 |
|  | 496313,787 | 8806725,580 |
|  | 496301,760 | 8806533,848 |
|  | 496515,853 | 8806491,071 |
|  | 496671,939 | 8806630,495 |
|  | 496799,892 | 8806720,484 |
|  | 497013,867 | 8806703,274 |
|  | 497055,144 | 8806564,643 |
|  | 497231,327 | 8806266,648 |
|  | 497101,586 | 8806164,782 |
|  | 497065,593 | 8806136,364 |
|  | 496984,202 | 8805919,073 |
|  | 496997,357 | 8805623,887 |
|  | 496832,822 | 8805703,493 |
|  | 496613,716 | 8806063,061 |
|  | 496537,016 | 8806057,892 |
|  | 496328,786 | 8805880,851 |
|  | 496203,888 | 8805640,252 |
|  | 496117,428 | 8805495,383 |
|  | 496526,548 | 8805635,412 |
|  | 496453,485 | 8805492,030 |
|  | 496369,896 | 8805361,880 |
|  | 496368,017 | 8805300,887 |
|  | 496469,727 | 8805126,271 |

|  |            |             |
|--|------------|-------------|
|  | 496846,427 | 8805007,674 |
|  | 496827,603 | 8804737,687 |
|  | 496818,263 | 8804420,742 |
|  | 496599,099 | 8804486,113 |
|  | 496086,313 | 8804424,676 |
|  | 495951,631 | 8804517,523 |
|  | 495843,704 | 8804398,682 |
|  | 495873,098 | 8804250,103 |
|  | 495972,557 | 8803776,006 |
|  | 495949,986 | 8803507,568 |
|  | 496099,666 | 8803260,313 |
|  | 496137,666 | 8803045,443 |
|  | 495999,540 | 8803062,989 |
|  | 495842,161 | 8803028,767 |
|  | 495624,530 | 8802912,386 |
|  | 495366,338 | 8802658,337 |
|  | 495210,974 | 8802616,324 |
|  | 495180,963 | 8802560,948 |
|  | 495225,239 | 8802438,851 |
|  | 495196,375 | 8802449,276 |

| Registry Document (Matrícula) | Y Coordinate (South) UTM 22S | X Coordinate (West) UTM 22S |
|-------------------------------|------------------------------|-----------------------------|
| 2036                          | 8802392,849                  | 491022,044                  |
|                               | 8802474,644                  | 494791,626                  |
|                               | 8802110,005                  | 494792,980                  |
|                               | 8802168,877                  | 494702,269                  |
|                               | 8802292,035                  | 494664,827                  |
|                               | 8802306,586                  | 494616,479                  |
|                               | 8802254,782                  | 494415,635                  |
|                               | 8802214,033                  | 494145,653                  |
|                               | 8802302,496                  | 494015,890                  |
|                               | 8802180,228                  | 493524,378                  |
|                               | 8802123,699                  | 493438,159                  |
|                               | 8801986,581                  | 493402,719                  |
|                               | 8801786,829                  | 493577,642                  |
|                               | 8801543,929                  | 493431,004                  |
|                               | 8801296,986                  | 493597,069                  |
|                               | 8800927,095                  | 493905,785                  |
|                               | 8800663,796                  | 493796,321                  |
|                               | 8800078,329                  | 493740,531                  |

|  |             |            |
|--|-------------|------------|
|  | 8800250,455 | 493636,804 |
|  | 8800276,982 | 493613,283 |
|  | 8800275,426 | 493536,636 |
|  | 8800067,910 | 493358,759 |
|  | 8799889,585 | 493176,972 |
|  | 8799630,061 | 492726,440 |
|  | 8799375,894 | 492530,190 |
|  | 8799295,450 | 492495,991 |
|  | 8798704,324 | 492448,505 |
|  | 8798591,026 | 492488,077 |
|  | 8798305,442 | 492318,126 |
|  | 8798144,730 | 492312,250 |
|  | 8798038,807 | 492508,744 |
|  | 8797734,829 | 492353,598 |
|  | 8797437,322 | 492267,167 |
|  | 8796989,825 | 492192,719 |
|  | 8796847,428 | 491789,169 |
|  | 8796831,675 | 491379,591 |
|  | 8796760,848 | 491137,544 |

| Registry Document (Matrícula) | Y Coordinate (South) UTM 22S | X Coordinate (West) UTM 22S |
|-------------------------------|------------------------------|-----------------------------|
| 5073                          | 8829930,424                  | 488343,824                  |
|                               | 8829983,121                  | 490661,776                  |
|                               | 8820946,464                  | 490870,504                  |
|                               | 8820923,822                  | 488550,209                  |
|                               | 8821925,791                  | 488521,669                  |
|                               | 8822924,970                  | 488491,114                  |
|                               | 8823924,306                  | 488458,839                  |
|                               | 8824923,725                  | 488423,704                  |
|                               | 8825922,287                  | 488399,481                  |
|                               | 8826921,836                  | 488384,023                  |
|                               | 8827915,385                  | 488370,253                  |
|                               | 8828931,296                  | 488356,377                  |
|                               | 8829216,668                  | 488352,367                  |

| Registry Document (Matrícula) | Y Coordinate (South) UTM 22S | X Coordinate (West) UTM 22S |
|-------------------------------|------------------------------|-----------------------------|
| 5074                          | 8830234,400                  | 497721,521                  |



|  |             |            |
|--|-------------|------------|
|  | 8821020,273 | 497796,107 |
|  | 8820946,464 | 490870,504 |
|  | 8829983,121 | 490661,776 |
|  | 8829930,424 | 488343,824 |

| Registry Document (Matrícula) | Y Coordinate (South) UTM 22S | X Coordinate (West) UTM 22S |
|-------------------------------|------------------------------|-----------------------------|
| 5075                          | 8821020,273                  | 497796,107                  |
|                               | 8819983,327                  | 497804,503                  |
|                               | 8815937,549                  | 497837,259                  |
|                               | 8815907,199                  | 497837,288                  |
|                               | 8815756,230                  | 498038,575                  |
|                               | 8815530,292                  | 498108,496                  |
|                               | 8815233,190                  | 497931,477                  |
|                               | 8814975,736                  | 497916,933                  |
|                               | 8814720,877                  | 497929,840                  |
|                               | 8814467,778                  | 498049,121                  |
|                               | 8814313,163                  | 498216,794                  |
|                               | 8814055,642                  | 498299,026                  |
|                               | 8813807,965                  | 498247,913                  |
|                               | 8813481,357                  | 498134,079                  |
|                               | 8813274,371                  | 498330,013                  |
|                               | 8813061,488                  | 498507,354                  |
|                               | 8812942,701                  | 498630,992                  |
|                               | 8812711,604                  | 498731,253                  |
|                               | 8812466,672                  | 498997,22                   |
|                               | 8812437,463                  | 498732,002                  |
|                               | 8812235,704                  | 498755,632                  |
|                               | 8812042,119                  | 498745,922                  |
|                               | 8811805,908                  | 498760,006                  |
|                               | 8811600,288                  | 491210,135                  |
|                               | 8820946,464                  | 490870,504                  |

| Registry Document (Matrícula) | Y Coordinate (South) UTM 22S | X Coordinate (West) UTM 22S |
|-------------------------------|------------------------------|-----------------------------|
| 5076                          | 8811600,288                  | 491210,135                  |
|                               | 8811805,908                  | 498760,005                  |
|                               | 8811742,516                  | 498763,785                  |

|  |             |            |
|--|-------------|------------|
|  | 8811651,625 | 498629,175 |
|  | 8811548,097 | 498478,114 |
|  | 8811564,713 | 498635,033 |
|  | 8811380,035 | 498760,359 |
|  | 8811153,044 | 498868,585 |
|  | 8811027,054 | 498969,177 |
|  | 8810838,394 | 499126,655 |
|  | 8810541,379 | 499209,461 |
|  | 8810353,857 | 499224,210 |
|  | 8810185,539 | 499240,516 |
|  | 8810160,205 | 499221,550 |
|  | 8810162,570 | 499168,290 |
|  | 8810249,900 | 497201,630 |
|  | 8810517,168 | 491188,009 |

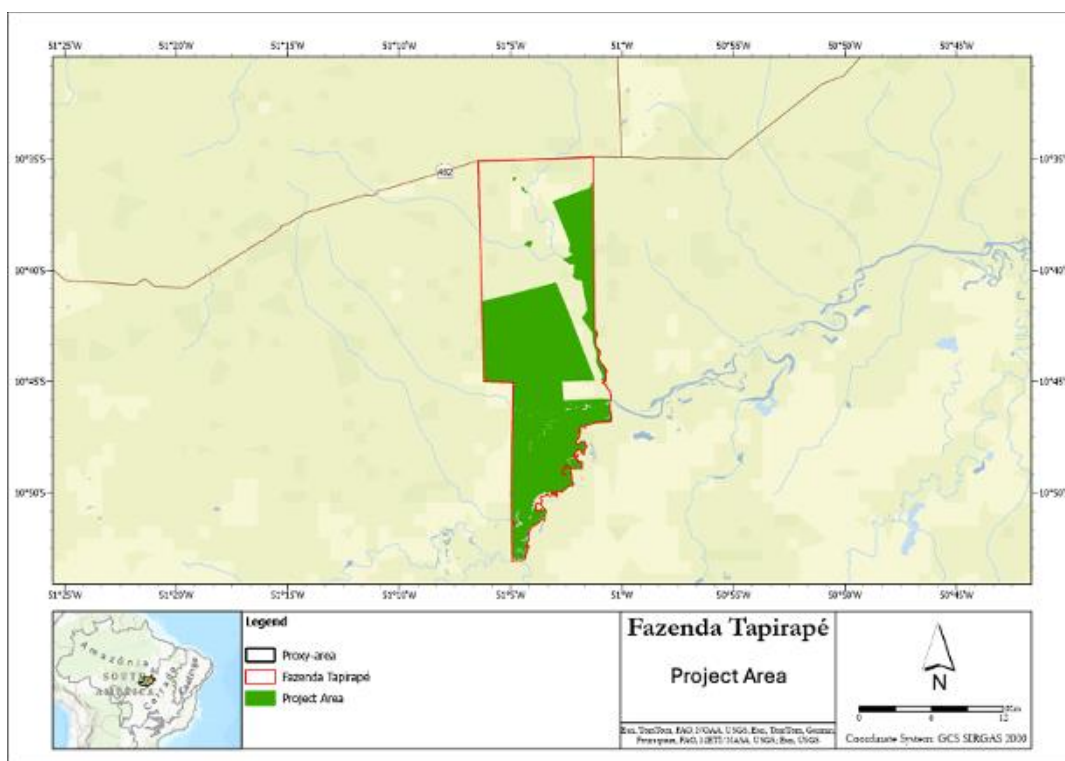
| Registry Document (Matrícula) | Y Coordinate (South) UTM 22S | X Coordinate (West) UTM 22S |
|-------------------------------|------------------------------|-----------------------------|
|                               | 8820923,822                  | 488550,209                  |
|                               | 8820946,464                  | 490870,504                  |
|                               | 8811600,288                  | 491210,135                  |
|                               | 8811657,759                  | 488784,908                  |
|                               | 8811993,681                  | 488777,874                  |
|                               | 8812938,034                  | 488756,412                  |
|                               | 8813939,516                  | 488733,203                  |
|                               | 8814968,644                  | 488708,000                  |
|                               | 8815919,633                  | 488683,027                  |
|                               | 8816912,902                  | 488657,131                  |
|                               | 8817916,143                  | 488628,411                  |
|                               | 8818910,207                  | 488601,939                  |
|                               | 8819936,654                  | 488577,033                  |
|                               | 8820923,822                  | 488550,209                  |

## Project Area

In accordance with VCS requirements, stipulated in Approved VCS Methodology VM0015, version 1.2, the project area may only include areas composed of “forest”<sup>7</sup> for a minimum of ten years prior to the project start date. Satellite images from 2013 and 2023 were analyzed and classified.

The remaining forest areas within ‘Tapirapé Farm’ that were defined as forest in 2023 and in 2013 were identified and utilized to compose the project area. In addition, some non-forest areas were also excluded, such as rivers, rocks, and non-forest vegetation.

As shown in the Figure below, the size of the areas that were considered “non-forest” (i.e. deforested areas, non-forest vegetation, hydrography or rock formations) and other uses within the total property area at the project start date was 24.100 ha. 10.174 ha were excluded from the initial area, resulting in 13.926 ha which was then defined as project area. The full contour coordinates of the project area are found previously on the Table 4.



*Image Source: the author – Project Area*

### Definition of the property boundaries

<sup>7</sup> Brazilian Forestry Service. Brazil adopts FAO forest definition: “Forest is an area of at least 0.05-1.0 ha with canopy cover (or equivalent density) of more than 10-30%, with trees with the potential to reach the minimum height of 2-5 meters at maturity *in situ*.” Available at: < <https://snif.florestal.gov.br/pt-br/conhecendo-sobre-florestas/167-definicao-de-floresta>>.

The project area borders used in the ARAGUAIA REDD+ BOCALON PROJECT were extracted from technical appraisals (Portuguese: laudos) registered at an official notary and at INCRA<sup>8</sup>. Vectorization – which is the process of converting the appraisal documents into digital shapefiles and polygons, being formats compatible with GIS.

The technical appraisals were double checked against the descriptive memorandum of the property, which shows the property limits through geographic coordinates. In addition, the hydrographic limits were also utilized because the rivers delineate the property in some locations.

#### *Historical reference period*

The historical reference period is the period in which analysis of LULUCF (land use, land use change and forest) within the reference region and project area is carried out. In accordance with the methodology, this period must not exceed 10 – 15 years in span and end as close as possible to the REDD project start date ( $\leq 2$  years). The project start date is JULY-2023 and the historical reference period comprised analysis of images from 2013 to 2023 in the project area and from 2008 to 2023 in the reference area. Sufficient satellite images referring to this period were available, with low presence of cloud cover.

#### *Project area time-series*

For the image classification of the project area, satellite images from Landsat 7 (Enhanced Thematic Mapper (ETM) sensor), and images from Landsat 8 (Operational Land Imager (OLI) sensor) were found for the 2008-2023 period (Table 5 below). One image was necessary to cover the whole Project Area, being the following orbit-point respectively: 224/67. A total of 16 satellite images were necessary to compose the Project Area during the historical reference period. These images were downloaded from U.S. Geological Survey (USGS)<sup>9</sup>.

**Table 5. Satellite images time-series used for PA classification**

| Satellite | Sensor | Resolution     |               | Coverage<br>(km <sup>2</sup> ) | Acquisition<br>date<br>DD/MM/YY | Scene             |                   |
|-----------|--------|----------------|---------------|--------------------------------|---------------------------------|-------------------|-------------------|
|           |        | Spatial<br>(m) | Spectral (μm) |                                |                                 | Path/<br>Latitude | Row/<br>Longitude |
| Landsat5  | TM     | 30             | 0.45 - 12.5   | 34,225                         | 17/07/2001                      | 224               | 67                |
| Landsat7  | ETM    | 30             | 0.45 - 12.5   | 34,225                         | 16/10/2002                      | 224               | 67                |
| Landsat7  | ETM    | 30             | 0.45 - 12.5   | 34,225                         | 07/07/2003                      | 224               | 67                |
| Landsat7  | ETM    | 30             | 0.45 - 12.5   | 34,225                         | 23/06/2004                      | 224               | 67                |

<sup>8</sup> Instituto Nacional de Colonização e Reforma Agrária (INCRA): <http://www.incra.gov.br/>

<sup>9</sup> U.S. Geological Survey. Available at: <<http://www.usgs.gov/>>

|           |     |    |             |        |            |     |    |
|-----------|-----|----|-------------|--------|------------|-----|----|
| Landsat7  | ETM | 30 | 0.45 - 12.5 | 34,225 | 28/07/2005 | 224 | 67 |
| Landsat7  | ETM | 30 | 0.45 - 12.5 | 34,225 | 31/07/2006 | 224 | 67 |
| Landsat7  | ETM | 30 | 0.45 - 12.5 | 34,225 | 10/07/2006 | 224 | 67 |
| Landsat5  | TM  | 30 | 0.45 - 12.5 | 34,225 | 16/06/2007 | 224 | 67 |
| Landsat5  | TM  | 30 | 0.45 - 12.5 | 34,225 | 04/07/2008 | 224 | 67 |
| Landsat5  | TM  | 30 | 0.45 - 12.5 | 34,225 | 07/07/2009 | 224 | 67 |
| Landsat7  | ETM | 30 | 0.45 - 12.5 | 34,225 | 26/07/2010 | 224 | 67 |
| Landsat5  | TM  | 30 | 0.45 - 12.5 | 34,225 | 27/06/2011 | 224 | 67 |
| Landsat7  | ETM | 30 | 0.45 - 12.5 | 34,225 | 07/07/2012 | 224 | 67 |
| Landsat 8 | OLI | 30 | 0.64 - 1.65 | 34,225 | 02/07/2013 | 224 | 67 |
| Landsat 8 | OLI | 30 | 0.64 - 1.65 | 34,225 | 05/07/2014 | 224 | 67 |
| Landsat 8 | OLI | 30 | 0.64 - 1.65 | 34,225 | 24/07/2015 | 224 | 67 |
| Landsat 8 | OLI | 30 | 0.64 - 1.65 | 34,225 | 11/08/2016 | 224 | 67 |
| Landsat 8 | OLI | 30 | 0.64 - 1.65 | 34,225 | 13/07/2017 | 224 | 67 |
| Landsat 8 | OLI | 30 | 0.64 - 1.65 | 34,225 | 16/07/2018 | 224 | 67 |
| Landsat 8 | OLI | 30 | 0.64 - 1.65 | 34,225 | 04/08/2019 | 224 | 67 |
| Landsat 8 | OLI | 30 | 0.64 - 1.65 | 34,225 | 06/08/2020 | 224 | 67 |
| Landsat 8 | OLI | 30 | 0.64 - 1.65 | 34,225 | 24/07/2021 | 224 | 67 |
| Landsat 8 | OLI | 30 | 0.64 - 1.65 | 34,225 | 27/07/2022 | 224 | 67 |

All the images were then processed through a band composition method (bands 1 – 5), since this composition enabled the mapping analysis in both real and false colors. The following bands and wavelengths were utilized:

- Band 1 – Visible (0.45 – 0.52  $\mu\text{m}$ ) = blue
- Band 2 – Visible (0.52 – 0.60  $\mu\text{m}$ ) = green
- Band 3 – Visible (0.63 – 0.69  $\mu\text{m}$ ) = red
- Band 4 – Near infrared (0.76 – 0.90  $\mu\text{m}$ )
- Band 5 – Mid infrared (1.55 – 1.75  $\mu\text{m}$ )

The first step of the visual image classification of land-use in the project area was done on ArcGIS Pro<sup>10</sup> software, using images from the Landsat 7 and Landsat 8 satellites, and in accordance with their minimal 30m resolution – the minimum mapping unit (MMU) was defined at 30x30m (0.09ha), therefore easily meeting the methodology requirement that the MMU cannot be larger than 1ha.

The supervised classification utilized the following three land use classes:

<sup>10</sup> Environmental Systems Research Institute, Inc. (Esri) ArcGIS Pro: <https://www.esri.com/en-us/arcgis/products/arcgis-pro/overview>

1- Forest;

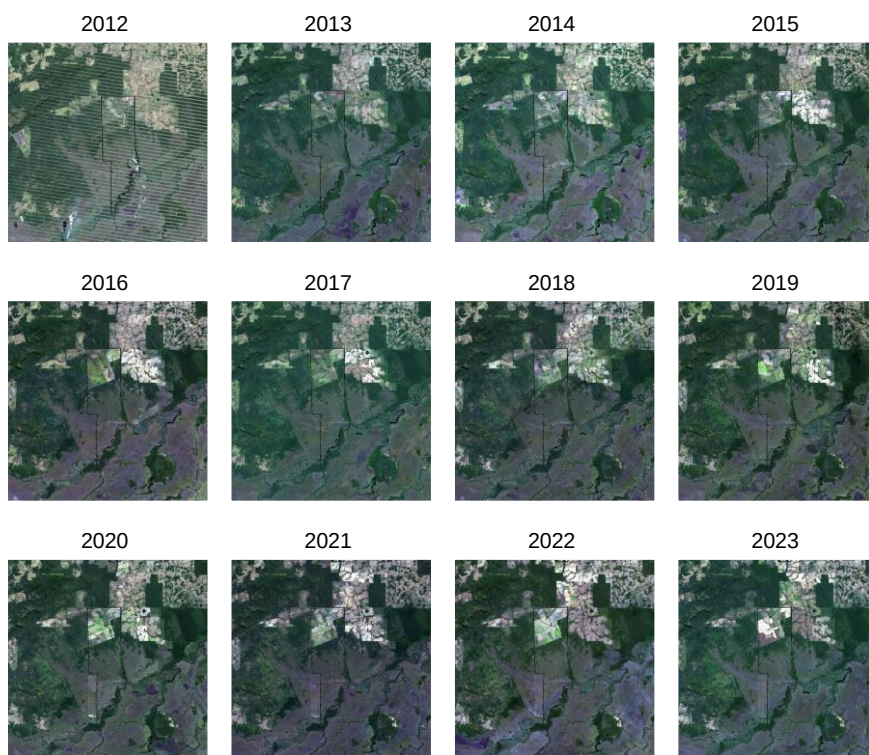
2- Non-Forest (exposed soil, agricultural lands, pasturelands, etc);

3- Hydrography.

The year 2013 was then utilized to exclude forests that are less than 10 years old at the project start date from the project area. Finally, the “hydrography” class was defined as constants throughout the historical reference period, and the same class was applied for each mapped year, as those values did not significantly vary across the analyzed period.

For the present project, the process of accumulating ‘non-Forest’ areas was adopted, in such a way that areas classified as ‘non-Forest’ in one year were necessarily included in the same category in the following year.

The project area contains only areas which were defined as “forest” in 2012 (more than 10 years prior to the project start date), as described in the forest cover benchmark maps in the figure below.



*Image Source: the author – Remaining Forest in the Project Area*

## The Reference Region

### *Reference region time-series*

For the image classification of the reference region, data from PRODES<sup>11</sup> (Project for Monitoring Deforestation in the Brazilian Amazon Forest by Satellite) Project were used.

PRODES uses satellite images of the LANDSAT class (20 to 30 meters of spatial resolution and 16-day revisit rate) in a combination that seeks to minimize the cloud cover problem and ensure interoperability criteria. The images of the American satellite LANDSAT-5/TM were historically the most used by the project, but the images of the CCD sensor on board CBERS-2/2B, satellites of the Sino-Brazilian remote sensing program, were widely used. It currently makes massive use of images from LANDSAT 8/OLI, CBERS 4 and IRS-2. Regardless of the instrument used, the minimum area mapped by PRODES is 6.25 hectares.

PRODES estimates are considered reliable by national and international scientists (KINTSCH, 2007<sup>12</sup>). This system has been shown to be of great importance for actions and planning of public policies in the Amazon. Recent results, from analyses conducted with independent experts, indicate a level of accuracy close to 95%.

The annual deforestation rate PRODES has been used as an indicator for the proposition of public policies and for the evaluation of the effectiveness of its implementations. PRODES spatial data are used in: (a) Certification of agribusiness production chains such as the Soy Moratorium and the Animal Husbandry Conduct Adjustment Term-TAC; (b) Intergovernmental agreements such as the United Nations Conference on Climate Change (COP 21) and the National Inventory Reports on Greenhouse Gas Emissions and (c) Monetary donations by the Amazon Fund, which use PRODES as a reference to deforestation activity in the Legal Amazon.

The data transparency policy of forest state monitoring adopted by INPE and the federal government since 2004 allows full access to all data generated by monitoring systems, independent evaluations by the user community, including the government in its spheres and instances, the academy, the citizen and the Brazilian civil society in its various institutional arrangements.

The Reference Region (RR) is an analytical domain through which information on rates, agents, drivers and underlying causes of land-use and land-cover (LU/LC) change are obtained, and subsequently used for future projection and monitoring.

The RR was defined in accordance with the following criteria:

- Conditions that determine the likelihood of deforestation within the project area are similar or expected to become similar to those found in the reference region, depending on

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<sup>11</sup> Instituto Nacional de Pesquisas Espaciais (INPE) / Coordenação Geral de Observação da Terra (OBT)  
<<http://www.obt.inpe.br/OBT/assuntos/programas/amazonia/prodes>>

<sup>12</sup> Kintisch, Eli. "Improved Monitoring of Rainforests Helps Pierce Haze of Deforestation." *Science* 316 (2007): 536 - 537.



landscape configuration and ecological conditions (elevation, slope, vegetation and precipitation) socio-economic and cultural conditions, agents and drivers of deforestation (groups of agents, infrastructure or other factors). The last condition was the most important to adjust the RR to represent the dynamics of land use more accurately.

Taking this into account, to define the boundaries of the Reference Region, a socioeconomic analysis of the region and the existence of government programs for forest preservation or even REDD projects was performed since these municipalities are within the limits of the "Amazon Arc of Deforestation" and agricultural expansion.

As detailed above, the project area is located at the northeastern region of Mato Grosso State, in the municipalities of Santa Terezinha and Porto Alegre do Norte that are in a region with large livestock expansion and history of increased deforestation rate.

According to the report Planning Regions of Mato Grosso<sup>13</sup> published by the Department of Planning of the Mato Grosso State (SEPLAG), deforestation at this region was 31.066 thousand km<sup>2</sup> until the year 2015, area that represents 8.52% of the space geographical state.

In terms of its economic performance, represented through Gross Domestic Product (GDP) and Gross Value Added (GVA) by sectors of economic activity of its municipalities, the region's GDP totaled R\$ 2.1 billion in 2014, representing 2.1% of Mato Grosso's GDP according to the report from SEPLAG. The cumulative nominal GDP growth of the region in the period from 2010 to 2014 registered (116.4%), or the equivalent of an Average Annual Growth Rate (TMCA) of 21.3%, 40% above the state average for the same period.

The analysis of the Gross Added Value (GVA) of 2014 indicated that the activity with the largest participation of the region is Agriculture with 37.7% of the total, where in the region no municipality has in Industry its main activity.

According to the Municipal Human Development Index (MHDI) calculated according to methodology adapted from the Global HDI by the United Nations Development Program in Brazil (UNDP Brazil) the report from SEPLAG showed a significant improvement in this indicator as in 2000, nine municipalities were classified as Very Low and four others as Low, but in 2010, all municipalities in the northeastern region obtained MHDI classified as Medium.

However, even considering that the average MHDI of the Region advanced from 0.490 in 2000 to 0.667 in 2010, when compared to the IDHM of Mato Grosso and Brazil, it was found that all municipalities in the northeastern region had lower indicators than the state (0.725) and the country (0.727), and the lowest IDHM in the region was presented by Santa Terezinha (0.609).

In terms of its demographic scenario, according to the report from SEPLAG, the northeastern region had a population of 113,442 inhabitants in 2015, with approximately 60% of the population in the urban area and 40% in the rural area. The population growth of the region in

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<sup>13</sup> Regiões de Planejamento de Mato Grosso: 2017 / Secretaria de Estado de Planejamento (SEPLAG) – Cuiabá, MT, 2017. Available at < <http://www.seplag.mt.gov.br/images/files/00seplan-5653-62d1bfa9829cd.pdf> >

the period 2010/2015 was 1.88% and reached a population density of 1.5 inhab./km<sup>2</sup>, representing less than 5% of the total population of the State of Mato Grosso.

The same report from SEPLAG also describes that the region has 3 conservation units, totaling 3,259 km<sup>2</sup>, which represents 8.3% of the conservation areas of Mato Grosso and 9 indigenous territories, which together reach 4,438 km<sup>2</sup>, representing 3.0% of the state's indigenous areas.

In terms of its geomorphological characterization, according to a Map of the Macro-regions published by the Mato-Grossense Institute of Agricultural Economics - IMEA<sup>14</sup>, the northeastern region of Mato Grosso State is part of the Araguaia Basin. While the southern part of the northeast region has savanna with characteristics suitable for the cultivation of perennial crops, the eastern part of the northeast region is formed by savannas susceptible to flooding and suitable for cattle production and the remaining area of the northeast region is formed mostly by forests of the Amazon biome, where livestock is the main economic activity.

The Map of the Macro-regions from IMEA also describes that to the west, the Xingu Indigenous Park prevents communication with the northern and middle-north regions, as there are no roads, while the southern part of the northeast region has good roads and has good traffic and trade with the southeast part of the northeast region. Although the distance limits traffic and trade between the two main economic poles of the northeast region, the livestock production system, which is the main economic activity of the region, is very similar in all its extension.

Thus, based on the project area definition, a Proxy-area was delimited encompassing the Reference Region the leakage belt. The criterion of the minimum area was adopted as mentioned in the applied methodology. A concentric circle was established around the project area with a radius of 100km, which resulted in a size 200 times greater than the project area. As the property area totals 24,100 ha, the reference area would need to have a limit of up to 4,817,728 ha. After this first delimitation, adjustments were made based on:

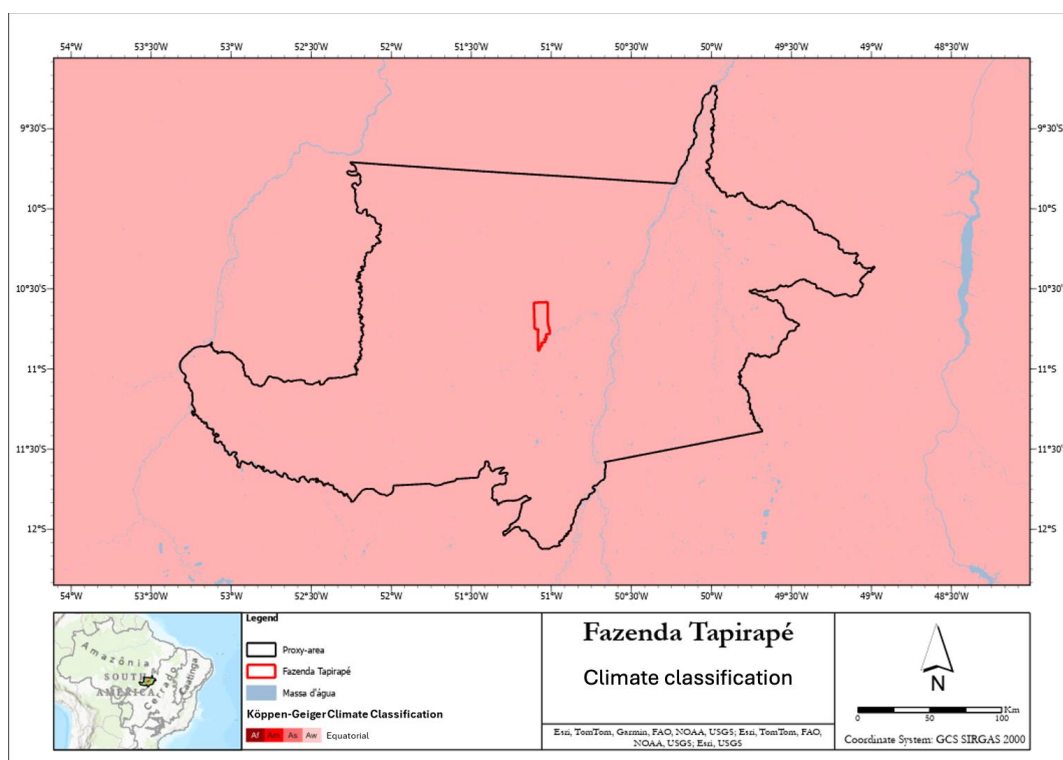
- Agents and drivers of deforestation expected to cause deforestation within the project area in absence of the proposed REDD project activity (i.e., cattle ranching and agriculture) caused deforestation in the reference region during the historical reference period;
- History of deforestation and economic growth of the region, expansion of the agricultural frontier, limits of the arc of deforestation, land conflicts, land structure in the region (presence of conservation units, indigenous lands and agrarian reform settlements);
- Existing infrastructure with the presence of navigable rivers, roads and highways and their respective production outflow routes;
- Socioeconomic characteristics of the municipalities involved, such as Gross Domestic Product (GDP), concentration of population in urban or rural areas and main economic products;

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<sup>14</sup> Instituto Mato-Grossense de Economia Agropecuária (IMEA) / Mapa das macrorregiões, 2017. Available at <<https://www.imea.com.br/imea-site/view/uploads/metodologia/justificativamapa.pdf>>

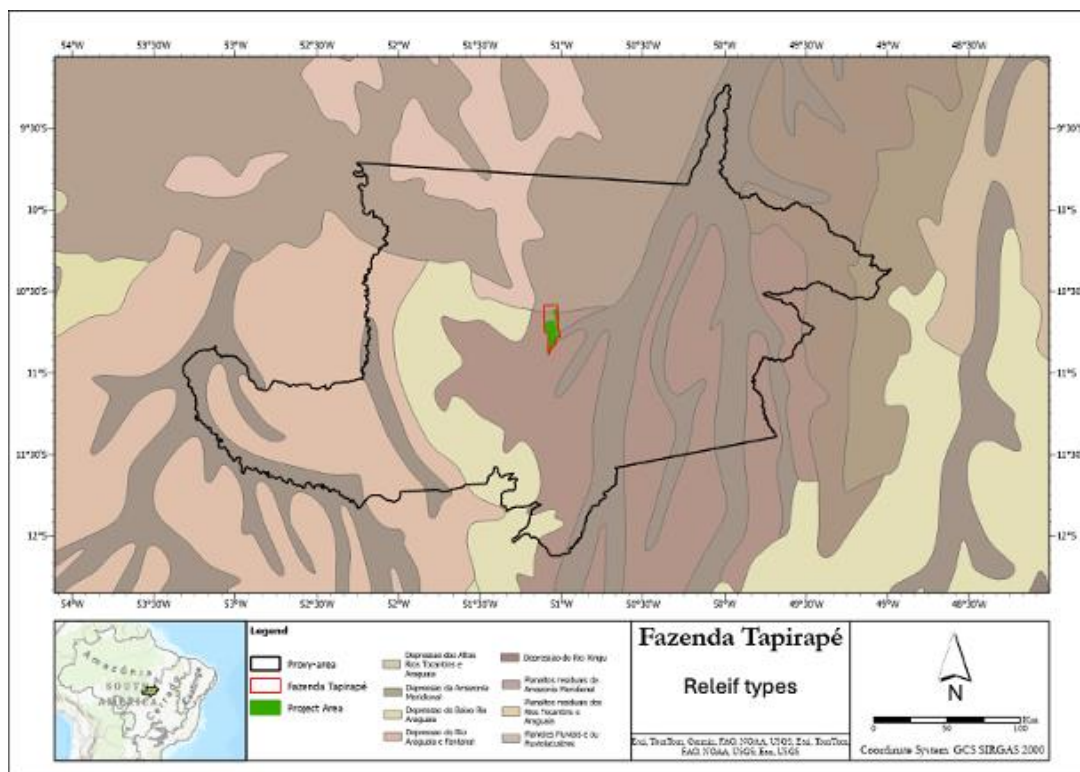
- The legal status of the land and the land tenure system found within the project area, i.e. private land, is also found in the reference region;
- Current and projected land use within the project area is found in the reference region;
- The project area is governed by the same policies, legislation and regulations that apply in the reference region;
- Elevation, Slope, Climate and Forest Classes characteristics in the Project Region: at least 90% of the project area have landscape configurations and ecological conditions that exist in at least 90% of the rest of the reference region.

Maps below show some of the criteria that were taken into account in order to adjust the ARAGUAIA REDD+ BOCALON PROJECT's Reference Region.



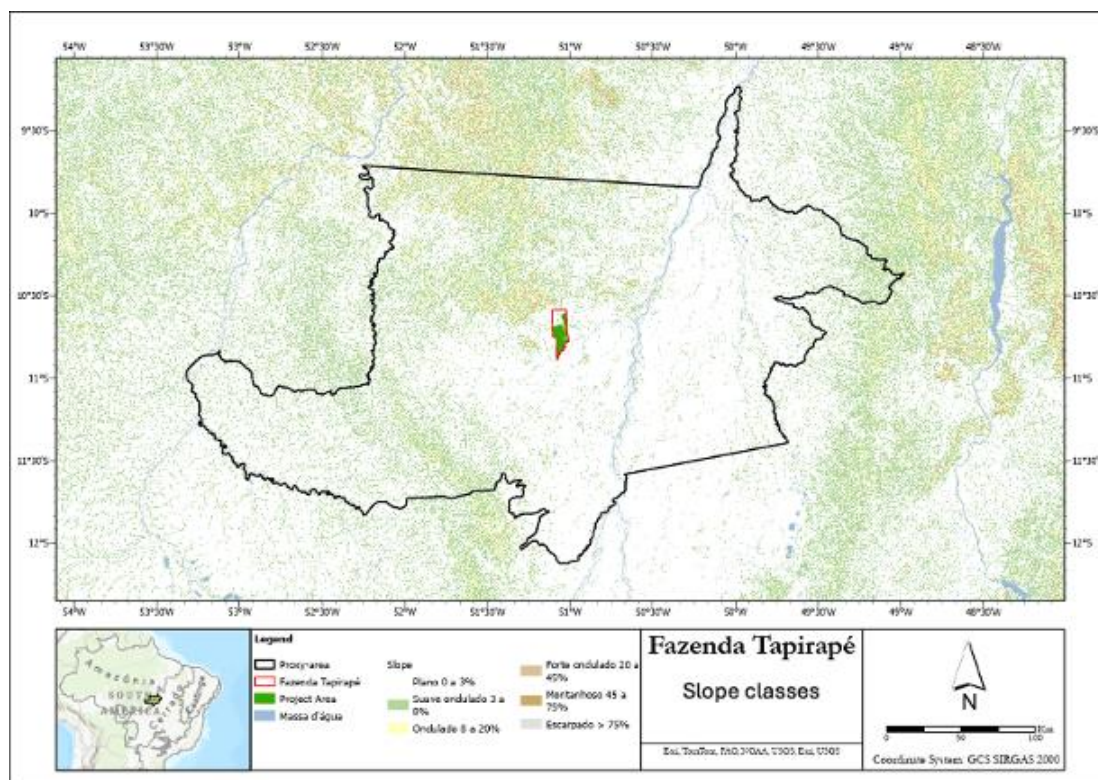
Map Source: the author – Climate conditions in the project region according to the Köppen-Geiger climate classification<sup>15</sup>, detail to the Project Area location, showing similar climate conditions in the project region.

<sup>15</sup> The Köppen-Geiger climate classification was developed based on the empirical relationship between climate and vegetation. The Köppen-Geiger system classifies climate into five main classes and 30 sub-types. The classification is based on threshold values and seasonality of monthly air temperature and precipitation. Considering vegetation as “crystallized, visible climate”, this classification aims to empirically map biome distributions around the world: different regions in a similar class share common vegetation characteristics. Source: <https://koeppen-geiger.vu-wien.ac.at/present.htm>



Map Source: the author – Relief types existing in the project region, showing similar relief conditions in the project region<sup>16</sup>

<sup>16</sup> The map of Relief Units of Brazil produced by the Brazilian Institute of Geography and Statistics - IBGE, on the scale of 1:5,000,000, aims to offer a synoptic view of the Brazilian relief, according to a systematic classification of geomorphological phenomena, adopting a taxonomy in which the identified units are part of a hierarchical organization based on temporal-spatial orders of magnitude, which include the Morpho-structural Domains, Geomorphological Regions, Relief Units and Modeled. Source: <https://www.ibge.gov.br/geociencias/informacoes-ambientais/geomorfologia/15827-unidades-de-relevo.html>

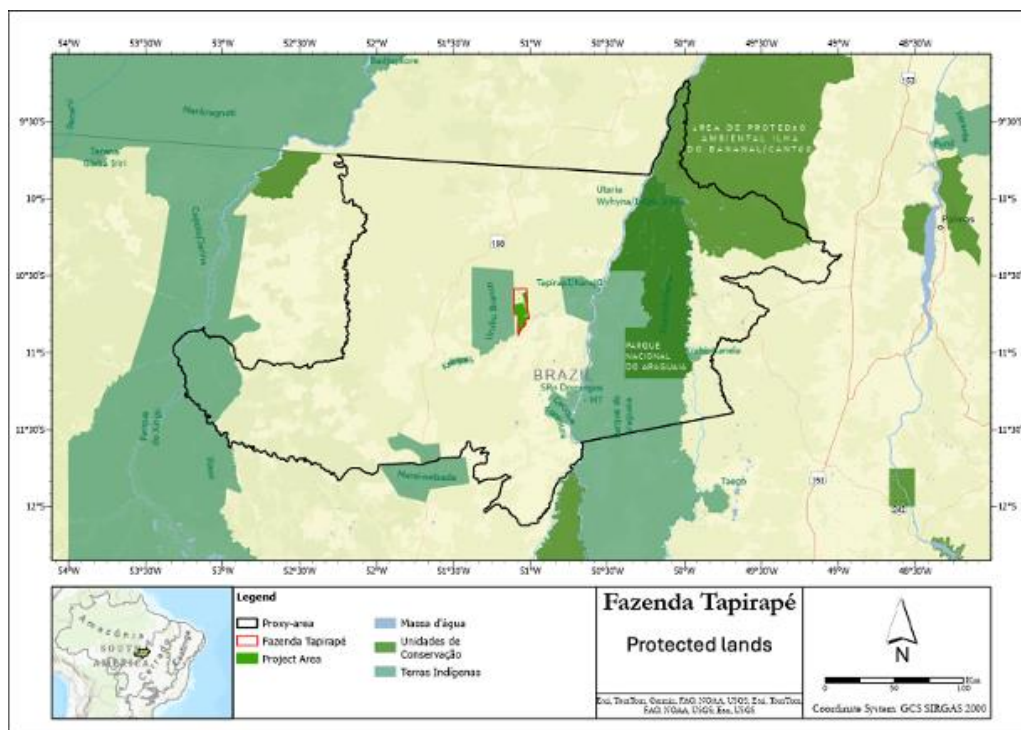


Map Source: the author – Slope map of the project region, showing similar slope classes in the project region<sup>17</sup>

<sup>17</sup> The slope map represents the slope of the terrain surface relative to the horizontal, that is, the relationship between the height difference between two points and the horizontal distance between these points. It is given by the angle of inclination (zenithal) of the surface of the terrain in relation to the horizontal. Slope values can range from 0 to 90 and can also be expressed as a percentage. Source: [https://www.dpi.inpe.br/Ambdata/declividade\\_gradiente.php](https://www.dpi.inpe.br/Ambdata/declividade_gradiente.php)







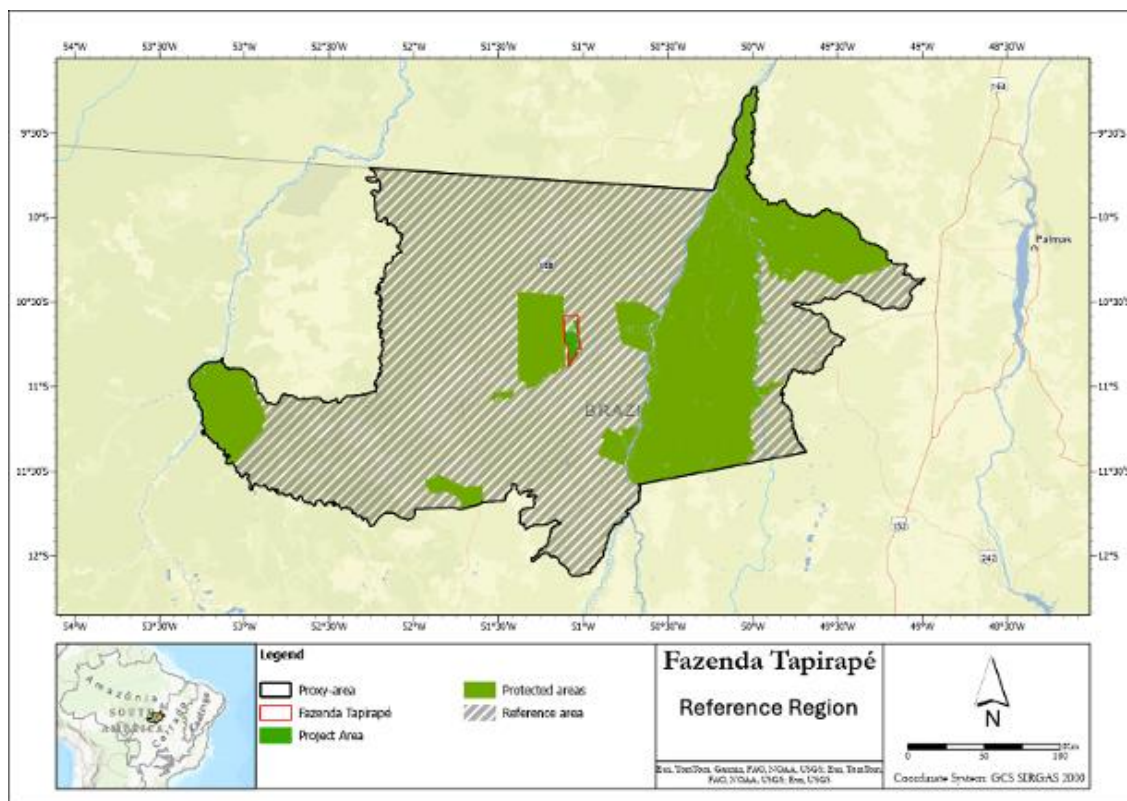
Map Source: the author – Public lands, such as Settlement Projects, Protected Areas and Indigenous Territories within the project region<sup>19</sup>

<sup>19</sup> The Conservation Unit map represents a territorial space and its environmental resources, including jurisdictional waters, with relevant natural characteristics, legally established by the government, with conservation objectives and defined limits, under special administration regime, to which appropriate safeguards of protection apply. Conservation units are protected areas by federal, state, district or municipal law, in order to conserve biodiversity and other natural attributes contained therein, with minimal impact. Source: <http://www.mma.gov.br/areas-protegidas.html>

The map of the Indigenous Lands represents the land traditionally occupied by Indians, defined as: those inhabited by them on a permanent basis, those used for their productive activities, those essential to the preservation of the environmental resources necessary for their wellbeing and necessary for their physical and cultural reproduction, being their uses, customs and traditions. Source: <https://www.gov.br/funai/pt-br>

The map of Community Use Areas may correspond to quilombola territories, extractive reserves, rural settlements or other local units. The category of Public Lands covers land acquired, expropriated, collected, recognized, donated, awarded, discriminated, incorporated, transferred, transferred, confiscated, reverted to domain, or in the process of obtaining, in which are installed settlement projects, in its most diverse typologies. Source: <https://www.gov.br/incra/pt-br>





Map Source: the author – Reference Region of ARAGUAIA REDD+ BOCALON PROJECT

The Reference Region sums to 4,817,728 ha, i.e. 200 times the Project Area, and is distributed over the area under administration of 9 municipalities: Lagoa da Confusão and Pium in the State of Tocantins, Canabrava do Norte, Confresa, Luciara, Porto Alegre do Norte, Santa Terezinha, São Félix do Araguaia e Vila Rica in the State of Mato Grosso.

The conservation areas and indigenous territories were extracted from the Proxy-area so that the Reference Area is standardized in relation to the legislation associated with land tenure.

### Leakage Belt

In accordance with the VM0015, the Leakage Belt (LK) was defined by means of opportunity cost analysis. The latter is applicable when at least 80% of deforested area in the reference region during the historical reference period occurred where deforestation was profitable for at least one product.

The analysis performed on the land-use/land-cover changes during the historical reference period allowed determining that cattle ranching and crop production are the main deforestation and degradation agents in the RR and the existing roads are the main drivers of deforestation.

At the same time, scientific studies in the Legal Amazon indicates that  $\frac{2}{3}$  of deforestation activities occurred in a 5,5 km range from main roads<sup>20,21,22,23,24,25</sup>

The Leakage Belt was defined considering this information through an analysis of each variable in a circular buffer of 25km and 50km from the center of the property. This approach was chosen considering the opportunity cost analysis performed, indicating that cattle ranching and crop production are very feasible activities on such regions.

The opportunity cost analysis was performed by Geoconsult (a service provider that was responsible for the following activities on the Project):

- Assessing the land-use and land-cover change during the historical reference period.
- Determining the agents and drivers of deforestation.
- Projecting the future deforestation.
- Defining the land-use and land-cover change component of the baseline.

Geoconsult, together with NOVAAGRI has determined the opportunity cost for cattle ranching and crop production based on data on the potential profitability associated to each activity. Profitability was determined using the Equation below:

$$PPx_i = S\$x - PCx_i - \sum_{v=1}^v (TDv * TCv)$$

Where:

|       |                                                                                   |
|-------|-----------------------------------------------------------------------------------|
| PPxi  | Potential profitability of product Px at location I (pixel or polygon);           |
| S\$xi | Selling price of product Px; \$/t                                                 |
| PCxi  | Average in situ production costs for one ton of product Px in stratum i;          |
| TCv   | Average transport cost per kilometer for one ton of product Px on road of type v; |

<sup>20</sup> Armenteras, D., Barreto, J. S., Tabor, K., Molowny-Horas, R., and Retana, J.: Changing patterns of fire occurrence in proximity to forest edges, roads and rivers between NW Amazonian countries, Biogeosciences, 14, 2755–2765.

<sup>21</sup> Roy, David P. and Kumar, S. S., "Multi-year MODIS Active Fire Type Classification Over the Brazilian Tropical Moist Forest Biome" (2017). GSCE Faculty Publications. 27.

<sup>22</sup> Kumar Sanath S., Roy David P., Cochrane Mark A., Souza Carlos M., Barber Chirstopher P., Boschetti L. (2014) A quantitative study of the proximity of satellite detected active fires to roads and rivers in the Brazilian tropical moist forest biome. International Journal of Wildland Fire 23, 532-543.

<sup>23</sup> Cano Crespo, Ana & Traxl, Dominik & Thonicke, Kirsten. (2021). Spatio-temporal patterns of extreme fires in Amazonian forests. The European Physical Journal Special Topics. 230.

<sup>24</sup> Dalton Cardoso et al. A relação entre áreas de exploração madeireira e estradas na Amazônia legal. In: Anais do XX Simpósio Brasileiro de Sensoriamento Remoto, 2023, Florianópolis. Anais eletrônicos. São José dos Campos, INPE, 2023.

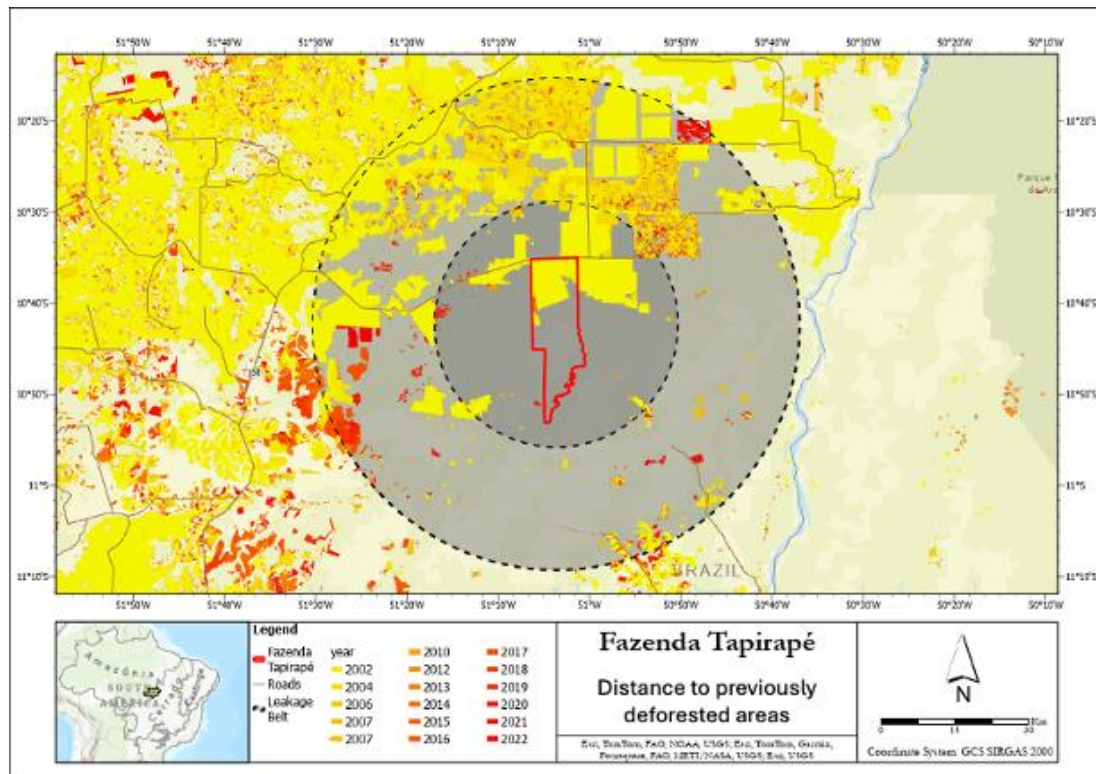
<sup>25</sup> Sathyachandran, S. K.. "Fire Type Classification in the Brazilian Tropical Moist Forest Biome." (2014).

**TDv** Transport distance on land, river or road of type v.

The formula above is used to calculate the potential profitability of a given product in a given location and the borders of the leakage belt correspond to the area where the profitability of at least one product is equal or above 1.

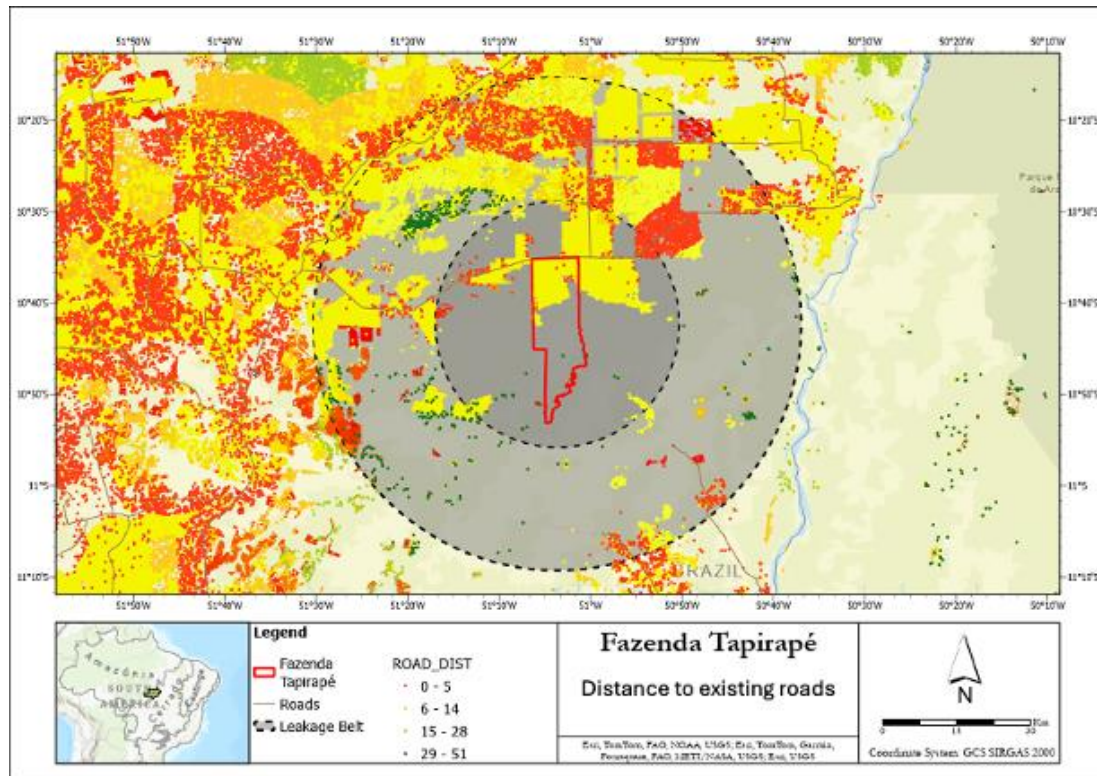
Therefore, to determine the potential profitability, the above parameters were identified considering current conditions and potential markets for each activity and product. This assessment allowed concluding both activities are potentially profitable, especially to the north and northeastern of the Project Area, along the roads MT-413 e MT-432.

The combinations of the maps for each variable helped produce the following maps that led to the determination of the Leakage Belt.

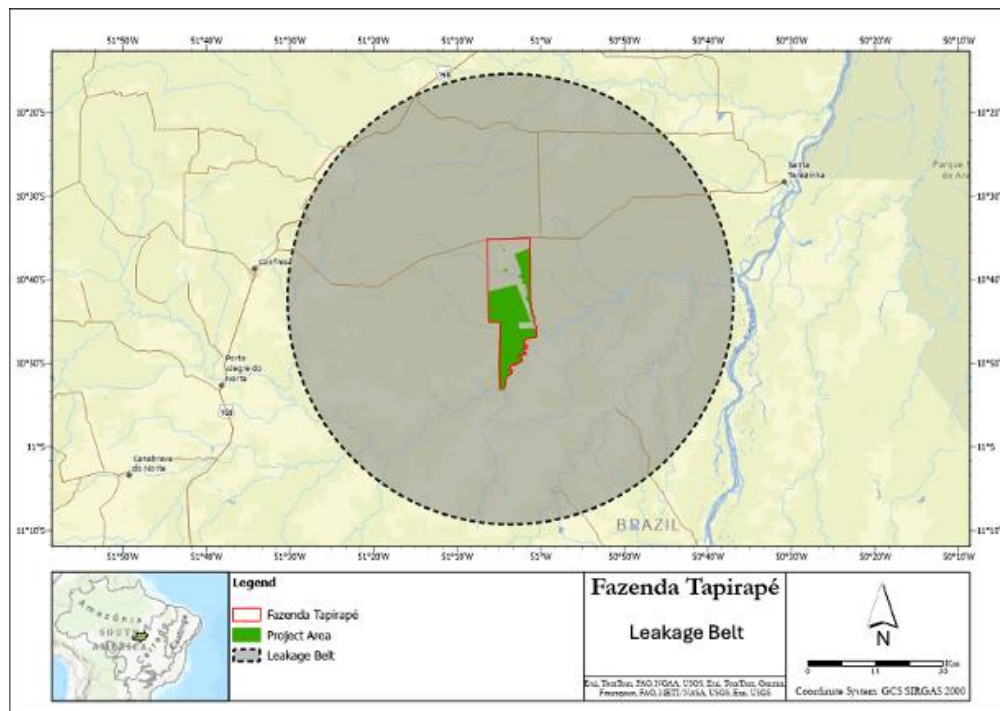


Map Source: the author – Attractiveness based on distance to previously deforested areas.





Map Source: the author – Attractiveness based on distance to existing roads.



Map Source: the author – Leakage belt as defined by attractiveness from deforestation drivers.

Therefore, the Leakage Belt was defined as an area of 759.879 ha of land located into a circular buffer of 50km from the center of the Project area, near the main deforestation driver.

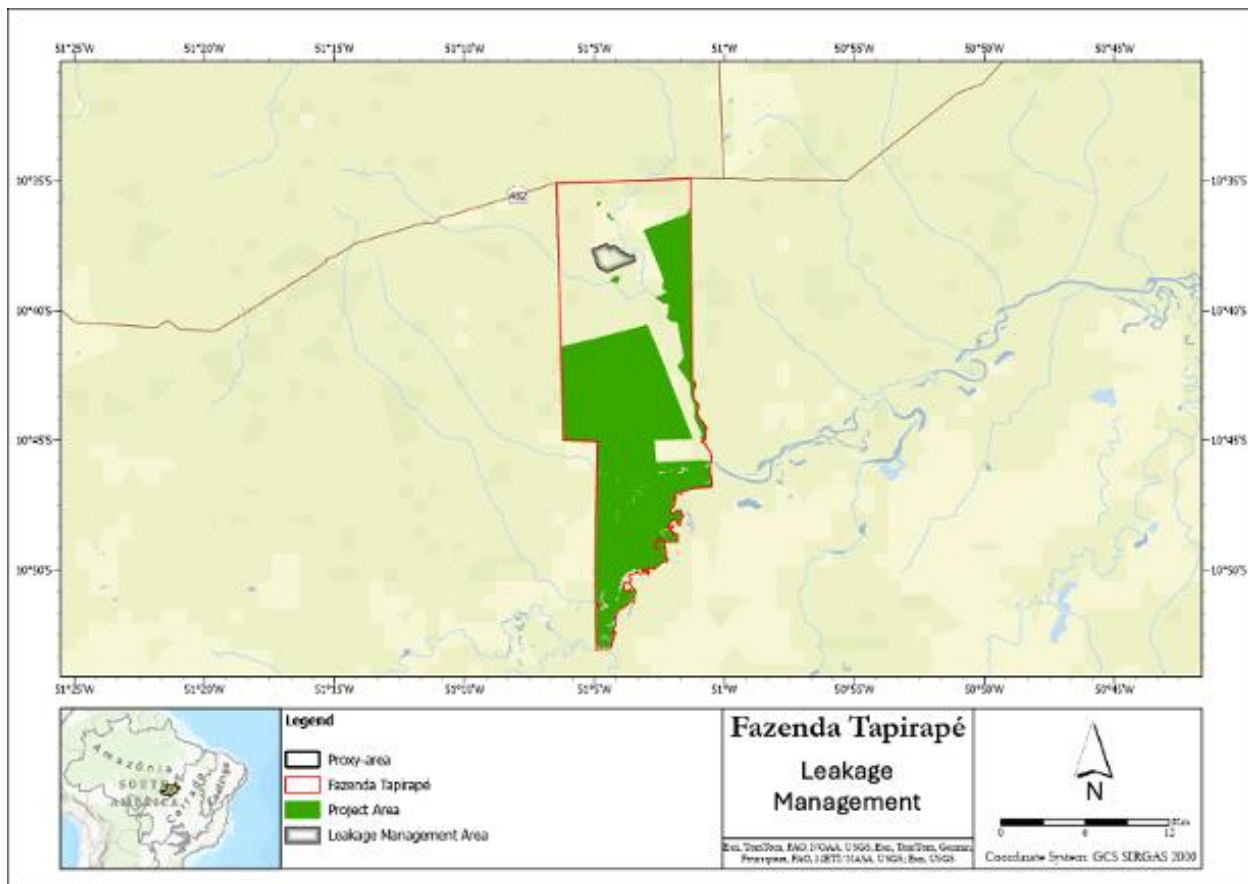
### Leakage Management Area

The Leakage Management Area (LMA) are places where the Project intends to exert the influence of its activities to reduce the risks of deforestation. These activities must include the agents of deforestation and involve seeking new sources of income which contribute to forest conservation.

This project aims to apply measures of environmental education and training, in addition to good agricultural and livestock practices in order to avoid leakage of deforestation. More details of the used arguments and details of the leakage management activities are available on the item 1.18 of this PD.

The leak management area is designed to involve communities located at key points of access to the project area and is thus located within the Tapirapé Farm, on a 387 hectares portion of non-forest cover within the Project Area.

Figure below shows the location of the Leakage Management Area within the Tapirapé Farm.



Map Source: the author – Leakage management area within the Project Area.

## 1.14 Conditions Prior to Project Initiation

### **Ecosystem Type**

The Amazon-Cerrado transition biome is characterized by a blend of Amazon rainforest and Cerrado savanna ecosystems, harboring a mix of flora and fauna from both regions, including endemic species. The biodiversity of eastern Mato Grosso is filled by its diverse habitats and abundant water sources, fostering a wide array of terrestrial animals and bird species, bringing richness to the ecosystem.

### **Current and historical land-use**

Beyond the project area, e.g. within a 50km buffer, the landscape is predominantly characterized by forested areas, agricultural lands, pasturelands, and water bodies, reflecting both anthropogenic and natural land uses. Despite significant areas of already consolidated deforestation, recent land clearing activities can be observed, particularly since the 2000s, indicating ongoing land use changes even after the implementation and reinforcement of environmental laws in the country.

The expansion of agriculture and livestock farming has been a primary driver behind the substantial changes in land use, with extensive clearing of forests and conversion of natural habitats to agricultural and pasture lands. These land use changes have had a profound impact on the local environment and biodiversity, highlighting the need for sustainable land management practices and conservation efforts to mitigate further degradation.

### **Climate**

According to the IBGE<sup>26</sup> (The Brazilian Institute of Geography and Statistics), the area is situated in a hot equatorial climate zone, with temperatures consistently exceeding 18 degrees Celsius throughout the year. It is characterized by high humidity, albeit experiencing a dry period lasting for three months.

Mato Grosso has a distinct rainy season, unlike some other parts of the country. The region experiences a historical average precipitation with 90% occurring in a concentrated 7-month period from October to April (considering the historical reference 1971–2000).<sup>27</sup> To achieve a comprehensive understanding, it is essential to study the climate effects not only within Mato Grosso but also in other regions, recognizing the interdependence of Brazilian climate systems, allowing insights from one region to inform and benefit others – for instance, the essential

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<sup>26</sup> Climate geospatial data from IBGE – <https://www.ibge.gov.br/geociencias/informacoes-ambientais/climatologia/15817-clima.html?=&t=downloads>

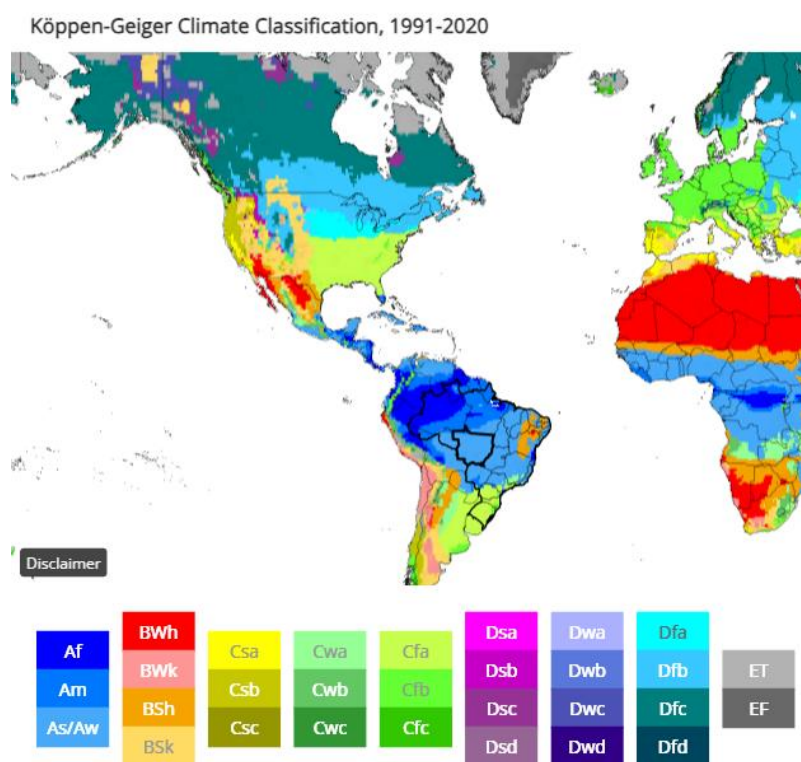
<sup>27</sup> Coe, M.T., Latrubesse, E.M., Ferreira, M.E. et al. The effects of deforestation and climate variability on the streamflow of the Araguaia River, Brazil. *Biogeochemistry* 105, 119–131 (2011). <https://doi.org/10.1007/s10533-011-9582-2>



influence of the Brazilian northern states and their vegetation for the micro-climate (especially rain) responses.

### *Köppen Climate Classification System*

Based on The Köppen climate classification system, one of the most common climate classification systems globally used to denote different climate regions on Earth based on local vegetation, the World Bank Group<sup>28</sup> shows that the Mato Grosso State falls into the As/Aw category, essentially the Savanna climate in tropical countries. It features a rainy season in the summer, from November to April, and a distinct dry season in the winter, from May to October (July being the driest month). The average temperature of the coldest month is above 18°C. Precipitation exceeds 750 mm annually, reaching 1,800 mm<sup>29</sup>.

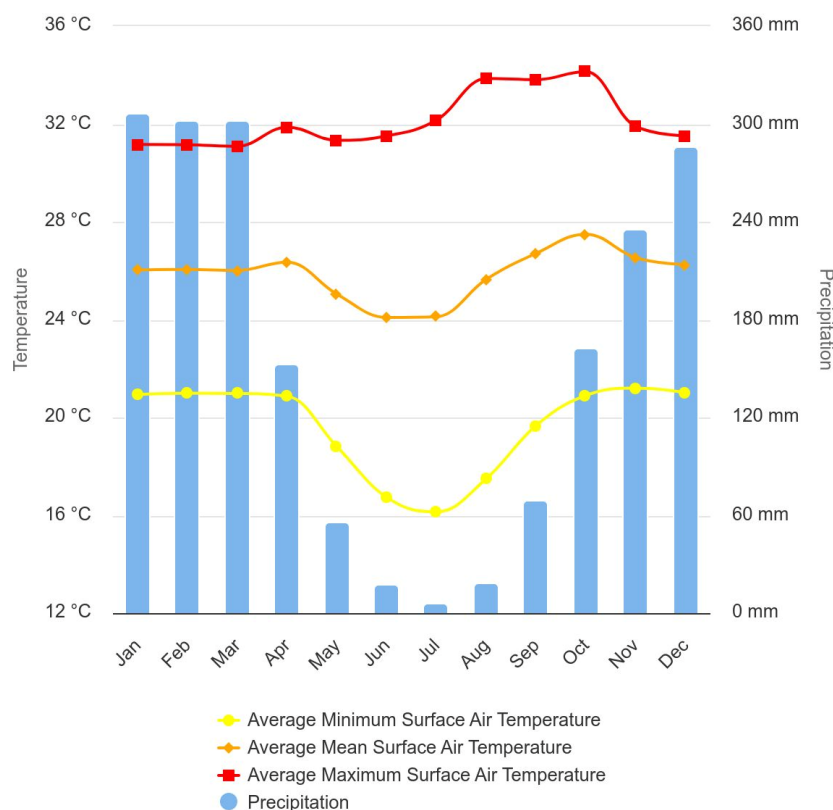


Map Source: The World Bank Group, *Climate Change Knowledge Portal – Mato Grosso/Brazil*<sup>30</sup>

<sup>28</sup> The World Bank Group, *Climate Change Knowledge Portal*, Köppen-Geiger Climate Classification, 1991-2020 – <https://climateknowledgeportal.worldbank.org/country/brazil>

<sup>29</sup> Pesquisa Embrapa, Clima – <https://www.cnpf.embrapa.br/pesquisa/efb/clima.htm>

<sup>30</sup> The World Bank Group, *Climate Change Knowledge Portal – Mato Grosso/Brazil* – <https://climateknowledgeportal.worldbank.org/country/brazil/climate-data-projections>



Graph Source: The World Bank Group, Climate Change Knowledge Portal – Temperature & Precipitation, Ref. 1991-2022, Mato Grosso/Brazil<sup>31</sup>

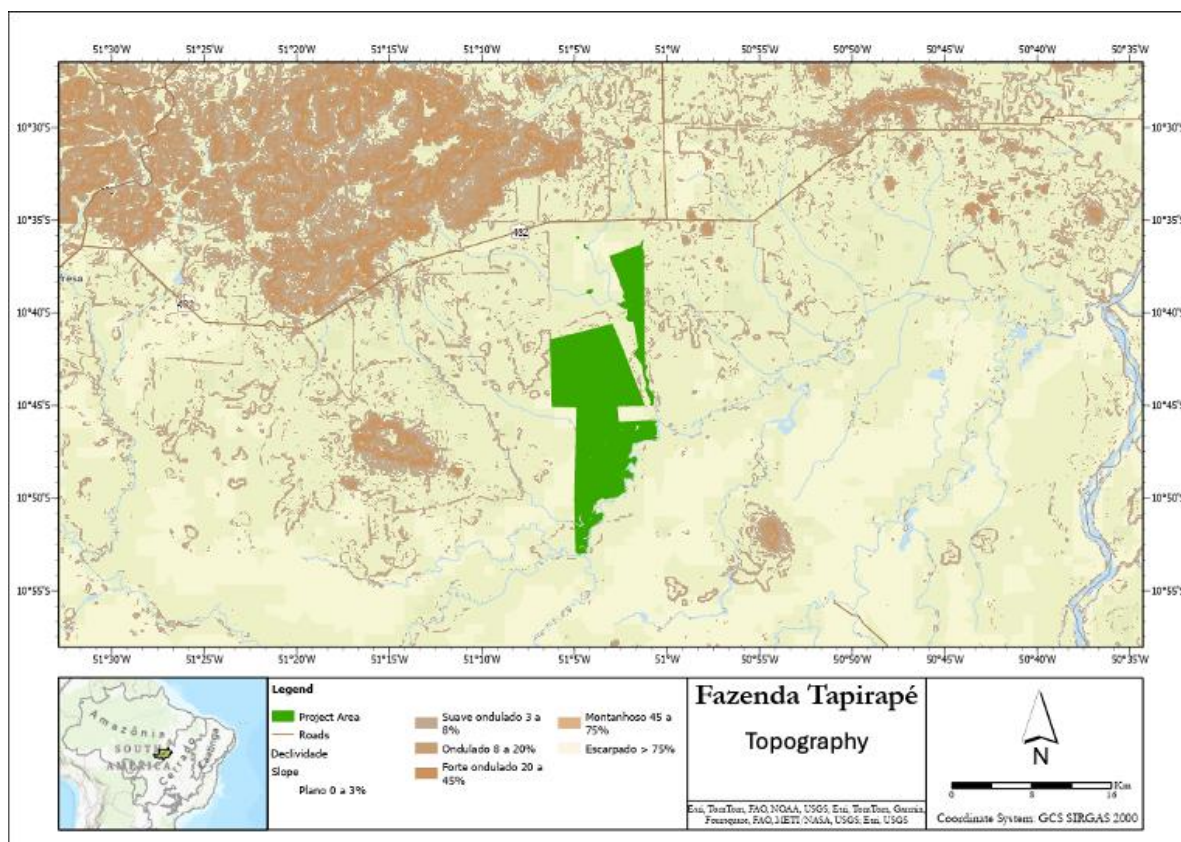
## Geology

The geology in the project area refers to the Phanerozoic Cenozoic Era, which is the current eon on the geologic time scale, spanning from about 66 million years ago to the present. This is marked by evolution and diversification of the rich and diverse flora that characterizes Brazil today. The tropical rainforests of the Amazon, the Cerrado savanna, and some other Brazilian ecosystems have their roots in the Cenozoic, with specific adaptations to changing climate conditions. The geological and climatic events of the Cenozoic Era have had a profound impact on the formation and distribution of vegetation in Brazil, contributing to the vast biodiversity that the country is known for.

## Topography

<sup>31</sup> The World Bank Group, Climate Change Knowledge Portal – Monthly Climatology of Average Minimum, Mean and Maximum Surface Temperature & Precipitation 1991-2022, Mato Grosso, Brazil.  
<https://climateknowledgeportal.worldbank.org/country/brazil/climate-data-projections>

The geological history has directly influenced the topography of Mato Grosso. The region includes plains, plateaus, and areas with gently rolling hills. The topography can influence water drainage, soil erosion, and agricultural suitability. The topographic map of the region is presented below:



Map Source: The World Bank Group, Climate Change Knowledge Portal – Temperature & Precipitation, Ref. 1991-2022, Mato Grosso/Brazil<sup>32</sup>

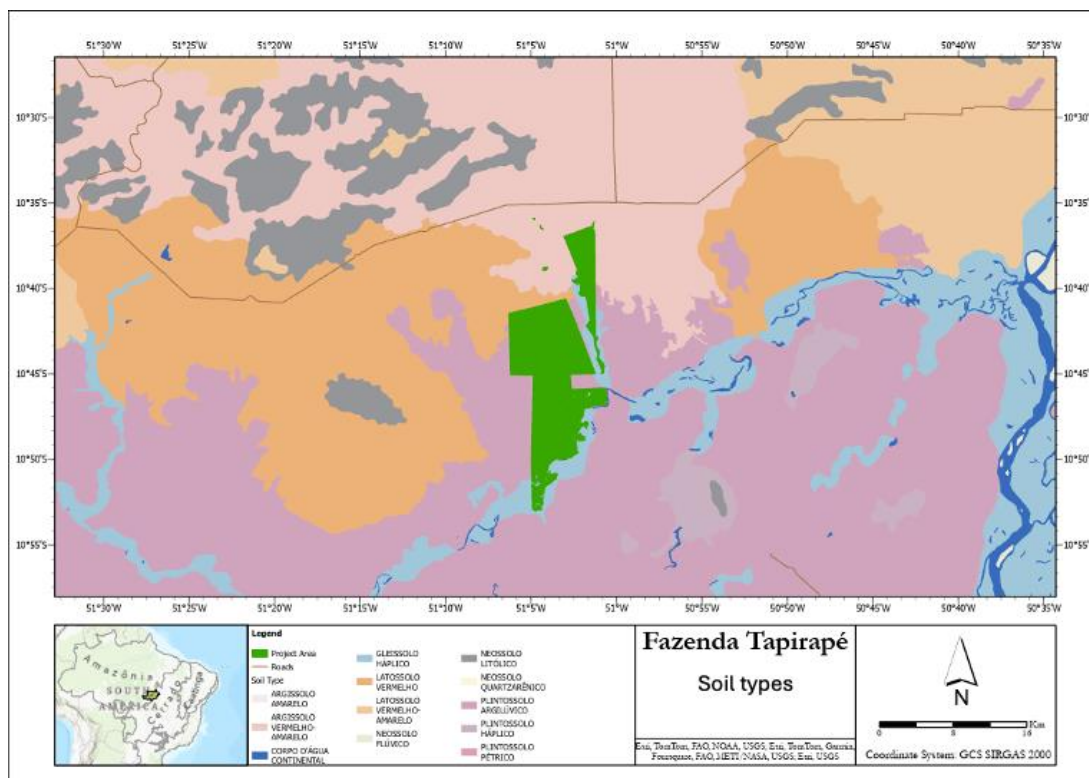
## Soil

The majority of the land/soil on which the project is situated consists of 'Plinthosols with a horizon or layer of clay accumulation', which exhibit variable drainage, leading to temporary water excesses and, at times, prolonged water excesses throughout the year. These plots are classified as 'typical,' meaning they do not exhibit any restrictive characteristics in terms of erosion. This parcel corresponds to a transition area from Amazon to Cerrado Savanna.<sup>33</sup> The other portion of the land, where a denser and more typical Amazonian vegetation is found, is composed of soils with a higher iron content, known as Oxisols (in Portuguese: Latossolos). These are non-

<sup>32</sup> The World Bank Group, Climate Change Knowledge Portal – Monthly Climatology of Average Minimum, Mean and Maximum Surface, Temperature & Precipitation 1991-2022, Mato Grosso, Brazil.  
<https://climateknowledgeportal.worldbank.org/country/brazil/climate-data-projections>

<sup>33</sup> EMBRAPA, Tropical Soils – “Plintossolos Argilúvicos” – <https://www.embrapa.br/agencia-de-informacao-tecnologica/tematicas/solos-tropicais/sibcs/chave-do-sibcs/plintossolos/plintossolos-argiluvicos>

hydromorphic and deep soils, typically exceeding 2 meters in depth. The colors of these soils range from very dark red to yellowish. Silica ( $\text{SiO}_2$ ) and other bases are leached from the system, leading to enrichment with iron and aluminum oxides, which act as binding agents, giving the soil mass a solid porous appearance. These soils have a silt content below 20% and clay ranging between 15% and 80%. They demonstrate high water permeability and can be worked across a wide range of moisture levels.<sup>34</sup> There is also a smallest portion of land classified in the northeast of the land, smaller, classified as 'Argissolo', which is moderately deep to deep, moderately drained, and reddish clayey soils. These soils develop from various source materials in areas with flat to hilly terrain.<sup>35</sup> All these information were taken from scientific studies made by Embrapa, and crossed with the IBGE shapefile database.<sup>36</sup>



Map Source: the author – Leakage management area within the Project Area

<sup>34</sup> EMBRAPA, Cerrado-Savanna Biome – “Latossolos” – <https://www.embrapa.br/agencia-de-informacao-tecnologica/tematicas/bioma-cerrado/solo/tipos-de-solo/latossolos>

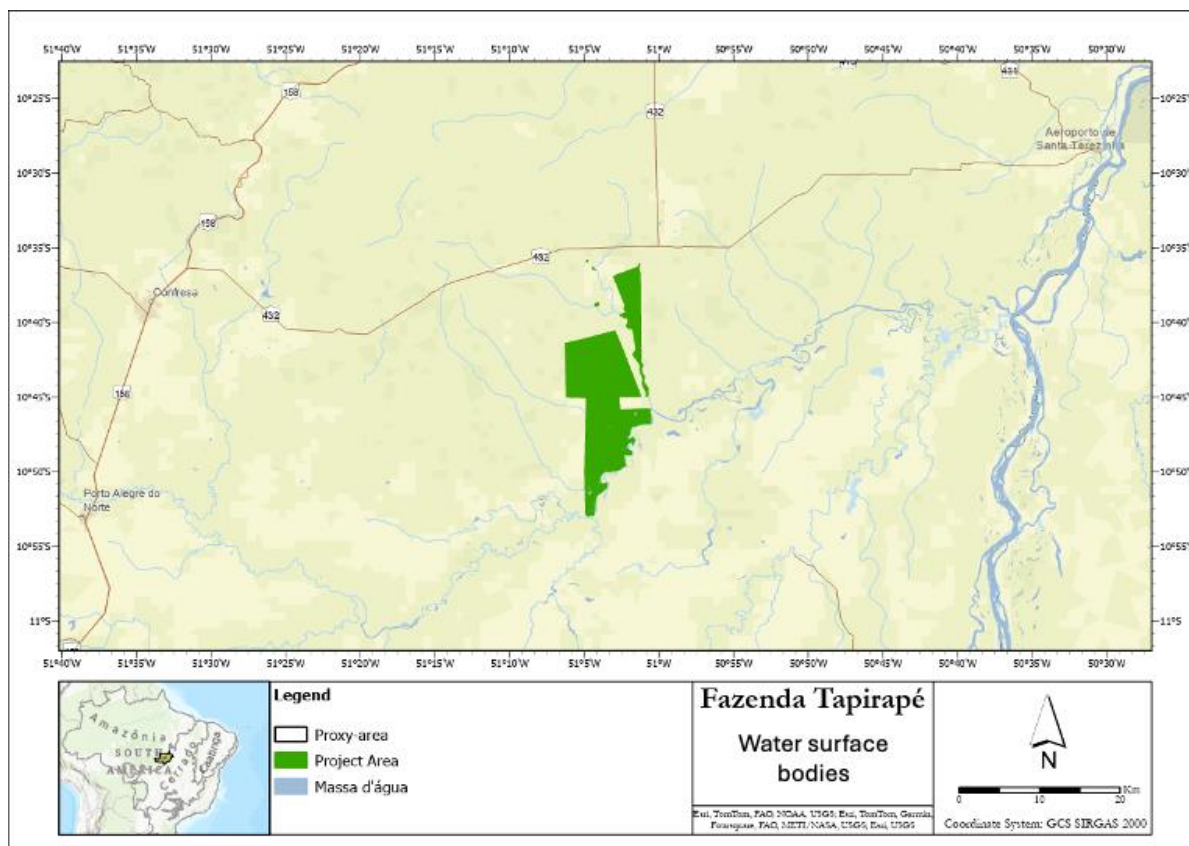
<sup>35</sup> EMBRAPA, Cerrado-Savanna Biome – “Argissolos” – <https://www.embrapa.br/agencia-de-informacao-tecnologica/tematicas/bioma-caatinga/solos/argissolos>

<sup>36</sup> IBGE, Downloads, GeoSciences – [https://geoftp.ibge.gov.br/informacoes\\_ambientais/pedologia/vetores/escala\\_250\\_mil/versao\\_2023/](https://geoftp.ibge.gov.br/informacoes_ambientais/pedologia/vetores/escala_250_mil/versao_2023/)



## Water

The project area is close to several water bodies, with the closest being the Tapirapé River, which directly intersects with the property terrain. Adjacent to this, there is also the Xavantino River. Both of them maintain a permanent flow and are part of one of the region's most significant basins, as they are connected to the Araguaia River. This covers a 385,000 km<sup>2</sup> watershed in eastern Amazonia and has an average annual discharge of approximately 6,500 m<sup>3</sup>/s (as an example, this might be comparable to the Danube).



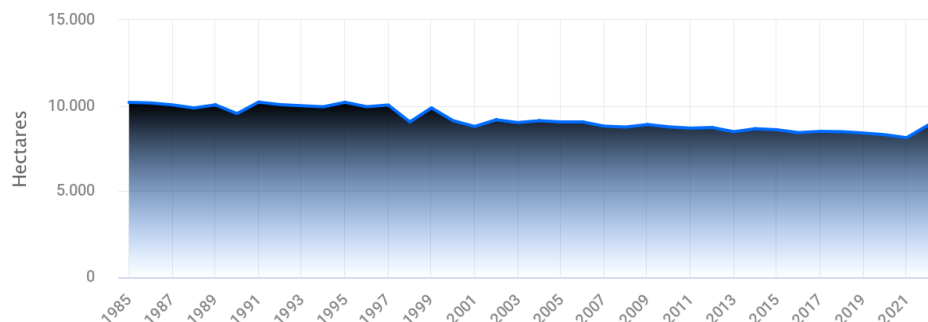
*Map Source: the author – Main water surface bodies on the surrounding reference*

According to M.T. COE (2011), deforestation in the Araguaia River watershed has led to substantial alterations in hydrological and morphological characteristics. The significant vegetation loss along the time has contributed to a 25% increase in observed discharge from the 1970s to the 1990s. Computer simulations indicate that two-thirds of this rise result from deforestation, with the remaining third attributed to climate variability. These findings underscore the extensive impact of anthropic actions on river systems in the region. In the context of REDD+ (Reducing Emissions from Deforestation and Forest Degradation), addressing and mitigating deforestation can play a crucial role in preserving hydrological balance and ecosystem health. The studies presented by the initiative MAPBIOMAS ÁGUA (WATER)<sup>37</sup> can corroborate with this

<sup>37</sup> MAPBIOMAS Water Methodology – <https://brasil.mapbiomas.org/metodo-mapbiomas-agua/>

study, showing that the amount of water volume has also suffered changes in the region along the time, in a municipality level analysis.

Série temporal da superfície d'água - Santa Terezinha



Map Source: MAPBIOMAS – Study of water volume change (municipality of Santa Terezinha – MT)

## Vegetation

For the vegetation study, we integrated information from the IBGE – Brazilian Institute of Geography and Statistics – vegetation database (details of which will be elucidated subsequently) with our project-specific databases concerning property and project area, as illustrated in the vegetation map below. Through this integration, we identified the primary Phyto physiognomies that constitute the natural environment of the designated area. Among the key and predominant ones are:

Fa - Alluvial Seasonal Semideciduous Forest (Floresta Estacional Semidecidual Aluvial)

Fs - Seasonal Semideciduous Forest (Floresta Estacional Semidecidual)

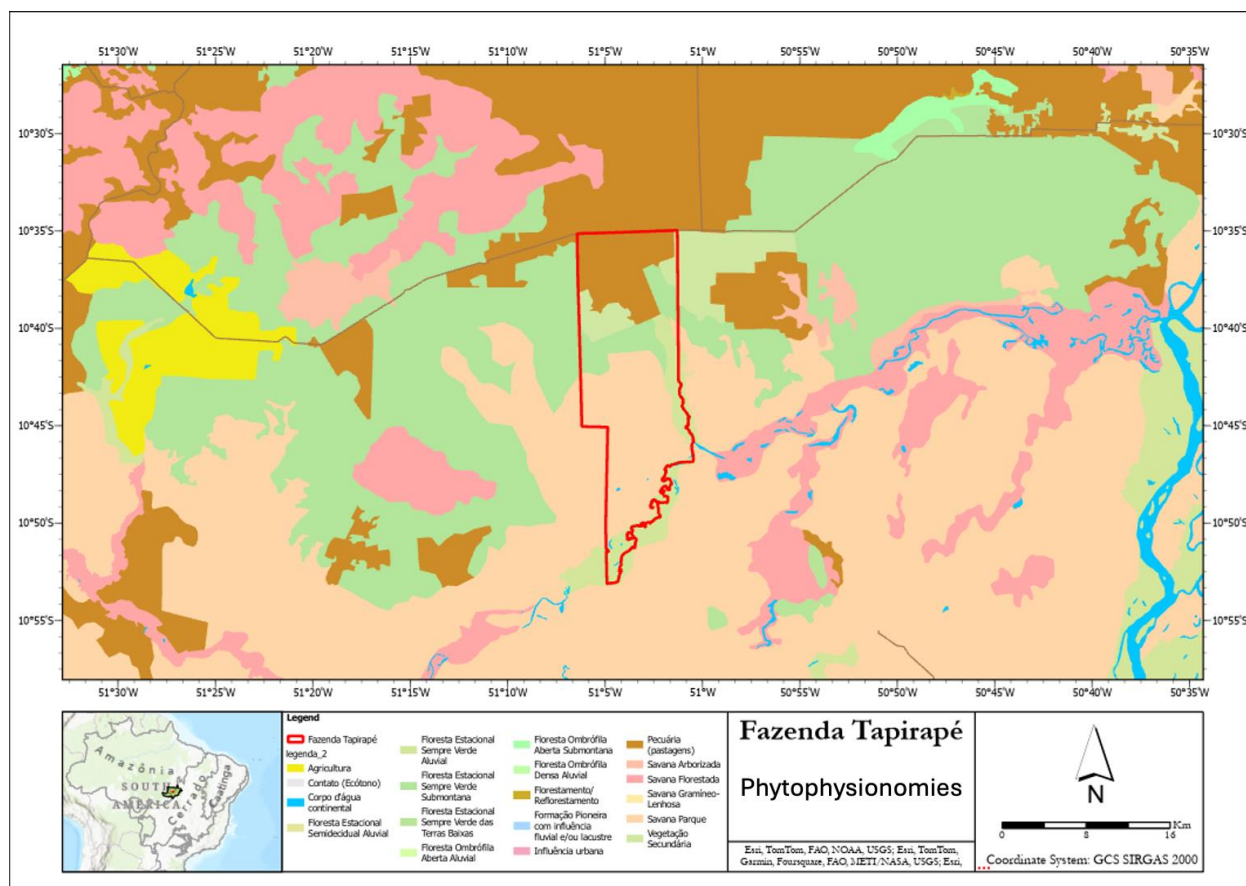
Sa - Wooded Savanna (Savana Arborizada)

Sp - Savannah Park (Savana Parque)

SN - Contact Savannah-Seasonal Forest (Contato Savana/Floresta Estacional)

These identified vegetation types serve as a foundational element for our carbon project, contributing essential insights into the ecosystem structure and facilitating a comprehensive understanding of the carbon dynamics within the studied area.





Map Source: the author – Phyto physiognomy classification in the project area and surroundings.

The study of vegetation was based on the Natural Resources Mapping of IBGE, an activity developed by the Environmental Coordination and the Environmental and Geography Management of the State Superintendencies of the Institute.

The classification of vegetation cover was performed at the 1:250,000 scale. At this level of precision, the mapping unit boundaries and their respective classification legends are visible.

To form the database with this level of precision, the institute used vegetation points, which consist of sample records from field expeditions. These expeditions involve physiognomic observations aiming to identify types of vegetation and anthropogenic areas. These records include forest inventories, botanical floristics, and other generic observations, as well as cross-referencing with other databases (e.g., soil, climate, etc.).

The study allows for the identification of attributes such as the type of cover class and predominant use (natural vegetation or predominant anthropogenic area with or without ecological tension); Phytogeographical regions and other areas; 1st, 2nd, and 3rd vegetation cover or other areas (predominant); vegetation class or natural area (when predominant); anthropogenic area class (when predominant); contact class (when existent); past vegetation (up to the identifiable level); among others.

It is important to mention that IBGE used its own methodology, detailed in the document "Technical Manual of Brazilian Vegetation"<sup>38</sup>. All this work has been carried out through the RADAMBRASIL Project since 1970, which is now part of the Ministry of Mines and Energy of the federal government and is managed by IBGE. The project has been conducting important mappings of Geology, Geomorphology, Pedology, Vegetation, and Potential Land Use through the analysis of radar aerial images in semi-controlled mosaics and field information collection expeditions.<sup>39</sup>

## Biodiversity

### *Flora (trees)*

The upland of the municipality of Santa Terezinha, where the more fertile loamy soils support denser forests, is naturally evergreen, with many species of native trees, as mentioned by EITEN (1985) on its very localized and specific observatory and photographic study. According to him, this upland naturally includes species such as:

Pau Brasil (*Caesalpinia echinata*, with white bark and dark red wood), Jatobá (*Hymenaea* sp.), Favela, Grapiá, "Pinho" (a dicot), Xixá (*Sterculia* sp.) Brejeiro, Quare-quare, Cacau Bravo, Mangueira, Tarumã (a dicot, not a palm), two species of Ipê (*Tabebuia*), Mirindiba, Sucupira Branca, Sucupira Preta, Cedro (*Cedrella* sp.), Angico (*Piptadenia* sens. lat. sp.), Murici da Mata (*Byrsonima* sp.), Piquiá (*Caryocar coriaceum* / forest form), and the tall palms Babaçu (*Orbignya* sp.), Naja cabeçudo (*Haximíana*), and Bacaba (*Oenocarpus* sp.). The giant herb, *Phenakospermum guianense*, occurs in the undergrowth. Almost all areas of the forest contain one or more of several species of canopy palms and in some portions a canopy palm species makes up half the trees, especially babaçu (*Orbignya* sp.). A large bamboo also occurs in the undergrowth. In that part of the area where forest covers the whole topographic profile, that portion bordering the rivers, even if not flooded, contains different tree species from those further upslope, or the same species in different proportions. (EITEN, 1985)<sup>40</sup>

For the area of the same municipality where the common Cerrado prevails, the following species are more present:

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<sup>38</sup> IBGE, Manual Técnico da Vegetação Brasileira / Coordenação de Recursos Naturais e Estudos Ambientais, 2nd edition, 2012.

<sup>39</sup> Banco de Dados e Informações Ambientais (BDiA) : Mapeamento de Recursos Naturais (MRN) : escala 1:250 000 : versão 2023 : nota metodológica / IBGE, Coordenação de Meio Ambiente <https://biblioteca.ibge.gov.br/index.php/biblioteca-catalogo?view=detalhes&id=2102042>

<sup>40</sup> GEORGE EITEN, VEGETATION NEAR SANTA TERESINHA, NE MATO GROSSO, 1985.

Connarus spp., Davilla Elliptica, Pau Terra (Qualea grandiflora, Q. multiflora, Q. parviflora), Murici (Byrsonima verbascifolia, B. coccolobifolia), Candeia, Araça (Psidium sp.), Bruto (Annona sp.), Puça Preto, Puça Branco (probably Mouriri spp.), oiti (probably Licania sp.) Macambeira (not the bromeliad macambira), and the palms Babaçu (Orbignya, the same species as in the upland forest), Pati (Syagrus sp.), Babão (Syagrus comosa) and Piaçaba (an acaulescent palm). (EITEN, 1985)

Although this might be an elderly study, this is a good representation of the native species in the region, which may have suffered small changes until then, but in a project scenario, where we analyze the historical land-use of soil from at least the last 20 years, we can consider those studies still valid, even though this might have traits of secondary vegetation.

#### *Inventoried flora species within the Project Area (Savanna)*

The forest inventory record for this project had some primary objectives: collecting field data to materialize information on local specific flora species for Savanna, as well as their phytosociological characteristics. The tree mapping was carried out mostly in the portion of Semi-deciduous and Cerrado forest reserves of the Project Area, as these substrates require more in-depth studies in light of existing literature. Thus, woody plants with a circumference  $\geq 15$  cm were identified, and their Diameter at Breast Height (DBH) was measured at 1.30 m above ground level using a tape measure in 10 fixed plots systematically distributed, totaling 10,000 m<sup>2</sup> in each stratum. The length of the plots was measured with the assistance of a Garmin Etrex 20 GPS (Global Positioning System). For the georeferencing of the plots, a Garmin Etrex 20 GPS was utilized. Individuals that bifurcated above ground level and below 1.30 m in height had their measurements recorded separately, considering them the same individual; trunks were identified by alphabetical letters (A, B, C, ...), and it is worth noting that each branch received the same numbering.



Picture Source: the author – Identification of tree sample

**Table 6. Arboreal species identified for the Savanna portion within the Project Area**

| Individual | Specie                   | Popular name (PT-BR) | Family           |
|------------|--------------------------|----------------------|------------------|
| 1          | Byrsonima cydoniifolia   | Murici do varjão     | Malpighiaceae    |
| 2          | Couepia grandiflora      | Oiti                 | Chrysobalanaceae |
| 3          | Curatella americana      | Lixeira              | Dilleniaceae     |
| 4          | Andira cujabensis        | Angelim do campo     | Fabaceae         |
| 5          | Hancornia speciosa       | Mangaba              | Apocynaceae      |
| 6          | Myrcia tortuosa          | Sem nome popular     | Myrtaceae        |
| 7          | Emmotum nitens           | Sobro                | Metteniusaceae   |
| 8          | Byrsonima coccolobifolia | Murici de porco      | Malpighiaceae    |
| 9          | Davilla elliptica        | Lixinha              | Dilleniaceae     |
| 10         | Myrcia multiflora        | Goiabinha            | Myrtaceae        |
| 11         | Myrcia splendens         | Grudento             | Myrtaceae        |
| 12         | Leptolobium dasycarpum   | Para tudo            | Fabaceae         |
| 13         | Caryocar brasiliense     | Pequi                | Caryocaraceae    |
| 14         | Bowdichia virgilioides   | Sucupira preta       | Fabaceae         |
| 15         | Vatairea macrocarpa      | Angelim margoso      | Fabaceae         |
| 16         | Erythroxylum suberosum   | Pimenta              | Erythroxylaceae  |
| 17         | Rourea induta            | Sem nome popular     | Connaraceae      |
| 18         | Aegiphila integrifolia   | Sem nome popular     | Lamiaceae        |
| 19         | Connarus suberosus       | Brinco de galinha    | Connaraceae      |
| 20         | Xylopia aromatica        | Pimenta de macaco    | Annonaceae       |
| 21         | Guatteria foliosa        | Envira               | Annonaceae       |
| 22         | Tapirira guianensis      | Pombeiro             | Anacardiaceae    |
| 23         | Byrsonima intermedia     | Murcizinho           | Malpighiaceae    |
| 24         | Casearia sylvestris      | Guaçatonga           | Salicaceae       |
| 25         | Aspidosperma multiflorum | Guatambuzinho        | Apocynaceae      |
| 26         | Guapira graciliflora     | Pau de ripa          | Nyctaginaceae    |
| 27         | Byrsonima crassifolia    | Murici               | Malpighiaceae    |
| 28         | Diospyros sericea        | Olho de boi          | Ebenaceae        |
| 29         | Virola sebifera          | Pau de sebo          | Myristicaceae    |
| 30         | Hirtella glandulosa      | Bosta de rato        | Chrysobalanaceae |
| 31         | Pouteria ramiflora       | Curriola             | Sapotaceae       |
| 32         | Anacardium occidentale   | Cajuzinho            | Anacardiaceae    |
| 33         | Handroanthus ochraceus   | Ipê amarelo do campo | Bignoniaceae     |
| 34         | Annona coriacea          | Bruto                | Annonaceae       |
| 35         | Maprounia guianensis     | Milho torrado        | Euphorbiaceae    |
| 36         | Moquilea egleri          | João duro            | Fabaceae         |

|    |                             |                          |                  |
|----|-----------------------------|--------------------------|------------------|
| 37 | Luetzelburgia praecox       | Guaiçara                 | Fabaceae         |
| 38 | Qualea parviflora           | Pau terra da folha miúda | Vochysiaceae     |
| 39 | Copaifera langsdorfii       | Pau d'óleo               | Fabaceae         |
| 40 | Euplassa inaequalis         | Faia                     | Proteaceae       |
| 41 | Tabebuia aurea              | Caraiba                  | Bignoniaceae     |
| 42 | Stryphnodendron adstringens | Barbatimão               | Fabaceae         |
| 43 | Vochysia pyramidalis        | Cambará vermelho         | Vochysiaceae     |
| 44 | Platymenia reticulata       | Vinhático                | Fabaceae         |
| 45 | Licania egleri              | João duro                | Chrysobalanaceae |
| 46 | Licania gardineri           | João duro                | Chrysobalanaceae |
| 47 | Guatteria sp.               | Envira                   | Annonaceae       |
| 48 | Syagrus comosa              | Gueirobinha              | Arecaceae        |
| 49 | Roupala montana             | Carne de vaca            | Proteaceae       |
| 50 | Andira cuyabensis           | Angelim do campo         | Fabaceae         |
| 51 | Eugenia dysenterica         | Cagaita                  | Myrtaceae        |
| 52 | Astronium fraxinifolium     | Gonçaleiro               | Anacardiaceae    |
| 53 | Ouratea castaneaefolia      | Vassoura de bruxa        | Ochnaceae        |
| 54 | Cordia glabrata             | Loro preto               | Boraginaceae     |
| 55 | Callisthene fasciculata     | Jacaré                   | Vochysiaceae     |

## Fauna

For a macro view of fauna biodiversity, official government literature on management plans in conservation units (state parks) within the reference area and adjacent to the project area was considered. It's important to note that many species utilize ecological corridors between forest fragments for survival, movement, and reproduction. As all parks are situated in the same ecosystem context (biological, vegetation, and climate), it is common for various species to coexist, even sharing spaces depending on their mobility.

The parks considered for this purpose include the Conservation Units of Rio Xingu (RESEX and FLONA), Araguaia National Park, and Cantão State Park.

The ICMBIO fauna inventory (ICMBIO, 2012)<sup>41</sup> for the Rio Xingu conservation units encompasses numerous species, many of them endangered, such as *Anodontites ensiformis* (estilete), *Anodontites soleniformes* (freshwater mussel), *Anodontites trapesialis* (soapdish mussel), *Anodontites elongatus* (pantanal mussel), *Ossubtus xinguense* (pacu-capivara), *Hypancistrus zebra* (zebra pleco), *Anodorhynchus hyacinthinus* (hyacinth macaw), *Guaruba guarouba* (golden parakeet), *Natalus espirotoantensis* (bat), *Priodontes maximus* (giant armadillo), *Myrmecophaga*

<sup>41</sup> ICMBIO, Sumário Executivo do Plano de Ação Nacional para a Conservação das Espécies Endêmicas e Ameaçadas de Extinção da Fauna da Região do Baixo e Médio Xingu, julho de 2012.

tridactyla (giant anteater), Ateles marginatus (white-cheeked spider monkey), Chiropotes utahickae (Uta-Hick's bearded saki), Speothos venaticus (bush dog), Leopardus wiedii (margay), Puma concolor (puma).

The inventory of Araguaia National Park (MMA, 2001)<sup>42</sup>, located in Tocantins (still within the reference area), is expressed in the Park's Management Plan. It includes species such as snakes - Atractus pantostictus, Clelia plumbea, Mastigodryas boddaerti, Spilotes pullatus, Tantilla melanocephala, Thamnodynastes strigilis, lizard - Hoplocercus spinosus, tortoise - Geochelone denticulata, birds (217 species) - Ramphocelus carbo, Pipra fasciicauda, Monasa nigrifrons, Platyrinchus mystaceus, Turdus leucomelas, Oryzoborus angolensis, medium and large mammals (18 species) - Gracilinanus sp.1, Gracilinanus sp.2, Marmosa cf murina, Calomys sp., Oecomys sp., Oryzomys megacephalus, Gracilinanus sp.1, Gracilinanus sp.2, Oryzomys megacephalus, Holochilus cf sciureus, Bolomys lasiurus, Calomys sp., Pseudoryzomys simplex, Oryzomys megacephalus, Oligoryzomys sp., Thrichomys apereoides. The boto, Inia geoffrensis, was observed several times in different locations along the Araguaia and Javaés rivers.

For Cantão State Park, the government's Management Plan (SEPLAN, 2001)<sup>43</sup> was also used for a fauna survey. It describes 44 mammal species, 316 bird species, 22 reptile species, 17 amphibian species, and 56 fish species. Birds show high habitat specificity, with distinct structures in each natural community, highlighting a heterogeneous environment organized into different systems. The abundance of species adapted to the annual flood cycle is notable, especially capybara, jaguar, tapir, and giant otter. River reptiles, especially alligators and turtles, are also abundant. The diverse and abundant ichthyofauna includes a high number of commercially valuable fish species. Cantão is likely a critical site for the feeding and reproduction of fish in the middle Araguaia.

#### *Observed fauna within the Project Area*

During the pre-project start field visits, some local species were observed, smoothly navigating between the forested and common areas of the property. Below are some photos documenting these sightings:

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<sup>42</sup> MMA/IBAMA. Parque Nacional do Araguaia – TO. Plano de Manejo, Brasília, julho, 2001.

<sup>43</sup> SEPLAN. Secretaria de Planejamento e Meio Ambiente, Governo do Estado do Tocantins, Banco Inter-Americano de Desenvolvimento Plano de Manejo Parque Estadual do Cantão, Janeiro, 2001.





*Picture Source: the author – Female ‘Veado-campeiro’ Ozotoceros bezoarticus (additional information on note<sup>44</sup>)*



*Picture Source: the author – ‘Jabuti-Piranga’ Chelonoidis carbonaria (additional information on note<sup>45</sup>)*

<sup>44</sup> Additional information reference on: <https://pantheon.ufrj.br/handle/11422/3047>. Pampas Deer Ozotoceros bezoarticus has most recently been assessed for The IUCN Red List of Threatened Species in 2015. <https://www.iucnredlist.org/species/15803/22160030>

<sup>45</sup> Additional information found on: Sumário Executivo do Plano de Ação Nacional para a Conservação dos Quelônios Amazônicos - [https://www.icmbio.gov.br/ran/images/stories/Downloads/PAN\\_Quelonios\\_Amazonicos\\_Sumario\\_Executivo.pdf](https://www.icmbio.gov.br/ran/images/stories/Downloads/PAN_Quelonios_Amazonicos_Sumario_Executivo.pdf)



Picture Source: the author – ‘Iguana-Verde’ *Iguana iguana* (note on IUCN Red List data<sup>46</sup>)



Picture Source: the author – ‘Anta-Brasileira’ *Tapirus terrestris* (note on IUCN Red List data<sup>47</sup>)

<sup>46</sup> Common Green Iguana *Iguana iguana* has most recently been assessed for The IUCN Red List of Threatened Species in 2018 - <https://www.iucnredlist.org/species/174481/218317281>

<sup>47</sup> Lowland Tapir *Tapirus terrestris* is listed as Vulnerable under criteria A2cde+3cde. <https://www.iucnredlist.org/species/21474/45174127>



*Picture Source: the author – ‘Anta-Brasileira’ Tapirus terrestris (note on IUCN Red List data)*



*Picture Source: the author – ‘Jibóia-constritora’ Boa constrictor (note on IUCN Red List data<sup>48</sup>)*

<sup>48</sup> Red-tailed Boa Boa constrictor has most recently been assessed for The IUCN Red List of Threatened Species in 2014.  
<https://www.iucnredlist.org/species/197462/2486405>





*Picture Source: the author – ‘Tuiúú’ Jabiru mycteria (note on IUCN Red List data<sup>49</sup>)*



*Picture Source: The author – Bird-egg snapshot*

After talking to the landowner, it was also reported that the area occasionally hosts animals of extreme importance in the Amazonian ecosystem, ranging from beautiful birds like the Blue Macaw ‘Arara Azul’ (*Anodorhynchus hyacinthinus*) to predatory mammals such as the Jaguar ‘Onça Pintada’ (*Panthera onca*). This highlights how crucial the project area is as an important ecological corridor, serving as a portion of the ecosystem that facilitates connectivity between fragmented areas in the Amazon’s borders. It plays a vital role in ensuring the movement of animals, making it an essential element for maintaining biodiversity. These corridors are instrumental in mitigating the impact of human activities in these regions, as habitat fragmentation can have drastic consequences for local fauna and flora, potentially leading to the extinction of species. Such corridors enhance the ability of species to navigate their environment, reduce population isolation, and foster connectivity between fragmented regions.

<sup>49</sup> Jabiru mycteria has most recently been assessed for The IUCN Red List of Threatened Species in 2019.  
<https://www.iucnredlist.org/species/22697710/163624043>

## Infrastructure

The infrastructure analysis for the project area draws upon data extracted from the IBGE studies (IBGE, 2014<sup>50</sup>). Proximity to various structures can significantly impact issues related to deforestation in the region, as observed in universal experiences. These structures encompass roads, railways, airports, ports, warehouses, among others.

Within the study area, several infrastructures were identified that could certainly influence and increase pressure for deforestation, particularly in support of agricultural activities' growth. Notably, there are key road sections connecting the region to the state's main urban hubs, called as 'Structuring Highways'. The structuring highway axis linking the roads MT-413 and MT-430, located approximately 40 km from the project area, plays a vital role in facilitating territorial articulation, and the flow of goods and people with other municipalities and Brazilian states.

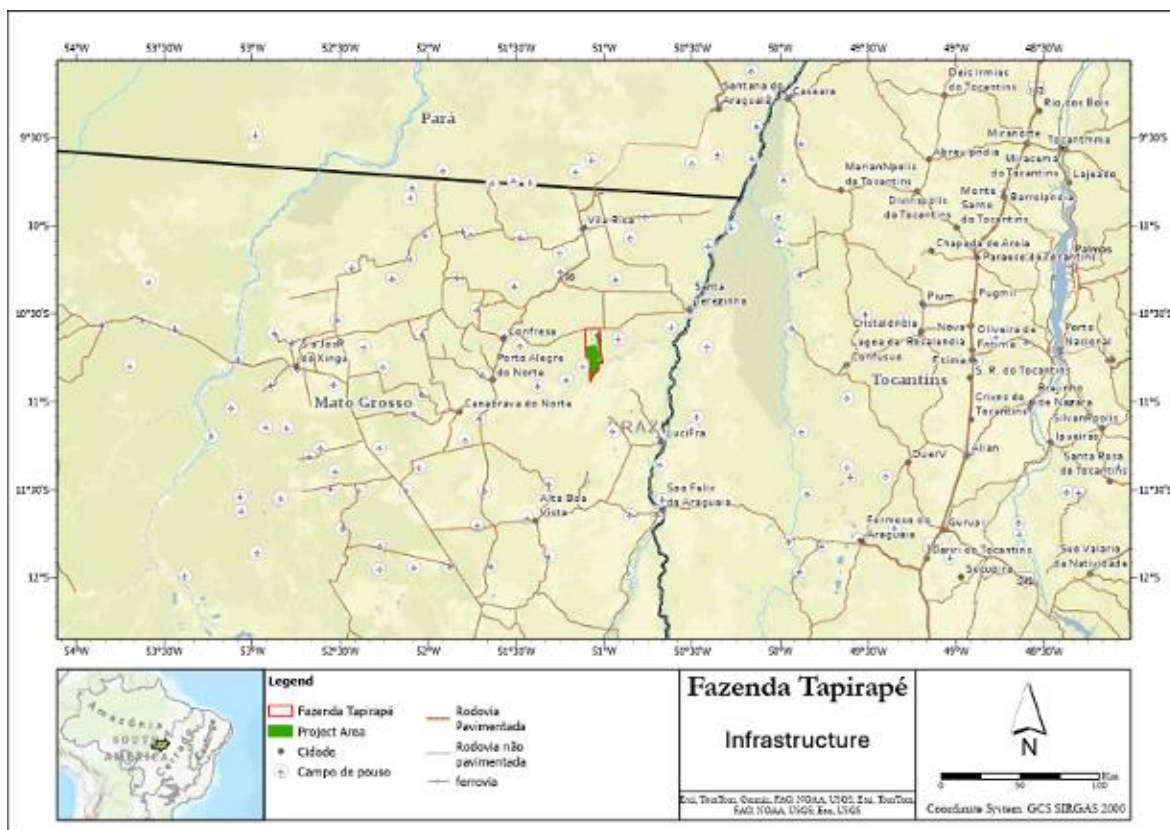
The MT-413 highway, mentioned before, traverses the project property, cutting through the terrain. This infrastructure might intensifies pressure on the area, primarily due to the ease of transporting and distributing products, inputs, natural resources, among others.

Another logistical structure facilitating access to the area is the airports. The airports of Confresa and São Felix do Araguaia are 50 km and 100 km away, respectively. The Palmas Airport in Tocantins State, a major logistics hub, is also in close proximity, approximately 600 km by road. The North-South Railway also crosses through the Airport area.

Several nearby warehouses, in addition to those not publicly registered, serve as facilitators for grain storage and distribution, with interconnected routes to the country's main ports. These elements collectively underscore the intricate relationship between infrastructure, logistical nodes, and the potential for environmental impact in the studied region.

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<sup>50</sup> IBGE – Logística dos Transportes no Brasil. <https://www.ibge.gov.br/geociencias/cartas-e-mapas/redes-geograficas/15793-logistica-dos-transportes.html>



Map Source: the author – Phyto physiognomy classification in the project area and surroundings.

## Lack of Preservation Actions Context

The conditions prevailing before the initiation of the carbon project reveal a landscape characterized by a lack of essential elements critical for effective environmental conservation and sustainable land management. These conditions are essential to understanding the baseline against which the project plans to unfold:

- The project area currently lacks a comprehensive near real-time satellite monitoring system.
- The current framework does not provide financial incentives for landowners to refrain from deforesting forested areas.
- At present, there are no financial returns or mechanisms tied to the maintenance of forests. This absence diminishes the economic value associated with forest conservation, limiting the potential for sustainable practices.
- The absence of structured financial returns from forest maintenance further translates into a deficit in resources for social responsibility initiatives. Consequently, there are limited avenues for converting forest preservation into tangible social and community benefits.



- There is a lack of targeted awareness and educational initiatives for both employees and the local community regarding the environmental significance, urgency, and future implications, particularly in the context of climate change.
- Enforcement of environmental laws is currently inadequate, contributing to a permissive environment for activities such as deforestation without stringent consequences.
- Over the history, inclusively recent years, the regulatory national/local framework have demonstrated limitations in effectively avoiding deforestation. The inefficiency of existing regulations poses a significant hurdle in preventing both legal and illegal deforestation activities, contributing to spreading uncontrolled devastation.

## 1.15 Compliance with Laws, Statutes and Other Regulatory Frameworks

### **National Laws regarding forest protection**

*Law n. 12.651, of 25th May 2012 Institutes the Forest Code:*

This law provides on protection on the vegetation, Permanent Preservation Areas, Legal Reserve; exploration of forests and succeeding formations, the supply of forest raw materials, control the origin of forest products and the prevention and control of forest fires, and provides economic and financial instruments for the achievement of its objectives, and makes other provisions.

*Complementary Law 140/2011:*

This Law establishes norms for the cooperation between the Union, the States, the Federal District and the Municipalities in the administrative actions arising from the exercise of the common competence regarding the protection of the remarkable natural landscapes, the protection of the environment, the fight against pollution in any of its forms and the preservation of forests, fauna and flora.

*Law 10.650/2003:*

This Law provides on public access to data and information in the institutes and entities that are members of Sisnama.

*Law n. 9,985 of 18th July 2000:*

This Law Institutes the National System of Conservation Units (SNUC), establishes criteria and norms for the creation, implementation, and management of protected areas and makes other provisions.

*Decree n. 4,340, of 22th August 2002:*

This Decree regulates Articles of Law n. 9,985, of 18 th July 2000, which provides on the National System of Conservation Units (SNUC) and makes other provisions. Provides on the creation of

Protected Areas, Management Plan, advisory management and authorization to explore goods and services.

*Normative Instruction 04/2009:*

This normative provides technical procedures for the use of vegetation of the Legal Reserve under a regime of sustainable forest management, and other measures.

*Normative Instruction n. 93, of 3th March 2006:*

This normative establishes technical norms for the presentation of maps and geo-referenced information about the localization of Legal Reserves and areas under forest management and respective subdivisions. Sustainable Forest Management Plans and authorization requests for alternative land use in the several Brazilian biomes, to be applied for at IBAMA and submitted to technical and juridical analysis, shall be accompanied by geo-referenced maps and forms prepared in accordance with technical norms and requirements set forth by this Normative Instruction.

## **Regulatory Market for Carbon Credits**

The law project 412/2022 (still in course for final approval, update on dec. 2023) establishes the Brazilian Emissions Trading System (SBCE) and regulates the carbon market in the country. The SBCE, a carbon commercialization system, aligns with global efforts to meet greenhouse gas reduction commitments under the United Nations Framework Convention on Climate Change and the Paris Agreement. The approved text indicates the adoption of cap-and-trade, setting emission limits overseen by a competent authority. Companies exceeding their allocated emission quotas may sell the surplus to those emitting above their quotas, ensuring the overall emission reduction targets are met within the established limits.

Companies with emission reduction obligations can demonstrate compliance by returning initially allocated quotas or purchasing carbon credits generated through accredited methodologies.

## **Regional Laws regarding environmental protection**

### **CONSTITUTION OF THE STATE OF MATO GROSSO**

*Article 1: ‘The State of Mato Grosso, as an integral part of the Federative Republic of Brazil, along with its municipalities and districts, proclaims and commits itself within the bounds of its autonomy and competence to ensure within its territory the values that underpin the existence and organization of the Brazilian state. This commitment extends to safeguarding the sovereignty of the nation and its people, the dignity of the human person, the social values of labor and free enterprise, and political pluralism—all in support of a democratic regime and the establishment of a just and solidarity-based society, free from arbitrary rule and prejudices of any kind.’*

Article 3: *'The currently existing Environmental Conservation Units are maintained, with the State promoting their demarcation, land regularization, and effective implementation within a two-year period. The necessary funds for these purposes shall be allocated in the upcoming budgets.'*

Article 273: *'The Pantanal, the Cerrado, and the Mato Grosso Amazon Rainforest shall constitute priority centers for environmental protection. Their utilization shall be carried out, in accordance with the law, under conditions that ensure the preservation of the environment, including the sustainable use of natural resources.'*

COMPLEMENTARY LAW No. 592, OF MAY 26, 2017 – Official Journal (in Brazil, called *Diário Oficial*): May 26, 2017.

Establishes the Environmental Regularization Program - PRA, regulates the Rural Environmental Registry (CAR), the Environmental Regularization of Rural Properties, and the Environmental Licensing of polluting activities or those using natural resources within the scope of the State of Mato Grosso, and provides for other measures.

Decree No. 1436 of July 18, 2022:

Provides for the State administrative process for investigating administrative infractions related to harmful conducts and activities to the environment, regulates the Environmental Fines Conversion Program, and establishes other provisions.

### Regulatory and controller Brazilian agencies

Brazil relies on a network of dedicated environmental control agencies. These agencies are entrusted with the responsibility of enforcing environmental regulations, managing protected forest areas, and preserving biodiversity. These agencies are crucial for the positive progress of this project, once they are the main instruments – based on major Law – to promoting the natural resources by penalizing and restricting the deforestation drivers indicated on this PDD.

At Federal level, the following agencies can be mentioned as basis:

**Table 7. Federal Agencies**

| Federal Agency | Attributes relative to Deforestation                                                                            |
|----------------|-----------------------------------------------------------------------------------------------------------------|
| IBAMA          | Inspection, Penalties and Embargo.                                                                              |
| CNAL           | Coordination of actions for the Amazon preservation and protection.                                             |
| SFB            | Coordination of the System of the CAR (Cadastro Ambiental Rural, in English: Rural Environmental Registration). |

| Federal Agency | Attributes relative to Deforestation                                                                                      |
|----------------|---------------------------------------------------------------------------------------------------------------------------|
| ICMBio         | Management, inspection, and protection of the 'Unidades de Conservação Federais', in English: Federal Conservation Units. |
| INCRA          | Management of settlements from Agrarian Reform and land regularization.                                                   |
| FUNAI          | Management and Protection of Indigenous lands.                                                                            |

## Labor and Human Rights

### *CLT – Consolidation of Labor Laws:*

The Consolidation of Labor Laws (CLT, in Portuguese), formalized by Decree-Law 5,452/1943, plays a crucial role in ensuring a range of fundamental rights for Brazilian workers. Among these rights are the maximum daily working hours, paid weekly rest, paid vacations, overtime pay, work in healthy environments, advance notice in case of dismissal, maternity, and paternity leave, 13th-month salary, protection against unjustified dismissal, and the benefit of unemployment insurance. The CLT is considered one of Brazil's first instruments of social inclusion. Its promulgation was a historic milestone that established essential rights for workers, significantly contributing to the construction of a more just and egalitarian society.

## Carbon and Climate Change specific Frameworks

In the project's commitment to advancing sustainable development and mitigating climate change impacts, this project is resolute in adhering to recognized Climate Change studies frameworks. By adopting these frameworks, the project aims to ensure the credibility and effectiveness of its initiatives while contributing to the global effort to combat climate change. In this regard, this project acknowledges the significance of staying abreast of the most recent and influential frameworks shaping the carbon credit market, considering both technical consensus and the relation with the VERRA's methodology. Some prominent frameworks which base this project are listed below:

### *Intergovernmental Panel on Climate Change (IPCC):*

As a leading authority on climate science, the IPCC's assessments and reports significantly influence global climate policies and carbon market mechanisms. By leveraging the IPCC's findings, this project ensures that initiatives are rooted in the latest scientific knowledge, contributing to a well-informed approach to addressing climate change.

### *Greenhouse Gas Protocol (GHG Protocol):*

The GHG Protocol is widely recognized as the most widely used framework for measuring and managing greenhouse gas emissions. Adopting the GHG Protocol enables us to comprehensively assess our projects' carbon footprints and implement effective emission reduction strategies.

### Recommended practices (PR) 2060 by ABNT:

This is a Brazilian technical standard developed by the Brazilian Association of Technical Standards (ABNT) specifically focused on carbon-related matters. This standard provides guidance for entities in Brazil to quantify, monitor, and report their greenhouse gas (GHG) emissions and removals. It aligns with international norms, including the ISO 14064 series, ensuring compatibility with global best practices in GHG accounting and reporting.

## 1.16 Double Counting and Participation under Other GHG Programs

### 1.16.1 No Double Issuance

Is the project receiving or seeking credit for reductions and removals from a project activity under another GHG program?

☐ Yes ☒ No

### 1.16.2 Registration in Other GHG Programs

Is the project registered or seeking registration under any other GHG programs?

☐ Yes ☒ No

### 1.16.3 Projects Rejected by Other GHG Programs

Has the project been rejected by any other GHG programs?

☐ Yes ☒ No

Our term of agreement alongside the landowner regards to not engaging in any other remunerated or non-remunerated activities in the Project Area, generating any credits/assets/values/rights, incurring to double-claiming and causing negative impacts on the project and parties' integrity and credibility, under penalty incurring fines.

## 1.17 Double Claiming, Other Forms of Credit, and Scope 3 Emissions

### 1.17.1 No Double Claiming with Emissions Trading Programs or Binding Emission Limits

Are project reductions and removals or project activities also included in an emissions trading program or binding emission limit? See the *VCS Program Definitions* for definitions of emissions trading program and binding emission limit.

☐ Yes ☒ No

### 1.17.2 No Double Claiming with Other Forms of Environmental Credit



Has the project activity sought, received, or is planning to receive credit from another GHG-related environmental credit system? See the *VCS Program Definitions* for definition of GHG-related environmental credit system.

☐ Yes ☒ No

### 1.17.3 Supply Chain (Scope 3) Emissions

Do the project activities specified in Section 1.12 affect the emissions footprint of any product(s) (goods or services) that are part of a supply chain?

☐ Yes ☒ No

In this case, the questions regarding Scope 3 Supply Chain emissions are not applicable. Neither the landowner nor retailer(s) of impacted goods and services, nor the project proponent, have posted any public statements indicating that 'VCUs may be issued for the greenhouse gas emission reductions and removals associated'. Additionally, our term of agreement alongside the landowner regards to not engaging in any other remunerated or non-remunerated activities in the Project Area, generating any credits/assets/values/rights, incurring to double-claiming, and causing negative impacts on the project and parties' integrity and credibility, under penalty incurring fines.

## 1.18 Sustainable Development Contributions

### 1.18.1 Sustainable Development Contributions Activity Description

The project aims to reduce deforestation and promote sustainable land use practices in Brazil's Amazon and Cerrado regions. Through REDD+ initiatives, it addresses climate change, biodiversity conservation, and socioeconomic challenges.

The project directly reduces greenhouse gas emissions by protecting forests and aligning with Brazil's climate targets. Conservation efforts preserve biodiversity, support local communities, and improve livelihoods. By generating employment, promoting inclusive practices, and fostering partnerships, the project contributes to economic growth and social well-being.

The company also is committed to allocating 5% of its own returns – generated from the sale of Verified Carbon Units (VCUs) – to fund and implement impactful social initiatives within the communities surrounding our project area. The social actions may include, for instance, initiatives such as educational support, infrastructure development, access to natural resources etc..

The project aligns with Brazil's goals of combating deforestation, conserving biodiversity, and promoting social inclusion. Monitoring and reporting mechanisms ensure compliance through using high-level technologies, a dedicated team specifically for the project, and a corporate governance structure.

### 1.18.2 Sustainable Development Contributions Activity Monitoring

**Table 8: Sustainable Development Contributions**

| Row number | SDG target | SDG indicator <sup>51</sup>                                                                                                                                     | Net impact on SDG indicator        | Current project contributions                                                                                                                                                                                                    | Contributions over project lifetime |
|------------|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|
| 1)         | 1.a        | 1.a.1: Total official development assistance grants from all donors that focus on poverty reduction as a share of the recipient country's gross national income | Implemented activities to increase | The proponent will relate all contributions to this official SDG target and indicator after the first crediting phase. As this is not a government project activity, the value will result from a project's self-defined measure | No historic yet                     |
| 2)         | 13.0       | Tonnes of greenhouse gas emissions avoided or removed                                                                                                           | Implemented activities to increase | 40,008                                                                                                                                                                                                                           | 40,008                              |
| 2)         | 15.1       | 15.1.1: Forest area as a proportion of total land area                                                                                                          | Implemented activities to maintain | 350 ha / 10,012 ha = 3,49 %                                                                                                                                                                                                      | 3,49%                               |

<sup>51</sup> SDG Indicators Metadata repository <https://unstats.un.org/sdgs/metadata/>

|    |      |                                                                                                                                                                  |                                    |                                                                                                                                                                                                                                  |                 |
|----|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| 4) | 17.7 | 17.7.1: Total amount of funding for developing countries to promote the development, transfer, dissemination and diffusion of environmentally sound technologies | Implemented activities to increase | The proponent will relate all contributions to this official SDG target and indicator after the first crediting phase. As this is not a government project activity, the value will result from a project's self-defined measure | No historic yet |
|----|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|

## 1.19 Additional Information Relevant to the Project

### Leakage Management

As expressed by the Methodology VM0015 v1.2, it is required to delineate '*leakage management zone*' as areas specifically designed to implement activities that reduce the risk of activity displacement leakage. This project has defined the productive land of the farmer proprietary as its 'Leakage Management' boundary for the following reasons:

1. **Economic Relevance:** The productive area, having historical deforestation, now holds significant economic importance for the region. The farm serves as a revenue-generating entity, contributing to the local GDP and, consequently, fostering positive externalities for the community. This dynamic ensures employment opportunities and enhances local infrastructure, for instance.
2. **Climatic and Activities Favorability:** The region's climate conditions and type of activities in the area allow for indefinite planting periods, providing substantial assurance that the area will not become unusable in the long term, as may be the case with other field activities or in other regions.
3. **Sustainable Agricultural Practices:** The agricultural activities in the area are further secured by serious soil management and sustainable grain cultivation. The farm employs rotating crop cycles, alternating between soybeans and corn, with straw cover between periods. This not only enhances carbon capture and storage in the soil but also ensures long-term soil protection and nourishment.
4. **Crop rotation between fertilized annual crops and pasture for livestock:** This refers to a system recommended for the chemical reconditioning of the soil and the production of grains and forages, which is crucial in an agricultural integration system. In addition to being an effective treatment against soil degradation, this system helps increase productive stability. Soil care is an essential factor for this (and any) AFOLU carbon project, which has an indefinite timeframe. In many cases, farmers and ranchers have to migrate periodically to other areas due to a permanent degradation of the soil. In these migrations, new forested areas need to be opened, promoting leakage to neighboring areas. The process of rotating between agricultural and livestock activities, using good practices, is a great ally to avoid the need to abandon a current productive area, as it will remain suitable for an indefinite period.
5. **Financial Feasibility:** As demonstrated throughout this PD, the financial returns from agricultural activities are highly profitable, ensuring the maintenance of economic standards for the landowner, employees, and the entire farm-related chain.
6. **Prevention of Landowner Practices:** The project explicitly restricts the landowner from using forested lands to compensate for legal reserves on other properties not registered in this project.

This measure aims to prevent leakage to remote areas that may go unnoticed due to geographical monitoring limitations.

These factors will be diligently monitored within the Leakage Management Plan. They are critical elements directly linked to activities conducted in the previously deforested portion of area, contributing to the prevention of the area's abandonment or leakage of new deforestation in the surroundings. In summary, these factors inhibit and restrict deforestation expansion within the reference area, especially.

### Commercially Sensitive Information

- Only a few pieces of information have been excluded from this public version of the PD due to commercially sensitive data that cannot be disclosed for business security reasons.
- It is important to note that none of this information is essential for building the baseline, demonstrating additionality, or estimating GHG reduction.
- The excluded information is specifically related to the profitability figures of conventional activities, potential gains from carbon credit revenue, and the comparison of these two for calculating NPV from short to long-term.
- Such information is referenced in the Cost Opportunity Costs section (Section 3.4) and in the Non-Permanence Risk Analysis of Section 4.4.
- The numerical information, along with the rationales, will be made available for validation by the designated Validation and Verification Body (VVB).

### Further Information

#### **Contractual Safeguards**

NovaAgri counts with a legal team and its own Policies to ensure the most appropriate execution of its carbon credit projects. As an essential step to firm a serious engagement between NovaAgri and the landowner is the agreement made between the parties, in a contractual formalization. This comprehensive agreement delineates the roles, responsibilities, general and specific commitments of both parties towards the development and execution of this VCS project.

The collaboration underscores the necessity for adherence to Novaagri's Compliance Policies. This includes transparency in all Project-related activities and the safeguarding of confidential information. Both parties are committed to upholding the highest ethical standards. The main three terms are summarized below:

#### *(A) Project Area integrity*

With the term, the landowner formally agrees to maintain the integrity of the area subject to the Project, making it free from deforestation, degradation, invasions, fires, or any other factors that

may pose a risk of damage or actual harm to the area. In addition, it includes adhering to the best and most correct soil management practices, aiming at the preservation and development of biodiversity, including fauna and flora. All this involves reporting any situation that deviates from normality promptly, from both parties, during the monitoring period.

*(B) No Double-counting Credits*

Another essential term of the agreement regards to not engaging in any other remunerated or non-remunerated activities in the Project Area, generating any credits/assets/values/rights, incurring to double-counting and causing negative impacts on the project and parties' integrity and credibility, under penalty incurring fines.

*(C) Ethical Governance*

Regarding the ethical governance aspects, the agreement includes compulsory the compliance with anti-corruption laws, regulations and policies, as well as determinations and rules with the aim of preventing corrupt practices, illegal expenses related to political activity, harmful acts, infractions or crimes against the financial system or public administration in order to avoid law laundering, hiding assets and financing illegal traffic and terrorism. Also, the parties agree not to do business with companies from countries that have suffered an international trade embargo according to the UN, OFAC, European Union or Japan list; and not make allowances for acts that result in discrimination based on race or gender, with disabled people or those with special needs, that involve child labor, slave labor or that characterize moral or sexual harassment or disrespect for human dignity.

## 2 SAFEGUARDS AND STAKEHOLDER ENGAGEMENT

### 2.1 Stakeholder Engagement and Consultation

*No Net Harm*

The REDD+ project led by NovaAgri aims primarily to tackle climate change by preventing carbon emissions resulting from deforestation/degradation. The positive impacts include contributions to global climate change mitigation, social livelihood benefits, biodiversity conservation, and support for sustainable ecosystem services. However, potential negative risks exist, including socioenvironmental and compliance risks, disruption of traditional economic practices in the region, challenges in land tenure clarity, cultural and educational aspects, and the risk of deforestation in unforeseen circumstances.

To address these risks, NovaAgri has implemented various measures. A thorough pre-project analysis was conducted, utilizing technology for a reliable assessment of socioenvironmental



compliance and relevant factors, alongside an Agreement Contract with the landowner. The project also considers the economic dependencies of local populations on agribusiness-related activities and demonstrates that existing cultivated lands are sufficient to sustain economic activities without compromising local jobs. Land ownership verification is prioritized to prevent phantom credits, involving a rigorous examination of possession documents, authentication in a notary office, and checks against the CAR document. The project addresses cultural and educational gaps through awareness initiatives, meetings, and distribution of informative brochures. The implementation of the MRV model and a dedicated department for carbon credit projects ensures monitoring, reporting, and verification aligned with the company's existing processes. Additionally, buffer credits are reserved to mitigate unforeseen circumstances. These measures aim to safeguard against potential risks and enhance the integrity of the project.

### 2.1.1 Stakeholder Identification

Find below the description of the stakeholder identification process.

#### Stakeholder Identification

*To identify the key stakeholders in this AFOLU project, extensive discussions were held with the Landowner and NovaAgri's regional team, which are physically present in the region. This approach ensured a more accurate understanding of the potentially impacted or related audiences in this project. In addition to these consultations with local teams, we enlisted support from NovaAgri's internal departments directly involved in credit, risk, sustainability, and compliance issues. This collaboration allowed us to comprehend which government institutions and departments could be involved in both the Carbon Project and the company's business. For instance, environmental entities regulating and overseeing rural areas of the state. Aligning the company's environmental compliance practices with the needs of the AFOLU Project was a crucial aspect. Finally, this information was validated through public and referenced sources, including IBGE census data and georeferenced information, such as updated vector files of indigenous lands (traditional communities), conservation units, infrastructure, water net flows, among others.*

*From this identification process, the following publics were identified: Brazilian Institute of the Environment and Renewable Natural Resources (IBAMA), Mato Grosso State Secretariat of the Environment (SEMA), farmers' representatives, the prefecture, the president of the*

|                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|----------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                    | <i>municipal chamber, municipal councilors, representatives from indigenous traditional communities, the local FUNAI (National Foundation for Indigenous Peoples), small-scale rural producers, and residents.</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Legal or customary tenure/access rights</b>     | <i>Following the consultation conducted and aligned with the internal due diligence process, no land ownership conflicts or land disputes involving the project area were identified. Moreover, the relationship between the landowner and neighboring indigenous lands is considered friendly. The landowner also demonstrates concern for all regularization issues with local and national environmental institutions, with no pending legal actions or other irregularities with the government.</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Stakeholder diversity and changes over time</b> | <p><i>Santa Terezinha – Mato Grosso is part of the 'Northeast of Mato Grosso's Mesoregion and the 'North Araguaia' Microregion, owing to its proximity to the Araguaia River, the main fluvial body in the region.</i></p> <p><i>1) Population: As of 2022, according to IBGE data, the population were represented by 7,596 inhabitants, with a population density of 1.17 inhabitants per square kilometer. In the population comparison with other municipalities in the state, it ranks 85th out of 141 municipalities. In comparison with municipalities nationwide, it holds the 3,497th position out of 5,570 Brazilian municipalities.</i></p> <p><i>2) Employment and Income: In 2021, the average monthly salary was 2.2 minimum wages. The proportion of employed individuals (with income) relative to the total population was only 5.8%. Considering households with monthly incomes of up to 1/2 minimum wage per person (about \$140/person/month in current values), 48.1% of the population fell into these conditions according to the 2010 census.</i></p> <p><i>3) Economy: The Municipal Human Development Index (Índice de Desenvolvimento Humano Municipal – IDHM, in Portuguese) was 0.609 according to the 2010 census. Although it is a medium rate, the municipality's GDP</i></p> |

|                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                              | <p><i>(Produto Interno Bruto – PIB, in Portuguese) ranks among the lowest in the state (95th out of 141 municipalities).</i></p> <p><i>4) Health and Environment: The most recent data (IBGE, 2020) show an children mortality rate of 8.33 deaths per thousand live births. The health-sanitarian situation is concerning, considering 2010 data, where only 43.3% of households had adequate sanitary sewage according to the census.</i></p> <p><i>5) The cultural relationship among local residents is already well-established in the project area’s surrounding. This is because the original landowners have been settled in that location for a long time, given the remote dates when forest clearings initially occurred. Local indigenous population, for example, have a certain familiarity with the current landowners, with no objections to the occupation of land neighboring the demarcated areas. On the contrary, indigenous people in the region already share cultural traits very similar to those of more recent residents.</i></p>       |
| <p><b>Expected changes in well-being</b></p> | <p><i>Firstly, the project directly contributes to combating climate change by reducing deforestation and forest degradation and promoting sustainable land use practices, which are relevant to the local population, once having better control of the micro-climate and future-climate is essential to guarantee predictability on any activities, including agriculture. Not only the climate-related changes, but also the natural resources’ preservation is essential to guarantee well-being for locals, such as good air/water/food quality.</i></p> <p><i>The implementation of this REDD+ project also generates a more rigorous level of employment. NovaAgri’s contractual agreement with the landowner explicitly incorporates clauses dedicated to upholding human rights. These provisions encompass a commitment to fair labor practices, non-discrimination, and the well-being of local communities.</i></p> <p><i>This project aims to bring substantial social benefits to local communities, addressing issues such as poverty, lack</i></p> |

|                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                 | <i>of infrastructure. By incentivizing social actions with part of the VCU's revenue, the project aims to provide economic opportunities, enhance livelihoods, and improve food security for communities that surrounds the project and reference areas.</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Location of stakeholders</b> | <i>The primary stakeholders directly impacted by the project activities are within the boundaries of the property, notably including the landowner itself and workers from the Tapirapé farm. However, in the PA's surroundings, which even coincide with the Leakage Belt, there are some stakeholders directly or indirectly affected, such as indigenous populations and residents of the municipality of Santa Terezinha. The Urubu Branco community, for instance, is adjacent to the project area, so any positive impacts will automatically extend to this area, as well as the allocation of social development revenues coming from the project – the same applies to local residents of the municipality. In this way, the relationship with the municipal council is important because it is the institution that can assist the identification of the actual social needs of the region.</i> |
| <b>Location of resources</b>    | <i>The only territory where the project activities will be implemented is the Tapirapé Farm, which is solely owned by the landowner mentioned at the beginning of this PD. All the forested area to be monitored, surveilled, and inspected is within the boundaries of this farm.</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

### 2.1.2 Stakeholder Consultation and Ongoing Communication

Find below the description of the process for and the outcomes from the stakeholder consultation conducted for the project.

|                                         |                                                                                                                                                                                                                                     |
|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Date of stakeholder consultation</b> | <i>07-February-2024</i>                                                                                                                                                                                                             |
| <b>Stakeholder engagement process</b>   | <i>In order to fulfil the proponent's commitment to transparency with its stakeholders, NovaAgri conducted visits to the localities surrounding the project area. These visits included numerous discussions with the landowner</i> |

as well as scheduled meetings with key representatives of each local stakeholder.

The main public disclosure and consultation meeting took place on February 7, 2024, in the afternoon (3:00 pm BRT), as previously agreed with the Santa Terezinha - MT prefecture. The municipality provided space within its municipal chamber for the meeting. Participation was open to all, including local citizens, as advertised via WhatsApp and social media platforms as showed on the flyer below.



Santa Terezinha-MT prefecture – meeting invitation

The meeting provided valuable insights as the project's general manager and regional trading manager, both representing NovaAgri, introduced the company, its current context, and the initiative to develop carbon projects. Emphasis was placed on climate issues, the challenges posed by carbon emissions and global climate change, and the efforts by both corporate and governmental sectors to mitigate the climate emergency. The aim was to raise awareness and disseminate information on a topic that remains largely overlooked in remote regions of Brazil. The list of present people are presented below, following this table.

Informative folders were distributed, detailing the purpose and key features of the NovaAgri's Carbon Project, presented in an accessible yet technically viewing manner. Furthermore, the environmental and social benefits of the project were highlighted for the company, landowner, and all directly and indirectly involved stakeholders, particularly the local communities, including traditional populations in the region.



Image Source: NovaAgri – Brochure of the Project ‘Did you know that preserving the forests and the native vegetation also ensures our future survival? (...)’

Additionally, a letter containing the link to the VERRA website page listing all recently listed projects in the pipeline was distributed to all present stakeholders, saying that this would be including NovaAgri's project, which would be listed furtherly.

#### Consultation outcome

At the end of the meeting, attendees were invited to the project's public consultation, where they could express their questions, opinions, suggestions, compliments, and criticisms.





*Meeting with local Stakeholders at Santa Terezinha prefecture hall*

*As a means of demonstrating transparency with the public, the event was livestreamed and publicly on the municipality's Facebook channel.*

*Below are some of the comments (questions and suggestions) that arose during the stakeholders' meeting, which we consider relevant to analyze for potentially shaping future project activities:*

*1. "Are there possibilities to develop something with small producers? What is the minimum area to generate Carbon credits?" - Through this question, we understand the curiosity and interest of small agricultural producers in participating in Carbon projects. We explained that we currently cannot make projects viable in very small areas due to financial feasibility and investment issues. However, we made it clear that the idea is to leave the door open for the inclusion of this audience, as future partnerships are very possible, including benefiting these groups through the proposed activities.*

*2. "It is important to emphasize education in the social actions that will be implemented, especially for young people, showing that it is possible to promote agriculture responsibly. Thus, the agribusiness sector can also change its image, which is often portrayed as a villain, as seen in schools."*

|                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                       | <p>3. "From what we understand, your idea is to prevent deforestation, right? We are concerned about this because it is an attempt to make those who have already deforested in the past does not deforest anymore."</p> <p>4. "We (the municipality) have the structure to help with the dissemination of campaigns. We have local media, which has a good reach. From March onwards, we also have a plan to bring cinema to the entire community. In this way, we could include the project's promotion."</p> <p>5. "Speaking of the municipality's social needs. Today, indigenous communities (pointing to indigenous leaders present) really need this investment. Investment, especially in sanitation, which is a priority, but also for cleaning or even acquiring machinery for small-scale agriculture. Another issue is the promotion of education in the short term. Training, for example. The idea of technical schools could also be interesting if there is a market to absorb the professionals. It's no use if the market is saturated and cannot absorb these recent graduates."</p> <p>6. "The municipality is in the process of implementing APAE (Association of Parents and Friends of the Exceptional), which will benefit between 40 to 50 children who need special care and treatment. Any support is welcome."</p> <p>7. "With this partnership, I think the municipality only has to gain."</p> |
| Ongoing communication | <p>The project establishes ways to maintain constant and meaningful communication among stakeholders. In addition to the direct contact the proponent maintains with the landowner, the company also relies on its regional trading team, which is physically close to the project territory and has commercial relationships beyond those of the carbon project. This proximity is crucial for maintaining a close relationship with our partner, as it enables periodic visits to the farm to monitor both production and preservation activities.</p> <p>Furthermore, the proponent is going to keep its internal channel open to all stakeholders. This channel aligns with</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

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|                                 | <p><i>the company's corporate governance of ethics and compliance, ensuring the highest level of business integrity in its relationships with stakeholders. The channel is indicated in section 2.1.4 of this PD.</i></p> <p><i>Additionally, we will provide regular and recurring communications on our social media channels for any project changes or updates, always following the format of a PD version, which will be publicly available on the VERRA Registry website. This transparency is already guaranteed by VCS projects to their stakeholders.</i></p> <p><i>Finally, the Carbon department's email addresses at NovaAgri will always be available to anyone wishing to express suggestions, questions, or general comments. The addresses are listed in section 2.1.5 of this PD.</i></p> |
| <p><b>Stakeholder input</b></p> | <p><i>The main inputs received from stakeholders during the first in-person public consultation were included in this initial version of the PD. These inputs are reflected, for example, in the monitoring plan, which is based on the intrinsic needs reported during the consultation. After the public consultation on the VERRA website, this PD may undergo further updates to address or clarify the comments received.</i></p>                                                                                                                                                                                                                                                                                                                                                                      |

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## Convite à Consulta Pública – Projeto de Carbono REDD+ NovaAgri

Prezado/a,

É com grande satisfação que a NovaAgri compartilha informações sobre nosso mais recente projeto de carbono, que representa os esforços contínuos da companhia para a promoção da sustentabilidade socioambiental nos locais em que atua.

**O projeto REDD+ (Redução de Emissões por Desmatamento e Degradação Florestal) tem como objetivo reduzir as emissões de CO<sub>2</sub> provenientes do desmatamento e da degradação florestal, oferecendo incentivos financeiros que garantem as atividades de preservação, além de retornar benefícios à comunidade e biodiversidade locais.**

Convidamos cordialmente a você participar da consulta pública para este projeto. Sua contribuição é de grande valor para nós. Junto a esta Carta estamos lhe entregando um Folder com as informações gerais do projeto, mas em breve o documento completo e detalhado estará disponível no site da Verra nos próximos meses. Para acessar, acesse o QR Code Abaixo ou acesse o site: <https://verra.org/consultations/projects-open-for-public-comment/>

Assim que divulgado, também comunicaremos em nossas redes sociais.



Caso você tenha sugestões, ou queira nos deixar qualquer comentário sobre o Projeto antes da abertura da consulta no site, entre em contato conosco pelo e-mail: [carbon.credit@novaagri.com.br](mailto:carbon.credit@novaagri.com.br)

*Contamos com sua contribuição para este projeto. A NovaAgri agradece seu apoio!*



Image source: the author – letter of invitation for local Stakeholders to participate on public consultation.



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 com ou sem fins lucrativos, no Brasil ou no Exterior.  
 Assino ciente abaixo, junto à lista de presença do atual encontro.

### LISTA DE PRESENÇA data 07/02/2024

| Nome                                 | Grupo             | Assinatura   | E-mail   |
|--------------------------------------|-------------------|--------------|----------|
| Daiana Silva Coimbra                 | Agricultura       | [Assinatura] | [E-mail] |
| Severina P. Barros                   | Camara            | [Assinatura] | [E-mail] |
| Mauricio J. da Silva                 | KAR AMBIENTAL     | [Assinatura] | [E-mail] |
| Hallan Thiago Cornel                 | SITIO 3DC         | [Assinatura] | [E-mail] |
| D. Maria Luiza Silva P. Antunes      | Camara            | [Assinatura] | [E-mail] |
| Antonio Rodrigues de Carvalho Junior | Camara            | [Assinatura] | [E-mail] |
| Walter Paulo Galvão                  | Embrapa           | [Assinatura] | [E-mail] |
| Edinaldo Lourenço de Jesus           | FUNRI/CT/STE      | [Assinatura] | [E-mail] |
| Thamara Karayo                       | Calique Alderikha | [Assinatura] | [E-mail] |
| TOAO EVILSON SANDES                  | VIP ARAGUAIA      | [Assinatura] | [E-mail] |
| EDIVAN TEIXEIRA DOS REIS             | PROD. RURAL       | [Assinatura] | [E-mail] |
| Edixon Pablo T. de la Cruz           | Camara            | [Assinatura] | [E-mail] |

Document source: the author – list of present people at the meeting (page1)



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 com ou sem fins lucrativos, no Brasil ou no Exterior.  
 Assino ciente abaixo, junto à lista de presença do atual encontro.

| Nome                        | Grupo  | Assinatura   | E-mail   |
|-----------------------------|--------|--------------|----------|
| Almirante Borges dos Santos | Camara | [Assinatura] | [E-mail] |
| FRANCISCO A. DE ANDRADE     | Camara | [Assinatura] | [E-mail] |
| Cláudia Maria de Souza      | Camara | [Assinatura] | [E-mail] |
|                             |        |              |          |
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Document source: the author – list of present people at the meeting (page2)

### 2.1.3 Free Prior and Informed Consent

|                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Obtaining consent</b> | <p>Consent was obtained from any key stakeholders. Especially the landowner, who is the primary stakeholder in terms of providing the land and facilitating the project implementation, is fully informed and in agreement regarding their participation, role, and responsibilities within the project. This was achieved by the proponent through multiple rounds of discussions and negotiations aimed at providing a general explanation of the project and its objectives, timeline, conditions, projections, and shared benefits. This stage ran parallel to all the documentary analyses mentioned earlier regarding property ownership and cross-referencing with all public boundaries to mitigate any risk of conflicts or false information.</p> |
| <b>Outcome of FPIC</b>   | <p>The entire consensus-building process between the landowner and the proponent resulted in a legally binding contract validated and signed by both parties, affirming through a formal document all the conditions accepted after the initial discussions. This project provides assurance that its implementation has not encroached on land, relocated people without consent, and forced physical or economic displacement, and all necessary evidence will be provided throughout the validation process alongside the VVB.</p>                                                                                                                                                                                                                       |

### 2.1.4 Grievance Redress Procedure

|                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Development process</b> | <p>For issues related exclusively to the relationship between the proponent and the landowner, the methods, tools, periods, and procedures for resolution are outlined in the legal binding contract signed between the parties. These issues are limited to pre-agreed matters, primarily concerning the rules for the use of property areas and forest preservation.</p> <p>For all other types of grievances, the proponent is providing internal channels for receiving comments, feedback, and suggestions. For inquiries, suggestions, and general feedback, NovaAgri offers a direct channel to its</p> |
|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|



|                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                    | Carbon department via email at <a href="mailto:carbon.credit@novaagri.com.br">carbon.credit@novaagri.com.br</a> . Furthermore, for compliance issues, the company provides its corporate channel accessible via phone at +55 0800 721 5954 or through the web portal: <a href="http://www.canaldedenuncias.com.br/novaagri">www.canaldedenuncias.com.br/novaagri</a> . The proponent will also make available a website portal exclusively to the NovaAgri's projects, which is previewed to be launched during the first period of monitoring. There will be specific boxes to contact the proponent about any grievance or project issues related. |
| <b>Grievance redress procedure</b> | <p>- Upon receiving any grievances from the landowners or any stakeholder related to them, the proponent will ensure to fulfill all conditions and deadlines outlined in the signed formal contract, with support of the proponent's legal team.</p> <p>- Upon receiving any grievances from other stakeholders, the proponent will take responsibility for providing a immediate response informing an appropriate timeframe to provide the best possible resolution, with all company departments available to provide this support.</p>                                                                                                           |

In case of any further grievances that might raise during the monitoring period, the proponent will report it in the table below, with the proper steps to give a proper resolution. This PD will update its version accordingly to new grievances received during each monitoring report.

| Grievances received                                                    | Resolution and outcome                                                                                |
|------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| <i>To summarize the grievance raised during the monitoring period.</i> | <i>To describe the steps taken to resolve the grievance including the outcomes of the resolution.</i> |

### 2.1.5 Public Comments

Verra encourages public comments on the compliance of projects with program requirements, and these comments will be publicly disclosed on the Verra Registry. All the projects may be found on the webpage: <https://verra.org/consultations/projects-open-for-public-comment/>

In addition to the public comment period on VERRA, this project maintains internal channels for receiving comments, feedback, and suggestions. For inquiries, suggestions, and general feedback, NovaAgri provides a direct channel to its Carbon department via email at

carbon.credit@novaagri.com.br. Furthermore, for matters related to compliance issues, the company offers its corporate channel accessible via phone at +55 0800 721 5954 or through the web portal: [www.canaldedenuncias.com.br/novaagri](http://www.canaldedenuncias.com.br/novaagri). This multi-faceted approach ensures transparency and accessibility for stakeholders to engage with the project.

All pertinent points and feedback raised through these channels will be thoroughly addressed and, when applicable, incorporated into this PD (following the table model below), verification reports or other suitable communication channels. NovaAgri is committed to keep its stakeholders informed of the project's progress and any adjustments made related to the Project. This collaborative approach, both internally and through the Verra public comment process, reinforces our commitment to accountability, integrity, and the overall success of the project.

**Table 9. Table model for comments received on Public Consulting**

| Comments received                  | Actions taken                                                                                                                             |
|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Summary of comment received</i> | <i>To be provided a summary of actions taken and any project design updates or justify why updates were not necessary or appropriate.</i> |

## 2.2 Risks to Stakeholders and the Environment

As part of NovaAgri's Sustainability Policy, and as standard procedure for its general businesses, the Credit Department conducts a meticulous analysis of sensitive and critical environmental points related to the products' origin areas. This includes assessing potential issues such as regarding IBAMA/SEMA Embargoes, debts to IBAMA, Soy Moratorium, and Conservation Units. These criteria are also replicated to the assessment and selection of viable landowners and its areas for a Carbon Project with NovaAgri, added to additional specific criteria (REDD+ and VCS/AFOLU criteria).

This comprehensive evaluation underscores our commitment to environmental responsibility, ensuring that the sourcing of grains aligns with legal, environmental, and conservation standards. Finally, all these criteria are replicated in the preliminary selection of eligible landowners entering a partnership with NovaAgri for Carbon projects. While some criteria are agribusiness specific, NovaAgri values their application for a secure and integral evaluation also in its carbon projects. Thus, the assessment is complemented with specific analyses, such as evaluations of boundaries' delimitation, phytophysiology types, ecosystem biodiversity, carbon stocks, higher resolution satellite analyses, and on-site assessments for data confirmation. NovaAgri remains dedicated to fostering a transparent, responsible, and environmentally conscious approach in all its services' endeavors and portfolios.

Table 10. Risks Identified for Stakeholders and the Environment

|                                    | Risks identified                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Mitigation or preventative measure taken                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Risks to stakeholder participation | <p>1) Risk of including properties and respective landowners with IBAMA/SEMA (environmental government institutions) Embargoes – which are primarily a penalty to halt the actual or potential degradation activity and, in the case of infractions against the flora, facilitate regeneration to the state prior to degradation.</p> <p>2) Risk of including or keeping properties and respective landowners with debts to IBAMA (environmental federal institution).</p> <p>3) Risk of including farmers with irregularities on Soy Moratorium Program – A environmental pact reached between the representative entities of soybean producers in Brazil, environmental NGOs, and later the government itself, to foresee the adoption of measures against Amazon deforestation.</p> <p>4) Risk of including stakeholders illegally</p> | <p>1) The proponent crosses the list of farmers´ IDs (in Brazil, called CPF/CNPJ) against the public list of embargoed areas provided by Federal environmental entities (IBAMA/SEMA). Then, spatial embargoes within the georeferenced coordinates of these areas are scrutinized. The evaluation criteria involve restricting grain purchases from farms within the same municipality or within a 100 km radius of the embargoed area, even if the CPF/CNPJ is not on the list. Ongoing monitoring and remote sensing further inform decision-making, with automatic internal system checks for real-time verification.</p> <p>2) To verify indebtedness, the proponent obtains a Negative Debts Certificate from IBAMA using the same ID (CPF/CNPJ) through the SICAFI system. Purchases or financing of products from farms with no indicated debts are approved. In cases of debt indication in a federative unit, the Credit Department investigates the severity and status of the process,</p> |

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <p>occupying Conservation Units public territories.</p> <p>5) Risks regarding disruption of traditional/conventional practices in the region – economic dependencies on agribusiness-related activities may be impacted, necessitating careful consideration of the socio-economic consequences.</p> <p>6) Risks related to losing project’s integrity and reputation, affecting the related stakeholders – implementing and enforcing conservation measures must have sufficient resources and capabilities for a long period of monitoring and reporting. Otherwise, this can lead to potential gaps in achieving the desired environmental outcomes or even causing greenwashing issues.</p> | <p>deciding on approval based on the gravity of the infraction.</p> <p>3) NovaAgri consults the ANEC (National Cereal Exporters Association, in Brazil) periodically released list to verify if the grain's origin area is listed under the Soy Moratorium. Cross-referencing the producer's and landowner(s)' names with the georeferenced shapefile of Moratorium areas prevents purchases from listed CPF/CNPJs or overlapping areas. Regular updates through the NovaAgri’s internal system ensure ongoing compliance.</p> <p>4) Geospatial analysis is conducted to identify restrictive conservation units within the georeferenced coordinates of the analyzed area. If overlaps occur with restrictive units, such as Biological Reserves, Ecological Stations, National and State Parks, Private Natural Heritage Reserves, National and State Forests, and Extractive Reserves, limits are imposed on CPF/CNPJ purchases only for the overlapping farm. Non-restrictive units, allowing sustainable agricultural use, are also considered.</p> |
|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

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|  |  | <p>5) In many rural areas, the livelihoods of local populations are intricately tied to the jobs generated by agricultural activities. This extends beyond the farms themselves, influencing the local economy broadly, including services, commerce, manufacturing of resources and equipment among others. In some cases, disrupting plans for opening new lands could potentially pose a threat to employment maintenance within the local chain. However, the data collected in this project demonstrates that the goods produced and the financial returns from consolidated cultivated lands are sufficient to sustain gainful activities for an extended period without compromising local jobs or the economy. This is because the established crop activities demonstrate to be very profitable. In addition, the activities conducted by the farmer (landowner) indicate responsible farming practices such as crop rotation and a good soil treatment. Such practices ensure enduring activities, eliminating the need for opening new lands periodically.</p> |
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|  |  | <p>Furthermore, this REDD+ Project itself aims to return new revenues for the farmer, allowing it to reinvest in their assets and workforce. This provides greater security for maintaining economic standards within the property. As a portion of the returns will be allocated to social initiatives, this commitment extends to the surrounding communities as well.</p> <p>Ultimately, the actions of this project safeguard against the 'leakage' of deforestation in the reference region. The activities alone prove to be sufficient on preserving the local economy, offering an extra protective shield against potential economic decline in the productive area.</p> <p>6) The expected results can only be guaranteed if the MRV (Monitoring, Reporting, and Verification) model is well-executed in the project. Ensuring a no-net-harm outcome for the environment and biodiversity depends on well-conducted monitoring, utilizing cutting-edge modern technology, aligned with strengthened on-site surveillance and local engagement. All of these</p> |
|--|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|



|                           |                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
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|                           |                    | <p>aspects will be provided through the proposed initiatives in this Project Design (PD) and the financial incentives derived from the generated credits.</p> <p>With a dedicated Carbon Credit Project development department within NovaAgri, a rigorous routine of monitoring and reporting will be established, aligning with the company's existing processes for its core business. Corporate goals will be assigned to the projects, along with verification and communication routines, and a technical and operational framework from various departments, providing constant support to the project, for instance the legal department. All of these measures ensure greater security and transparency, enabling the project to be conducted in the most accurate and refined manner over time.</p> |
| <b>Working conditions</b> | No risk identified | <p>The project does not involve any risk conditions for workers directly or indirectly related to the project, except for the risks associated with the routine conventional work of each party's main business, which are minimized through training,</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

|                                                                                                          |                    |                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|----------------------------------------------------------------------------------------------------------|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                          |                    | guidance, and continuous monitoring.                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Safety of women and girls</b>                                                                         | No risk identified | The project does not involve any activities that expose women of any age to any type of risk. As previously mentioned, the proponent takes proper care regarding publicly disclosed lists of worker mistreatment by the government and also maintains a very serious internal policy on this topic. Moreover, the proponent provides its channels for reporting any kind of issues, concerns, and complaints.                       |
| <b>Safety of minority and marginalized groups, including children</b>                                    | No risk identified | The project does not involve any activities that expose vulnerable groups of people of any age to any type of risk. As previously mentioned, the proponent takes proper care regarding publicly disclosed lists of worker mistreatment by the government and also maintains a very serious internal policy on this topic. Moreover, the proponent provides its channels for reporting any kind of issues, concerns, and complaints. |
| <b>Pollutants (air, noise, discharges to water, generation of waste, release of hazardous materials)</b> | No risk identified | The project does not involve any activities that generate or exacerbate environmental issues related to pollutants of any kind, as all implemented tools and methods do not involve the use of combustion                                                                                                                                                                                                                           |

|  |  |                                           |
|--|--|-------------------------------------------|
|  |  | of fuels, waste generation, or effluents. |
|--|--|-------------------------------------------|

## 2.3 Respect for Human Rights and Equity

### 2.3.1 Labor and Work

|                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
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| Discrimination and sexual harassment | <p><i>NovaAgri, as the proponent and developer of the project, adheres to a rigorous Compliance Policy expressed through its LATAM Code of Ethics from its Group. This project will be entirely governed by this instrument, and therefore, in all its activities, it will not tolerate any form of discrimination.</i></p> <p><i>Also, NovaAgri counts on a <b>Moral Harassment and Sexual Harassment Booklet</b> to help understanding what harassment is and thus make each of us a multiplier of awareness-raising actions on the topic. The purpose is to continuously train each of the direct and indirect employees, to not only prevent situations of moral and sexual harassment from occurring within stakeholders, but also to help friends and family members who may be suffering from it. The official channels for reporting any cases of harassment are presented on this booklet.</i></p> |
| Management experience                | <p><i>NovaAgri has involved its corporate business departments to support the technical team with a right approach recommendation, as well as to elaborate specific clauses within the agreement between the proponent and the landowner explicitly incorporating human rights concerns.</i></p> <p><i>These provisions encompass a commitment to fair labor practices, non-discrimination, and the well-being of local communities. NovaAgri ensures compliance with local labor laws and international human rights standards, fostering an environment where all individuals involved in the project, including the landowner's employees are treated with dignity and respect.</i></p>                                                                                                                                                                                                                  |

|                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|---------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                         | <p><i>The communication department has also contributed with a proper approach incorporated on any graphical material, such as posts on LinkedIn, or paper brochures.</i></p> <p><i>The technical teams also contemplate experienced professionals with sustainability background, which is an important factor when conducting in-person (e.g. meetings) or remote activities (e.g. monitoring, reporting etc.) that need a proper approach among different publics, such as local communities, government agencies, and other stakeholders.</i></p> <p><i>All teams are aware and concerned to considering all aspects of a good and respectful partnership.</i></p> |
| <b>Gender equity in labor and work</b>                  | <p><i>The global policy from NovaAgri's Group has expressed DE&amp;I (Diversity, Equity &amp; Inclusion) efforts centered on the following three themes<sup>52</sup>. 1. Opportunities for diverse human capital; 2. Work-style improvements and work-life balance; 3. Fostering a corporate culture that leverages diversity and changes people's mindset. This reflects directly on the daily actions and decisions of the proponent<sup>53</sup>, including the activities and management of this carbon project, and shared with the landowner – which is the main stakeholder of this partnership, which also employs dozens of workers.</i></p>                  |
| <b>Human trafficking, forced labor, and child labor</b> | <p><i>As part of NovaAgri's Sustainability Policy, and in alignment with its standard operating procedures across its diverse business operations, the Credit Department meticulously analyzes sensitive and critical social factors related to the origin areas of its products.</i></p> <p><i>This comprehensive assessment encompasses potential issues related to Indigenous Areas and the Registry of Employers who have subjected workers to conditions analogous to slavery. These criteria are also extended to the evaluation and selection of viable landowners and</i></p>                                                                                  |

<sup>52</sup> <https://www.toyota-tsusho.com/english/sustainability/ttc/business/case06.html>

<sup>53</sup> [https://www.linkedin.com/posts/novaagri\\_8-de-mar%C3%A7o-dia-da-mulher-o-dia-internacional-activity-7171914789990850560-63Ky?utm\\_source=share&utm\\_medium=member\\_desktop](https://www.linkedin.com/posts/novaagri_8-de-mar%C3%A7o-dia-da-mulher-o-dia-internacional-activity-7171914789990850560-63Ky?utm_source=share&utm_medium=member_desktop)

their areas for a Carbon Project with NovaAgri, supplemented by additional specific criteria in accordance with REDD+ and VCS/AFOLU standards.

*Indigenous Lands:* As a strategy for verification, the Credit Department utilizes a geospatial information system to check the presence of official indigenous lands within the georeferenced coordinates of the analyzed areas. In the event of intersections with indigenous lands, the purchase limit for that area's owner is denied.

*Registry of Employers who have subjected workers to conditions analogous to slavery:*

During the registration process, the Credit Department checks the name and ID (In Brazil, called CPF/CNPJ) of the producer against the list of employers who have subjected workers to conditions analogous to slavery, sporadically released by the Brazilian Ministry of Labor. Approval decisions are based on the following criteria:

- If the name and CPF/CNPJ are on the list, the registration is denied.
- If the CPF/CNPJ is listed in the automatic system check, every time a contract is generated.

This comprehensive evaluation underscores our commitment to social responsibility, ensuring that the sourcing of grains aligns with social standards and good practices. Finally, all these criteria are replicated in the preliminary selection of eligible landowners entering a partnership with NovaAgri for Carbon projects. Thus, the assessment is complemented with specific analyses, such as local public consultation, engagement and educational actions, interview with local employees, study of local investments' possibilities, among others. NovaAgri remains dedicated to fostering a transparent, responsible, and socially conscious approach in all its services' endeavors and portfolios.

## 2.3.2 Human Rights

Our term of agreement properly signed with the landowner regards to recognize, respect, and promote the protection of stakeholders' rights line with applicable international conventions and law. Find below specific clauses from that agreement:

(...)

3.1. *It is the responsibility of the Parties together to:*

(i) *Comply with NOVAAGRI's Compliance policies and law related to anti-corruption, the anti-trust laws, trade embargoes, laws that prohibit slave labor, child labor, any type of discrimination, money laundering, terrorism, unfair competition, and breach of human rights.*

(...)

3.2. *When marketing carbon credits to third parties, the Parties shall pay attention to the status, practices, and reputation of the buyer in order to preserve the reputation of the Parties, as well as of the Project, and shall verify in advance whether the buyer:*

(i) *Complies with anti-corruption laws and regulations, as well as determinations and rules to prevent corrupt practices, illegal expenses related to political activity, harmful acts, breaches or crimes against the financial system or public administration, in order to avoid money laundering, concealments of goods and financing of illegal traffic and terrorism;*

(ii) *does not intend to do business with companies from countries that have suffered an international trade embargo according to the UN, OFAC, European Union or Japan list; and*

(iii) *does not respond for acts that matter in discrimination of race or gender, disabled people or people with special needs, is involved in exploitation of child labor, slave labor or attitude that characterize moral or sexual harassment and disrespect for human dignity.*

Also, the proponent's LATAM Code of Ethics ("Code") – which applies to all its employees, directors, advisors, service providers, agents, and suppliers, regardless of hierarchical level – has a main concern on the Principle of Inviolability of Personal Information and Human Dignity, which involves:

- respecting individuals, therefore, not tolerating any abuse of human rights and preserving the inviolability of information and personal data of clients, suppliers, employees, visitors, and any individuals that the Company may have access to in the course of its activities.
- not engaging in or being involved in any business that goes against human dignity.
- not discriminating against any individual based on race, color, sex, religion, or nationality.
- not engaging in activities that exploit child labor, human trafficking, or any other form of forced or involuntary labor.



- not tolerating any form of harassment.
- not dealing with any individuals engaged or involved in any violations of human rights.

All these aspects and rules are going to be directly internalized into the ARAGUAIA REDD+ BOCALON PROJECT, extending the initial public to other stakeholders eventually involved with the project activities.

### 2.3.3 Indigenous Peoples and Cultural Heritage

As a strategy for verification, the proponent's Credit/Risk and Sustainability Department utilizes a geospatial information system to check the presence of official indigenous lands within the georeferenced coordinates of the analyzed areas. In the event of intersections with indigenous lands, the business and/or commercial permission for that property is denied. This control is carried out through geo-monitoring systems, by using satellite images, applied to 100% of the commercial contracts carried out at NovaAgri. In addition to the use of geospatial technology, a series of documentary analyses are also conducted because it is essential to understand whether such areas are subject to complaints, embargoes, or any other type of irregularity with government institutions. Once any critical cases are identified, the project is immediately stopped, and the proponent does not proceed to any subsequent phases. These actions are part of the NovaAgri's rigorous Socioenvironmental Policy, aligned with other company's policies, such as the Ethic Code.

In addition, the proponent considers direct engagement with the traditional communities in the region as essential. As mentioned earlier in this section, contact with indigenous populations in the area is established directly, both through the landowner and NovaAGri's regional team in Mato Grosso, as well as through the project's technical teams during occasional on-site visits. This contact is crucial for fostering partnerships in protecting the project area and leakage belt, exchanging knowledge about climate-related issues and forest preservation, and especially for understanding the main social or environmental needs that could be addressed or alleviated through potential income generation coming from the revenues generated by the respective VCUs. All forms of communication are guided to occur based on the company's ethical principles and formal policies, but after all in a respectful manner, without infringing upon any history, ancestral heritage, or intrinsic culture of a people.

### 2.3.4 Property Rights

#### Rights to territories and resources

*The right to access the property and its forested areas is fully granted to the project proponent, in accordance with the landowner through a legally binding contract, where official documents, the project area in hectares, and other pertinent information regarding the land use for project activities implementation are specified, including*

|                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                    | <i>permission for full access by the proponent for visits and/or remote monitoring.</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Respect for property rights</b> | <p><i>The accurate verification of land ownership for the project is of priority importance, as the accounting and claiming of credits must be meticulously tied to genuine land ownership in order to prevent the issuance of phantom credits or credits attributed to public areas. This risk is effectively mitigated through a rigorous examination of the owner's possession documents, which are signed and authenticated in a notary (registry) office, complete with coordinates and various other official documentation. This assessment is conducted through an internal audit at NovaAgri. In addition, the CAR document (Rural Environmental Registry, in English) is also requested and verified. Although it is self-declared, CAR is a legal requirement imposed by the government, serving the specific purpose of registering and monitoring the environmental aspects of the property in question. Once done, the area can proceed to subsequent evaluations.</i></p> <p><i>Finally, a contractual binding agreement is established alongside the landowner, reaffirming its commitment to providing accurate information regarding land ownership and ensuring the periodic update of this information with NovaAgri. This comprehensive approach ensures the credibility and integrity of the land tenure information throughout the project.</i></p> |

### 2.3.5 Benefit Sharing

|                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|--------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Process used to design the benefit sharing plan</b> | <p><i>The process was based on several steps that occurred prior to choosing the partnering landowner for the project. NovaAgri, the proponent, held internal meetings within the company's board to define the strategy for sharing future revenues coming from the carbon projects. It was determined that NovaAgri would not retain the majority of these revenues since its strategic business vision focuses on adding value to its products and services rather than solely profiting from credit sales. As the landowner is the primary strategic partner in the project, being the holder of</i></p> |
|--------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|----------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                    | <p>forested areas, it was decided that they would receive the largest portion of this division. Another group of stakeholders of essential importance are the local community members in the surrounding areas, to whom a fixed and guaranteed fraction of this division will be allocated for the purpose of returning social development benefits and well-being to those who are part of the regional context.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Summary of the benefit sharing plan                | <p>The first step in executing the plan was to ensure specific clauses regarding the distribution of carbon credits within the legally binding contract. Thus, for this project specifically, it was defined and signed with the landowner as follows:</p> <ul style="list-style-type: none"> <li>(i) The OWNER will be entitled to seventy percent (70%) of the remaining carbon credits after discounting the BUFFER.</li> <li>(ii) NOVAAGRI will be entitled to twenty-five percent (25%) of the remaining carbon credits after discounting the BUFFER. <ul style="list-style-type: none"> <li>a. The percentage of NOVAAGRI above is the number of carbon credits that can be traded.</li> </ul> </li> <li>(iii) The remaining five percent (5%) of carbon credits after discounting the BUFFER will be intended for the development of a social project to be jointly defined by the Parties.</li> </ul> |
| Approval and dissemination of benefit sharing plan | <p>As the sharing agreements involve formal negotiation issues, all relations regarding the divisions were defined in a formal manner through a signed legal binding contract. However, it is understood between the parties that the revenues generated should preferably be reinvested within the project activities, as the idea is that the project returns benefits in other ways, such as reputation and added value to the businesses, without causing financial harm to any of the parties. Regarding third-party stakeholders, such as local communities, specific numerical values were not initially communicated, as a more concrete scenario would only be more concrete after the first crediting</p>                                                                                                                                                                                           |

|                                                     |                                                                                                                                                                                                                                                                                                                                                                                               |
|-----------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                     | <i>period. This is a way to avoid generating expectations that are not yet factual, although feasible. However, throughout the visits and public consultation, the social benefits were presented transparently and directly to every present group.</i>                                                                                                                                      |
| <b>Benefit sharing during the monitoring period</b> | <i>The expectation is that the division of social benefits will be applied only after the first crediting period and also after the initial negotiations of VCUs. However, during the monitoring period, efforts will continue to establish partnerships with individuals and groups which directly represent local communities, especially those in conditions of greater vulnerability.</i> |

## 2.4 Ecosystem Health

In the context of AFOLU-Specific Safeguards, this project focusing on REDD+ seeks mainly to mitigate climate change by preventing carbon emissions resulting from deforestation. Thus, the main impact represents positive externalities when considering the whole Net system. However, like any large-scale project, it might comprehend a spectrum of potential impacts, encompassing environmental and socio-economic dimensions.

The positive aspects of the project are evident in its contribution to global climate change mitigation, social livelihood benefits (and consequent inequality reduction), and other positive externalities not directly observed. By conserving native forests, the initiative helps sequester carbon dioxide, fostering biodiversity conservation and supporting sustainable ecosystem services. Local communities may benefit from strengthened livelihoods through sustainable forest management practices and consequent financial returns from that. Additionally, the project has the potential to bolster Brazil's commitment to international climate goals.

**Table 11. Risks Identified for the Ecosystem**

|                                               | <b>Risks identified</b>                                                                                                                                             | <b>Mitigation or preventative measure taken</b>                                                                                                                                                                                         |
|-----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Impacts on biodiversity and ecosystems</b> | 1) Risks regarding areas with vegetation being deforested or degraded without motivation from the activities or owners of this Project, meaning it may occur due to | 1) For occurrences of unforeseen circumstances, the Project will set aside Buffer credits based on the risk percentages assessed in the VERRA's 'Non-Permanence Risks' tool. The pooled buffer account functions as a reservoir of non- |

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <p>unforeseen circumstances. This includes, for instance, natural wildfires, neighboring land fires that extend into the project area, or even environmental crimes occasioned by external agents, that are challenging to prevent or mitigate.</p> <p>2) Risks regarding cultural aspects about an actual conservation as a challenge, once there are no deep discussions or educational activities about these topics, especially in rural areas.</p> | <p>tradable credits, serving as a type of insurance reserve for all VCS AFOLU projects. In the event of unforeseen circumstances such as fire, disease, or external encroachment leading to the loss of carbon stocks in an individual project, buffer credits may be deducted from the buffer account to offset the loss. Over time, credits from the buffer pool may be reinstated to projects where risks have been effectively mitigated, promoting improved management practices. This procedure and the results of the tool are detailed in subsequent sections of this Project Design (PD).</p> <p>2) Cultural and educational gaps pose a distinct challenge across various classes, regions, and scales in Brazil. However, when it comes to climate awareness, this seems to be an even more distant topic, especially in rural areas. As the project's activities are intrinsically connected to climate issues, it is crucial that those involved in the project are guided to act in accordance with correct biases and appropriate environmental principles. For this reason, NovaAgri encourages farm audiences and other local stakeholders, such as the local community, through project conversations/presentations and environmental awareness initiatives. In addition, to strengthen engagement in the</p> |
|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|                                          |                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|------------------------------------------|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                          |                    | project, informative brochures with simple and accessible language have been created, explaining climate emergencies, Carbon definitions, REDD+, and other project details. These brochures have been distributed to farm and NovaAgri workers, as well as the local population, during in-person meetings.                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Soil degradation and soil erosion</b> | No risk identified | <p>No risks of soil degradation or erosion resulting from the project activities were identified. Some evidence for this absence includes:</p> <p>1) Activities are based on forest conservation, so there will be no invasive methods in the field, and the monitoring activities mostly utilize technology.</p> <p>2) The area is quite flat, as observed in the elevation map presented earlier, and does not pose risks of erosion or critical landslides.</p> <p>3) The productive areas used for agriculture and livestock follow sustainable practices, such as activity rotation (agriculture-pasture) and crop rotation (for example, soybean-corn-straw), which determine the proper land use and mitigate the risks of degradation affecting the project area.</p> |
| <b>Water consumption and stress</b>      | No risk identified | No risks involving water stress were identified, as the region benefits from an abundance of water without extensive periods of drought, as presented in section 1.14 (Conditions Prior to Project Initiation) under 'Water'.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |



|                      |                    |                                                                                                                                                                                                                                                                                                                                                        |
|----------------------|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Usage of fertilizers | No risk identified | No direct risks involving fertilizers were identified, as project activities are based on conservation rather than carbon capture through perennial crops or any other types of planting. It is important to mention that activities may occasionally use fertilizers, but with methods of targeted applications that do not extend to forested areas. |
|----------------------|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|



## Sobre o Projeto

Você sabia que ao **preservar as florestas e a vegetação nativa**, estamos também garantindo a nossa sobrevivência no futuro?

Isto porque as florestas desempenham um papel crucial na regulação do clima devido à sua grande capacidade de absorver e estocar Carbono – que, quando liberado na atmosfera, causa o que chamamos de Efeito Estufa: o principal vilão do Aquecimento Global.

No contexto dos **projetos REDD+ (Redução de Emissões por Desmatamento e Degradação Florestal)**, o objetivo é reduzir as emissões provenientes do desmatamento e da degradação florestal, oferecendo incentivos financeiros que garantem os processos de preservação, além de retornar benefícios à comunidade e biodiversidade locais.

Dessa forma, os projetos REDD+ não apenas contribuem para a mitigação das mudanças climáticas, mas também promovem o **desenvolvimento sustentável e a preservação da rica diversidade biológica que as florestas abrigam**. Investir na conservação das florestas é, portanto, um investimento no nosso próprio futuro, bem-estar local e equilíbrio ambiental do planeta.

## Qual o padrão de certificação utilizado?

**Verified Carbon Standard**  
A VERRA STANDARD

A NovaAgri adota o padrão de certificação VERRA, a maior e mais consolidada certificadora do mundo, além de contar com um idôneo processo de Auditoria. Tudo isso reflete o compromisso da NovaAgri com a excelência em suas práticas de sustentabilidade, garantindo a integridade de seus projetos e serviços.

**NOVAAGRI**  
TOYOTA TSUSHO GROUP

## PROJETO DE CARBONO REDD+

NOVAAGRI & FAZENDA BOCALON

**Região:** Mato Grosso

**Município:** Santa Terezinha e arredores

**Preservação de:** +17 mil hectares de vegetação

**Biotopo:** Transição Amazônia/Cerrado

**Mais de:** 30 mil toneladas de CO<sub>2</sub> a menos na atmosfera por ano

Source: NovaAgri – Content of the brochure (part1)



Source: NovaAgri – Content of the brochure (part2)

## 2.4.1 Rare, Threatened, and Endangered species

Is the project located in or adjacent to habitats for rare, threatened, or endangered species?

☒ Yes

☐ No

|                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Species and habitat                                               | As presented in the Section 1.14 (Conditions Prior to Project Initiation) of this PD, the project area is located in an important region of rich biodiversity and rare characteristics of ecological corridors, which allows permanence and facilitation for a series of fauna and flora species. This can be demonstrated that the project will not adversely impact those habitats since there are no invasive, degrading, or any malefic externalities coming from its activities – as also shown in the Section 1.12 (Description of the Project Activity). At the very contrary, the project aims to bring factual benefits for the ecosystem and any species living in it. |
| Amazon Forest (biome)                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Cerrado Savanna (biome)                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Endemic, endangered, threatened or rare species (flora and fauna) |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

### 2.4.2 Introduction of species

This section is N/A because this project does not have any planting or species introduction.

### 2.4.3 Ecosystem conversion

The project area contains only areas which were defined as “forest” in 2012 (more than 10 years prior to the project start date), as described in the forest cover benchmark maps in the section 1.13 (Project Location). That means, the project area’s ecosystem has been preserved as its natural state, with no conversion activities.

## 3 APPLICATION OF METHODOLOGY

### 3.1 Title and Reference of Methodology

**Table 12. Methodology's References**

| Type<br>(methodology,<br>tool or module). | Reference ID, if<br>applicable | Title                                                                                                                                       | Version                                                                                 |
|-------------------------------------------|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| <i>Methodology</i>                        | VM0015                         | <i>Methodology for Avoided Unplanned Deforestation</i>                                                                                      | <i>version 1.2.<br/>published<br/>on 18-<br/>December-<br/>2023</i>                     |
| <i>Tool</i>                               | VT0001                         | <i>Demonstration and Assessment of<br/>Additionality in VCS Agriculture, Forestry<br/>and Other Land Use (AFOLU) Project<br/>Activities</i> | <i>v3.0,<br/>Sectoral<br/>Scope 14,<br/>published<br/>on 01-<br/>February-<br/>2012</i> |
| <i>Tool</i>                               | -                              | <i>AFOLU Non-Permanence Risk Tool</i>                                                                                                       | <i>VCS Version<br/>4.2,<br/>published<br/>on 12-</i>                                    |

october-  
2023

This project applies the VM0015 v1.2, methodology by VCS (VERRA), specifically categorized as AFOLU REDD+. Adhering to the robust standards of the VERRA certifier, in its latest version, the project ensures heightened integrity and credibility.

Furthermore, it aligns with the recent updates of the "New Era for VERRA" regarding REDD+ methodologies, positioning itself within a thoroughly revised framework implemented by a body of qualified scientists and responsive to diverse feedback from stakeholders globally. The adoption of the VM0015 in its updated version as of December 2023 is also an important transitional step towards full compliance with the newly launched Consolidated VCS REDD+ methodology, which ensures higher integrity, usability, and consistency with market best practices (including the ICVCM's Core Carbon Principles [CCPs]) as well as latest scientific evidence. In addition, the selection and definition of the baseline are constructed to align with a jurisdictional baseline framework, (e.g. in a State level in Brazil).

The project's interests are also linked to the goals of aligning and amplifying global and national emission reduction targets, for instance contributing to the fulfillment of nationally established objectives (NDCs).

### 3.2 Applicability of Methodology

The project's alignment with the applicability criteria underscores its commitment to preventing unplanned deforestation (AUD) and its associated GHG emissions. The applicability conditions of the methodology VM0015 and their respective relation to the project are detailed below:

**Table 13. Methodology's Applicability**

| Methodology ID | Applicability condition                                                                                                                                                                                                                                               | Justification of compliance                                                                                                                                                                                                                                             |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| VM0015         | <i>Baseline activities may include planned or unplanned logging for timber, fuelwood collection, charcoal production, agricultural and grazing activities as long as the category is unplanned deforestation according to the most recent VCS AFOLU requirements.</i> | <i>The project applies to preventing unplanned deforestation caused by agricultural activities, in accordance with the most recent VCS AFOLU guidelines. The project is situated within the dynamic MATOPIBA region, encompassing the Brazilian states of Maranhão,</i> |

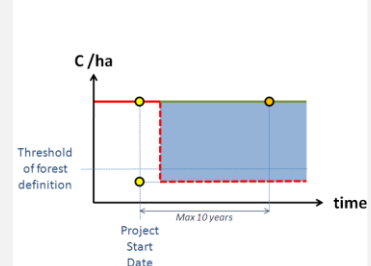
|        |                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|        |                                                                                                                                                                                 | <p><i>Tocantins, and Bahia. This region, known for its burgeoning agricultural activity, is emblematic of the intricate interplay between agricultural expansion and environmental preservation. The historical baseline for deforestation within this context is deeply rooted in the pursuit of economic gains through monoculture crops, prominently soybeans and corn, representing the broader agricultural trends observed across the North of Brazil.</i></p>                                                                                                                                                                              |
| VM0015 | <p><i>Project activities may include one or a combination of the eligible categories defined in the description of the scope of the methodology (table 1 and figure 2).</i></p> | <p><i>This project is defined within the scope A, thus considering Old growth without logging baseline and Protection without logging, fuel wood collection or charcoal production.</i></p> <p><i>The Project activities are essentially based on i) designing financial incentives for local communities and landowners who commit to conserving forests by generating commercial VCUs, where participants receive compensation for maintaining intact forests that sequester carbon; ii) developing robust monitoring and reporting systems that track changes in forest cover, land use and land cover, and other relevant indicators.</i></p> |



These systems are taken with satellite technologies with a high spatial and temporal resolution, providing near-real-time data for assessing the effectiveness of REDD+ actions; iii) Engaging local communities in the design and implementation of this REDD+ project. Involving them in a public discussion and encouraging participation in forest conservation; iv) Allocating a portion of the revenue generated from REDD+ initiatives to fund social benefit programs aimed at enhancing the well-being of local communities. This could lead to a holistic development of the region, by directly improving the lives of those residing in the reference region.

*Note: This project operates with activities which do not involve logging, fuel wood collection, or charcoal*

**A - Avoided Deforestation without Logging**



production within its operational scope. This approach is in line with the project's commitment to the



|        |                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|        |                                                                                                                                                                                                                                                                         | goals of the REDD+ initiatives.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| VM0015 | The project area can include different types of forest, such as, but not limited to, old growth forest, degraded forest, secondary forests, planted forests and agro-forestry systems meeting the definition of “forest”.                                               | <p>The project area includes old-growth forest, in line with the IPCC guidelines definition (Kyoto Forest):</p> <p>Minimum Canopy Cover: The IPCC criteria suggest a minimum tree canopy cover of 10-30% for a vegetation area to be classified as a forest.</p> <p>Minimum Tree Crown Cover: Land of 0.05-1.0 hectares with tree crown cover (or equivalent stocking level).</p> <p>Minimum Tree Height: Trees within the area should typically reach a minimum height, which can vary but is often around 2-5 meters at maturity in situ.</p> |
| VM0015 | At project commencement, the project area shall include only land qualifying as “forest” for a minimum of 10 years prior to the project start date.                                                                                                                     | This project area consists exclusively of land that has met the criteria for being classified as a “forest” for a minimum of 10 years before the project's initiation.                                                                                                                                                                                                                                                                                                                                                                          |
| VM0015 | The project area can include forested wetlands (such as bottomland forests, floodplain forests, mangrove forests) as long as they do not grow on peat. Peat shall be defined as organic soils with at least 65% organic matter and a minimum thickness of 50 cm. If the | <p>The project area does not include any forested area growing on peat. Eventually there are some lakes and other water bodies in the property. However, these areas are not permanent underwater sports.</p> <p>Otherwise, they would not be</p>                                                                                                                                                                                                                                                                                               |

|  |                                                                                                                               |                                          |
|--|-------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|
|  | <i>project area includes forested wetlands growing on peat (e.g. peat swamp forests), this methodology is not applicable.</i> | <i>included within the project area.</i> |
|--|-------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|

### 3.3 Project Boundary

The project area is part of the 'Tapirapé Farm' with total area of 13,925.78 ha. The area is officially registered in the CAR (Rural Socioenvironmental Registry), which can be consulted publicly in the national website. Thus, the property has 1 CAR and 7 property registration documents (called 'matrículas' in Portuguese), that register all the exact coordinates and officialize the legal ownership in compliance with the environmental and land legislation in Brazil. Some of these documents are presented on the APPENDIX 2 of this PD.

The Leakage Belt was defined as an area of 759,878.56 ha of land located into a circular buffer of 50km from the center of the Project area, near the main deforestation driver (road).

The property comprising the project area – defined in accordance with the methodology's rules governing the latter – as well as the size of the leakage belt, are displayed in Table 14 below:

**Table 14. Project Area and Leakage Belt Boundaries**

| Name         | Area (ha)     |
|--------------|---------------|
| Project Area | 13,925.78 ha  |
| Leakage Belt | 759,878.56 ha |

The applied Methodology considers the six carbon pools listed in Table 15 below. Their inclusion or exclusion within the boundary of the proposed AUD project activity, as well as the respective justification/explanation, are described in the Table below.

**Table 15. Carbon Pool Inclusion/Exclusion**

| Carbon Pool  | Included?          | Justification/Explanation of choice                    |
|--------------|--------------------|--------------------------------------------------------|
| Above-ground | TREE: Included     | Carbon stock change in this pool is always significant |
|              | Non-Tree: Excluded | No existence of perennial crops as final class         |
| Below-ground | Included           | Stock change in this pool is significant               |

| Carbon Pool              | Included? | Justification/Explanation of choice                                                                                |
|--------------------------|-----------|--------------------------------------------------------------------------------------------------------------------|
| Dead wood                | Excluded  | Excluded for simplification. This exclusion is conservative.                                                       |
| Harvested wood products+ | Excluded  | Not significant                                                                                                    |
| Litter                   | Excluded  | Not to be measured according to VCS (AFOLU) Requirements, v3.6.                                                    |
| Soil organic carbon+     | Excluded  | Excluded when the ground cover is grassland in the baseline scenario, according to VCS (AFOLU) Requirements, v3.6. |

In accordance with the Methodology, approximately 1/10 of the carbon stock in the below-ground pool of the initial “forest” class will be released in a ten-year interval. This is further discussed in Section 3.4 – baseline emissions.

**Table 16. Sources and GHG included or excluded within the boundary of the proposed AUD project activity (Table 4 of Methodology VM0015)**

| Source   |                     | Gas              | Included? | Justification/Explanation                                                                                                                                                                                           |
|----------|---------------------|------------------|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Baseline | BIOMASS BURNING     | CO <sub>2</sub>  | Excluded  | Counted as carbon stock change                                                                                                                                                                                      |
|          |                     | CH <sub>4</sub>  | Excluded  | Excluded as recommended by the applied methodology. This exclusion is conservative and does not lead to a significant over-estimation of the net anthropogenic GHG emission reductions of the AUD project activity. |
|          |                     | N <sub>2</sub> O | Excluded  | Considered insignificant according to VCS (AFOLU) Requirements, v3.6.                                                                                                                                               |
|          |                     | Other            | Excluded  | No other gases were not considered in this Scope project                                                                                                                                                            |
|          | LIVESTOCK EMISSIONS | CO <sub>2</sub>  | Excluded  | Not a significant source                                                                                                                                                                                            |
|          |                     | CH <sub>4</sub>  | Excluded  | Not a significant source                                                                                                                                                                                            |
|          |                     | N <sub>2</sub> O | Excluded  | Not a significant source                                                                                                                                                                                            |

| Source | Gas   | Included? | Justification/Explanation                                |
|--------|-------|-----------|----------------------------------------------------------|
|        | Other | Excluded  | No other gases were not considered in this Scope project |

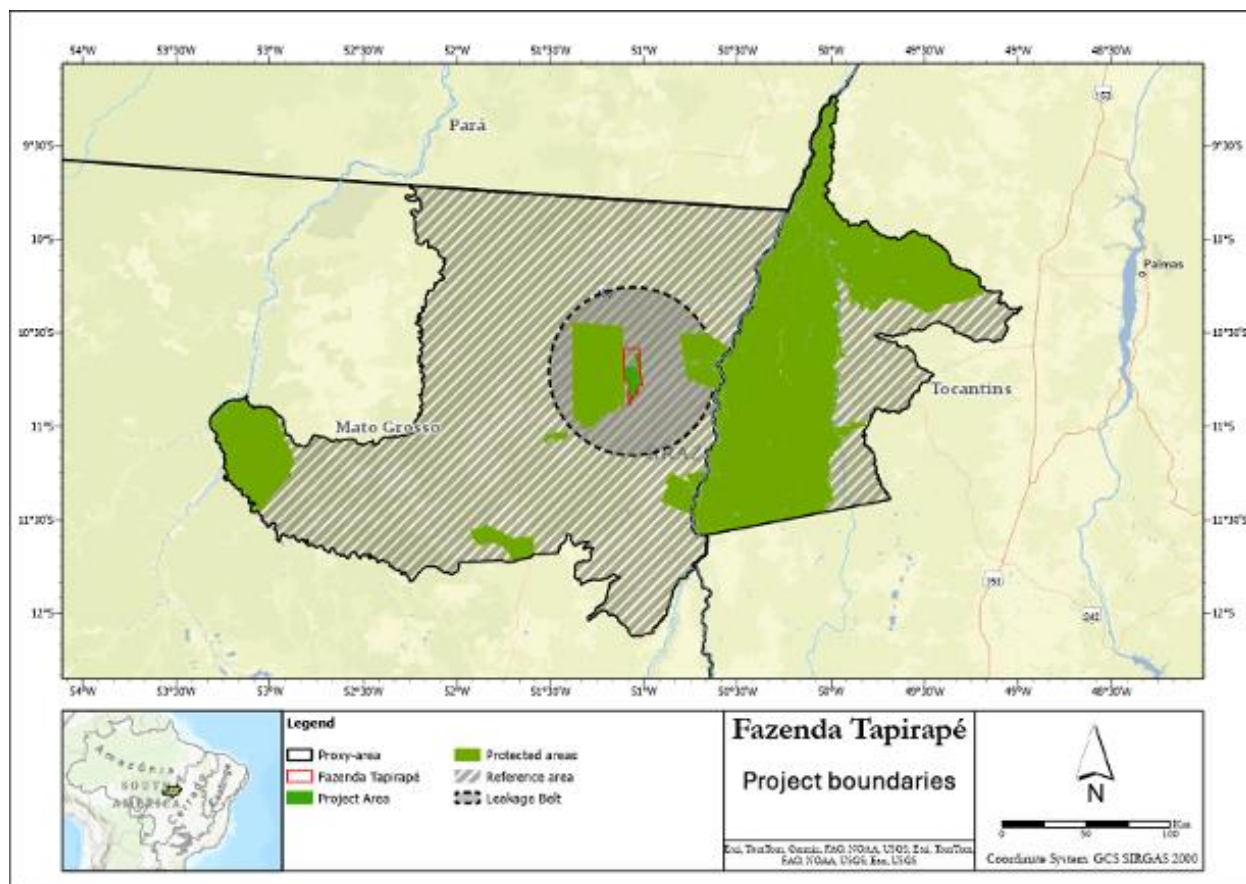


Image Source: the author – Fazenda Tapirapé project boundaries.

### 3.4 Baseline Scenario

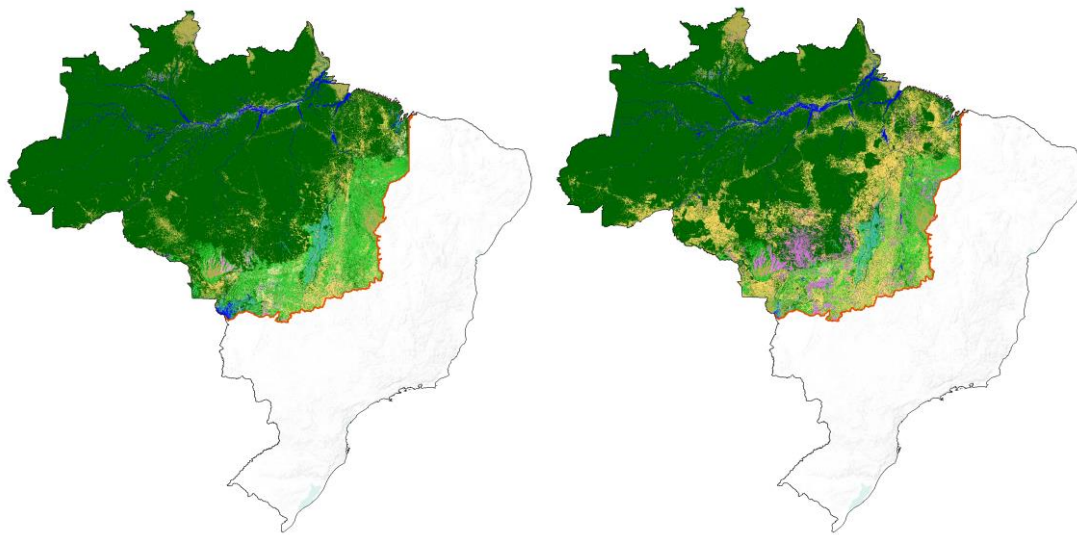
In the baseline scenario, forest is expected to be converted to non-forest by the agents of deforestation acting in the reference region, project area and leakage belt, as described below. Therefore, the project falls into the AFOLU-REDD category, specifically: Avoided unplanned deforestation (AUD). The revenue from the present REDD project is essential to maintain this area as standing forest, as described under Additionality (section 3.5), as well as to carry out the present project's leakage management activities.

Degradation was not considered in the present REDD project, in accordance with methodology requirements, which define “forest” and “non-forest” as the minimum land-use and land-cover classes.

### Contextualization of LULC changes in the Brazilian Amazon macro territory

An analysis conducted by Mapbiomas<sup>54</sup> over the historical period between 1985 and 2021 in the Brazilian Legal Amazon territory reveals notable trends. The extent of forested areas decreased from 473 million hectares to 411 million hectares, signifying a substantial decline of 12%. Similarly, savanna formations experienced a reduction of 13 million hectares, reflecting a significant decline of 26%.

Turning to the realm of human-induced land use changes, there was a remarkable increase from 36.4 million hectares to 98.4 million hectares. This highlights a marked shift in land utilization dynamics. Of particular significance is the expansion of areas dedicated to agriculture, which exhibited a momentous expansion, increasing by 15 million hectares, an astonishing surge of 1,024% during this analysis period.



1958 to 2021 LULC comparison in the amazon national territory (MAPBIOMAS, 2022)

### Historical baseline for the reference region

For the mapping of the changes in the classes of use and soil cover, data from the PRODES Digital program (INPE, 2023) were used in vector format (shapefile) with spatial resolution of

<sup>54</sup> Mapbiomas Amazônia (2022). Colección 4.0 de mapas anuales de cobertura y uso del suelo en la Amazonía. Accedido en 2023, através del enlace: [https://s3.amazonaws.com/amazonia.mapbiomas.org/destaques/AmazoniaC4\\_FactSheet\\_Regional.pdf](https://s3.amazonaws.com/amazonia.mapbiomas.org/destaques/AmazoniaC4_FactSheet_Regional.pdf)

30 meters. A total of 11 Landsat satellite images were used to map forest, hydrography, and anthropogenic vegetation (deforestation) (Table 17).

**Table 17. Data used to identify and map historical LU/LC change analysis in ARAGUAIA BOCALON REDD+ Project**

| Vector<br>(Satellite) | Sensor   | Resolution |                | Coverage     | Acquisition<br>Date | Scene or Point<br>Identifier |                    |
|-----------------------|----------|------------|----------------|--------------|---------------------|------------------------------|--------------------|
|                       |          | Spatial    | Spectral       | (km²)        | (DD/MM/AAA)         | Path /<br>Latitude           | Row /<br>Longitude |
| Satellite             | Landsat7 | 30 x 30 m  | 0,45 – 2,35 µm | 185 x 185 km | 16/10/2002          | 224/67                       |                    |
| Satellite             | Landsat7 | 30 x 30 m  | 0,45 – 2,35 µm | 185 x 185 km | 23/06/2004          | 224/67                       |                    |
| Satellite             | Landsat7 | 30 x 30 m  | 0,45 – 2,35 µm | 185 x 185 km | 31/07/2006          | 224/67                       |                    |
| Satellite             | Landsat7 | 30 x 30 m  | 0,45 – 2,35 µm | 185 x 185 km | 26/07/2010          | 224/67                       |                    |
| Satellite             | Landsat8 | 30 x 30 m  | 0,45 – 2,35 µm | 185 x 185 km | 05/07/2014          | 224/67                       |                    |
| Satellite             | Landsat8 | 30 x 30 m  | 0,45 – 2,35 µm | 185 x 185 km | 24/07/2015          | 224/67                       |                    |
| Satellite             | Landsat8 | 30 x 30 m  | 0,45 – 2,35 µm | 185 x 185 km | 11/08/2016          | 224/67                       |                    |
| Satellite             | Landsat8 | 30 x 30 m  | 0,45 – 2,35 µm | 185 x 185 km | 16/07/2018          | 224/67                       |                    |
| Satellite             | Landsat8 | 30 x 30 m  | 0,45 – 2,35 µm | 185 x 185 km | 04/08/2019          | 224/67                       |                    |
| Satellite             | Landsat8 | 30 x 30 m  | 0,45 – 2,35 µm | 185 x 185 km | 06/08/2020          | 224/67                       |                    |
| Satellite             | Landsat8 | 30 x 30 m  | 0,45 – 2,35 µm | 185 x 185 km | 27/07/2022          | 224/67                       |                    |

Deforestation data provided by PRODES were used to analyze the historical baseline for the reference region. The main activities carried out by the PRODES Project to monitor the forest cover loss of the Brazilian Amazon and Cerrado Biomes are detailed below.

### Pre-processing

The procedures of imagery preprocessing performed by the PRODES Project are constituted in the following steps (Almeida, C. A. et al., 2021<sup>55</sup>):

- Selection of images with lower cloud cover and acquisition date closer to dry season in the region and with adequate radiometric quality.
- Once selected, images are highlighted to highlight areas of clear-cutting by applying a contrast enhancement operation.

<sup>55</sup> Almeida, C. A. et al. Metodologia para monitoramento da floresta usada nos projetos PRODES e DETER. São José dos Campos: INPE, versão: 2021-01-26. Available at <<http://mtc-m21c.sid.inpe.br/col/sid.inpe.br/mtc-m21c/2021/01.25.18.31/doc/publicacao.pdf>>



## Interpretation and classification

The identification of deforestation is made by photointerpretation of images, performed by trained experts, that delimit the deforestation polygons directly on the computer screen. These experts identify the pattern of change of forest cover for shallow felling based on the main elements that are observable in images: hue, color, shape, texture and context.

These elements are analyzed comparatively considering images of the subsequent year, in addition to the pictures of the historical series. This procedure is performed for all images selected for PRODES year mapping.

An audit of all the scenes and 100% of the mapped area is performed by INPE's experts, in order to avoid the generation of false positives of deforestation and to reduce omissions.

## Post-processing

According to VM0015, the post-processing step includes the use of non-spectral information for the stratification of the carbon density of the land cover classes. This information was generated implicitly during the next steps.

## Map accuracy assessment

PRODES is validated by the Temporal Visual Inspection (TVI) tool, an open-source tool that simplifies the visualization of points in Landsat images for long time series and allows to perform analysis quickly, practically and with simultaneous supervision (Parente et al., 2021<sup>56</sup>; Nogueira et al., 2017<sup>57</sup>).

The accuracy assessment (global and by class) of the mapping of PRODES data is done with the random sampling procedure stratified by class that considers the classes of natural vegetation and deforestation of each year.

The sample size (n) is defined by the equation:

$$n = \left( \frac{\sum_{i=1}^H W_i * S_i}{s(\hat{O})} \right)^2$$

Where:

$W_i$  is the proportion of stratum  $i$  in the target population.

<sup>56</sup> Parente, L., Nogueira, S., Baumann, L., Almeida, C., Maurano, L., Affonso, A. G., & Ferreira, L.. Quality assessment of the PRODES Cerrado deforestation data. Remote Sensing Applications: Society and Environment, 21, 100444. 2021.

<sup>57</sup> Nogueira, S. H. M, Parente, L. L, Ferreira, L.G. Temporal visual inspection: uma ferramenta destinada à inspeção visual de pontos em séries históricas de imagens de sensoriamento remoto. In: Anais Do XXVII Congresso Brasileiro de Cartografia e XXVI Exposicarta. Rio de Janeiro, Brazil, pp. 624–628. 2017.

$$S_i = \sqrt{U_i * (1 - U_i)};$$

$U_i$  is the expected value of the population proportion, and the values for distant forest (Forest > 5km) is 0.9, for nearby forest (Forest 5km) 0.7 and deforestation 0.7

$s(\hat{\theta})$  is the standard error expected from sampling and was considered equal to 0.01.

Power allocation was used to determine the number of samples per stratum ( $n_i$ ):

$$n_i = n * \frac{(t_{xi})^a * \sqrt{(1 - U_i)/U_i}}{\sum_i^H (t_{xi})^a * \sqrt{(1 - U_i)/U_i}}$$

Where:

$$t_{xi} = U_i * N_i$$

$a = 1/3$  (Särndal et al. (1992) recommends the use of  $a = 1/2$  or  $1/3$ ).

Based on the confusion matrix constructed from the cross tabulation of the sampling frequencies of the mapped and reference classes it was possible to obtain the values of global accuracy, of the producer and the user (Stehman, 2012<sup>58</sup>; Olofson et al. 2014<sup>59</sup>).

The evaluation performed by INPE showed that the overall accuracy of PRODES CERRADO 2022<sup>60</sup> in the Cerrado was 94.3%.

| PRODES 2022     |           |                           |
|-----------------|-----------|---------------------------|
| Global Accuracy | Index (%) | Confidence Interval (CI±) |
|                 | 94,3      | 1,4                       |

| PRODES 2022     |           |     |             |     |
|-----------------|-----------|-----|-------------|-----|
| Global Accuracy | Natural   |     | Suppression |     |
|                 | Index (%) | CI± | Index (%)   | CI± |

<sup>58</sup> Stehman, S. v. (2012). Impact of sample size allocation when using stratified random sampling to estimate accuracy and area of land-cover change. Remote Sensing Letters,3(2), 111–120.

<sup>59</sup> Olofsson, P., Foody, G. M., Herold, M., Stehman, S. v., Woodcock, C. E., Wulder, M. A. (2014). Good practices for estimating area and assessing accuracy of land change. Remote Sensing of Environment, 148, 42–57.

<sup>60</sup> Avaliação da acurácia do mapeamento do PRODES 2022. Available at <[http://terrabilis.dpi.inpe.br/download/terrabilis/technicalnotes/nota\\_tecnica\\_avaliacao\\_PRODES2022.pdf](http://terrabilis.dpi.inpe.br/download/terrabilis/technicalnotes/nota_tecnica_avaliacao_PRODES2022.pdf)>

|  |      |      |      |     |
|--|------|------|------|-----|
|  | 94,4 | 1,4  | 91,9 | 3   |
|  | 99,9 | 0,03 | 15,3 | 3,3 |

### Definition of classes of Land-Use and Land-Cover

The classes of LU/LC were defined as “forest” and “non forest” in accordance with the procedures described above. These classes are the minimum classes to be considered in the present REDD project as stipulated by the Methodology. As such, degradation was not a factor.

Stratification was not carried out in either class, and therefore the categories “forest” and “non forest” have homogenous carbon stocks.

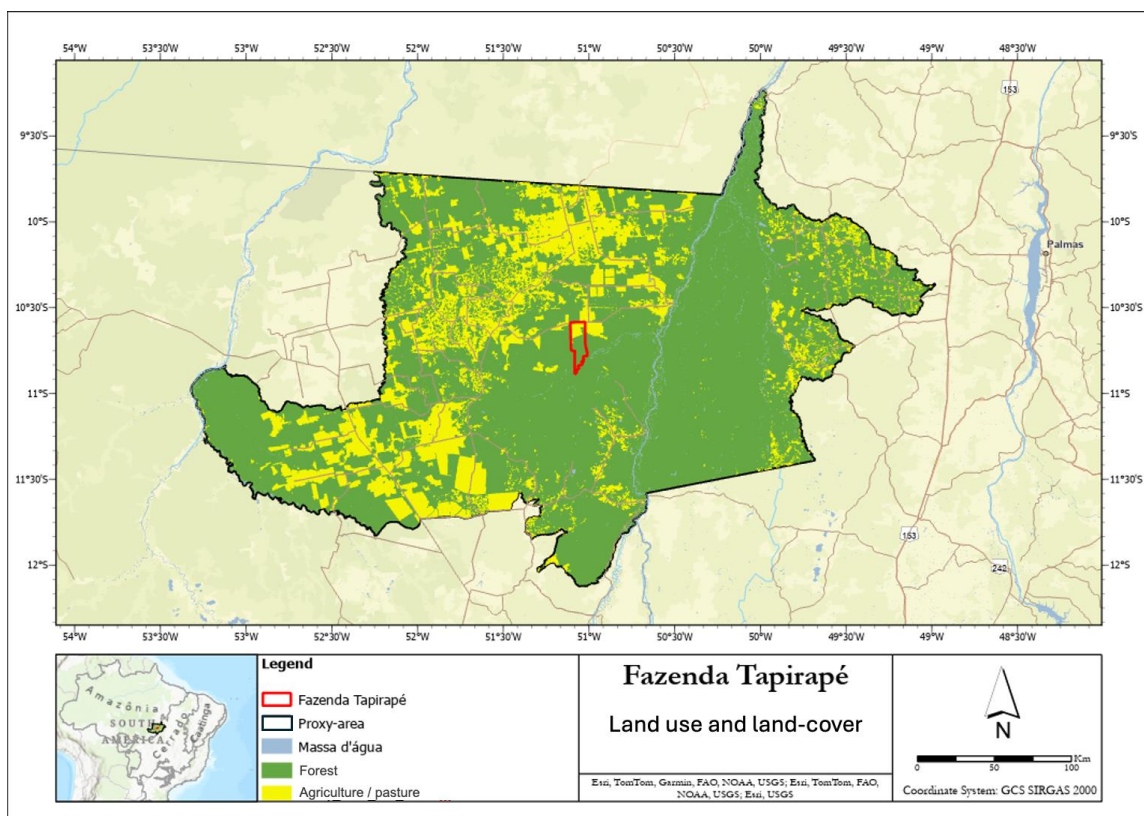
The following are the description the classes used in the Project and its area at the beginning of the historical period (2000):

The following are the description the classes used in the Project and its area at the beginning of the historical period (2000):

- **Forest (13,926 ha):** area of forest remnant belonging to different phytophysiognomies of the Contact Savannah-Seasonal Forest;
- **Hydrography (2 ha):** water bodies (rivers, lakes, streams, among others);
- **Anthropogenic Vegetation in equilibrium and other uses (10,172 ha):** area where there was forest, but that was removed through the clear-cutting process (totally removal of forest cover).

### Analysis of the historical Land-Use and Land-Cover change

According to PRODES data, up until 2000 the deforestation rate is unknown, however, there is a large proportion of deforested areas - about 41% – in the 2000 land-use and land-cover map below.



Map Source: the author – 2000 land use and land-cover maps at the reference region

Between 2001 and 2008 the rate of deforestation was very high and therefore there is a large proportion of deforested areas in this period. After 2009 the deforestation rate decreased significantly in the reference region of the ARAGUAIA REDD+ BOCALON PROJECT.

The 2001 – 2008 period displayed an annual average deforestation of 2.97% per year as depicted in Table 18 below (applying  $r$ : annual rate of change of forest cover<sup>61</sup>, which was also used in deforestation rates below). This is approximately 3,5 times greater than average annual deforestation from 2009 – 2022 (0.85% per year).

**Table 18. Annual deforestation, cumulative deforestation and R in the reference region during historical reference period.**

| YEAR | Dominant Vegetation in Ecological Tension (ha) | Annual deforestation RR (ha) | Cumulative deforestation (ha) | R: annual rate of forest cover change |
|------|------------------------------------------------|------------------------------|-------------------------------|---------------------------------------|
| 2000 | 3.391.836,72                                   | -                            | -                             | -                                     |

<sup>61</sup> Puyravaud, J.-P. (2003), "Standardizing the calculation of the annual rate of deforestation." *Forest Ecology and Management*, 177: 593-596

|      |              |            |              |       |
|------|--------------|------------|--------------|-------|
| 2001 | 3.278.413,97 | 113.422,76 | 113.422,76   | 3,46% |
| 2002 | 3.164.991,21 | 113.422,76 | 226.845,51   | 3,58% |
| 2003 | 3.065.872,86 | 99.118,35  | 325.963,86   | 3,23% |
| 2004 | 2.966.754,51 | 99.118,35  | 425.082,21   | 3,34% |
| 2005 | 2.901.333,58 | 65.420,93  | 490.503,15   | 2,25% |
| 2006 | 2.835.912,64 | 65.420,93  | 555.924,08   | 2,31% |
| 2007 | 2.760.457,95 | 75.454,70  | 631.378,78   | 2,73% |
| 2008 | 2.685.003,25 | 75.454,70  | 706.833,47   | 2,81% |
| 2009 | 2.670.343,00 | 14.660,25  | 721.493,72   | 0,55% |
| 2010 | 2.655.682,75 | 14.660,25  | 736.153,98   | 0,55% |
| 2011 | 2.640.832,87 | 14.849,88  | 751.003,86   | 0,56% |
| 2012 | 2.625.982,99 | 14.849,88  | 765.853,74   | 0,57% |
| 2013 | 2.596.345,18 | 29.637,81  | 795.491,54   | 1,14% |
| 2014 | 2.574.104,48 | 22.240,70  | 817.732,24   | 0,86% |
| 2015 | 2.544.150,20 | 29.954,28  | 847.686,52   | 1,18% |
| 2016 | 2.513.402,05 | 30.748,15  | 878.434,68   | 1,22% |
| 2017 | 2.497.375,69 | 16.026,36  | 894.461,04   | 0,64% |
| 2018 | 2.474.293,20 | 23.082,49  | 917.543,53   | 0,93% |
| 2019 | 2.456.824,89 | 17.468,31  | 935.011,83   | 0,71% |
| 2020 | 2.435.756,27 | 21.068,62  | 956.080,45   | 0,86% |
| 2021 | 2.408.831,94 | 26.924,33  | 983.004,79   | 1,12% |
| 2022 | 2.385.567,79 | 23.264,14  | 1.006.268,93 | 0,98% |

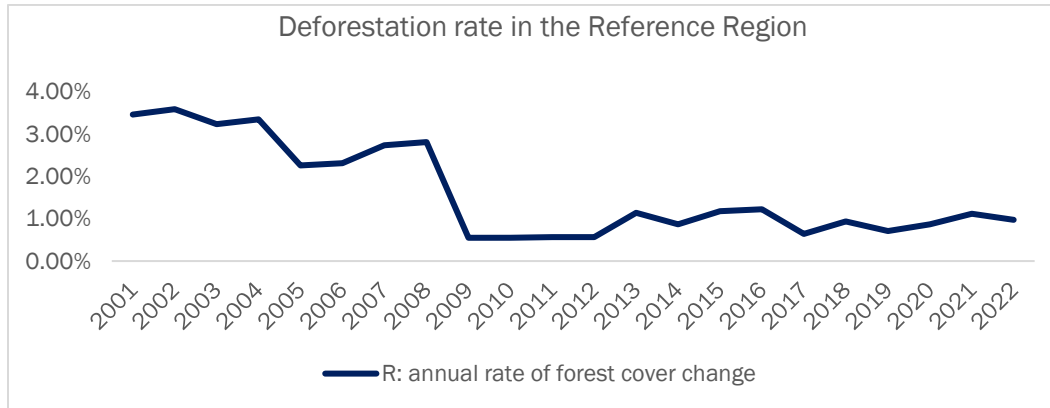
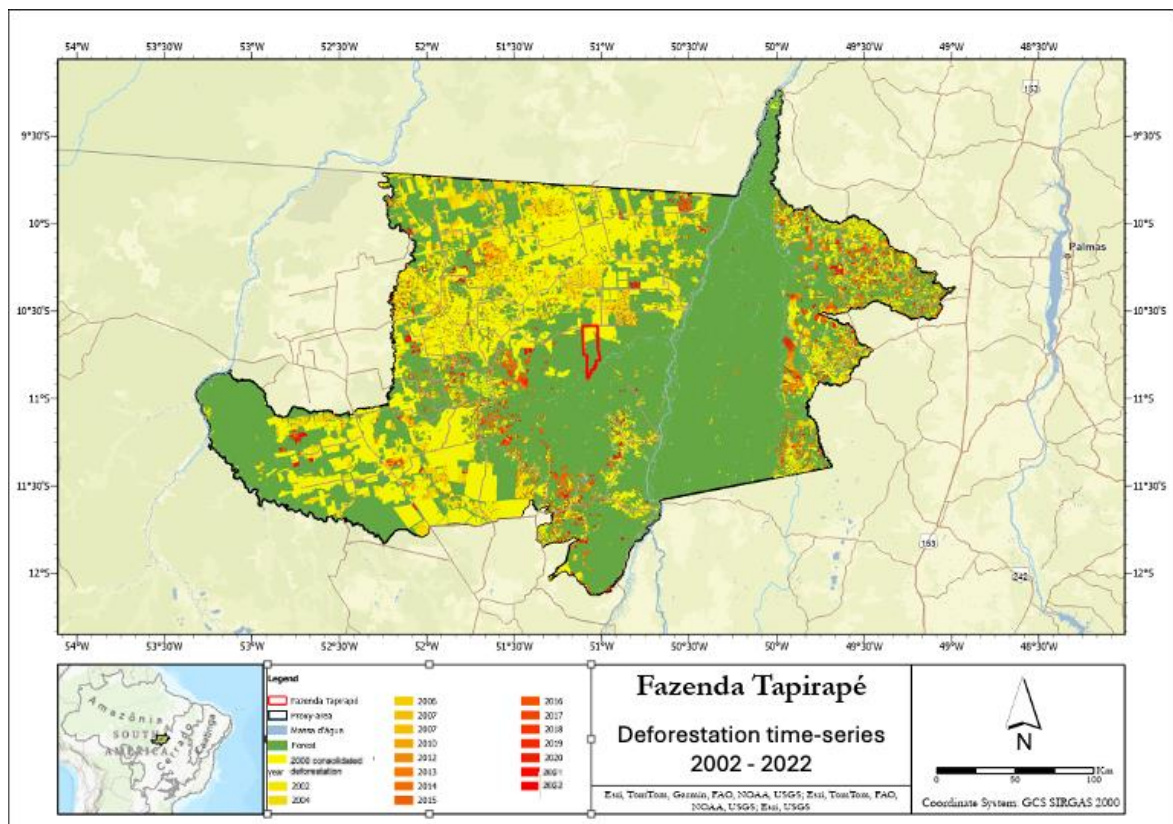


Chart Source: the author – "r" – annual rate of forest cover change in the reference region for 2001 – 2022

The annual deforestation of years analyzed within the historical reference period are also represented in the deforestation map below.



Map Source: the author – Deforestation map of within reference region over the historical reference period



## Results in change history analysis in Land-Use and Land-Cover

Based on the data obtained in the previous steps, the analysis of the historical change in land cover between 2008 and 2023 was carried out in the Reference Region of the Fazenda Tapirapé Project Area. The subtraction map analysis resulted in a deforested area between 2008 and 2022 of approximately 1,006,269 ha (30% of forest remnant in 2008).

Table 19 shows the changes occurring from the Forest class to the Deforestation class, with a decrease in the carbon stock. The Map presented previously (see ‘Deforestation map of within reference region over the historical reference period’) shows the annual deforestation that occurred between 2008 and 2022 in the Reference Region.

**Table 19. Potential land-use and land-cover change matrix in the Reference Region between 2008 and 2023**

| Initial class (2001)     |    |               |           |             |                        |            |
|--------------------------|----|---------------|-----------|-------------|------------------------|------------|
| ID <sub>cl</sub>         |    | Name          | Forest    | Hydrography | Anthropized vegetation | Total (ha) |
|                          |    |               | L1        | L2          | L3                     |            |
| Class LU/LC Final (2022) | F1 | Forest        | 4,812,696 | 0           | 0                      | 4,812,318  |
|                          | F2 | Hydrography   | 0         | 378         | 0                      | 378        |
|                          | F3 | Deforestation | 1,006,269 | 0           | 1,420,481              | 2,426,750  |
| Total (ha)               |    |               | 3,806,427 | 378         | 1,420,481              | 2,385,568  |

## Specifications of an alternative methodology

Methodological procedures for acquisition, pre-processing, classification, post-classification and evaluation of the accuracy of remote sensing images for analysis of changes in land use and land cover during the duration of the Project.

The official monitoring of the Brazilian Amazon conducted annually by PRODES was used for the development of the baseline and will be used to monitor the Project Area and Leakage Belt. In case PRODES data are unavailable or new sensors with better resolution are available, the following procedures will be used to maintain consistency in monitoring using remote sensing:

- Data sources:** images of satellites of optical sensors should be used. The optical images should be multispectral with a spectral resolution between 0.45 and 2.35  $\mu\text{m}$ . For the mapping of forest cover and land use, images with spatial resolution equal to or greater than 30 meters should be used. The data acquisition period should be in the period of low

- cloudiness and rainfall in the region for the monitoring of forest cover in the Project Area and the Leakage Belt;
- b) **Pre-processing:** the images must be geometrically corrected by means of georeferencing or orthorectification. The RMS error of the georeferencing must be less than one pixel;
  - c) **Data classification:** The identification of deforestation should be made by visual interpretation of images, carried out by trained specialists, able to delimit the deforestation polygons directly on the computer screen. These experts will identify the pattern of change from forest cover to shallow felling based on the main observable elements in optical images: hue, color, shape, texture and context;
  - d) **Post-Processing:** the result of the classification in vector format should be used for auditing the classification accuracy;
  - e) **Classification accuracy assessment:** should be performed through the analysis of the general accuracy and the kappa index obtained from a confusion matrix. The minimum accuracy of classification mapping is 80%.

### **Analysis of agents, drivers and underlying causes of deforestation**

As specified in the methodology, the analysis of deforestation agents is important for two reasons: i) estimating the quantity and location of future deforestation; and ii) Designing effective measures to address deforestation, including leakage prevention methods.

In recent years, the project region has been deforested for the expansion of agricultural and livestock activities, mainly due to the advancement of the arc of deforestation from the south of the Amazon biome. This pressure is expected to continue, given the globalization of markets in the Amazon region and international development policies planned for the region<sup>62</sup>.

### **Agents of deforestation**

In terms of analysis of deforestation patterns, deforestation agents are not considered separately, but as being spatially overlapping and forming a single dynamic within Homogeneous Deforestation Areas (HDAs).

Thus, deforestation activity is not modeled in terms of the individual action of each agent, but homogenized in terms of GIS analysis, since in the delimitation of HDAs considers the main aspects related to land tenure, geomorphological and social characteristics that contribute to the underlying causes of deforestation and its likely future development.

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<sup>62</sup> Nepstad, D. C.; C. M. Stickler e O. T. Almeida. 2006. Globalization of the Amazon Soy and Beef Industries: Opportunities for Conservation. *Conservation Biology* 20(6):1595-1603.

The analysis performed on the historical land-use/land-cover changes within the HDAs during the historical reference period allowed conclusions to be drawn related to the correlation between deforestation and the main deforestation agents in the reference region.

### **Deforestation drivers**

In addition, the main drivers affecting the location of deforestation in the reference region are:

#### **1- Distance from deforested areas**

Forested areas are influenced by their proximity to areas that have already been deforested. The distance from previously deforested areas is one of the major causes of forest degradation in the Amazon biome and their spatio-temporal dynamics are highly influenced by annual deforestation patterns. In addition, forest fragmentation results from deforestation and disturbance, with subsequent edge effects extending deep into remaining forest areas.

As described in the "future deforestation location projection" section below, a regression between the currently deforested areas and future deforestation was performed, which achieved a significant result. A road distance factor was then applied to capture the direct influence of the flow vectors around the Project Area.

#### **2- Distance from nearest roads**

Roads are means of transport that impact the spatial distribution of land uses through the influence on fragmentation, population densities, agriculture and pastureland. The possible creation of new roads increases anthropogenic pressure and, consequently, the intensity of deforestation<sup>63,64,65</sup>.

In addition, the spatial-temporal dynamics of deforestation in the reference region during the historical period clearly indicates that the existence of access routes increased anthropogenic pressure and, consequently, the intensity of deforestation in the vicinity of these roads.

### **Conclusion of the analysis of agents, drivers and underlying causes of deforestation and their likely future development**

Most of the deforestation occurred in the Amazon region is related to the implementation of infrastructure projects, population increase and agricultural and cattle ranching activities, mainly in the region known as the "Brazilian arc of deforestation", where the present project is located.

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<sup>63</sup> Hawbaker, T.J.; Radeloff, V.C.; Hammer, R.B.; Clayton, M.K. Road density and landscape pattern in relation to housing density, land ownership, land cover, and soils. *Landscape Ecology*, v.20, p.609-625, 2004.

<sup>64</sup> Geneletti, D. Biodiversity Impact Assessment of roads: an approach based on ecosystem rarity. *Environmental Impact Assessment Review*, v.23, n.3, p.343-365, 2003.

<sup>65</sup> Fearnside, P.M. e P.M.L.A. Graça. 2006. BR-319: Brazil's Manaus-Porto Velho Highway and the Potential Impact of Linking the Arc of Deforestation to Central Amazonia. *Environmental Management* 38:705-716.

The historical deforestation that has been occurring over the past 15 years within the reference region has followed this same pattern.

According to Razera (2005)<sup>66</sup>, the absence of economic alternatives has created the conditions for livestock farming to establish itself as the primary land occupation activities in deforested areas in Amazonia, being primarily concentrated in the arc of deforestation, where preferably there are some means of access to the rest of the country.

The land-uses that stimulate deforestation within the project reference region have been the same ever since the era of expansion of the Brazilian national territory into the north of the country. The activities at that time included: clearing military era roads; infrastructure projects, such as Avanço Brasil and the Program for Acceleration of Growth; and the belief that forests are a non-productive domain and need to be replaced by productive areas, such as agriculture and livestock, especially in areas considered to hinder regional development or social and economic evolution. These State policies led to the clearing of new roads in the region, allowing the implementation of colonization programs as an incentive to immigration, primarily in the South and Southeast of the country, creating a great social diversity in the region.

The future trend of these underlying causes of deforestation will tend to remain as verified in the past years, as Brazilian government policies and recent changes in Brazilian environment laws still prioritize large agricultural and ranching companies<sup>67</sup>. This is a gap that needs to be addressed by projects seeking to develop alternative sources of income as well as ongoing training, being from private companies, NGOs, or donors.

Finally, global demand for agricultural commodities has increased the opportunity cost of conserving areas of standing forests in the Amazon and the Cerrado biomes.

Based on the information gathered above and on the baseline analysis, it is possible to find conclusive evidence that deforestation pattern explained by the relations among the agents, drivers and underlying causes present in the region during the historical period will most likely continue to cause deforestation in the future, frequently followed this order of events:

- Roads and infrastructure projects reach remote forested areas.
- Timber logging seeks the most valuable commercial species.
- Deforestation for livestock production followed by crop production.

Furthermore, it is expected that agents and drivers of deforestation may increase because of a possible population growth in the reference region, which is also influenced by the prices of crop

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<sup>66</sup> Razera, Allan. Dinâmica do desmatamento em uma nova fronteira do sul do Amazonas: uma análise da pecuária de corte no município do Apuí. 2005. 109 f. MSc. Thesis - Curso de Biologia, Universidade Federal do Amazonas - UFAM, Amazônia, 2005.

<sup>67</sup> Moutinho, P. et al. REDD no Brasil: um enfoque amazônico: fundamentos, critérios e estruturas institucionais para um regime nacional de Redução de Emissões por Desmatamento e Degradação Florestal – REDD. Brasília, DF: Instituto de Pesquisa Ambiental da Amazônia, 2011.

commodities and livestock arroba, thus intensifying the demand to convert forested areas to productive areas in the region.

### **Description of baseline scenario adopted**

The analysis of the chain events that led to deforestation in the reference region was based on the relations between main deforestation agents and the drivers present in each HDA that contribute to the underlying causes of deforestation and its probable development future.

Based on the historical evidence collected, the relationships between agents, key drivers and underlying causes of deforestation that typically lead to and will most likely lead to deforestation within HDAs and therefore can be demonstrated by the analysis of the historical series of deforestation as a function of time and distance to the nearest road.

### **Projection of Future Deforestation**

As the Methodology stipulates, the aim of this step is to locate in space and time the baseline deforestation in the project area, reference region and leakage belt.

### **Selection of Baseline Approach**

Establishing the baseline is an exercise of predicting the future value of a socioeconomic variable, i.e., the future value of the deforestation extension. The baseline is counterfactual, that is, something that would happen in the absence of the project implementation and above all, the baseline must be credible, identifiable, plausible and relevant to the project area.

The methodology VM0015 suggests the use of three approaches to forecast the amount of future deforestation: (1) historical average of deforestation; (2) deforestation as a function of time; (3) modeling the rate of deforestation.

As shown in the Map previously showed (see '*Deforestation map of within reference region over the historical reference period*'), there is a clear decreasing trend in deforestation during the historical reference period within the reference region is present. As explained earlier in section 3.4 – Baseline Scenario, conclusive evidence from the analysis of the deforestation dynamic was found to suggest that this trend would continue in the future.

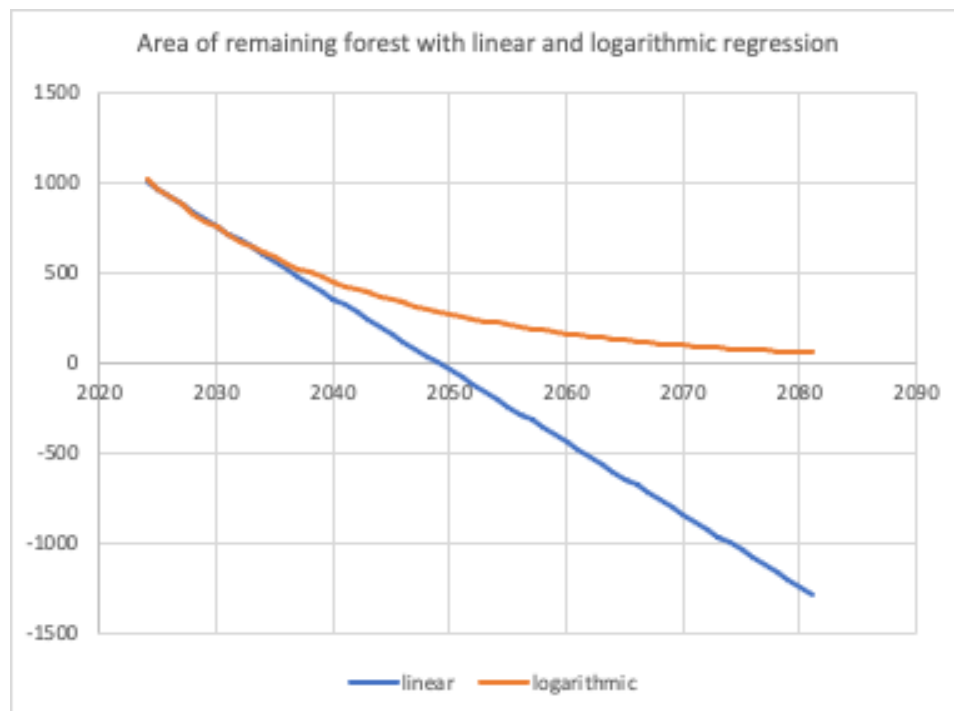
Overall, project proponents and specially governments (e.g. Brazilian Forest Reference Level – FREL) predict deforestation from the historical rate of deforestation and often use linear regression (i.e. average) to extrapolate the rate of deforestation into the future.

This approach, however, when analyzed from the perspective of forest cover loss (the opposite of deforestation) – without proper statistical procedure often results in a prediction of a negative value, beyond the remaining forest extension.

In addition, an inspection, albeit superficial, of the history of deforestation in the Amazon Biome shows that, in general, as deforestation advances in a given region, its speed over time decreases, because, among other reasons, the amount of forest remaining becomes increasingly scarce.

The simplest mathematical representation of this fact, which also avoids the non-plausible or credible negative values of forest cover is given by logarithmic regression of forest cover loss.

The graph below highlights the two different statistical treatments when establishing the deforestation trend, using both logarithmic and linear regression.



In red, the regression of the logarithm of the area of remaining forest.  
In blue, the linear regression of the rate of deforestation.

## Logistic Regression

The logistic regression on the other hand, also often used by project proponents, is a statistical model that allows us to estimate the probability associated with the occurrence of a given event in the face of a set of explanatory variables.

Although logistic regression is being used in studies on the various causes of deforestation, this is based on the correlation between several external factors (predictive variables) and the rate of deforestation (explanatory variable) and, therefore, as a technique for predicting the future, suffers from the insufficiency of predictive variables that are globally comparable to feed this statistical model.



In summary, the practical application of this approach depends on the availability of the independent variables, which explain the dependent variable. The difficulty in homogenizing the predictive variables make it impossible to apply this technique on a large scale to estimate the rate of deforestation.

The method developed by Geoconsult does not necessitate to identify agents of deforestation but is based on the simple yet powerful assumption that all factors influencing deforestation are represented within the defined areas of homogeneous deforestation - HDAs.

An analysis of all overlapping municipality districts is undertaken to consider land tenure and its geomorphology, by which areas are individualized and defined as indigenous lands, conservation units, settlements, type of soil, climate and declivity, among other aspects, which may influence the homogeneity of the deforestation.

The HDAs are formed by the intersection of these boundaries and, therefore, formed by a combination of both their legal and geomorphological identities. Each HDA is then used to estimate a baseline, from a historical dataset of deforestation distributed in space and time.

For these reasons, approach (2) deforestation as a function of time was adopted to create the baseline. With this approach, the rate of deforestation will be estimated using a model that expresses baseline deforestation as a function of the observed time-series of deforestation multiplied by a road-distance factor, which is detailed below.

### **Logarithmic Regression analysis**

The annual deforestation rate is considered a predictor of the likelihood of future deforestation and the distance from deforested areas to the nearest road a driver of deforestation.

Thus, the correlation between these two variables was analyzed, namely: i) observed annual rate of deforestation and ii) distance from deforested areas to the nearest road. The analysis of these variables generated data on, respectively: i) annual rate of deforestation projected for the future; and ii) multiplier factor of road distance.

It should be noted that the annual rate of deforestation is equal to the annual decrease in the remaining forest area, that is, it is equal to the time derived from the remaining forest area and that the cumulative difference in forest loss is a consequence of the annual rate of deforestation.

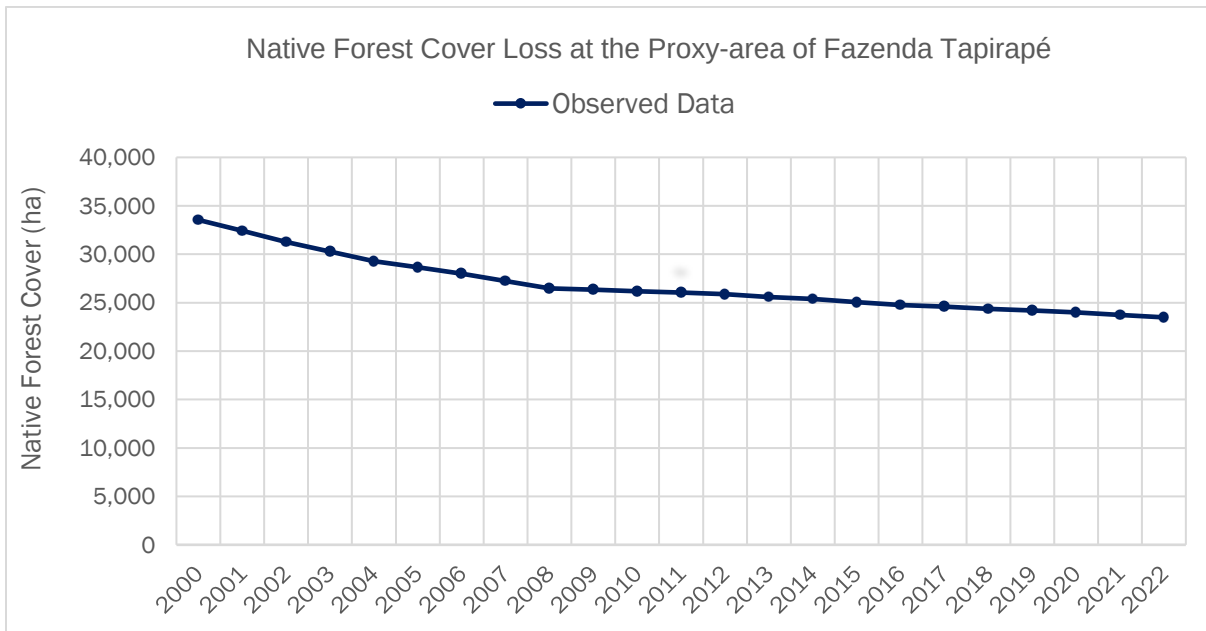


Chart Source: the author – Cumulative difference of forest loss in the Reference Region between 2000 and 2022

A simple visual inspection of the deforestation graph shows that time series of forest cover loss are not composed of independent random values. In other words, the decision to deforest is not taken every year, regardless of what happened in the previous year or in recent years.

It should be noted that in recent decades there have been important economic and political changes that have somehow affected deforestation. Thus, it can be concluded that economic changes and relevant public policies, if they affect the deforestation process, this will happen slowly, with effect in a period of four or five years.

This conclusion is because deforestation observed data follow a Markov series of 4 or 5, that is, one-year deforestation is correlated with what occurred one year earlier and two years earlier, with a lower correlation, and so on up to four or five years ago.

### Markov Chains contextualization

Any real system always operates in environments where uncertainty prevails, especially when the system involves, by its nature, unpredictable human actions.

Deterministic models certainly contribute to the understanding, at a basic level, of the dynamic behavior of a system. However, because they cannot deal with uncertainty, it ends up being insufficient in decision-making processes.

Thus, stochastic processes are used as a way of quantitatively treating these phenomena, taking advantage of certain regularity characteristics that they present to be described by probabilistic

models. A stochastic process is therefore a mathematical model that evolves over time in a probabilistic way.

Markov Chains constitute a special type of stochastic process that has the property that the probabilities associated with the process at a given time in the future depend on the present state and are therefore independent of events in the past.

Although the Markov Chain is characterized by what is called “lack of memory”<sup>68</sup>, some Markov processes have a finite memory<sup>69</sup>, or finite range, and the determination of the order of the Markov chain makes explicit the magnitude of this range.

Therefore, if a dataset comes from a Markov chain, it is essential to determine the order of this dependence, that is, how much a given information, at a given instant, depends on previous information in the process.

### **Deforestation activity data**

Earth observation data via remote sensing satellite, provides information about deforestation over a given period of time. This is the source of information that is used to create a time series of deforestation, or when considered from a different perspective, it also demonstrates the loss of forest cover in space and time.

Data from PRODES<sup>70</sup> produced based on satellite images with sufficient spatial resolution to identify deforestation size from 1ha were used to create an annual time series with at least 20 years of remaining forest area observations within the reference area.

The logarithm of the observed values of the remaining forest area is the object of a linear regression that is then extrapolated into the future. The extrapolated values are then the object of an exponential function, which result in the predicted values of the remaining forest area.

The difference from year to year of the predicted values of remaining forest area, is the actual predicted annual forest cover loss. When applied to a specific project reference area, these values are then multiplied by the road proximity factor to obtain the baseline rate of deforestation for the project.

### **Road-proximity factor**

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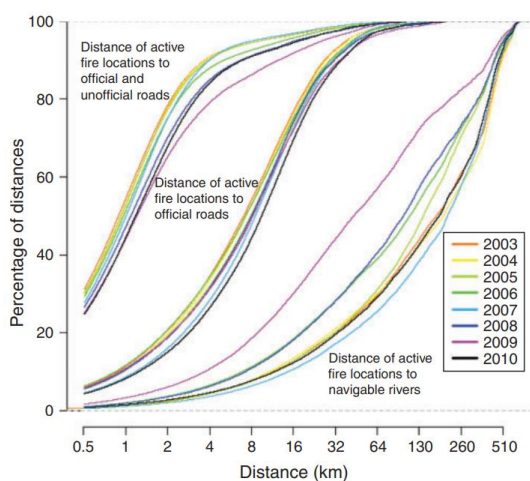
<sup>68</sup> Transition probability is defined as the one that characterizes the transition to the state at instant  $t$  given that the previous state was  $t-1$ . This is the definition of what is called a Markov chain of first order.

<sup>69</sup> These models have as their basic element the Markov Chain, which is a discrete-time stochastic process in which the transition to a state at instant  $t$  depends on the last  $k$  states visited. In this case the Markov chain is called  $k$  order.

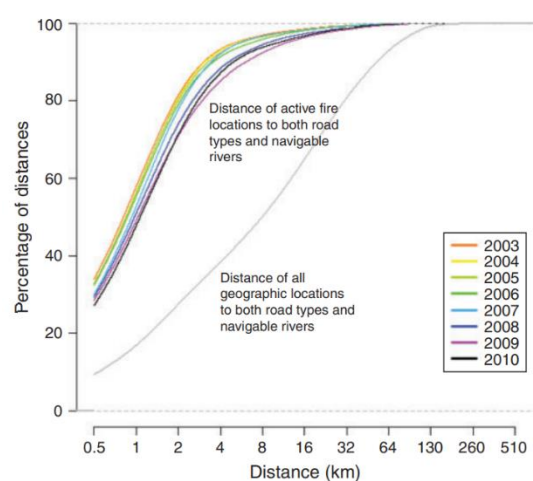
<sup>70</sup> Instituto Nacional de Pesquisas Espaciais. Coordenação Geral de Observação da Terra. Programa de Monitoramento da Amazônia e demais Biomas. Desmatamento – Amazônia Legal – Disponível em: <http://terrabrasilis.dpi.inpe.br/downloads/>.

The available scientific literature<sup>71,72,73,74,75,76</sup> agrees on the influence of roads on deforestation, and shows that there is an exponential decay with the distance to the nearest road. In the Amazonia biome in South America the parameter of this exponential decay is 5.5 km. This means that the deforested area in any year is proportional to the exponential of the negative value of the distance to the nearest road divided by the decay parameter:  $\text{Exp}(-\text{road distance} / 5.5 \text{ km})$ .

The following graphs summarize how the decay factor of annual deforestation is determined for the Amazonia biome of South America, based on multispectral observations of the annual decrease in forest cover and its correlation with transport vectors of deforestation products (roads and rivers).



**Fig. 5.** Cumulative frequency of the closest distances from each MODIS active fire detection to the transportation networks. Results shown for each year of MODIS active fire detections (colour coded) for the Brazilian tropical moist forest biome study area.



**Fig. 8.** Cumulative frequency of the closest distances from each MODIS active fire detection (coloured by year) and of the geographic locations (grey) to the transportation networks. Roads are official and unofficial roads, rivers are the navigable rivers, geographic locations spaced every 0.5 km over the Brazilian tropical moist forest biome study area.

For the purpose of applying the road proximity factor, an area of direct influence of the project was established based on a circle of 25 km from the center of the property. This distance is equivalent, according to scientific references, to approximately 90% of deforestation sites.

<sup>71</sup> Armenteras, D., Barreto, J. S., Tabor, K., Molowny-Horas, R., and Retana, J.: Changing patterns of fire occurrence in proximity to forest edges, roads and rivers between NW Amazonian countries, *Biogeosciences*, 14, 2755–2765.

<sup>72</sup> Roy, David P. and Kumar, S. S., "Multi-year MODIS Active Fire Type Classification Over the Brazilian Tropical Moist Forest Biome" (2017). GSCE Faculty Publications. 27.

<sup>73</sup> Kumar Sanath S., Roy David P., Cochrane Mark A., Souza Carlos M., Barber Chirstopher P., Boschetti L. (2014) A quantitative study of the proximity of satellite detected active fires to roads and rivers in the Brazilian tropical moist forest biome. *International Journal of Wildland Fire* 23, 532-543.

<sup>74</sup> Cano Crespo, Ana & Traxl, Dominik & Thonicke, Kirsten. (2021). Spatio-temporal patterns of extreme fires in Amazonian forests. *The European Physical Journal Special Topics*. 230.

<sup>75</sup> Dalton Cardoso et al. A relação entre áreas de exploração madeireira e estradas na Amazônia legal. In: *Anais do XX Simpósio Brasileiro de Sensoriamento Remoto*, 2023, Florianópolis. Anais eletrônicos. São José dos Campos, INPE, 2023.

<sup>76</sup> Sathyachandran, S. K.. "Fire Type Classification in the Brazilian Tropical Moist Forest Biome." (2014).

Within this area is then calculated the distance from each site of deforestation to the nearest road, in each year of the historical series of deforestation and once the proximity factor is calculated, this factor is used to multiply the annual deforestation rate of the baseline, causing the annual deforestation rate at the project baseline to be corrected by the proximity factor.

### **Quantitative projection of future deforestation**

As stated earlier, demonstrate the baseline scenario is difficult because it implies predicting what the rate of deforestation will be in the future, in the absence of the project. This is what in logic is called counterfactual, which is the prediction of a future that will not come true.

To demonstrate a baseline scenario, it is necessary to consider what occurred in the recent past and what occurs in the vicinity of the project (i.e. reference area) so that the predicted future in the baseline is credible and, among reasonable scenarios, that is the most likely.

As previously referenced above, the general orientation of this methodological standards is that the reference areas should represent the characteristics of the region, without any bias in relation to its delimitation to estimate the deforestation trend in the most plausible way, and that is why the reference area should be several times greater than the project area.

For this reason, the deforestation baseline is calculated based on a historical time-series of deforestation compiled within the HDAs that belong to the reference area, where regulatory and geomorphological characteristics (i.e. vegetation, soil, climate and slope) are similar.

An analysis of the projection of future deforestation across the project crediting period was subsequently carried out, using the logarithmic regression explained above. This demonstrated that only 00% of the landscape underwent change from 2023 to 2063. It was observed that the annual rates of change were practically constant from year to year.

When the LU/LC-change was analyzed at ten-year intervals, it was observed that the rate was approximately 0,0%, always being a transition from "forest" to "non-forest". The quantity of baseline LU/LC-change was projected throughout the crediting period, in the reference region, project area, and leakage belt in each stratum.

This is in accordance with step 5 of the Methodology - Definition of The Land-Use and Land-Cover Change Component of The Baseline. The baseline deforestation within the reference region per stratum is provided in Table 20 below. The only "forest" stratum used consisted of Dominant Vegetation in Ecological Tension<sup>77</sup>, which is represented by stratum i:

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<sup>77</sup> IBGE. Manual Técnico da Vegetação Brasileira. Brasília: IBGE, 2 ed. 2012. 271p - Transition System (Ecological Tension): regions of contact between two or more vegetational typologies where the floras interpenetrate, forming undifferentiated communities.

**Table 20. Annual areas of baseline deforestation in the reference region across the project monitoring period**

| Project year t | Stratum i in the<br>reference region<br>(ha) | Total (ha)                    |                      |
|----------------|----------------------------------------------|-------------------------------|----------------------|
|                | ABSLRR <sub>i,t</sub>                        | Annual<br>ABSLRR <sub>t</sub> | Cumulative<br>ABSLRR |
| 2023           | 32.030                                       | 32.030                        | 32.030               |
| 2024           | 31.402                                       | 31.402                        | 63.432               |
| 2025           | 30.789                                       | 30.789                        | 94.221               |
| 2026           | 30.190                                       | 30.190                        | 124.411              |
| 2027           | 29.605                                       | 29.605                        | 154.016              |
| 2028           | 29.034                                       | 29.034                        | 183.050              |
| 2029           | 28.475                                       | 28.475                        | 211.526              |
| 2030           | 27.930                                       | 27.930                        | 239.455              |
| 2031           | 27.397                                       | 27.397                        | 266.852              |
| 2032           | 26.876                                       | 26.876                        | 293.728              |
| 2033           | 26.367                                       | 26.367                        | 320.095              |
| 2034           | 25.869                                       | 25.869                        | 345.965              |
| 2035           | 25.383                                       | 25.383                        | 371.348              |
| 2036           | 24.908                                       | 24.908                        | 396.256              |
| 2037           | 24.443                                       | 24.443                        | 420.699              |
| 2038           | 23.989                                       | 23.989                        | 444.689              |
| 2039           | 23.545                                       | 23.545                        | 468.234              |
| 2040           | 23.111                                       | 23.111                        | 491.345              |
| 2041           | 22.686                                       | 22.686                        | 514.031              |
| 2042           | 22.271                                       | 22.271                        | 536.302              |
| 2043           | 21.865                                       | 21.865                        | 558.168              |
| 2044           | 21.468                                       | 21.468                        | 579.636              |



|      |        |        |           |
|------|--------|--------|-----------|
| 2045 | 21.080 | 21.080 | 600.716   |
| 2046 | 20.700 | 20.700 | 621.416   |
| 2047 | 20.329 | 20.329 | 641.745   |
| 2048 | 19.965 | 19.965 | 661.710   |
| 2049 | 19.610 | 19.610 | 681.320   |
| 2050 | 19.262 | 19.262 | 700.582   |
| 2051 | 18.921 | 18.921 | 719.503   |
| 2052 | 18.588 | 18.588 | 738.091   |
| 2053 | 18.263 | 18.263 | 756.354   |
| 2054 | 17.944 | 17.944 | 774.298   |
| 2055 | 17.631 | 17.631 | 791.929   |
| 2056 | 17.326 | 17.326 | 809.255   |
| 2057 | 17.027 | 17.027 | 826.282   |
| 2058 | 16.734 | 16.734 | 843.016   |
| 2059 | 16.448 | 16.448 | 859.464   |
| 2060 | 16.167 | 16.167 | 875.631   |
| 2061 | 15.892 | 15.892 | 891.523   |
| 2062 | 15.623 | 15.623 | 907.147   |
| 2063 | 15.360 | 15.360 | 922.507   |
| 2064 | 15.102 | 15.102 | 937.609   |
| 2065 | 14.850 | 14.850 | 952.459   |
| 2066 | 14.602 | 14.602 | 967.061   |
| 2067 | 14.360 | 14.360 | 981.421   |
| 2068 | 14.123 | 14.123 | 995.544   |
| 2069 | 13.890 | 13.890 | 1.009.435 |
| 2070 | 13.663 | 13.663 | 1.023.098 |
| 2071 | 13.440 | 13.440 | 1.036.537 |
| 2072 | 13.221 | 13.221 | 1.049.758 |
| 2073 | 13.007 | 13.007 | 1.062.765 |

|      |        |        |           |
|------|--------|--------|-----------|
| 2074 | 12.797 | 12.797 | 1.075.563 |
| 2075 | 12.592 | 12.592 | 1.088.154 |
| 2076 | 12.390 | 12.390 | 1.100.544 |
| 2077 | 12.193 | 12.193 | 1.112.737 |
| 2078 | 11.999 | 11.999 | 1.124.736 |
| 2079 | 11.809 | 11.809 | 1.136.545 |
| 2080 | 11.623 | 11.623 | 1.148.169 |
| 2081 | 11.441 | 11.441 | 1.159.610 |
| 2082 | 11.262 | 11.262 | 1.170.872 |

Table 21 below shows the projected annual deforestation in the sole stratum of “forest” in the project area across the project crediting period, represented by the variable ABSLPA.

**Table 21. Annual areas of baseline deforestation in the project area**

| Project year t | Stratum i in the<br>reference region (ha) | Total (ha)                 |                      |
|----------------|-------------------------------------------|----------------------------|----------------------|
|                | ABSLPA <sub>i,t</sub>                     | Annual ABSLPA <sub>t</sub> | Cumulative<br>ABSLPA |
| 2023           | 350                                       | 350                        | 350                  |
| 2024           | 339                                       | 339                        | 689                  |
| 2025           | 328                                       | 328                        | 1.017                |
| 2026           | 318                                       | 318                        | 1.336                |
| 2027           | 308                                       | 308                        | 1.644                |
| 2028           | 299                                       | 299                        | 1.943                |
| 2029           | 290                                       | 290                        | 2.233                |
| 2030           | 281                                       | 281                        | 2.514                |
| 2031           | 273                                       | 273                        | 2.787                |
| 2032           | 264                                       | 264                        | 3.051                |
| 2033           | 257                                       | 257                        | 3.308                |

|      |     |     |       |
|------|-----|-----|-------|
| 2034 | 249 | 249 | 3.556 |
| 2035 | 241 | 241 | 3.798 |
| 2036 | 234 | 234 | 4.032 |
| 2037 | 227 | 227 | 4.260 |
| 2038 | 221 | 221 | 4.481 |
| 2039 | 214 | 214 | 4.695 |
| 2040 | 208 | 208 | 4.903 |
| 2041 | 202 | 202 | 5.105 |
| 2042 | 196 | 196 | 5.302 |
| 2043 | 191 | 191 | 5.493 |
| 2044 | 185 | 185 | 5.678 |
| 2045 | 180 | 180 | 5.858 |
| 2046 | 175 | 175 | 6.033 |
| 2047 | 170 | 170 | 6.203 |
| 2048 | 165 | 165 | 6.369 |
| 2049 | 161 | 161 | 6.530 |
| 2050 | 156 | 156 | 6.686 |
| 2051 | 152 | 152 | 6.838 |
| 2052 | 148 | 148 | 6.986 |
| 2053 | 144 | 144 | 7.130 |
| 2054 | 140 | 140 | 7.270 |
| 2055 | 136 | 136 | 7.407 |
| 2056 | 133 | 133 | 7.539 |
| 2057 | 129 | 129 | 7.668 |
| 2058 | 126 | 126 | 7.794 |
| 2059 | 122 | 122 | 7.916 |
| 2060 | 119 | 119 | 8.035 |
| 2061 | 116 | 116 | 8.151 |

|      |     |     |        |
|------|-----|-----|--------|
| 2062 | 113 | 113 | 8.264  |
| 2063 | 110 | 110 | 8.374  |
| 2064 | 107 | 107 | 8.482  |
| 2065 | 104 | 104 | 8.586  |
| 2066 | 102 | 102 | 8.688  |
| 2067 | 99  | 99  | 8.787  |
| 2068 | 97  | 97  | 8.884  |
| 2069 | 94  | 94  | 8.978  |
| 2070 | 92  | 92  | 9.070  |
| 2071 | 90  | 90  | 9.160  |
| 2072 | 87  | 87  | 9.247  |
| 2073 | 85  | 85  | 9.332  |
| 2074 | 83  | 83  | 9.416  |
| 2075 | 81  | 81  | 9.497  |
| 2076 | 79  | 79  | 9.576  |
| 2077 | 77  | 77  | 9.653  |
| 2078 | 75  | 75  | 9.728  |
| 2079 | 74  | 74  | 9.802  |
| 2080 | 72  | 72  | 9.874  |
| 2081 | 70  | 70  | 9.944  |
| 2082 | 68  | 68  | 10.012 |

Table 22 below shows the projected annual deforestation in the sole stratum of “forest” in the leakage belt across the project crediting period, represented by the variable ABSLLK.

Table 22. Annual areas of baseline deforestation in the leakage belt

| Project year t | Stratum i in the<br>reference region<br>(ha) | Total (ha)                    |                      |
|----------------|----------------------------------------------|-------------------------------|----------------------|
|                | ABSLLK <sub>i,t</sub>                        | Annual<br>ABSLLK <sub>t</sub> | Cumulative<br>ABSLLK |
| 2023           | 16.059                                       | 16.059                        | 16.059               |
| 2024           | 15.472                                       | 15.472                        | 31.531               |
| 2025           | 14.910                                       | 14.910                        | 46.441               |
| 2026           | 14.371                                       | 14.371                        | 60.812               |
| 2027           | 13.854                                       | 13.854                        | 74.666               |
| 2028           | 13.358                                       | 13.358                        | 88.024               |
| 2029           | 12.882                                       | 12.882                        | 100.906              |
| 2030           | 12.425                                       | 12.425                        | 113.331              |
| 2031           | 11.987                                       | 11.987                        | 125.317              |
| 2032           | 11.566                                       | 11.566                        | 136.883              |
| 2033           | 11.162                                       | 11.162                        | 148.045              |
| 2034           | 10.774                                       | 10.774                        | 158.819              |
| 2035           | 10.401                                       | 10.401                        | 169.221              |
| 2036           | 10.043                                       | 10.043                        | 179.264              |
| 2037           | 9.700                                        | 9.700                         | 188.964              |
| 2038           | 9.369                                        | 9.369                         | 198.333              |
| 2039           | 9.052                                        | 9.052                         | 207.384              |
| 2040           | 8.746                                        | 8.746                         | 216.131              |
| 2041           | 8.453                                        | 8.453                         | 224.583              |
| 2042           | 8.170                                        | 8.170                         | 232.754              |
| 2043           | 7.899                                        | 7.899                         | 240.652              |
| 2044           | 7.638                                        | 7.638                         | 248.290              |

|      |       |       |         |
|------|-------|-------|---------|
| 2045 | 7.387 | 7.387 | 255.677 |
| 2046 | 7.145 | 7.145 | 262.822 |
| 2047 | 6.912 | 6.912 | 269.734 |
| 2048 | 6.688 | 6.688 | 276.422 |
| 2049 | 6.473 | 6.473 | 282.894 |
| 2050 | 6.265 | 6.265 | 289.159 |
| 2051 | 6.065 | 6.065 | 295.224 |
| 2052 | 5.872 | 5.872 | 301.097 |
| 2053 | 5.687 | 5.687 | 306.783 |
| 2054 | 5.508 | 5.508 | 312.291 |
| 2055 | 5.336 | 5.336 | 317.627 |
| 2056 | 5.170 | 5.170 | 322.797 |
| 2057 | 5.010 | 5.010 | 327.807 |
| 2058 | 4.855 | 4.855 | 332.662 |
| 2059 | 4.707 | 4.707 | 337.369 |
| 2060 | 4.563 | 4.563 | 341.932 |
| 2061 | 4.425 | 4.425 | 346.356 |
| 2062 | 4.291 | 4.291 | 350.647 |
| 2063 | 4.162 | 4.162 | 354.809 |
| 2064 | 4.038 | 4.038 | 358.847 |
| 2065 | 3.918 | 3.918 | 362.765 |
| 2066 | 3.802 | 3.802 | 366.567 |
| 2067 | 3.690 | 3.690 | 370.256 |
| 2068 | 3.582 | 3.582 | 373.838 |
| 2069 | 3.478 | 3.478 | 377.316 |
| 2070 | 3.377 | 3.377 | 380.693 |
| 2071 | 3.279 | 3.279 | 383.972 |
| 2072 | 3.185 | 3.185 | 387.157 |
| 2073 | 3.094 | 3.094 | 390.251 |



|      |       |       |         |
|------|-------|-------|---------|
| 2074 | 3.006 | 3.006 | 393.258 |
| 2075 | 2.921 | 2.921 | 396.179 |
| 2076 | 2.839 | 2.839 | 399.019 |
| 2077 | 2.760 | 2.760 | 401.778 |
| 2078 | 2.683 | 2.683 | 404.461 |
| 2079 | 2.609 | 2.609 | 407.070 |
| 2080 | 2.537 | 2.537 | 409.607 |
| 2081 | 2.467 | 2.467 | 412.074 |
| 2082 | 2.400 | 2.400 | 414.475 |

### Calculation of baseline activity data per forest class

The following is in accordance with step 5.1 of the Methodology: “Calculation of baseline activity data per forest class”, in which it is stipulated that the previously created maps of annual baseline deforestation and LU/LC map be combined, producing a map showing deforestation per class in the baseline case. The number of hectares deforested in each forest class, within the reference region, project area and leakage belt are found in tables 23 – 25 below.

According to the baseline projections of the present project, accumulated deforestation from 2023 – 2082 in the Project Area will sum to 1,170,872 ha (Table 23 below). The LU/LC-change within the project crediting period, caused by baseline deforestation, consisted of the initial class of dominant vegetation in ecological tension being converted to the final LU/LC class of “non-forest”.

**Table 23. Annual areas deforested per forest class *icl* within the reference region in the baseline case (baseline activity data per forest class)**

| Area deforested per forest class <i>icl</i><br>within the reference region |                                                 | Total baseline deforestation in<br>the reference region |                                           |
|----------------------------------------------------------------------------|-------------------------------------------------|---------------------------------------------------------|-------------------------------------------|
| <i>IDicl</i>                                                               | 1                                               |                                                         |                                           |
| Name                                                                       | Dominant<br>Vegetation in<br>Ecological Tension | Annual<br>ABSLRR <sub>t</sub><br>(ha)                   | ABSLRR <sub>t</sub><br>cumulative<br>(ha) |
| Project year <i>t</i>                                                      | ABSLRR <sub>t,t</sub>                           |                                                         |                                           |
| 2023                                                                       | 32.030                                          | 32.030                                                  | 32.030                                    |
| 2024                                                                       | 31.402                                          | 31.402                                                  | 63.432                                    |
| 2025                                                                       | 30.789                                          | 30.789                                                  | 94.221                                    |
| 2026                                                                       | 30.190                                          | 30.190                                                  | 124.411                                   |
| 2027                                                                       | 29.605                                          | 29.605                                                  | 154.016                                   |
| 2028                                                                       | 29.034                                          | 29.034                                                  | 183.050                                   |
| 2029                                                                       | 28.475                                          | 28.475                                                  | 211.526                                   |
| 2030                                                                       | 27.930                                          | 27.930                                                  | 239.455                                   |
| 2031                                                                       | 27.397                                          | 27.397                                                  | 266.852                                   |
| 2032                                                                       | 26.876                                          | 26.876                                                  | 293.728                                   |
| 2033                                                                       | 26.367                                          | 26.367                                                  | 320.095                                   |
| 2034                                                                       | 25.869                                          | 25.869                                                  | 345.965                                   |
| 2035                                                                       | 25.383                                          | 25.383                                                  | 371.348                                   |
| 2036                                                                       | 24.908                                          | 24.908                                                  | 396.256                                   |
| 2037                                                                       | 24.443                                          | 24.443                                                  | 420.699                                   |
| 2038                                                                       | 23.989                                          | 23.989                                                  | 444.689                                   |
| 2039                                                                       | 23.545                                          | 23.545                                                  | 468.234                                   |
| 2040                                                                       | 23.111                                          | 23.111                                                  | 491.345                                   |
| 2041                                                                       | 22.686                                          | 22.686                                                  | 514.031                                   |

|      |        |        |           |
|------|--------|--------|-----------|
| 2042 | 22.271 | 22.271 | 536.302   |
| 2043 | 21.865 | 21.865 | 558.168   |
| 2044 | 21.468 | 21.468 | 579.636   |
| 2045 | 21.080 | 21.080 | 600.716   |
| 2046 | 20.700 | 20.700 | 621.416   |
| 2047 | 20.329 | 20.329 | 641.745   |
| 2048 | 19.965 | 19.965 | 661.710   |
| 2049 | 19.610 | 19.610 | 681.320   |
| 2050 | 19.262 | 19.262 | 700.582   |
| 2051 | 18.921 | 18.921 | 719.503   |
| 2052 | 18.588 | 18.588 | 738.091   |
| 2053 | 18.263 | 18.263 | 756.354   |
| 2054 | 17.944 | 17.944 | 774.298   |
| 2055 | 17.631 | 17.631 | 791.929   |
| 2056 | 17.326 | 17.326 | 809.255   |
| 2057 | 17.027 | 17.027 | 826.282   |
| 2058 | 16.734 | 16.734 | 843.016   |
| 2059 | 16.448 | 16.448 | 859.464   |
| 2060 | 16.167 | 16.167 | 875.631   |
| 2061 | 15.892 | 15.892 | 891.523   |
| 2062 | 15.623 | 15.623 | 907.147   |
| 2063 | 15.360 | 15.360 | 922.507   |
| 2064 | 15.102 | 15.102 | 937.609   |
| 2065 | 14.850 | 14.850 | 952.459   |
| 2066 | 14.602 | 14.602 | 967.061   |
| 2067 | 14.360 | 14.360 | 981.421   |
| 2068 | 14.123 | 14.123 | 995.544   |
| 2069 | 13.890 | 13.890 | 1.009.435 |

|      |        |        |           |
|------|--------|--------|-----------|
| 2070 | 13.663 | 13.663 | 1.023.098 |
| 2071 | 13.440 | 13.440 | 1.036.537 |
| 2072 | 13.221 | 13.221 | 1.049.758 |
| 2073 | 13.007 | 13.007 | 1.062.765 |
| 2074 | 12.797 | 12.797 | 1.075.563 |
| 2075 | 12.592 | 12.592 | 1.088.154 |
| 2076 | 12.390 | 12.390 | 1.100.544 |
| 2077 | 12.193 | 12.193 | 1.112.737 |
| 2078 | 11.999 | 11.999 | 1.124.736 |
| 2079 | 11.809 | 11.809 | 1.136.545 |
| 2080 | 11.623 | 11.623 | 1.148.169 |
| 2081 | 11.441 | 11.441 | 1.159.610 |
| 2082 | 11.262 | 11.262 | 1.170.872 |

**Table 24. Annual areas deforested per forest class *icl* within the project area in the baseline case (baseline activity data per forest class)**

| Area deforested per forest class <i>icl</i><br>within the project area |                                                 | Total baseline deforestation in<br>the project area |                               |
|------------------------------------------------------------------------|-------------------------------------------------|-----------------------------------------------------|-------------------------------|
| <i>IDicl</i>                                                           | 1                                               |                                                     |                               |
| Name                                                                   | Dominant<br>Vegetation in<br>Ecological Tension | Annual<br>ABSLPAt<br>(ha)                           | ABSLPAt<br>cumulative<br>(ha) |
| Project year <i>t</i>                                                  | ABSLPA <sub>i,t</sub>                           |                                                     |                               |
| 2023                                                                   | 350                                             | 350                                                 | 350                           |
| 2024                                                                   | 339                                             | 339                                                 | 689                           |
| 2025                                                                   | 328                                             | 328                                                 | 1.017                         |
| 2026                                                                   | 318                                             | 318                                                 | 1.336                         |
| 2027                                                                   | 308                                             | 308                                                 | 1.644                         |

|      |     |     |       |
|------|-----|-----|-------|
| 2028 | 299 | 299 | 1.943 |
| 2029 | 290 | 290 | 2.233 |
| 2030 | 281 | 281 | 2.514 |
| 2031 | 273 | 273 | 2.787 |
| 2032 | 264 | 264 | 3.051 |
| 2033 | 257 | 257 | 3.308 |
| 2034 | 249 | 249 | 3.556 |
| 2035 | 241 | 241 | 3.798 |
| 2036 | 234 | 234 | 4.032 |
| 2037 | 227 | 227 | 4.260 |
| 2038 | 221 | 221 | 4.481 |
| 2039 | 214 | 214 | 4.695 |
| 2040 | 208 | 208 | 4.903 |
| 2041 | 202 | 202 | 5.105 |
| 2042 | 196 | 196 | 5.302 |
| 2043 | 191 | 191 | 5.493 |
| 2044 | 185 | 185 | 5.678 |
| 2045 | 180 | 180 | 5.858 |
| 2046 | 175 | 175 | 6.033 |
| 2047 | 170 | 170 | 6.203 |
| 2048 | 165 | 165 | 6.369 |
| 2049 | 161 | 161 | 6.530 |
| 2050 | 156 | 156 | 6.686 |
| 2051 | 152 | 152 | 6.838 |
| 2052 | 148 | 148 | 6.986 |
| 2053 | 144 | 144 | 7.130 |
| 2054 | 140 | 140 | 7.270 |
| 2055 | 136 | 136 | 7.407 |

|      |     |     |        |
|------|-----|-----|--------|
| 2056 | 133 | 133 | 7.539  |
| 2057 | 129 | 129 | 7.668  |
| 2058 | 126 | 126 | 7.794  |
| 2059 | 122 | 122 | 7.916  |
| 2060 | 119 | 119 | 8.035  |
| 2061 | 116 | 116 | 8.151  |
| 2062 | 113 | 113 | 8.264  |
| 2063 | 110 | 110 | 8.374  |
| 2064 | 107 | 107 | 8.482  |
| 2065 | 104 | 104 | 8.586  |
| 2066 | 102 | 102 | 8.688  |
| 2067 | 99  | 99  | 8.787  |
| 2068 | 97  | 97  | 8.884  |
| 2069 | 94  | 94  | 8.978  |
| 2070 | 92  | 92  | 9.070  |
| 2071 | 90  | 90  | 9.160  |
| 2072 | 87  | 87  | 9.247  |
| 2073 | 85  | 85  | 9.332  |
| 2074 | 83  | 83  | 9.416  |
| 2075 | 81  | 81  | 9.497  |
| 2076 | 79  | 79  | 9.576  |
| 2077 | 77  | 77  | 9.653  |
| 2078 | 75  | 75  | 9.728  |
| 2079 | 74  | 74  | 9.802  |
| 2080 | 72  | 72  | 9.874  |
| 2081 | 70  | 70  | 9.944  |
| 2082 | 68  | 68  | 10.012 |



**Table 25. Annual areas deforested per forest class icl within the leakage belt in the baseline case (baseline activity data per forest class)**

| Area deforested per forest class icl within the leakage belt |                                           | Total baseline deforestation in the leakage belt |                         |
|--------------------------------------------------------------|-------------------------------------------|--------------------------------------------------|-------------------------|
| IDicl                                                        | 1                                         |                                                  |                         |
| Name                                                         | Dominant Vegetation in Ecological Tension | Annual ABSLLKt (ha)                              | ABSLLKt cumulative (ha) |
| Project year $t$                                             | ABSLLK <sub>i,t</sub>                     |                                                  |                         |
| 2023                                                         | 16.059                                    | 16.059                                           | 16.059                  |
| 2024                                                         | 15.472                                    | 15.472                                           | 31.531                  |
| 2025                                                         | 14.910                                    | 14.910                                           | 46.441                  |
| 2026                                                         | 14.371                                    | 14.371                                           | 60.812                  |
| 2027                                                         | 13.854                                    | 13.854                                           | 74.666                  |
| 2028                                                         | 13.358                                    | 13.358                                           | 88.024                  |
| 2029                                                         | 12.882                                    | 12.882                                           | 100.906                 |
| 2030                                                         | 12.425                                    | 12.425                                           | 113.331                 |
| 2031                                                         | 11.987                                    | 11.987                                           | 125.317                 |
| 2032                                                         | 11.566                                    | 11.566                                           | 136.883                 |
| 2033                                                         | 11.162                                    | 11.162                                           | 148.045                 |
| 2034                                                         | 10.774                                    | 10.774                                           | 158.819                 |
| 2035                                                         | 10.401                                    | 10.401                                           | 169.221                 |
| 2036                                                         | 10.043                                    | 10.043                                           | 179.264                 |
| 2037                                                         | 9.700                                     | 9.700                                            | 188.964                 |
| 2038                                                         | 9.369                                     | 9.369                                            | 198.333                 |
| 2039                                                         | 9.052                                     | 9.052                                            | 207.384                 |
| 2040                                                         | 8.746                                     | 8.746                                            | 216.131                 |
| 2041                                                         | 8.453                                     | 8.453                                            | 224.583                 |

|      |       |       |         |
|------|-------|-------|---------|
| 2042 | 8.170 | 8.170 | 232.754 |
| 2043 | 7.899 | 7.899 | 240.652 |
| 2044 | 7.638 | 7.638 | 248.290 |
| 2045 | 7.387 | 7.387 | 255.677 |
| 2046 | 7.145 | 7.145 | 262.822 |
| 2047 | 6.912 | 6.912 | 269.734 |
| 2048 | 6.688 | 6.688 | 276.422 |
| 2049 | 6.473 | 6.473 | 282.894 |
| 2050 | 6.265 | 6.265 | 289.159 |
| 2051 | 6.065 | 6.065 | 295.224 |
| 2052 | 5.872 | 5.872 | 301.097 |
| 2053 | 5.687 | 5.687 | 306.783 |
| 2054 | 5.508 | 5.508 | 312.291 |
| 2055 | 5.336 | 5.336 | 317.627 |
| 2056 | 5.170 | 5.170 | 322.797 |
| 2057 | 5.010 | 5.010 | 327.807 |
| 2058 | 4.855 | 4.855 | 332.662 |
| 2059 | 4.707 | 4.707 | 337.369 |
| 2060 | 4.563 | 4.563 | 341.932 |
| 2061 | 4.425 | 4.425 | 346.356 |
| 2062 | 4.291 | 4.291 | 350.647 |
| 2063 | 4.162 | 4.162 | 354.809 |
| 2064 | 4.038 | 4.038 | 358.847 |
| 2065 | 3.918 | 3.918 | 362.765 |
| 2066 | 3.802 | 3.802 | 366.567 |
| 2067 | 3.690 | 3.690 | 370.256 |
| 2068 | 3.582 | 3.582 | 373.838 |
| 2069 | 3.478 | 3.478 | 377.316 |

|      |       |       |         |
|------|-------|-------|---------|
| 2070 | 3.377 | 3.377 | 380.693 |
| 2071 | 3.279 | 3.279 | 383.972 |
| 2072 | 3.185 | 3.185 | 387.157 |
| 2073 | 3.094 | 3.094 | 390.251 |
| 2074 | 3.006 | 3.006 | 393.258 |
| 2075 | 2.921 | 2.921 | 396.179 |
| 2076 | 2.839 | 2.839 | 399.019 |
| 2077 | 2.760 | 2.760 | 401.778 |
| 2078 | 2.683 | 2.683 | 404.461 |
| 2079 | 2.609 | 2.609 | 407.070 |
| 2080 | 2.537 | 2.537 | 409.607 |
| 2081 | 2.467 | 2.467 | 412.074 |
| 2082 | 2.400 | 2.400 | 414.475 |

### Calculation of baseline activity data per post-deforestation forest class

The following is in accordance with step 5.2 of the Methodology: “Calculation of baseline activity data per postdeforestation forest class.” As all of the initial classes represented in the tables above were transformed into non-forest (final post-deforestation class) in the considered baseline, the annual values corresponding to the final classes are the same as those as the initial class in tables 26 – 28 below, which depict baseline activity data per post-deforestation forest class in the reference region, project area, and leakage belt, respectively.

**Table 26. Annual areas deforested in each zone within the reference region in the baseline case (baseline activity data zone)**

| Area established after deforestation<br>per zone within the reference region |                       | Total baseline deforestation in<br>the reference region |                       |
|------------------------------------------------------------------------------|-----------------------|---------------------------------------------------------|-----------------------|
| <i>IDct</i>                                                                  | 2                     |                                                         |                       |
| Name                                                                         | Non forest            | ABSLRRt<br>Annual                                       | ABSLRRt<br>cumulative |
| Project year <i>t</i>                                                        | ABSLRR <sub>i,t</sub> | (ha)                                                    | (ha)                  |

|      |        |        |         |
|------|--------|--------|---------|
| 2023 | 32.030 | 32.030 | 32.030  |
| 2024 | 31.402 | 31.402 | 63.432  |
| 2025 | 30.789 | 30.789 | 94.221  |
| 2026 | 30.190 | 30.190 | 124.411 |
| 2027 | 29.605 | 29.605 | 154.016 |
| 2028 | 29.034 | 29.034 | 183.050 |
| 2029 | 28.475 | 28.475 | 211.526 |
| 2030 | 27.930 | 27.930 | 239.455 |
| 2031 | 27.397 | 27.397 | 266.852 |
| 2032 | 26.876 | 26.876 | 293.728 |
| 2033 | 26.367 | 26.367 | 320.095 |
| 2034 | 25.869 | 25.869 | 345.965 |
| 2035 | 25.383 | 25.383 | 371.348 |
| 2036 | 24.908 | 24.908 | 396.256 |
| 2037 | 24.443 | 24.443 | 420.699 |
| 2038 | 23.989 | 23.989 | 444.689 |
| 2039 | 23.545 | 23.545 | 468.234 |
| 2040 | 23.111 | 23.111 | 491.345 |
| 2041 | 22.686 | 22.686 | 514.031 |
| 2042 | 22.271 | 22.271 | 536.302 |
| 2043 | 21.865 | 21.865 | 558.168 |
| 2044 | 21.468 | 21.468 | 579.636 |
| 2045 | 21.080 | 21.080 | 600.716 |
| 2046 | 20.700 | 20.700 | 621.416 |
| 2047 | 20.329 | 20.329 | 641.745 |
| 2048 | 19.965 | 19.965 | 661.710 |
| 2049 | 19.610 | 19.610 | 681.320 |
| 2050 | 19.262 | 19.262 | 700.582 |

|      |        |        |           |
|------|--------|--------|-----------|
| 2051 | 18.921 | 18.921 | 719.503   |
| 2052 | 18.588 | 18.588 | 738.091   |
| 2053 | 18.263 | 18.263 | 756.354   |
| 2054 | 17.944 | 17.944 | 774.298   |
| 2055 | 17.631 | 17.631 | 791.929   |
| 2056 | 17.326 | 17.326 | 809.255   |
| 2057 | 17.027 | 17.027 | 826.282   |
| 2058 | 16.734 | 16.734 | 843.016   |
| 2059 | 16.448 | 16.448 | 859.464   |
| 2060 | 16.167 | 16.167 | 875.631   |
| 2061 | 15.892 | 15.892 | 891.523   |
| 2062 | 15.623 | 15.623 | 907.147   |
| 2063 | 15.360 | 15.360 | 922.507   |
| 2064 | 15.102 | 15.102 | 937.609   |
| 2065 | 14.850 | 14.850 | 952.459   |
| 2066 | 14.602 | 14.602 | 967.061   |
| 2067 | 14.360 | 14.360 | 981.421   |
| 2068 | 14.123 | 14.123 | 995.544   |
| 2069 | 13.890 | 13.890 | 1.009.435 |
| 2070 | 13.663 | 13.663 | 1.023.098 |
| 2071 | 13.440 | 13.440 | 1.036.537 |
| 2072 | 13.221 | 13.221 | 1.049.758 |
| 2073 | 13.007 | 13.007 | 1.062.765 |
| 2074 | 12.797 | 12.797 | 1.075.563 |
| 2075 | 12.592 | 12.592 | 1.088.154 |
| 2076 | 12.390 | 12.390 | 1.100.544 |
| 2077 | 12.193 | 12.193 | 1.112.737 |
| 2078 | 11.999 | 11.999 | 1.124.736 |
| 2079 | 11.809 | 11.809 | 1.136.545 |

|      |        |        |           |
|------|--------|--------|-----------|
| 2080 | 11.623 | 11.623 | 1.148.169 |
| 2081 | 11.441 | 11.441 | 1.159.610 |
| 2082 | 11.262 | 11.262 | 1.170.872 |

**Table 27. Annual areas deforested in each zone within the project area in the baseline case (baseline activity data zone)**

| Area established after deforestation<br>per zone within the project area |                       | Total baseline deforestation in<br>the project area |                                           |
|--------------------------------------------------------------------------|-----------------------|-----------------------------------------------------|-------------------------------------------|
| IDct                                                                     | 2                     |                                                     |                                           |
| Name                                                                     | Non forest            |                                                     |                                           |
| Project year $t$                                                         | ABSLPA <sub>i,t</sub> | ABSLPA <sub>t</sub><br>Annual (ha)                  | ABSLPA <sub>t</sub><br>cumulative<br>(ha) |
| 2023                                                                     | 350                   | 350                                                 | 350                                       |
| 2024                                                                     | 339                   | 339                                                 | 689                                       |
| 2025                                                                     | 328                   | 328                                                 | 1.017                                     |
| 2026                                                                     | 318                   | 318                                                 | 1.336                                     |
| 2027                                                                     | 308                   | 308                                                 | 1.644                                     |
| 2028                                                                     | 299                   | 299                                                 | 1.943                                     |
| 2029                                                                     | 290                   | 290                                                 | 2.233                                     |
| 2030                                                                     | 281                   | 281                                                 | 2.514                                     |
| 2031                                                                     | 273                   | 273                                                 | 2.787                                     |
| 2032                                                                     | 264                   | 264                                                 | 3.051                                     |
| 2033                                                                     | 257                   | 257                                                 | 3.308                                     |
| 2034                                                                     | 249                   | 249                                                 | 3.556                                     |
| 2035                                                                     | 241                   | 241                                                 | 3.798                                     |
| 2036                                                                     | 234                   | 234                                                 | 4.032                                     |
| 2037                                                                     | 227                   | 227                                                 | 4.260                                     |
| 2038                                                                     | 221                   | 221                                                 | 4.481                                     |

|      |     |     |       |
|------|-----|-----|-------|
| 2039 | 214 | 214 | 4.695 |
| 2040 | 208 | 208 | 4.903 |
| 2041 | 202 | 202 | 5.105 |
| 2042 | 196 | 196 | 5.302 |
| 2043 | 191 | 191 | 5.493 |
| 2044 | 185 | 185 | 5.678 |
| 2045 | 180 | 180 | 5.858 |
| 2046 | 175 | 175 | 6.033 |
| 2047 | 170 | 170 | 6.203 |
| 2048 | 165 | 165 | 6.369 |
| 2049 | 161 | 161 | 6.530 |
| 2050 | 156 | 156 | 6.686 |
| 2051 | 152 | 152 | 6.838 |
| 2052 | 148 | 148 | 6.986 |
| 2053 | 144 | 144 | 7.130 |
| 2054 | 140 | 140 | 7.270 |
| 2055 | 136 | 136 | 7.407 |
| 2056 | 133 | 133 | 7.539 |
| 2057 | 129 | 129 | 7.668 |
| 2058 | 126 | 126 | 7.794 |
| 2059 | 122 | 122 | 7.916 |
| 2060 | 119 | 119 | 8.035 |
| 2061 | 116 | 116 | 8.151 |
| 2062 | 113 | 113 | 8.264 |
| 2063 | 110 | 110 | 8.374 |
| 2064 | 107 | 107 | 8.482 |
| 2065 | 104 | 104 | 8.586 |
| 2066 | 102 | 102 | 8.688 |



|      |    |    |        |
|------|----|----|--------|
| 2067 | 99 | 99 | 8.787  |
| 2068 | 97 | 97 | 8.884  |
| 2069 | 94 | 94 | 8.978  |
| 2070 | 92 | 92 | 9.070  |
| 2071 | 90 | 90 | 9.160  |
| 2072 | 87 | 87 | 9.247  |
| 2073 | 85 | 85 | 9.332  |
| 2074 | 83 | 83 | 9.416  |
| 2075 | 81 | 81 | 9.497  |
| 2076 | 79 | 79 | 9.576  |
| 2077 | 77 | 77 | 9.653  |
| 2078 | 75 | 75 | 9.728  |
| 2079 | 74 | 74 | 9.802  |
| 2080 | 72 | 72 | 9.874  |
| 2081 | 70 | 70 | 9.944  |
| 2082 | 68 | 68 | 10.012 |

**Table 28. Annual areas deforested in each zone within the leakage belt in the baseline case (baseline activity data zone)**

| Area established after deforestation<br>per zone within the leakage belt |                       | Total baseline deforestation in<br>the leakage belt |                               |
|--------------------------------------------------------------------------|-----------------------|-----------------------------------------------------|-------------------------------|
| <i>IDct</i>                                                              | 2                     |                                                     |                               |
| Name                                                                     | Non forest            |                                                     |                               |
| Project year <i>t</i>                                                    | ABSLLK <sub>i,t</sub> | ABSLLKt<br>Annual (ha)                              | ABSLLKt<br>cumulative<br>(ha) |
| 2023                                                                     | 16.059                | 16.059                                              | 16.059                        |
| 2024                                                                     | 15.472                | 15.472                                              | 31.531                        |
| 2025                                                                     | 14.910                | 14.910                                              | 46.441                        |
| 2026                                                                     | 14.371                | 14.371                                              | 60.812                        |

|      |        |        |         |
|------|--------|--------|---------|
| 2027 | 13.854 | 13.854 | 74.666  |
| 2028 | 13.358 | 13.358 | 88.024  |
| 2029 | 12.882 | 12.882 | 100.906 |
| 2030 | 12.425 | 12.425 | 113.331 |
| 2031 | 11.987 | 11.987 | 125.317 |
| 2032 | 11.566 | 11.566 | 136.883 |
| 2033 | 11.162 | 11.162 | 148.045 |
| 2034 | 10.774 | 10.774 | 158.819 |
| 2035 | 10.401 | 10.401 | 169.221 |
| 2036 | 10.043 | 10.043 | 179.264 |
| 2037 | 9.700  | 9.700  | 188.964 |
| 2038 | 9.369  | 9.369  | 198.333 |
| 2039 | 9.052  | 9.052  | 207.384 |
| 2040 | 8.746  | 8.746  | 216.131 |
| 2041 | 8.453  | 8.453  | 224.583 |
| 2042 | 8.170  | 8.170  | 232.754 |
| 2043 | 7.899  | 7.899  | 240.652 |
| 2044 | 7.638  | 7.638  | 248.290 |
| 2045 | 7.387  | 7.387  | 255.677 |
| 2046 | 7.145  | 7.145  | 262.822 |
| 2047 | 6.912  | 6.912  | 269.734 |
| 2048 | 6.688  | 6.688  | 276.422 |
| 2049 | 6.473  | 6.473  | 282.894 |
| 2050 | 6.265  | 6.265  | 289.159 |
| 2051 | 6.065  | 6.065  | 295.224 |
| 2052 | 5.872  | 5.872  | 301.097 |
| 2053 | 5.687  | 5.687  | 306.783 |
| 2054 | 5.508  | 5.508  | 312.291 |

|      |       |       |         |
|------|-------|-------|---------|
| 2055 | 5.336 | 5.336 | 317.627 |
| 2056 | 5.170 | 5.170 | 322.797 |
| 2057 | 5.010 | 5.010 | 327.807 |
| 2058 | 4.855 | 4.855 | 332.662 |
| 2059 | 4.707 | 4.707 | 337.369 |
| 2060 | 4.563 | 4.563 | 341.932 |
| 2061 | 4.425 | 4.425 | 346.356 |
| 2062 | 4.291 | 4.291 | 350.647 |
| 2063 | 4.162 | 4.162 | 354.809 |
| 2064 | 4.038 | 4.038 | 358.847 |
| 2065 | 3.918 | 3.918 | 362.765 |
| 2066 | 3.802 | 3.802 | 366.567 |
| 2067 | 3.690 | 3.690 | 370.256 |
| 2068 | 3.582 | 3.582 | 373.838 |
| 2069 | 3.478 | 3.478 | 377.316 |
| 2070 | 3.377 | 3.377 | 380.693 |
| 2071 | 3.279 | 3.279 | 383.972 |
| 2072 | 3.185 | 3.185 | 387.157 |
| 2073 | 3.094 | 3.094 | 390.251 |
| 2074 | 3.006 | 3.006 | 393.258 |
| 2075 | 2.921 | 2.921 | 396.179 |
| 2076 | 2.839 | 2.839 | 399.019 |
| 2077 | 2.760 | 2.760 | 401.778 |
| 2078 | 2.683 | 2.683 | 404.461 |
| 2079 | 2.609 | 2.609 | 407.070 |
| 2080 | 2.537 | 2.537 | 409.607 |
| 2081 | 2.467 | 2.467 | 412.074 |
| 2082 | 2.400 | 2.400 | 414.475 |

## Estimation of baseline carbon stock changes and non-CO2 emissions

The following is in accordance with step 6 of the methodology, specifically, 6.1.1 Estimation of the average carbon stocks of each LU/LC class, the goal of which is to finalize the baseline assessment by calculating the baseline carbon stock changes.

The estimation of the carbon stored in the vegetation present in the project area was based on information from the Fourth National Communication of Brazil to the United Nations Framework Convention on Climate Change and the vegetation map available at System of National Emissions Registration - SIRENE<sup>78</sup>.

The Table 29 below shows the density of carbon for each type of vegetation present in the ARAGUAIA REDD+ BOCALON Project area.

**Table 29. Carbon Stocks for Each Vegetation Class**

| Phytophysionomies                               | AGB<br>(tC/ha) | BGB<br>(tC/ha) | DW<br>(tC/ha) | LT<br>(tC/ha) | Ctotal<br>(tC/ha) | Scientific sources                                                                                                                              |
|-------------------------------------------------|----------------|----------------|---------------|---------------|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| Fa - Alluvial Semi-deciduous Seasonal Forest    | 121,9          | 29,3           | 13,4          | 2,9           | 167,5             | Wittman et al., 2008 (above-ground biomass); IPCC, 2006 (below-ground biomass); IPCC, 2003 (dead wood); Moreira-Burger & Delitti, 1999 (litter) |
| Fs - Sub-montane Semi-deciduous Seasonal Forest | 61,8           | 14,8           | 6,8           | 4,1           | 87,6              | Scolforo et al., 2008c (above-ground biomass); IPCC, 2006 (below-ground biomass); IPCC, 2003 (dead wood); Amaro et al., 2003 (litter)           |
| Sa - Wooded Savanna                             | 13,6           | 22,6           | 0,3           | 3,4           | 39,9              | Ottmar et al., 2001 (above-ground biomass, dead wood, litter); Miranda et al., 2014 (below-ground biomass)                                      |

<sup>78</sup> The Brazilian government established the National Emissions Registration System - SIRENE as an MRV system (measurement, reporting and verification) for emissions, to give the continuity and accessibility to the results of the National Inventory of Anthropogenic Emissions by Sources and Removals by Sinks of Greenhouse Gases not Controlled by the Montreal Protocol. The methodological basis for calculating emissions are the guidance documents and methodological guidelines prepared by the Intergovernmental Panel on Climate Change - IPCC. Currently, SIRENE can access the time series of emissions referring to the latest results published in the National Inventory, as part of the Fourth National Communication of Brazil to the United Nations Framework Convention on Climate Change. Access link: <http://sirene.mctic.gov.br/>

|                                         |      |      |     |     |       |                                                                                                            |
|-----------------------------------------|------|------|-----|-----|-------|------------------------------------------------------------------------------------------------------------|
| Sp - Savanna Park                       | 11,0 | 18,3 | 0,1 | 2,4 | 24,7  | Ottmar et al., 2001 (above-ground biomass, dead wood, litter); Miranda et al., 2014 (below-ground biomass) |
| SNs - Contact Savanna / Seasonal Forest | 73,2 | 17,5 | 6,1 | 8,6 | 105,5 | Modified version of the National Forest Reference Emission Level (FREL), 2023                              |

This presents the forest phytophysiognomies considered in the Cerrado biome, for the construction of the FREL (Forest Reference Emission Level), and the respective carbon stocks for each carbon pool. For each type of forest phytophysiognomies, the total stock (Ctotal) corresponds to the sum of the individual stocks of the four carbon pools included: above ground biomass (AGB), below ground biomass (BGB), dead wood (DW) and litter (LI).

Table 30 below shows the carbon density identified in the vegetation maps provided by SIRENE and the result of the weighting made based on the area of each type of vegetation present in the project area.

**Table 30. Carbon Density present in the project area, the respective areas, and the result of the weighting.**

| Vegetation acronym | AGB (tC/ha) | BGB (tC/ha) | Area (ha) | weighted AGB (tC/ha) | weighted BGB (tC/ha) |
|--------------------|-------------|-------------|-----------|----------------------|----------------------|
| Fa                 | 121,9       | 29,3        | 1369      | 107,3                | 70,6                 |
| Fs                 | 62,2        | 14,6        | 9719      |                      |                      |
| Sa                 | 13,6        | 22,6        | 350       |                      |                      |
| Sp                 | 11,0        | 18,3        | 9737      |                      |                      |
| SNs                | 73,2        | 17,5        | 31        |                      |                      |

Thus, the Carbon stocks per hectare of initial forest classes icl existing in the project area and leakage belt are found in tables 31 and 32 below. To conclude this step, the area-weighted average carbon stocks of the post deforestation LU/LC classes existing within each zone are displayed in table 33.

Table 31. Carbon stocks per hectare of initial forest classes (icl) existing in the project area

| Project year | Initial class                              |                       |                                    |                       |                    |                       |
|--------------|--------------------------------------------|-----------------------|------------------------------------|-----------------------|--------------------|-----------------------|
|              | Name                                       |                       | Riparian Dense Tropical Rainforest |                       |                    |                       |
|              | IDcl                                       |                       | 1                                  |                       |                    |                       |
|              | Average carbon stock per hectare +- 90% CI |                       |                                    |                       |                    |                       |
|              | Cabicl                                     |                       | Cbbicl                             |                       | Ctoticl            |                       |
|              | C stock                                    | ±90%                  | C stock                            | ±90%                  | C stock            | ±90%                  |
|              | tCO <sub>2</sub> e                         | tCO <sub>2</sub> e/ha | tCO <sub>2</sub> e                 | tCO <sub>2</sub> e/ha | tCO <sub>2</sub> e | tCO <sub>2</sub> e/ha |
| 2023         | 37.538                                     | 107,29                | 2.471                              | 70,61                 | 40.008             | 180,94                |
| 2024         | 36.366                                     | 107,29                | 4.864                              | 70,61                 | 41.230             | 180,94                |
| 2025         | 35.236                                     | 107,29                | 7.183                              | 70,61                 | 42.419             | 180,94                |
| 2026         | 34.146                                     | 107,29                | 9.430                              | 70,61                 | 43.576             | 180,94                |
| 2027         | 33.095                                     | 107,29                | 11.609                             | 70,61                 | 44.703             | 180,94                |
| 2028         | 32.081                                     | 107,29                | 13.720                             | 70,61                 | 45.801             | 180,94                |
| 2029         | 31.103                                     | 107,29                | 15.767                             | 70,61                 | 46.870             | 180,94                |
| 2030         | 30.159                                     | 107,29                | 17.752                             | 70,61                 | 47.911             | 180,94                |
| 2031         | 29.248                                     | 107,29                | 19.677                             | 70,61                 | 48.925             | 180,94                |
| 2032         | 28.368                                     | 107,29                | 21.544                             | 70,61                 | 49.912             | 180,94                |
| 2033         | 27.519                                     | 107,29                | 20.885                             | 70,61                 | 48.404             | 180,94                |
| 2034         | 26.700                                     | 107,29                | 20.249                             | 70,61                 | 46.948             | 180,94                |
| 2035         | 25.908                                     | 107,29                | 19.635                             | 70,61                 | 45.543             | 180,94                |
| 2036         | 25.144                                     | 107,29                | 19.042                             | 70,61                 | 44.186             | 180,94                |
| 2037         | 24.406                                     | 107,29                | 18.470                             | 70,61                 | 42.876             | 180,94                |
| 2038         | 23.693                                     | 107,29                | 17.918                             | 70,61                 | 41.611             | 180,94                |
| 2039         | 23.004                                     | 107,29                | 17.385                             | 70,61                 | 40.389             | 180,94                |
| 2040         | 22.338                                     | 107,29                | 16.871                             | 70,61                 | 39.209             | 180,94                |
| 2041         | 21.695                                     | 107,29                | 16.373                             | 70,61                 | 38.068             | 180,94                |

|      |        |        |        |       |        |        |
|------|--------|--------|--------|-------|--------|--------|
| 2042 | 21.073 | 107,29 | 15.893 | 70,61 | 36.966 | 180,94 |
| 2043 | 20.472 | 107,29 | 15.429 | 70,61 | 35.901 | 180,94 |
| 2044 | 19.890 | 107,29 | 14.981 | 70,61 | 34.872 | 180,94 |
| 2045 | 19.328 | 107,29 | 14.548 | 70,61 | 33.876 | 180,94 |
| 2046 | 18.785 | 107,29 | 14.130 | 70,61 | 32.914 | 180,94 |
| 2047 | 18.259 | 107,29 | 13.725 | 70,61 | 31.984 | 180,94 |
| 2048 | 17.750 | 107,29 | 13.334 | 70,61 | 31.084 | 180,94 |
| 2049 | 17.258 | 107,29 | 12.956 | 70,61 | 30.214 | 180,94 |
| 2050 | 16.782 | 107,29 | 12.590 | 70,61 | 29.372 | 180,94 |
| 2051 | 16.322 | 107,29 | 12.236 | 70,61 | 28.558 | 180,94 |
| 2052 | 15.876 | 107,29 | 11.894 | 70,61 | 27.770 | 180,94 |
| 2053 | 15.444 | 107,29 | 11.564 | 70,61 | 27.008 | 180,94 |
| 2054 | 15.026 | 107,29 | 11.243 | 70,61 | 26.269 | 180,94 |
| 2055 | 14.621 | 107,29 | 10.934 | 70,61 | 25.555 | 180,94 |
| 2056 | 14.230 | 107,29 | 10.634 | 70,61 | 24.863 | 180,94 |
| 2057 | 13.850 | 107,29 | 10.344 | 70,61 | 24.194 | 180,94 |
| 2058 | 13.482 | 107,29 | 10.063 | 70,61 | 23.545 | 180,94 |
| 2059 | 13.126 | 107,29 | 9.791  | 70,61 | 22.917 | 180,94 |
| 2060 | 12.781 | 107,29 | 9.527  | 70,61 | 22.309 | 180,94 |
| 2061 | 12.447 | 107,29 | 9.272  | 70,61 | 21.719 | 180,94 |
| 2062 | 12.123 | 107,29 | 9.025  | 70,61 | 21.148 | 180,94 |
| 2063 | 11.809 | 107,29 | 8.786  | 70,61 | 20.595 | 180,94 |
| 2064 | 11.504 | 107,29 | 8.554  | 70,61 | 20.059 | 180,94 |
| 2065 | 11.209 | 107,29 | 8.330  | 70,61 | 19.539 | 180,94 |
| 2066 | 10.923 | 107,29 | 8.112  | 70,61 | 19.035 | 180,94 |
| 2067 | 10.645 | 107,29 | 7.901  | 70,61 | 18.547 | 180,94 |
| 2068 | 10.376 | 107,29 | 7.697  | 70,61 | 18.073 | 180,94 |
| 2069 | 10.115 | 107,29 | 7.499  | 70,61 | 17.614 | 180,94 |



|      |       |        |       |       |        |        |
|------|-------|--------|-------|-------|--------|--------|
| 2070 | 9.862 | 107,29 | 7.306 | 70,61 | 17.168 | 180,94 |
| 2071 | 9.616 | 107,29 | 7.120 | 70,61 | 16.736 | 180,94 |
| 2072 | 9.378 | 107,29 | 6.939 | 70,61 | 16.317 | 180,94 |
| 2073 | 9.146 | 107,29 | 6.764 | 70,61 | 15.910 | 180,94 |
| 2074 | 8.922 | 107,29 | 6.594 | 70,61 | 15.516 | 180,94 |
| 2075 | 8.703 | 107,29 | 6.429 | 70,61 | 15.133 | 180,94 |
| 2076 | 8.492 | 107,29 | 6.269 | 70,61 | 14.761 | 180,94 |
| 2077 | 8.286 | 107,29 | 6.114 | 70,61 | 14.400 | 180,94 |
| 2078 | 8.087 | 107,29 | 5.963 | 70,61 | 14.050 | 180,94 |
| 2079 | 7.893 | 107,29 | 5.817 | 70,61 | 13.710 | 180,94 |
| 2080 | 7.705 | 107,29 | 5.675 | 70,61 | 13.380 | 180,94 |
| 2081 | 7.522 | 107,29 | 5.537 | 70,61 | 13.059 | 180,94 |
| 2082 | 7.344 | 107,29 | 5.403 | 70,61 | 12.747 | 180,94 |

**Table 32. Carbon stocks per hectare of initial forest classes icl existing in the leakage belt**

| Project year | Initial class                              |                       |                                    |                       |                    |                       |
|--------------|--------------------------------------------|-----------------------|------------------------------------|-----------------------|--------------------|-----------------------|
|              | Name                                       |                       | Riparian Dense Tropical Rainforest |                       |                    |                       |
|              | IDcl                                       |                       | 1                                  |                       |                    |                       |
|              | Average carbon stock per hectare +- 90% CI |                       |                                    |                       |                    |                       |
|              | Cabicl                                     |                       | Cbbicl                             |                       | Ctoticl            |                       |
|              | C stock                                    | ±90%                  | C stock                            | ±90%                  | C stock            | ±90%                  |
|              | tCO <sub>2</sub> e                         | tCO <sub>2</sub> e/ha | tCO <sub>2</sub> e                 | tCO <sub>2</sub> e/ha | tCO <sub>2</sub> e | tCO <sub>2</sub> e/ha |
| 2023         | 1.722.928                                  | 107,29                | 113.396                            | 70,61                 | 1.836.324          | 180,94                |
| 2024         | 1.660.004                                  | 107,29                | 222.651                            | 70,61                 | 1.882.655          | 180,94                |
| 2025         | 1.599.676                                  | 107,29                | 327.935                            | 70,61                 | 1.927.612          | 180,94                |
| 2026         | 1.541.827                                  | 107,29                | 429.412                            | 70,61                 | 1.971.239          | 180,94                |
| 2027         | 1.486.344                                  | 107,29                | 527.237                            | 70,61                 | 2.013.582          | 180,94                |
| 2028         | 1.433.122                                  | 107,29                | 621.560                            | 70,61                 | 2.054.682          | 180,94                |

|      |           |        |         |       |           |        |
|------|-----------|--------|---------|-------|-----------|--------|
| 2029 | 1.382.059 | 107,29 | 712.522 | 70,61 | 2.094.581 | 180,94 |
| 2030 | 1.333.059 | 107,29 | 800.258 | 70,61 | 2.133.318 | 180,94 |
| 2031 | 1.286.031 | 107,29 | 884.900 | 70,61 | 2.170.930 | 180,94 |
| 2032 | 1.240.886 | 107,29 | 966.570 | 70,61 | 2.207.456 | 180,94 |
| 2033 | 1.197.542 | 107,29 | 931.991 | 70,61 | 2.129.533 | 180,94 |
| 2034 | 1.155.920 | 107,29 | 898.814 | 70,61 | 2.054.734 | 180,94 |
| 2035 | 1.115.944 | 107,29 | 866.977 | 70,61 | 1.982.920 | 180,94 |
| 2036 | 1.077.542 | 107,29 | 836.419 | 70,61 | 1.913.962 | 180,94 |
| 2037 | 1.040.647 | 107,29 | 807.085 | 70,61 | 1.847.733 | 180,94 |
| 2038 | 1.005.193 | 107,29 | 778.921 | 70,61 | 1.784.114 | 180,94 |
| 2039 | 971.117   | 107,29 | 751.874 | 70,61 | 1.722.991 | 180,94 |
| 2040 | 938.361   | 107,29 | 725.897 | 70,61 | 1.664.258 | 180,94 |
| 2041 | 906.868   | 107,29 | 700.942 | 70,61 | 1.607.810 | 180,94 |
| 2042 | 876.584   | 107,29 | 676.965 | 70,61 | 1.553.549 | 180,94 |
| 2043 | 847.458   | 107,29 | 653.924 | 70,61 | 1.501.382 | 180,94 |
| 2044 | 819.441   | 107,29 | 631.778 | 70,61 | 1.451.219 | 180,94 |
| 2045 | 792.486   | 107,29 | 610.489 | 70,61 | 1.402.976 | 180,94 |
| 2046 | 766.548   | 107,29 | 590.021 | 70,61 | 1.356.569 | 180,94 |
| 2047 | 741.586   | 107,29 | 570.338 | 70,61 | 1.311.924 | 180,94 |
| 2048 | 717.557   | 107,29 | 551.407 | 70,61 | 1.268.964 | 180,94 |
| 2049 | 694.424   | 107,29 | 533.196 | 70,61 | 1.227.620 | 180,94 |
| 2050 | 672.149   | 107,29 | 515.675 | 70,61 | 1.187.824 | 180,94 |
| 2051 | 650.697   | 107,29 | 498.815 | 70,61 | 1.149.512 | 180,94 |
| 2052 | 630.035   | 107,29 | 482.588 | 70,61 | 1.112.623 | 180,94 |
| 2053 | 610.129   | 107,29 | 466.968 | 70,61 | 1.077.097 | 180,94 |
| 2054 | 590.949   | 107,29 | 451.930 | 70,61 | 1.042.879 | 180,94 |
| 2055 | 572.466   | 107,29 | 437.449 | 70,61 | 1.009.915 | 180,94 |
| 2056 | 554.652   | 107,29 | 423.503 | 70,61 | 978.154   | 180,94 |

|      |         |        |         |       |         |        |
|------|---------|--------|---------|-------|---------|--------|
| 2057 | 537.479 | 107,29 | 410.069 | 70,61 | 947.548 | 180,94 |
| 2058 | 520.921 | 107,29 | 397.127 | 70,61 | 918.049 | 180,94 |
| 2059 | 504.955 | 107,29 | 384.657 | 70,61 | 889.612 | 180,94 |
| 2060 | 489.556 | 107,29 | 372.640 | 70,61 | 862.196 | 180,94 |
| 2061 | 474.703 | 107,29 | 361.056 | 70,61 | 835.759 | 180,94 |
| 2062 | 460.373 | 107,29 | 349.890 | 70,61 | 810.263 | 180,94 |
| 2063 | 446.545 | 107,29 | 339.124 | 70,61 | 785.669 | 180,94 |
| 2064 | 433.201 | 107,29 | 328.741 | 70,61 | 761.942 | 180,94 |
| 2065 | 420.321 | 107,29 | 318.728 | 70,61 | 739.049 | 180,94 |
| 2066 | 407.888 | 107,29 | 309.068 | 70,61 | 716.956 | 180,94 |
| 2067 | 395.883 | 107,29 | 299.749 | 70,61 | 695.632 | 180,94 |
| 2068 | 384.291 | 107,29 | 290.756 | 70,61 | 675.047 | 180,94 |
| 2069 | 373.095 | 107,29 | 282.078 | 70,61 | 655.173 | 180,94 |
| 2070 | 362.280 | 107,29 | 273.701 | 70,61 | 635.981 | 180,94 |
| 2071 | 351.832 | 107,29 | 265.614 | 70,61 | 617.446 | 180,94 |
| 2072 | 341.737 | 107,29 | 257.806 | 70,61 | 599.543 | 180,94 |
| 2073 | 331.981 | 107,29 | 250.266 | 70,61 | 582.246 | 180,94 |
| 2074 | 322.551 | 107,29 | 242.983 | 70,61 | 565.534 | 180,94 |
| 2075 | 313.436 | 107,29 | 235.949 | 70,61 | 549.384 | 180,94 |
| 2076 | 304.623 | 107,29 | 229.152 | 70,61 | 533.775 | 180,94 |
| 2077 | 296.102 | 107,29 | 222.585 | 70,61 | 518.687 | 180,94 |
| 2078 | 287.861 | 107,29 | 216.238 | 70,61 | 504.099 | 180,94 |
| 2079 | 279.889 | 107,29 | 210.104 | 70,61 | 489.993 | 180,94 |
| 2080 | 272.178 | 107,29 | 204.174 | 70,61 | 476.352 | 180,94 |
| 2081 | 264.717 | 107,29 | 198.440 | 70,61 | 463.157 | 180,94 |
| 2082 | 257.498 | 107,29 | 192.896 | 70,61 | 450.394 | 180,94 |

Table 15b. of the methodology, which is related to the above tables was not created for the following reason: This table is not applicable due to no necessary discounts for uncertainties.

**Table 33. Long-term (60-years) area weighted average carbon stock per zone**

| Post – deforestation LU/LC-classes fcl                   |                                                          |
|----------------------------------------------------------|----------------------------------------------------------|
| Name                                                     | Non-Forest                                               |
| IDfcl                                                    | 2                                                        |
| Cabfcl<br>C stock<br>tCO <sub>2</sub> e ha <sup>-1</sup> | Cbbfcl<br>C stock<br>tCO <sub>2</sub> e ha <sup>-1</sup> |
| 0                                                        | 0                                                        |

### 3.5 Additionality

#### 3.5.1 Regulatory Surplus

Is the project registered or seeking registration in an UNFCCC Annex 1 or Non-Annex 1 country?

☐ Annex 1 country

☒ Non-Annex 1 country



## Brazil

Type of Party:  
**Non-Annex I**

Ratification status:  
**Paris Agreement**  
Date of signature: 22 April 2016  
Date of ratification: 21 September 2016

**Kyoto Protocol**  
Date of signature: 29 April 1998  
Date of ratification: 23 August 2002

Source: UNFCC<sup>79</sup>

Are the project activities mandated by any law, statute, or other regulatory framework?

☐ Yes

☒ No

If the project is located inside a Non-Annex 1 country and the project activities are mandated by a law, statute, or other regulatory framework, are such laws, statutes, or regulatory frameworks systematically enforced?

☐ Yes

☒ No

<sup>79</sup> <https://unfccc.int/node/180413>

### **Non-enforcement scenario (Brazilian GHG regulatory framework)**

Brazil is not part of the UNFCCC Annex 1, and even though it has agreed GHG targets in the global community, there is currently no compulsory reduction imposed on sectors. There is an initial law discussion under consideration - at the time when the first version of this PD is being written - for the regulation of a local carbon market. However, the project proponent understands that it will not be approved anytime soon, and even if it is approved, 1) there will be a significant delay before it is effectively implemented and 2) companies/rural producers engaged in agricultural activities are not likely to be included initially, based on discussions in parliament. The activities from this project, therefore, must be prioritized, anticipating government directives, as it understands that the climate scenario is urgent and needs to be addressed as a top priority. However, any advances in legislation will be adapted to the scope of this project when required and applicable.

### **Project Activities Surplus**

The activities of this project are considered as additional surplus to the activities regularly consolidated in the regulatory case because:

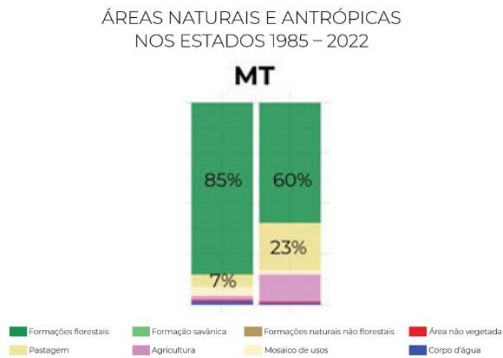
- 1) Although there are regulatory instruments for forest protection governed by the Brazilian Forest Code, data<sup>80</sup> shows that there is little effectiveness in terms of monitoring, penalties, or sanctions. In this case, the activities of this project ensure that, through the strong partnership between the private landowner and NovaAgri, which has been in place since the commercial relationship, there will be a more efficient approach to voluntary protection, based on good relations, environmental awareness, mobilization for social change, all tied to a legal binding contract that formalizes this commitment.
- 2) Another essential point of additional surplus is related to the intensive investment in technology, involving a dynamic system with deforestation alerts, fire and wildfire hotspots, and the use of state-of-the-art satellite images with excellent resolution and periodicity. All this investment is only possible due to NovaAgri's commitment to its strategic planning and relationship with its stakeholders. This investment would hardly be made to the same extent by the public sphere.
- 3) It is understood that forest legislation is very comprehensive regarding possible mechanisms for displacing deforestation in forested areas. For example, ensuring that an area has a legal reserve to be guaranteed by law today does not mean that this area cannot be deforested. This is because there are mechanisms that allow the purchase of other forested areas in other regions, as long as they are registered under the same owner, to compensate for a deforested legal reserve area. Among the activities of this project is the signing of a legal binding contract,

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<sup>80</sup> MAPBIOMAS. RAD 2022 – Relatório anual do desmatamento no Brasil (Junho, 2023).  
<https://alerta.mapbiomas.org/relatorio>

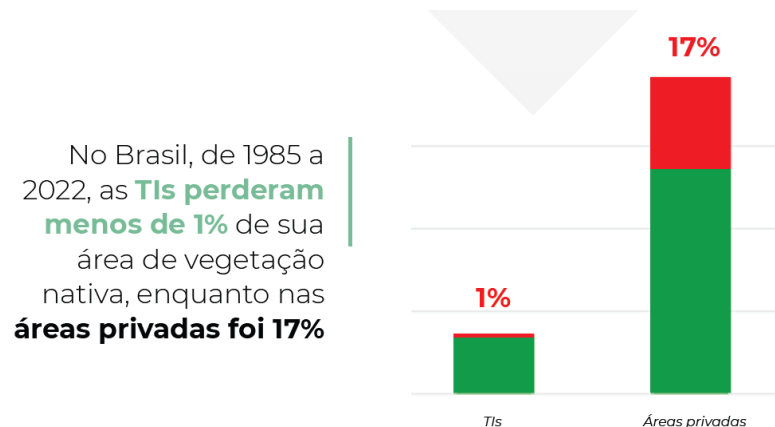
with clear clauses regarding the restriction of this type of action. Thus, it is avoided not only the protection of the project area but also the forested areas of other regions, thus favouring non-leakage. Moreover, the project area has additional areas beyond what is required by forest law, being an important additional reserve for carbon stock preservation.

4) The revenues from the sale of VCUs cannot overcome the margins of conventional agricultural and livestock activities because they are highly profitable activities in the country, especially at the reference region, as shown on the image below.



Source: Mapbiomas – Expansion of anthropical areas in the MT state between 1985-2022

Also, there is clear evidence that public protected areas, such as delimited indigenous areas, have a proven efficiency against the deforestation, when comparing to private areas.



Source: Mapbiomas<sup>81</sup> – ‘In Brazil, from 1985 to 2022, the indigenous land lost less than 1% of its native vegetation, while the private areas lost 17%’

<sup>81</sup> Projeto MapBiomas – Mapeamento anual de cobertura e uso da terra no Brasil entre 1985 a 2022 – Coleção 8, acessado em 12/2023 através do link: [https://brasil.mapbiomas.org/wp-content/uploads/sites/4/2023/08/FACT\\_MapBiomas\\_Mapeamento-Anual-Cobertura\\_Colecao8\\_31.pdf](https://brasil.mapbiomas.org/wp-content/uploads/sites/4/2023/08/FACT_MapBiomas_Mapeamento-Anual-Cobertura_Colecao8_31.pdf)

Therefore, there are many arguments which come from large-scale rural producer to carry out forest openings and put regulatory issues in the second plan. The activities of this project aim to a different strategy, through the good commercial relationship between the parties, socio-environmental awareness, and legal agreement for no deforestation. This is a way to replicate as maximum as possible the protection structure of the public lands to the private lands.

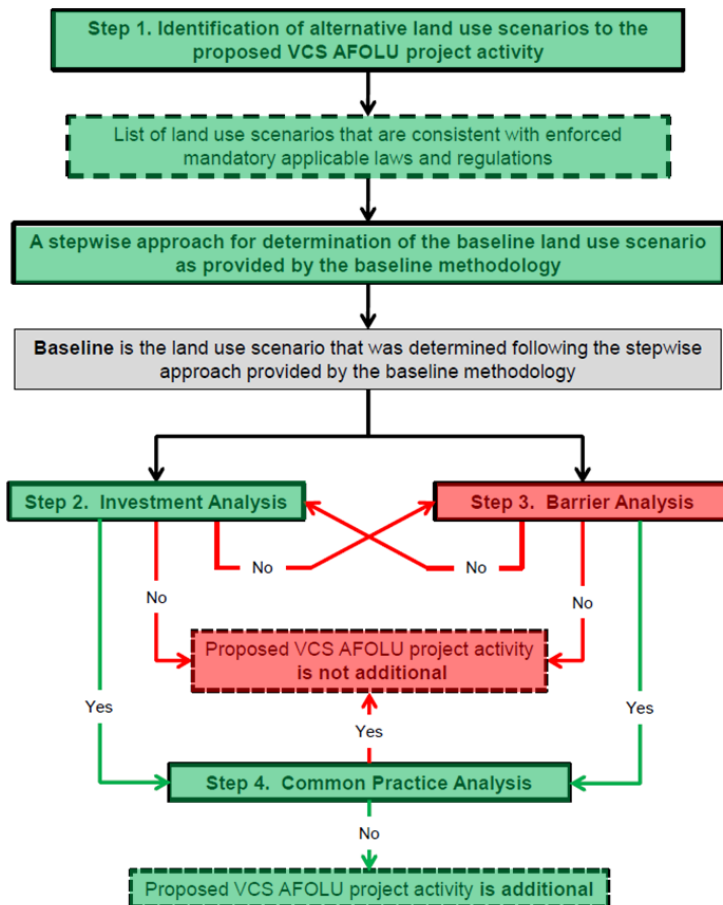
5) Even if there were a Regulated Carbon Market today, there is no guarantee that the structure of this system would ensure that the financial resources (raised through carbon credits) would be reinvested in social investment. The activities of this project compulsorily establish this commitment with society, especially with the most vulnerable communities present in the surrounding area of the project, which can be considered an important additional surplus.

6) Even though the area destined for preservation was indicated in the baseline as 10,012 ha, the total area subject to monitoring and protection totals 13,926 ha. This shows that the proponent is allocating project activities against deforestation in additional portions of forested areas, which can be seen as an additional surplus.

### 3.5.2 Additionality Methods

As required in the VERRA Additionality tool VT0001 (v3), the additionality analysis for this project is applied following the steps below:





VT001 Indicative flowchart of the Tool for the Demonstration and Assessment of Additionality in VCS AFOLU Project Activities.

### STEP 1: Identification of alternative land use scenarios to the proposed VCS AFOLU project activity

This step serves to identify alternative land use scenarios to the proposed VCS AFOLU project activity(s) that could be the baseline scenario.

**Sub-step 1a.** This step aims to Identify credible alternative land use scenarios to the proposed VCS AFOLU project activity.

The project has identified realistic and credible land-use scenarios that could/might occur on the land encompassed by the project boundary. These scenarios have been aligned with the project area's historical land uses, practices, economic trends, and relevant national and sectoral policies. The identified land use scenarios encompass:

#### **Scenario A) The continuation of the pre-project land use.**

In the absence of the REDD+ AUD project's intervention, the area would be susceptible to the continuation of conventional agricultural practices and potential deforestation, particularly

considering the already high deforestation rates in the reference region. Agricultural and cattle ranching expansion driven by conventional practices could exacerbate these rates, leading to the conversion of forested land into agricultural fields for monoculture crops such as soybeans and corn, and cattle ranching activities.

The region's alarming deforestation rates accentuate the urgency of taking proactive measures. The pressures for land conversion, coupled with high economic incentives, could escalate deforestation even further, resulting in the loss of vital biodiversity, and ecosystem services. Furthermore, the prevailing financial motivation for farmers to engage in crop cultivation rather than forest preservation amplifies the risk of escalating deforestation because the prospect of immediate profit often shades the long-term environmental consequences.

The project area is included on the transition of Amazon and *Cerrado Savanna* biomes. Historically known for its significant agricultural expansion, especially in soybean and cattle ranching, the state of Mato Grosso experienced a surge in deforestation rates during the early 2000s. If considering the historical baseline deforestation, the entire project area could be vanished from short to medium term. Even though there are legal instruments to prevent the illegal deforestation in such biomes, for instance the Forest Code (Law n. 12.651, 2012). The fact is that illegal deforestation in this biome is a common practice within legal and non-legal reserve areas. About the impunity, the legal actions and embargoes carried out by IBAMA and ICMBio (by May 2023) in Brazil reached only 10.2% of the illegal deforested area identified from 2019 to 2022. That is a clear and worrying signal that the law against deforestation in Brazil is far away to be effective itself. It is important to mention that, until now, the Brazilian government has not compromised to implement a fully national monitoring system with the same precision and periodicity proposed by this project. Therefore, a feasible implementation and law enforcement is still an issue with many challenges to overcome, especially in national levels.

As a conclusion, once the Project Area is situated within the *Cerrado Savanna*, an environment where prevailing regulations permit legal deforestation of a significant portion of the land – the government mandates a legal reserve area for *Cerrado Savanna* areas of only 35%, allowing for extensive land use change. Without the implementation of this REDD+ project, the area would face a worrying threat of conversion, primarily driven by the lucrative prospects offered by agribusiness activities – the profitability associated with agricultural expansion poses a substantial risk to the preservation of natural ecosystems in the absence of sustainable alternatives. Moreover, even the portion of the area that cannot be legally deforested, as stipulated by law, remains at risk due to challenges related to national enforcement and the prevalence of impunity. This poses a serious threat to the integrity of the legally protected zones within the transitional *Cerrado Savanna*, which also have great portions of Amazon forests, as presented before on this PD.

***Scenario B) The project activity conducted within the project boundary, excluding registration under the VCS AFOLU project.***

In a scenario where REDD+ activities are introduced to the project boundary without being officially registered under a VCS AFOLU project, there would be an absence of a structured MRV framework. Without this, there might be limited assurance in the land protection, and consequently in the integrity of carbon sequestration and emission reduction efforts.

Additionally, the lack of formal registration might hinder the access to monetary incentives associated with certified carbon credits. The absence of a clear financial return for implementing REDD+ actions could lead some landowners to reconsider their participation in the project. This could potentially result in a farmer opting out of the project, choosing instead to pursue other land uses that offer more immediate financial benefits.

As showed in the Opportunity Cost analysis, even though traditional activities seem to be more profitable, a project of preservation under VCS criteria can benefit the rancher/farmer with a financial incentive, even in a smaller scale. Thus, without ensuring a financial return for the taken preservation activities, the initiatives would likely not to be sustained for a long time.

In this scenario, even though forest preservation activities could be implemented, the activities that could potentially benefit the people in the communities might not be put into practice or managed properly, as is the case in VCS projects. Economic incentives and effective management of activities are essential factors for positive and meaningful impacts to occur.

In summary, without the formal recognition and oversight provided by VCS registration, there could be challenges in tracking and attributing the environmental, social, and financial co-benefits of the REDD+ activities. This could also lead to a diminished ability to quantify and report the broader positive impacts on biodiversity, local communities, and ecosystem services.

#### **Sub-step 1b. Consistency of credible land use scenarios with enforced mandatory applicable laws and regulations**

##### ***Scenario A) The continuation of the pre-project land use.***

###### *Agriculture activities*

This scenario is credible once there is no restriction of mandatory applicable laws and regulations for the agricultural (soybeans and corn crops) activities.

###### *Deforestation activities*

Brazil possesses a distinctive forest law compared to other forest-developing countries. However, the country faces intrinsic difficulties in tackling deforestation due to geographical, cultural, and historical factors. Despite the existence of the 'Forest Code law' which has prevented a more significant advance of deforestation, vegetation suppression remains pervasive in Brazil, with thousands of hectares being cleared across its main biomes. In other words, even legally protected reserve areas, which should not be subject to deforestation, suffer due to a lack of oversight, weak public policies, non-compliance with laws, and a failure to enforce penalties - as mentioned earlier, the legal actions and embargoes executed by IBAMA and ICMBio (as of May

2023) only covered 10.2% of the identified deforested area from 2019 to 2022 (MAPBIOMAS, 2023)<sup>82</sup>. As conclusion, the forests' clearing activities might not comply with the current forest legislation, but this scenario is still credible, once the clearings are widespread and prevalent on great part of the Reference Area, which include non-compliance deforestation.

However, the forest law in Brazil is not completely inflexible. There are legal instruments that permits that, even if a landowner exceeds their legal reserve through deforestation, this can offset that by reforesting, preserving, or generating/buying credits of another forested area elsewhere. In this way, adhering to the current *modus operandi* without a carbon project or other financial incentive, the landowner could clear an area without violating the law, as long as this can be justified by this instrument. Even though reforestation plans might be important somehow, this is not the case, once secondary forests are not valorous as a native / original forest. In terms of biodiversity this can also be an issue, once there are species dependent of that ecosystem which was changed. The conclusion is that the continuation of the pre-project land use can infringe the regulatory scenario but also can be compensated by other legal instruments – Both cases cause irreversible impacts. Thus, the land use and conditions for this scenario is credible and likely possible to occur.

***Scenario B) The project activity conducted within the project boundary, excluding registration under the VCS AFOLU project.***

Similarly, this scenario is also credible once there is no restriction of mandatory applicable laws and regulations for AFOLU activities (e.g. private monitoring in rural areas). It is important to mention that the project area is not encompassed by any jurisdictional VCS project area or demarcated area by the government, such as indigenous parks or conservation units.

### **Sub-step 1c. Selection of the baseline scenario**

The most plausible scenario for the baseline construction is the Scenario A 'The continuation of the pre-project land use'.

The scenario described has been chosen as the most plausible baseline for the Carbon AFOLU VCS project due to:

- **Current Deforestation Trends:** The scenario acknowledges the alarming deforestation rates in the reference region. These existing trends provide a baseline that reflects the current trajectory of land-use change. By highlighting the ongoing deforestation, the baseline scenario accurately represents the business-as-usual situation without project intervention.
- **Potential for Escalation:** The potential for escalation in deforestation due to various factors such as economic incentives for agriculture and the risk of immediate profit overshadowing long-term environmental consequences.

<sup>82</sup> MAPBIOMAS. RAD 2022 – Relatório anual do desmatamento no Brasil (Junho, 2023).  
<https://alerta.mapbiomas.org/relatorio>

- **Historical Deforestation Data:** Referring to historical data on the reference area's forest loss rate between specific years adds empirical evidence to the baseline construction.
- **Legal Framework and Enforcement Challenges:** The acknowledgement of challenges in enforcing existing legal instruments, such as the Forest Code, contributes to the baseline's realism. The common practice of illegal deforestation despite regulatory measures suggests that, without the project, the legal framework alone may not be sufficient to curb deforestation.
- **Lack of National Monitoring System:** The absence of a fully implemented monitoring system with precision and periodicity adds another argument to the baseline construction. It underscores the uncertainty and challenges in accurately tracking and enforcing land-use changes, supporting the argument that deforestation could proceed unpredicted without the project.

In summary, it has been assessed that scenario B – even though it is credible, is unlikely to occur. This is primarily due to the absence of any types of incentives in Brazil for maintaining forests untouched. As a remote possibility, the area might be designated as part of a jurisdictional private preservation zone, by determination of the government – This is unlikely to happen given the area is part of a well-established private productive area in the region. Additionally, it is recognized that the legislation is fragile, and any gap found on it could lead to a tendency toward scenario A.

Further details will be complemented on section 4.1, by analyzing the historical data and processing the respective deforestation images.

## **Step 2. Investment analysis**

### **Sub-step 2a. Determine appropriate analysis method**

The VCS AFOLU project generates no revenues other than VCU-related income. Thus, the analysis method applied was the 'simple cost analysis', once the sole income generated by keeping the native vegetation intact is the selling of VCU credits – when the project is fully implemented and verified. There are no plans, for instance, for activities such as agroforestry production, ecosystem service management for income generation, or the generation of credits of any other nature apart from VCU credits.

Furthermore, for the project's implementation, the proponent will allocate investments that would not materialize without the execution of this VCS project. Below are the very approximate values of these investments<sup>83</sup>:

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<sup>83</sup> Note: in order to protect sensible commercial or strategic information, the values presented above do not represent the actual numbers – all the items were shown in a range basis just to demonstrate the magnitude order of those investments.

**Table 34. Investments Estimation Analysis**

| Activity / System / Utility               | Type of Investment<br>(Initial or Periodic) | Investment<br>(approx. in kR\$) |
|-------------------------------------------|---------------------------------------------|---------------------------------|
| Feasibility Analysis System               | Initial                                     | 200                             |
| Consulting                                | Initial                                     | 100                             |
| Certifier <sup>84</sup>                   | Initial                                     | 20                              |
|                                           | Periodic                                    | 2.5                             |
| Validation                                | Initial                                     | 200                             |
| Verification                              | Periodic                                    | 100                             |
| Travels                                   | Initial                                     | 30                              |
| Other Expenses (technology, systems etc.) | Initial                                     | 200                             |
|                                           | Periodic                                    | 200                             |
| <b>Total</b>                              | Initial                                     | >750                            |
|                                           | Periodic                                    | >300                            |

#### Step 4. Common practice analysis

As of the writing of this PDD (end of 2023), only 22 REDD projects are effectively registered on the VERRA VCS platform, considering solely the Brazilian scenario. This indicates that, although the practice of offsetting through forest preservation is on an upward trend, it is still quite incipient and not widely spread in private areas. What is more common in Brazil are public conservation areas, established through legal demarcations such as indigenous lands, conservation units, and forest parks. Nevertheless, even in benchmarked public preservation areas, signs of deforestation can be observed, as seen for instance in the adjacent indigenous area to this project area, where several clearings have been identified by satellite imagery.

When we narrow down this scenario to Brazil, we find that only 1 out of these 22 projects is located in the State of MT (Mato Grosso). Moreover, it is situated in a very distant location, to the northwest of the state, while our project is located on the opposite end. Therefore, there is ample room for action within the state, which historically exhibits a harm dynamic to forests. As mentioned earlier, the state is within the Arc of Deforestation and is a region of extreme

<sup>84</sup> Note: not considering costs for VCUs emission

biodiversity richness, carrying significant ecological corridors and a rare biome transition ecosystem (Amazon and *Cerrado* Savanna).

Consequently, it can be asserted that the project meets strong additionality criteria, considering that voluntary forest preservation practices in private areas within the state are quite rare. What is predominantly observed as instruments to combat deforestation are punitive actions under Brazilian forest law. However, as mentioned earlier, even these instruments prove to be fragile in face of the various challenges the country has to confront.

**Table 35. Common Practice (REDD+ VCS) in Mato Grosso**

| VCS REDD ID<br>(registered in Brazil –<br>green highlight: MT)<br>by DEC.2023 |
|-------------------------------------------------------------------------------|
| 2566                                                                          |
| 2558                                                                          |
| 2551                                                                          |
| 2539                                                                          |
| 2508                                                                          |
| 2373                                                                          |
| 1953                                                                          |
| 1686                                                                          |
| 1654                                                                          |
| 1147                                                                          |
| 1118                                                                          |
| 1113                                                                          |
| 1112                                                                          |
| 1571                                                                          |
| 875                                                                           |
| 1094                                                                          |
| 1115                                                                          |
| 1503                                                                          |
| 963                                                                           |
| 832                                                                           |
| 1382                                                                          |
| 1329                                                                          |

Another crucial additional point is the project's identity within this context. This might be the first REDD+ project in Mato Grosso (or in Brazil) developed by a private company whose core business is not centered around carbon credit projects. Its investment purpose is much more aligned with minimizing environmental impacts and achieving climate targets than revenue generation, because NovaAgri understands the necessity to address the negative impacts within its Value



Chain. Therefore, the proposing company is fully committed to ensuring a high-integrity project through a well-crafted Project Design (PD) and a rigorously executed Monitoring, Reporting, and Verification (MRV) process, utilizing an in-depth study for the Baseline, and high-modern technology for monitoring.

In light of these considerations, we conclude that Step 4 is deemed satisfied, i.e. similar activities can be observed and essential distinctions between the proposed VCS AFOLU project activity and similar activities can be made, hence, it is additional.

### 3.6 Methodology Deviations

There are no methodology deviations applied in this project, only complementary rationales and processes which are not related to monitoring, and are entirely showed and discussed along this PD.

## 4 IMPLEMENTATION STATUS

### 4.1 Implementation Status of the Project Activity

At the time of the first validation, the initial verification monitoring is already underway, hence this Joint PD version. The monitoring techniques used followed the same tools and procedures used for the baseline survey (PRODES data).

For future verifications, additional complementary tools will be applied to achieve a more precise determination of potential deforested areas. These tools are those expressed in Section 1.12 (Description of the Project Activity) of this PD. For this reason, there will indeed be a change in project activities from this first verification to the next ones, primarily based on the enhancement of monitoring techniques, using more accurate images and an exclusive dashboard for alerts (fire and deforestation), especially for the project area. However, assessing public data will also continue to be an important item as part of the assessment strategy.

For the project area itself, which is the focus of AUD analysis, during the first crediting period (July 2023 to June 2024), there was no loss of vegetation, resulting in an effective reduction of 40,008 tCO<sub>2</sub>e during this monitoring year and no buffer decrease was required.

All other project activities will continue to be executed as planned in this PD. Those KPIs whose results depend directly on the first sale of VCU will be applied only after the second monitoring period.

# 5 QUANTIFICATION OF ESTIMATED GHG EMISSION REDUCTIONS AND REMOVALS

## 5.1 Baseline Emissions

The total average biomass stock per hectare ( $\text{Mg ha}^{-1}$ ) was converted to  $\text{tCO}_2\text{e}$  using the following equations:

$$Cabicl = ab \times CF \times 44/12$$

Where,

*Cabicl* Average carbon stock per hectare in the above-ground biomass carbon pool of initial forest class *icl*;  $\text{tCO}_2\text{e ha}^{-1}$

*ab* Average biomass stock per hectare in the above-ground biomass pool of initial forest class *icl*;  $\text{Mg ha}^{-1}$

*CF* Default value of carbon fraction in biomass

*44/12* Ratio converting C to  $\text{CO}_2\text{e}$

$$Cbbicl = bb \times CF \times 44/12$$

Where,

*Cbbicl* Average carbon stock per hectare in the below-ground biomass carbon pool of initial forest class *icl*;  $\text{tCO}_2\text{e ha}^{-1}$

*bb* Average biomass stock per hectare in the below-ground biomass pool of initial forest class *icl*;  $\text{Mg ha}^{-1}$

*CF* Default value of carbon fraction in biomass

*44/12* Ratio converting C to  $\text{CO}_2\text{e}$

The total baseline carbon stock change in the project area at year *t* is calculated as follows:

$$CBSLPAt = CabBSLPAicl, t + CbbBSLPAicl, t$$

Where,

*CBSLPAt* Total baseline carbon stock changes in the project area at year *t*;  $\text{tCO}_2\text{e}$

|                     |                                                                                                                                          |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| $CabBSLPA_{icl, t}$ | Total baseline carbon stock change for the above-ground biomass pool in the project area for initial forest class at year $t$ ; $tCO_2e$ |
| $CbbBSLPA_{icl, t}$ | Total baseline carbon stock change for the below-ground biomass pool in the project area for initial forest class at year $t$ ; $tCO_2e$ |

$$CabBSLPA_{icl, t} = ABSLPA_{icl, t} \times Cab_{icl}$$

Where,

|                     |                                                                                                                                          |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| $CabBSLPA_{icl, t}$ | Total baseline carbon stock change for the above-ground biomass pool in the project area for initial forest class at year $t$ ; $tCO_2e$ |
| $ABSLPA_{icl, t}$   | Area of initial forest class $icl$ deforested at time $t$ within the project area in the baseline case; ha                               |
| $Cab_{icl}$         | Average carbon stock change factor per hectare in the above-ground biomass carbon pool of initial forest class $icl$ ; $tCO_2e\ ha^{-1}$ |

$$CbbBSLPA_{icl, t} = ABSLPA_{icl, t} \times Cbb_{icl}$$

Where,

|                     |                                                                                                                                          |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| $CbbBSLPA_{icl, t}$ | Total baseline carbon stock change for the below-ground biomass pool in the project area for initial forest class at year $t$ ; $tCO_2e$ |
| $ABSLPA_{icl, t}$   | Area of initial forest class $icl$ deforested at time $t$ within the project area in the baseline case; ha                               |
| $Cbb_{icl}$         | Average carbon stock change factor per hectare in the below-ground biomass carbon pool of category $icl$ ; $tCO_2e\ ha^{-1}$             |

The Methodology step 6.1.3 stipulates that various change factors must be applied to the baseline case initial and post-deforestation classes in above-ground and below ground biomass. The carbon stocks in various pools are stipulated in Section 3.3 of the present VCS-PD. As such, Tables 36 and 37 below show carbon stock change factors for initial forest classes in above and below-ground carbon pools, which were then applied to calculate baseline carbon stock changes in various classes and pools shown in sequence.

**Table 36. Carbon stock change factors for initial forest classes (icl) in below-ground carbon stocks (Method 1)**

| Year after deforestation |       | $\Delta C_{bbicl,t}$  |
|--------------------------|-------|-----------------------|
|                          |       | tCO <sub>2</sub> e/ha |
| 1                        | t*    | -7.06                 |
| 2                        | t*+1  | -7.06                 |
| 2                        | t*+2  | -7.06                 |
| 4                        | t*+3  | -7.06                 |
| 5                        | t*+4  | -7.06                 |
| 6                        | t*+5  | -7.06                 |
| 7                        | t*+6  | -7.06                 |
| 8                        | t*+7  | -7.06                 |
| 9                        | t*+8  | -7.06                 |
| 10                       | t*+9  | -7.06                 |
| 11                       | t*+10 | 0                     |
| 12                       | t*+11 | 0                     |
| 13                       | t*+12 | 0                     |
| 14                       | t*+13 | 0                     |
| 15                       | t*+14 | 0                     |
| 16                       | t*+15 | 0                     |
| 17                       | t*+16 | 0                     |
| 18                       | t*+17 | 0                     |
| 19                       | t*+18 | 0                     |
| 20                       | t*+19 | 0                     |
| 21                       | t*+20 | 0                     |
| 22                       | t*+21 | 0                     |
| 23                       | t*+22 | 0                     |
| 24                       | t*+23 | 0                     |

|    |            |   |
|----|------------|---|
| 25 | $t^{*}+24$ | 0 |
| 26 | $t^{*}+25$ | 0 |
| 27 | $t^{*}+26$ | 0 |
| 28 | $t^{*}+27$ | 0 |
| 29 | $t^{*}+28$ | 0 |
| 30 | $t^{*}+29$ | 0 |

**Table 37. Carbon stock change factors for initial forest classes (icl) in above-ground carbon stocks (Method 1)**

| Year after deforestation |            | $\Delta C_{bi,cl,t}$<br>tCO <sub>2</sub> e/ha |
|--------------------------|------------|-----------------------------------------------|
| 1                        | $t^{*}$    | -107.29                                       |
| 2                        | $t^{*}+1$  | 0                                             |
| 2                        | $t^{*}+2$  | 0                                             |
| 4                        | $t^{*}+3$  | 0                                             |
| 5                        | $t^{*}+4$  | 0                                             |
| 6                        | $t^{*}+5$  | 0                                             |
| 7                        | $t^{*}+6$  | 0                                             |
| 8                        | $t^{*}+7$  | 0                                             |
| 9                        | $t^{*}+8$  | 0                                             |
| 10                       | $t^{*}+9$  | 0                                             |
| 11                       | $t^{*}+10$ | 0                                             |
| 12                       | $t^{*}+11$ | 0                                             |
| 13                       | $t^{*}+12$ | 0                                             |
| 14                       | $t^{*}+13$ | 0                                             |
| 15                       | $t^{*}+14$ | 0                                             |
| 16                       | $t^{*}+15$ | 0                                             |
| 17                       | $t^{*}+16$ | 0                                             |

|    |            |   |
|----|------------|---|
| 18 | $t^{*}+17$ | 0 |
| 19 | $t^{*}+18$ | 0 |
| 20 | $t^{*}+19$ | 0 |
| 21 | $t^{*}+20$ | 0 |
| 22 | $t^{*}+21$ | 0 |
| 23 | $t^{*}+22$ | 0 |
| 24 | $t^{*}+23$ | 0 |
| 25 | $t^{*}+24$ | 0 |
| 26 | $t^{*}+25$ | 0 |
| 27 | $t^{*}+26$ | 0 |
| 28 | $t^{*}+27$ | 0 |
| 29 | $t^{*}+28$ | 0 |
| 30 | $t^{*}+29$ | 0 |

**Table 38. Carbon stock change factors for final forest classes (fcl) or zones z (Method 1)**

| Year after deforestation |           | $\Delta C_{bicl,t}$<br>tCO <sub>2</sub> e/ha | $\Delta C_{bbicl,t}$<br>tCO <sub>2</sub> e/ha |
|--------------------------|-----------|----------------------------------------------|-----------------------------------------------|
| 1                        | $t^{*}$   | 0                                            | 0                                             |
| 2                        | $t^{*}+1$ | 0                                            | 0                                             |
| 2                        | $t^{*}+2$ | 0                                            | 0                                             |
| 4                        | $t^{*}+3$ | 0                                            | 0                                             |
| 5                        | $t^{*}+4$ | 0                                            | 0                                             |
| 6                        | $t^{*}+5$ | 0                                            | 0                                             |
| 7                        | $t^{*}+6$ | 0                                            | 0                                             |
| 8                        | $t^{*}+7$ | 0                                            | 0                                             |
| 9                        | $t^{*}+8$ | 0                                            | 0                                             |
| 10                       | $t^{*}+9$ | 0                                            | 0                                             |

|    |       |   |   |
|----|-------|---|---|
| 11 | t*+10 | 0 | 0 |
| 12 | t*+11 | 0 | 0 |
| 13 | t*+12 | 0 | 0 |
| 14 | t*+13 | 0 | 0 |
| 15 | t*+14 | 0 | 0 |
| 16 | t*+15 | 0 | 0 |
| 17 | t*+16 | 0 | 0 |
| 18 | t*+17 | 0 | 0 |
| 19 | t*+18 | 0 | 0 |
| 20 | t*+19 | 0 | 0 |
| 21 | t*+20 | 0 | 0 |
| 22 | t*+21 | 0 | 0 |
| 23 | t*+22 | 0 | 0 |
| 24 | t*+23 | 0 | 0 |
| 25 | t*+24 | 0 | 0 |
| 26 | t*+25 | 0 | 0 |
| 27 | t*+26 | 0 | 0 |
| 28 | t*+27 | 0 | 0 |
| 29 | t*+28 | 0 | 0 |
| 30 | t*+29 | 0 | 0 |

The resulting changes in carbon stock for initial (pre-deforestation) forest classes for the reference region, project area and leakage belt are shown in tables 39 – 46 below.



**Table 39. Baseline carbon stock change in the above-ground biomass in the reference region**

| Carbon stock change in the above-ground biomass per initial forest class |                                           |                                                                                                       |                                                            |
|--------------------------------------------------------------------------|-------------------------------------------|-------------------------------------------------------------------------------------------------------|------------------------------------------------------------|
| IDcl                                                                     | 1                                         | Total carbon stock change in the above-ground biomass of initial forest class in the reference region |                                                            |
| Name                                                                     | Dominant Vegetation in Ecological Tension |                                                                                                       |                                                            |
| Project year                                                             | tCO <sub>2e</sub>                         | $\Delta C_{abBSLRR}/cl, t$<br>Annual (tCO <sub>2e</sub> )                                             | $\Delta C_{abBSLRR}/cl$<br>Cumulative (tCO <sub>2e</sub> ) |
| 2023                                                                     | 3.436.403                                 | 3.436.403                                                                                             | 3.436.403                                                  |
| 2024                                                                     | 3.369.064                                 | 3.369.064                                                                                             | 6.805.467                                                  |
| 2025                                                                     | 3.303.292                                 | 3.303.292                                                                                             | 10.108.759                                                 |
| 2026                                                                     | 3.239.046                                 | 3.239.046                                                                                             | 13.347.805                                                 |
| 2027                                                                     | 3.176.286                                 | 3.176.286                                                                                             | 16.524.091                                                 |
| 2028                                                                     | 3.114.975                                 | 3.114.975                                                                                             | 19.639.065                                                 |
| 2029                                                                     | 3.055.074                                 | 3.055.074                                                                                             | 22.694.139                                                 |
| 2030                                                                     | 2.996.547                                 | 2.996.547                                                                                             | 25.690.686                                                 |
| 2031                                                                     | 2.939.358                                 | 2.939.358                                                                                             | 28.630.044                                                 |
| 2032                                                                     | 2.883.474                                 | 2.883.474                                                                                             | 31.513.518                                                 |
| 2033                                                                     | 2.828.860                                 | 2.828.860                                                                                             | 34.342.379                                                 |
| 2034                                                                     | 2.775.485                                 | 2.775.485                                                                                             | 37.117.864                                                 |
| 2035                                                                     | 2.723.316                                 | 2.723.316                                                                                             | 39.841.180                                                 |
| 2036                                                                     | 2.672.323                                 | 2.672.323                                                                                             | 42.513.502                                                 |
| 2037                                                                     | 2.622.476                                 | 2.622.476                                                                                             | 45.135.978                                                 |
| 2038                                                                     | 2.573.745                                 | 2.573.745                                                                                             | 47.709.723                                                 |
| 2039                                                                     | 2.526.104                                 | 2.526.104                                                                                             | 50.235.827                                                 |
| 2040                                                                     | 2.479.523                                 | 2.479.523                                                                                             | 52.715.350                                                 |
| 2041                                                                     | 2.433.977                                 | 2.433.977                                                                                             | 55.149.327                                                 |

|      |           |           |             |
|------|-----------|-----------|-------------|
| 2042 | 2.389.440 | 2.389.440 | 57.538.767  |
| 2043 | 2.345.886 | 2.345.886 | 59.884.653  |
| 2044 | 2.303.291 | 2.303.291 | 62.187.944  |
| 2045 | 2.261.631 | 2.261.631 | 64.449.575  |
| 2046 | 2.220.882 | 2.220.882 | 66.670.457  |
| 2047 | 2.181.023 | 2.181.023 | 68.851.480  |
| 2048 | 2.142.031 | 2.142.031 | 70.993.511  |
| 2049 | 2.103.885 | 2.103.885 | 73.097.396  |
| 2050 | 2.066.563 | 2.066.563 | 75.163.959  |
| 2051 | 2.030.047 | 2.030.047 | 77.194.006  |
| 2052 | 1.994.315 | 1.994.315 | 79.188.321  |
| 2053 | 1.959.350 | 1.959.350 | 81.147.672  |
| 2054 | 1.925.132 | 1.925.132 | 83.072.803  |
| 2055 | 1.891.643 | 1.891.643 | 84.964.447  |
| 2056 | 1.858.866 | 1.858.866 | 86.823.312  |
| 2057 | 1.826.783 | 1.826.783 | 88.650.095  |
| 2058 | 1.795.377 | 1.795.377 | 90.445.472  |
| 2059 | 1.764.633 | 1.764.633 | 92.210.105  |
| 2060 | 1.734.535 | 1.734.535 | 93.944.640  |
| 2061 | 1.705.066 | 1.705.066 | 95.649.706  |
| 2062 | 1.676.213 | 1.676.213 | 97.325.919  |
| 2063 | 1.647.960 | 1.647.960 | 98.973.878  |
| 2064 | 1.620.293 | 1.620.293 | 100.594.171 |
| 2065 | 1.593.198 | 1.593.198 | 102.187.369 |
| 2066 | 1.566.663 | 1.566.663 | 103.754.032 |
| 2067 | 1.540.673 | 1.540.673 | 105.294.705 |
| 2068 | 1.515.215 | 1.515.215 | 106.809.920 |
| 2069 | 1.490.279 | 1.490.279 | 108.300.198 |

|      |           |           |             |
|------|-----------|-----------|-------------|
| 2070 | 1.465.850 | 1.465.850 | 109.766.048 |
| 2071 | 1.441.918 | 1.441.918 | 111.207.966 |
| 2072 | 1.418.470 | 1.418.470 | 112.626.436 |
| 2073 | 1.395.496 | 1.395.496 | 114.021.933 |
| 2074 | 1.372.985 | 1.372.985 | 115.394.918 |
| 2075 | 1.350.926 | 1.350.926 | 116.745.844 |
| 2076 | 1.329.309 | 1.329.309 | 118.075.153 |
| 2077 | 1.308.122 | 1.308.122 | 119.383.275 |
| 2078 | 1.287.358 | 1.287.358 | 120.670.633 |
| 2079 | 1.267.005 | 1.267.005 | 121.937.638 |
| 2080 | 1.247.055 | 1.247.055 | 123.184.693 |
| 2081 | 1.227.499 | 1.227.499 | 124.412.192 |
| 2082 | 1.208.327 | 1.208.327 | 125.620.519 |

**Table 40. Baseline carbon stock change in the below-ground biomass in the reference region**

| Carbon stock change in the below-ground biomass per initial forest class |                                           |                                                                                                       |                                                            |
|--------------------------------------------------------------------------|-------------------------------------------|-------------------------------------------------------------------------------------------------------|------------------------------------------------------------|
| IDcl                                                                     | 1                                         | Total carbon stock change in the below-ground biomass of initial forest class in the reference region |                                                            |
| Name                                                                     | Dominant Vegetation in Ecological Tension |                                                                                                       |                                                            |
| Project year                                                             | tCO <sub>2</sub> e                        | $\Delta C_{bbBSLRR}/cl,t$<br>Annual (tCO <sub>2</sub> e)                                              | $\Delta C_{bbBSLRR}/cl$<br>Cumulative (tCO <sub>2</sub> e) |
| 2023                                                                     | 226.170                                   | 226.170                                                                                               | 226.170                                                    |
| 2024                                                                     | 447.909                                   | 447.909                                                                                               | 674.079                                                    |
| 2025                                                                     | 665.318                                   | 665.318                                                                                               | 1.339.397                                                  |
| 2026                                                                     | 878.499                                   | 878.499                                                                                               | 2.217.896                                                  |
| 2027                                                                     | 1.087.550                                 | 1.087.550                                                                                             | 3.305.446                                                  |
| 2028                                                                     | 1.292.565                                 | 1.292.565                                                                                             | 4.598.011                                                  |

|      |           |           |            |
|------|-----------|-----------|------------|
| 2029 | 1.493.638 | 1.493.638 | 6.091.648  |
| 2030 | 1.690.858 | 1.690.858 | 7.782.507  |
| 2031 | 1.884.315 | 1.884.315 | 9.666.822  |
| 2032 | 2.074.094 | 2.074.094 | 11.740.916 |
| 2033 | 2.034.108 | 2.034.108 | 13.775.024 |
| 2034 | 1.995.041 | 1.995.041 | 15.770.065 |
| 2035 | 1.956.869 | 1.956.869 | 17.726.934 |
| 2036 | 1.919.570 | 1.919.570 | 19.646.503 |
| 2037 | 1.883.120 | 1.883.120 | 21.529.624 |
| 2038 | 1.847.499 | 1.847.499 | 23.377.122 |
| 2039 | 1.812.684 | 1.812.684 | 25.189.806 |
| 2040 | 1.778.655 | 1.778.655 | 26.968.461 |
| 2041 | 1.745.393 | 1.745.393 | 28.713.855 |
| 2042 | 1.712.878 | 1.712.878 | 30.426.733 |
| 2043 | 1.681.090 | 1.681.090 | 32.107.823 |
| 2044 | 1.650.013 | 1.650.013 | 33.757.836 |
| 2045 | 1.619.626 | 1.619.626 | 35.377.462 |
| 2046 | 1.589.914 | 1.589.914 | 36.967.376 |
| 2047 | 1.560.860 | 1.560.860 | 38.528.236 |
| 2048 | 1.532.446 | 1.532.446 | 40.060.682 |
| 2049 | 1.504.657 | 1.504.657 | 41.565.339 |
| 2050 | 1.477.478 | 1.477.478 | 43.042.817 |
| 2051 | 1.450.893 | 1.450.893 | 44.493.709 |
| 2052 | 1.424.887 | 1.424.887 | 45.918.597 |
| 2053 | 1.399.447 | 1.399.447 | 47.318.043 |
| 2054 | 1.374.558 | 1.374.558 | 48.692.602 |
| 2055 | 1.350.207 | 1.350.207 | 50.042.808 |
| 2056 | 1.326.380 | 1.326.380 | 51.369.189 |

|      |           |           |            |
|------|-----------|-----------|------------|
| 2057 | 1.303.066 | 1.303.066 | 52.672.255 |
| 2058 | 1.280.250 | 1.280.250 | 53.952.505 |
| 2059 | 1.257.922 | 1.257.922 | 55.210.427 |
| 2060 | 1.236.069 | 1.236.069 | 56.446.496 |
| 2061 | 1.214.680 | 1.214.680 | 57.661.177 |
| 2062 | 1.193.744 | 1.193.744 | 58.854.921 |
| 2063 | 1.173.250 | 1.173.250 | 60.028.171 |
| 2064 | 1.153.186 | 1.153.186 | 61.181.357 |
| 2065 | 1.133.544 | 1.133.544 | 62.314.901 |
| 2066 | 1.114.312 | 1.114.312 | 63.429.214 |
| 2067 | 1.095.482 | 1.095.482 | 64.524.695 |
| 2068 | 1.077.043 | 1.077.043 | 65.601.738 |
| 2069 | 1.058.986 | 1.058.986 | 66.660.724 |
| 2070 | 1.041.302 | 1.041.302 | 67.702.026 |
| 2071 | 1.023.983 | 1.023.983 | 68.726.008 |
| 2072 | 1.007.019 | 1.007.019 | 69.733.027 |
| 2073 | 990.403   | 990.403   | 70.723.430 |
| 2074 | 974.126   | 974.126   | 71.697.556 |
| 2075 | 958.181   | 958.181   | 72.655.737 |
| 2076 | 942.559   | 942.559   | 73.598.296 |
| 2077 | 927.253   | 927.253   | 74.525.549 |
| 2078 | 912.257   | 912.257   | 75.437.806 |
| 2079 | 897.562   | 897.562   | 76.335.368 |
| 2080 | 883.162   | 883.162   | 77.218.529 |
| 2081 | 869.049   | 869.049   | 78.087.579 |
| 2082 | 855.219   | 855.219   | 78.942.797 |

**Table 41. Baseline carbon stock change in the above-ground biomass in the project area**

| Carbon stock change in the above-ground biomass per initial forest class |                                           |                                                                                                   |                                                                      |
|--------------------------------------------------------------------------|-------------------------------------------|---------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| IDcl                                                                     | 1                                         | Total carbon stock change in the above-ground biomass of initial forest class in the project area |                                                                      |
| Name                                                                     | Dominant Vegetation in Ecological Tension |                                                                                                   |                                                                      |
| Project year                                                             | tCO <sub>2</sub> e                        | $\Delta\text{CabBSLPA}/\text{cl}, t$<br>Annual (tCO <sub>2</sub> e)                               | $\Delta\text{CabBSLPA}/\text{cl}$<br>Cumulative (tCO <sub>2</sub> e) |
| 2023                                                                     | 37.538                                    | 37.538                                                                                            | 37.538                                                               |
| 2024                                                                     | 36.366                                    | 36.366                                                                                            | 73.904                                                               |
| 2025                                                                     | 35.236                                    | 35.236                                                                                            | 109.139                                                              |
| 2026                                                                     | 34.146                                    | 34.146                                                                                            | 143.285                                                              |
| 2027                                                                     | 33.095                                    | 33.095                                                                                            | 176.380                                                              |
| 2028                                                                     | 32.081                                    | 32.081                                                                                            | 208.461                                                              |
| 2029                                                                     | 31.103                                    | 31.103                                                                                            | 239.563                                                              |
| 2030                                                                     | 30.159                                    | 30.159                                                                                            | 269.722                                                              |
| 2031                                                                     | 29.248                                    | 29.248                                                                                            | 298.970                                                              |
| 2032                                                                     | 28.368                                    | 28.368                                                                                            | 327.338                                                              |
| 2033                                                                     | 27.519                                    | 27.519                                                                                            | 354.857                                                              |
| 2034                                                                     | 26.700                                    | 26.700                                                                                            | 381.557                                                              |
| 2035                                                                     | 25.908                                    | 25.908                                                                                            | 407.466                                                              |
| 2036                                                                     | 25.144                                    | 25.144                                                                                            | 432.610                                                              |
| 2037                                                                     | 24.406                                    | 24.406                                                                                            | 457.016                                                              |
| 2038                                                                     | 23.693                                    | 23.693                                                                                            | 480.709                                                              |
| 2039                                                                     | 23.004                                    | 23.004                                                                                            | 503.713                                                              |
| 2040                                                                     | 22.338                                    | 22.338                                                                                            | 526.051                                                              |
| 2041                                                                     | 21.695                                    | 21.695                                                                                            | 547.745                                                              |

|      |        |        |         |
|------|--------|--------|---------|
| 2042 | 21.073 | 21.073 | 568.818 |
| 2043 | 20.472 | 20.472 | 589.290 |
| 2044 | 19.890 | 19.890 | 609.180 |
| 2045 | 19.328 | 19.328 | 628.509 |
| 2046 | 18.785 | 18.785 | 647.293 |
| 2047 | 18.259 | 18.259 | 665.552 |
| 2048 | 17.750 | 17.750 | 683.303 |
| 2049 | 17.258 | 17.258 | 700.561 |
| 2050 | 16.782 | 16.782 | 717.344 |
| 2051 | 16.322 | 16.322 | 733.665 |
| 2052 | 15.876 | 15.876 | 749.541 |
| 2053 | 15.444 | 15.444 | 764.985 |
| 2054 | 15.026 | 15.026 | 780.011 |
| 2055 | 14.621 | 14.621 | 794.632 |
| 2056 | 14.230 | 14.230 | 808.862 |
| 2057 | 13.850 | 13.850 | 822.712 |
| 2058 | 13.482 | 13.482 | 836.194 |
| 2059 | 13.126 | 13.126 | 849.321 |
| 2060 | 12.781 | 12.781 | 862.102 |
| 2061 | 12.447 | 12.447 | 874.549 |
| 2062 | 12.123 | 12.123 | 886.672 |
| 2063 | 11.809 | 11.809 | 898.480 |
| 2064 | 11.504 | 11.504 | 909.985 |
| 2065 | 11.209 | 11.209 | 921.194 |
| 2066 | 10.923 | 10.923 | 932.117 |
| 2067 | 10.645 | 10.645 | 942.762 |
| 2068 | 10.376 | 10.376 | 953.139 |
| 2069 | 10.115 | 10.115 | 963.254 |



|      |       |       |           |
|------|-------|-------|-----------|
| 2070 | 9.862 | 9.862 | 973.116   |
| 2071 | 9.616 | 9.616 | 982.732   |
| 2072 | 9.378 | 9.378 | 992.109   |
| 2073 | 9.146 | 9.146 | 1.001.255 |
| 2074 | 8.922 | 8.922 | 1.010.177 |
| 2075 | 8.703 | 8.703 | 1.018.880 |
| 2076 | 8.492 | 8.492 | 1.027.372 |
| 2077 | 8.286 | 8.286 | 1.035.658 |
| 2078 | 8.087 | 8.087 | 1.043.745 |
| 2079 | 7.893 | 7.893 | 1.051.638 |
| 2080 | 7.705 | 7.705 | 1.059.342 |
| 2081 | 7.522 | 7.522 | 1.066.864 |
| 2082 | 7.344 | 7.344 | 1.074.208 |

**Table 42. Baseline carbon stock change in the below-ground biomass in the project area**

| Carbon stock change in the below-ground biomass per initial forest class |                                           |                                                                                                   |                                                         |
|--------------------------------------------------------------------------|-------------------------------------------|---------------------------------------------------------------------------------------------------|---------------------------------------------------------|
| IDcl                                                                     | 1                                         | Total carbon stock change in the below-ground biomass of initial forest class in the project area |                                                         |
| Name                                                                     | Dominant Vegetation in Ecological Tension |                                                                                                   |                                                         |
| Project year                                                             | tCO <sub>2</sub> e                        | $\Delta C_{bbBSLPA} / \Delta t$<br>Annual (tCO <sub>2</sub> e)                                    | $\Delta C_{bbBSLPA}$<br>Cumulative (tCO <sub>2</sub> e) |
| 2023                                                                     | 2.471                                     | 2.471                                                                                             | 2.471                                                   |
| 2024                                                                     | 4.864                                     | 4.864                                                                                             | 7.335                                                   |
| 2025                                                                     | 7.183                                     | 7.183                                                                                             | 14.518                                                  |
| 2026                                                                     | 9.430                                     | 9.430                                                                                             | 23.948                                                  |
| 2027                                                                     | 11.609                                    | 11.609                                                                                            | 35.557                                                  |
| 2028                                                                     | 13.720                                    | 13.720                                                                                            | 49.277                                                  |

|      |        |        |         |
|------|--------|--------|---------|
| 2029 | 15.767 | 15.767 | 65.044  |
| 2030 | 17.752 | 17.752 | 82.796  |
| 2031 | 19.677 | 19.677 | 102.473 |
| 2032 | 21.544 | 21.544 | 124.017 |
| 2033 | 20.885 | 20.885 | 144.902 |
| 2034 | 20.249 | 20.249 | 165.150 |
| 2035 | 19.635 | 19.635 | 184.785 |
| 2036 | 19.042 | 19.042 | 203.827 |
| 2037 | 18.470 | 18.470 | 222.297 |
| 2038 | 17.918 | 17.918 | 240.216 |
| 2039 | 17.385 | 17.385 | 257.601 |
| 2040 | 16.871 | 16.871 | 274.471 |
| 2041 | 16.373 | 16.373 | 290.845 |
| 2042 | 15.893 | 15.893 | 306.738 |
| 2043 | 15.429 | 15.429 | 322.168 |
| 2044 | 14.981 | 14.981 | 337.149 |
| 2045 | 14.548 | 14.548 | 351.697 |
| 2046 | 14.130 | 14.130 | 365.827 |
| 2047 | 13.725 | 13.725 | 379.552 |
| 2048 | 13.334 | 13.334 | 392.886 |
| 2049 | 12.956 | 12.956 | 405.841 |
| 2050 | 12.590 | 12.590 | 418.432 |
| 2051 | 12.236 | 12.236 | 430.668 |
| 2052 | 11.894 | 11.894 | 442.562 |
| 2053 | 11.564 | 11.564 | 454.126 |
| 2054 | 11.243 | 11.243 | 465.369 |
| 2055 | 10.934 | 10.934 | 476.303 |
| 2056 | 10.634 | 10.634 | 486.937 |

|      |        |        |         |
|------|--------|--------|---------|
| 2057 | 10.344 | 10.344 | 497.280 |
| 2058 | 10.063 | 10.063 | 507.343 |
| 2059 | 9.791  | 9.791  | 517.134 |
| 2060 | 9.527  | 9.527  | 526.661 |
| 2061 | 9.272  | 9.272  | 535.934 |
| 2062 | 9.025  | 9.025  | 544.959 |
| 2063 | 8.786  | 8.786  | 553.745 |
| 2064 | 8.554  | 8.554  | 562.300 |
| 2065 | 8.330  | 8.330  | 570.629 |
| 2066 | 8.112  | 8.112  | 578.742 |
| 2067 | 7.901  | 7.901  | 586.643 |
| 2068 | 7.697  | 7.697  | 594.340 |
| 2069 | 7.499  | 7.499  | 601.838 |
| 2070 | 7.306  | 7.306  | 609.145 |
| 2071 | 7.120  | 7.120  | 616.265 |
| 2072 | 6.939  | 6.939  | 623.204 |
| 2073 | 6.764  | 6.764  | 629.969 |
| 2074 | 6.594  | 6.594  | 636.563 |
| 2075 | 6.429  | 6.429  | 642.992 |
| 2076 | 6.269  | 6.269  | 649.261 |
| 2077 | 6.114  | 6.114  | 655.376 |
| 2078 | 5.963  | 5.963  | 661.339 |
| 2079 | 5.817  | 5.817  | 667.156 |
| 2080 | 5.675  | 5.675  | 672.831 |
| 2081 | 5.537  | 5.537  | 678.368 |
| 2082 | 5.403  | 5.403  | 683.772 |

Table 43. Total baseline carbon stock change of initial forest class in project area

| Carbon stock change per initial forest class |                                                          |                                                                       |                          |                              |
|----------------------------------------------|----------------------------------------------------------|-----------------------------------------------------------------------|--------------------------|------------------------------|
| IDcl                                         | 1                                                        | Total carbon stock change of initial forest class in the project area |                          |                              |
| Name                                         | Forest                                                   |                                                                       |                          |                              |
| Project year                                 | $\Delta C_{abBSLPAicl,t}$<br>Annual (tCO <sub>2e</sub> ) | $\Delta C_{bbBSLPAicl,t}$<br>Annual (tCO <sub>2e</sub> )              | Annual tCO <sub>2e</sub> | Cumulative tCO <sub>2e</sub> |
| 2023                                         | 37.538                                                   | 2.471                                                                 | 40.008                   | 40.008                       |
| 2024                                         | 36.366                                                   | 4.864                                                                 | 41.230                   | 81.238                       |
| 2025                                         | 35.236                                                   | 7.183                                                                 | 42.419                   | 123.657                      |
| 2026                                         | 34.146                                                   | 9.430                                                                 | 43.576                   | 167.233                      |
| 2027                                         | 33.095                                                   | 11.609                                                                | 44.703                   | 211.937                      |
| 2028                                         | 32.081                                                   | 13.720                                                                | 45.801                   | 257.738                      |
| 2029                                         | 31.103                                                   | 15.767                                                                | 46.870                   | 304.607                      |
| 2030                                         | 30.159                                                   | 17.752                                                                | 47.911                   | 352.518                      |
| 2031                                         | 29.248                                                   | 19.677                                                                | 48.925                   | 401.442                      |
| 2032                                         | 28.368                                                   | 21.544                                                                | 49.912                   | 451.355                      |
| 2033                                         | 27.519                                                   | 20.885                                                                | 48.404                   | 499.759                      |
| 2034                                         | 26.700                                                   | 20.249                                                                | 46.948                   | 546.707                      |
| 2035                                         | 25.908                                                   | 19.635                                                                | 45.543                   | 592.250                      |
| 2036                                         | 25.144                                                   | 19.042                                                                | 44.186                   | 636.437                      |
| 2037                                         | 24.406                                                   | 18.470                                                                | 42.876                   | 679.313                      |
| 2038                                         | 23.693                                                   | 17.918                                                                | 41.611                   | 720.924                      |
| 2039                                         | 23.004                                                   | 17.385                                                                | 40.389                   | 761.314                      |
| 2040                                         | 22.338                                                   | 16.871                                                                | 39.209                   | 800.522                      |
| 2041                                         | 21.695                                                   | 16.373                                                                | 38.068                   | 838.590                      |
| 2042                                         | 21.073                                                   | 15.893                                                                | 36.966                   | 875.556                      |
| 2043                                         | 20.472                                                   | 15.429                                                                | 35.901                   | 911.457                      |

|      |        |        |        |           |
|------|--------|--------|--------|-----------|
| 2044 | 19.890 | 14.981 | 34.872 | 946.329   |
| 2045 | 19.328 | 14.548 | 33.876 | 980.206   |
| 2046 | 18.785 | 14.130 | 32.914 | 1.013.120 |
| 2047 | 18.259 | 13.725 | 31.984 | 1.045.104 |
| 2048 | 17.750 | 13.334 | 31.084 | 1.076.188 |
| 2049 | 17.258 | 12.956 | 30.214 | 1.106.403 |
| 2050 | 16.782 | 12.590 | 29.372 | 1.135.775 |
| 2051 | 16.322 | 12.236 | 28.558 | 1.164.333 |
| 2052 | 15.876 | 11.894 | 27.770 | 1.192.103 |
| 2053 | 15.444 | 11.564 | 27.008 | 1.219.111 |
| 2054 | 15.026 | 11.243 | 26.269 | 1.245.380 |
| 2055 | 14.621 | 10.934 | 25.555 | 1.270.935 |
| 2056 | 14.230 | 10.634 | 24.863 | 1.295.799 |
| 2057 | 13.850 | 10.344 | 24.194 | 1.319.992 |
| 2058 | 13.482 | 10.063 | 23.545 | 1.343.537 |
| 2059 | 13.126 | 9.791  | 22.917 | 1.366.454 |
| 2060 | 12.781 | 9.527  | 22.309 | 1.388.763 |
| 2061 | 12.447 | 9.272  | 21.719 | 1.410.482 |
| 2062 | 12.123 | 9.025  | 21.148 | 1.431.631 |
| 2063 | 11.809 | 8.786  | 20.595 | 1.452.226 |
| 2064 | 11.504 | 8.554  | 20.059 | 1.472.284 |
| 2065 | 11.209 | 8.330  | 19.539 | 1.491.823 |
| 2066 | 10.923 | 8.112  | 19.035 | 1.510.858 |
| 2067 | 10.645 | 7.901  | 18.547 | 1.529.405 |
| 2068 | 10.376 | 7.697  | 18.073 | 1.547.478 |
| 2069 | 10.115 | 7.499  | 17.614 | 1.565.092 |
| 2070 | 9.862  | 7.306  | 17.168 | 1.582.260 |
| 2071 | 9.616  | 7.120  | 16.736 | 1.598.997 |

|      |       |       |        |           |
|------|-------|-------|--------|-----------|
| 2072 | 9.378 | 6.939 | 16.317 | 1.615.314 |
| 2073 | 9.146 | 6.764 | 15.910 | 1.631.224 |
| 2074 | 8.922 | 6.594 | 15.516 | 1.646.740 |
| 2075 | 8.703 | 6.429 | 15.133 | 1.661.872 |
| 2076 | 8.492 | 6.269 | 14.761 | 1.676.634 |
| 2077 | 8.286 | 6.114 | 14.400 | 1.691.034 |
| 2078 | 8.087 | 5.963 | 14.050 | 1.705.084 |
| 2079 | 7.893 | 5.817 | 13.710 | 1.718.794 |
| 2080 | 7.705 | 5.675 | 13.380 | 1.732.173 |
| 2081 | 7.522 | 5.537 | 13.059 | 1.745.232 |
| 2082 | 7.344 | 5.403 | 12.747 | 1.757.980 |

**Table 44. Baseline carbon stock change in the above-ground biomass in the leakage belt area**

| Carbon stock change in the above-ground biomass per initial forest class |                   |                                                                                                   |                                                            |
|--------------------------------------------------------------------------|-------------------|---------------------------------------------------------------------------------------------------|------------------------------------------------------------|
| IDcl                                                                     | 1                 | Total carbon stock change in the above-ground biomass of initial forest class in the leakage belt |                                                            |
| Name                                                                     | Forest            |                                                                                                   |                                                            |
| Project year                                                             | tCO <sub>2e</sub> | $\Delta C_{abBSLLKicl,t}$<br>Annual (tCO <sub>2e</sub> )                                          | $\Delta C_{abBSLLKicl}$<br>Cumulative (tCO <sub>2e</sub> ) |
| 2023                                                                     | 1.722.928         | 1.722.928                                                                                         | 1.722.928                                                  |
| 2024                                                                     | 1.660.004         | 1.660.004                                                                                         | 3.382.932                                                  |
| 2025                                                                     | 1.599.676         | 1.599.676                                                                                         | 4.982.608                                                  |
| 2026                                                                     | 1.541.827         | 1.541.827                                                                                         | 6.524.435                                                  |
| 2027                                                                     | 1.486.344         | 1.486.344                                                                                         | 8.010.779                                                  |
| 2028                                                                     | 1.433.122         | 1.433.122                                                                                         | 9.443.902                                                  |
| 2029                                                                     | 1.382.059         | 1.382.059                                                                                         | 10.825.961                                                 |
| 2030                                                                     | 1.333.059         | 1.333.059                                                                                         | 12.159.020                                                 |
| 2031                                                                     | 1.286.031         | 1.286.031                                                                                         | 13.445.051                                                 |

|      |           |           |            |
|------|-----------|-----------|------------|
| 2032 | 1.240.886 | 1.240.886 | 14.685.937 |
| 2033 | 1.197.542 | 1.197.542 | 15.883.479 |
| 2034 | 1.155.920 | 1.155.920 | 17.039.398 |
| 2035 | 1.115.944 | 1.115.944 | 18.155.342 |
| 2036 | 1.077.542 | 1.077.542 | 19.232.885 |
| 2037 | 1.040.647 | 1.040.647 | 20.273.532 |
| 2038 | 1.005.193 | 1.005.193 | 21.278.725 |
| 2039 | 971.117   | 971.117   | 22.249.842 |
| 2040 | 938.361   | 938.361   | 23.188.203 |
| 2041 | 906.868   | 906.868   | 24.095.071 |
| 2042 | 876.584   | 876.584   | 24.971.655 |
| 2043 | 847.458   | 847.458   | 25.819.114 |
| 2044 | 819.441   | 819.441   | 26.638.555 |
| 2045 | 792.486   | 792.486   | 27.431.041 |
| 2046 | 766.548   | 766.548   | 28.197.590 |
| 2047 | 741.586   | 741.586   | 28.939.175 |
| 2048 | 717.557   | 717.557   | 29.656.732 |
| 2049 | 694.424   | 694.424   | 30.351.156 |
| 2050 | 672.149   | 672.149   | 31.023.305 |
| 2051 | 650.697   | 650.697   | 31.674.002 |
| 2052 | 630.035   | 630.035   | 32.304.037 |
| 2053 | 610.129   | 610.129   | 32.914.166 |
| 2054 | 590.949   | 590.949   | 33.505.115 |
| 2055 | 572.466   | 572.466   | 34.077.582 |
| 2056 | 554.652   | 554.652   | 34.632.233 |
| 2057 | 537.479   | 537.479   | 35.169.712 |
| 2058 | 520.921   | 520.921   | 35.690.633 |
| 2059 | 504.955   | 504.955   | 36.195.588 |



|      |         |         |            |
|------|---------|---------|------------|
| 2060 | 489.556 | 489.556 | 36.685.145 |
| 2061 | 474.703 | 474.703 | 37.159.848 |
| 2062 | 460.373 | 460.373 | 37.620.220 |
| 2063 | 446.545 | 446.545 | 38.066.766 |
| 2064 | 433.201 | 433.201 | 38.499.967 |
| 2065 | 420.321 | 420.321 | 38.920.288 |
| 2066 | 407.888 | 407.888 | 39.328.176 |
| 2067 | 395.883 | 395.883 | 39.724.059 |
| 2068 | 384.291 | 384.291 | 40.108.350 |
| 2069 | 373.095 | 373.095 | 40.481.444 |
| 2070 | 362.280 | 362.280 | 40.843.724 |
| 2071 | 351.832 | 351.832 | 41.195.556 |
| 2072 | 341.737 | 341.737 | 41.537.293 |
| 2073 | 331.981 | 331.981 | 41.869.273 |
| 2074 | 322.551 | 322.551 | 42.191.824 |
| 2075 | 313.436 | 313.436 | 42.505.260 |
| 2076 | 304.623 | 304.623 | 42.809.884 |
| 2077 | 296.102 | 296.102 | 43.105.985 |
| 2078 | 287.861 | 287.861 | 43.393.846 |
| 2079 | 279.889 | 279.889 | 43.673.735 |
| 2080 | 272.178 | 272.178 | 43.945.913 |
| 2081 | 264.717 | 264.717 | 44.210.631 |
| 2082 | 257.498 | 257.498 | 44.468.128 |

**Table 45. Baseline carbon stock change in the below-ground biomass in the leakage belt area**

| Carbon stock change in the below-ground biomass per initial forest class |                   |                                                                                                   |                                                            |
|--------------------------------------------------------------------------|-------------------|---------------------------------------------------------------------------------------------------|------------------------------------------------------------|
| IDcl                                                                     | 1                 | Total carbon stock change in the below-ground biomass of initial forest class in the leakage belt |                                                            |
| Name                                                                     | Forest            |                                                                                                   |                                                            |
| Project year                                                             | tCO <sub>2e</sub> | $\Delta C_{bbBSLLKicl,t}$<br>Annual (tCO <sub>2e</sub> )                                          | $\Delta C_{bbBSLLKicl}$<br>Cumulative (tCO <sub>2e</sub> ) |
| 2023                                                                     | 113.396           | 113.396                                                                                           | 113.396                                                    |
| 2024                                                                     | 222.651           | 222.651                                                                                           | 336.047                                                    |
| 2025                                                                     | 327.935           | 327.935                                                                                           | 663.983                                                    |
| 2026                                                                     | 429.412           | 429.412                                                                                           | 1.093.395                                                  |
| 2027                                                                     | 527.237           | 527.237                                                                                           | 1.620.632                                                  |
| 2028                                                                     | 621.560           | 621.560                                                                                           | 2.242.192                                                  |
| 2029                                                                     | 712.522           | 712.522                                                                                           | 2.954.714                                                  |
| 2030                                                                     | 800.258           | 800.258                                                                                           | 3.754.972                                                  |
| 2031                                                                     | 884.900           | 884.900                                                                                           | 4.639.872                                                  |
| 2032                                                                     | 966.570           | 966.570                                                                                           | 5.606.441                                                  |
| 2033                                                                     | 931.991           | 931.991                                                                                           | 6.538.432                                                  |
| 2034                                                                     | 898.814           | 898.814                                                                                           | 7.437.246                                                  |
| 2035                                                                     | 866.977           | 866.977                                                                                           | 8.304.223                                                  |
| 2036                                                                     | 836.419           | 836.419                                                                                           | 9.140.642                                                  |
| 2037                                                                     | 807.085           | 807.085                                                                                           | 9.947.728                                                  |
| 2038                                                                     | 778.921           | 778.921                                                                                           | 10.726.648                                                 |
| 2039                                                                     | 751.874           | 751.874                                                                                           | 11.478.523                                                 |
| 2040                                                                     | 725.897           | 725.897                                                                                           | 12.204.419                                                 |
| 2041                                                                     | 700.942           | 700.942                                                                                           | 12.905.361                                                 |
| 2042                                                                     | 676.965           | 676.965                                                                                           | 13.582.326                                                 |

|      |         |         |            |
|------|---------|---------|------------|
| 2043 | 653.924 | 653.924 | 14.236.250 |
| 2044 | 631.778 | 631.778 | 14.868.028 |
| 2045 | 610.489 | 610.489 | 15.478.518 |
| 2046 | 590.021 | 590.021 | 16.068.539 |
| 2047 | 570.338 | 570.338 | 16.638.877 |
| 2048 | 551.407 | 551.407 | 17.190.284 |
| 2049 | 533.196 | 533.196 | 17.723.480 |
| 2050 | 515.675 | 515.675 | 18.239.155 |
| 2051 | 498.815 | 498.815 | 18.737.970 |
| 2052 | 482.588 | 482.588 | 19.220.558 |
| 2053 | 466.968 | 466.968 | 19.687.526 |
| 2054 | 451.930 | 451.930 | 20.139.455 |
| 2055 | 437.449 | 437.449 | 20.576.904 |
| 2056 | 423.503 | 423.503 | 21.000.407 |
| 2057 | 410.069 | 410.069 | 21.410.476 |
| 2058 | 397.127 | 397.127 | 21.807.603 |
| 2059 | 384.657 | 384.657 | 22.192.260 |
| 2060 | 372.640 | 372.640 | 22.564.900 |
| 2061 | 361.056 | 361.056 | 22.925.956 |
| 2062 | 349.890 | 349.890 | 23.275.846 |
| 2063 | 339.124 | 339.124 | 23.614.970 |
| 2064 | 328.741 | 328.741 | 23.943.711 |
| 2065 | 318.728 | 318.728 | 24.262.439 |
| 2066 | 309.068 | 309.068 | 24.571.507 |
| 2067 | 299.749 | 299.749 | 24.871.256 |
| 2068 | 290.756 | 290.756 | 25.162.012 |
| 2069 | 282.078 | 282.078 | 25.444.090 |
| 2070 | 273.701 | 273.701 | 25.717.791 |

|      |         |         |            |
|------|---------|---------|------------|
| 2071 | 265.614 | 265.614 | 25.983.405 |
| 2072 | 257.806 | 257.806 | 26.241.212 |
| 2073 | 250.266 | 250.266 | 26.491.477 |
| 2074 | 242.983 | 242.983 | 26.734.461 |
| 2075 | 235.949 | 235.949 | 26.970.409 |
| 2076 | 229.152 | 229.152 | 27.199.561 |
| 2077 | 222.585 | 222.585 | 27.422.146 |
| 2078 | 216.238 | 216.238 | 27.638.384 |
| 2079 | 210.104 | 210.104 | 27.848.488 |
| 2080 | 204.174 | 204.174 | 28.052.662 |
| 2081 | 198.440 | 198.440 | 28.251.102 |
| 2082 | 192.896 | 192.896 | 28.443.998 |

**Table 46. Total baseline carbon stock change of initial forest class in leakage belt area**

| Carbon stock change per initial forest class |                                                          |                                                                       |                          |                              |
|----------------------------------------------|----------------------------------------------------------|-----------------------------------------------------------------------|--------------------------|------------------------------|
| IDcl                                         | 1                                                        | Total carbon stock change of initial forest class in the leakage belt |                          |                              |
| Name                                         | Forest                                                   |                                                                       |                          |                              |
| Project year                                 | $\Delta C_{abBSLPAicl,t}$<br>Annual (tCO <sub>2e</sub> ) | $\Delta C_{bbBSLPAicl,t}$<br>Annual (tCO <sub>2e</sub> )              | Annual tCO <sub>2e</sub> | Cumulative tCO <sub>2e</sub> |
| 2023                                         | 1.722.928                                                | 113.396                                                               | 1.836.324                | 1.836.324                    |
| 2024                                         | 1.660.004                                                | 222.651                                                               | 1.882.655                | 3.718.979                    |
| 2025                                         | 1.599.676                                                | 327.935                                                               | 1.927.612                | 5.646.591                    |
| 2026                                         | 1.541.827                                                | 429.412                                                               | 1.971.239                | 7.617.830                    |
| 2027                                         | 1.486.344                                                | 527.237                                                               | 2.013.582                | 9.631.412                    |
| 2028                                         | 1.433.122                                                | 621.560                                                               | 2.054.682                | 11.686.094                   |
| 2029                                         | 1.382.059                                                | 712.522                                                               | 2.094.581                | 13.780.675                   |
| 2030                                         | 1.333.059                                                | 800.258                                                               | 2.133.318                | 15.913.992                   |

|      |           |         |           |            |
|------|-----------|---------|-----------|------------|
| 2031 | 1.286.031 | 884.900 | 2.170.930 | 18.084.923 |
| 2032 | 1.240.886 | 966.570 | 2.207.456 | 20.292.378 |
| 2033 | 1.197.542 | 931.991 | 2.129.533 | 22.421.911 |
| 2034 | 1.155.920 | 898.814 | 2.054.734 | 24.476.645 |
| 2035 | 1.115.944 | 866.977 | 1.982.920 | 26.459.565 |
| 2036 | 1.077.542 | 836.419 | 1.913.962 | 28.373.527 |
| 2037 | 1.040.647 | 807.085 | 1.847.733 | 30.221.259 |
| 2038 | 1.005.193 | 778.921 | 1.784.114 | 32.005.373 |
| 2039 | 971.117   | 751.874 | 1.722.991 | 33.728.364 |
| 2040 | 938.361   | 725.897 | 1.664.258 | 35.392.622 |
| 2041 | 906.868   | 700.942 | 1.607.810 | 37.000.432 |
| 2042 | 876.584   | 676.965 | 1.553.549 | 38.553.982 |
| 2043 | 847.458   | 653.924 | 1.501.382 | 40.055.364 |
| 2044 | 819.441   | 631.778 | 1.451.219 | 41.506.583 |
| 2045 | 792.486   | 610.489 | 1.402.976 | 42.909.559 |
| 2046 | 766.548   | 590.021 | 1.356.569 | 44.266.128 |
| 2047 | 741.586   | 570.338 | 1.311.924 | 45.578.052 |
| 2048 | 717.557   | 551.407 | 1.268.964 | 46.847.016 |
| 2049 | 694.424   | 533.196 | 1.227.620 | 48.074.636 |
| 2050 | 672.149   | 515.675 | 1.187.824 | 49.262.460 |
| 2051 | 650.697   | 498.815 | 1.149.512 | 50.411.972 |
| 2052 | 630.035   | 482.588 | 1.112.623 | 51.524.595 |
| 2053 | 610.129   | 466.968 | 1.077.097 | 52.601.692 |
| 2054 | 590.949   | 451.930 | 1.042.879 | 53.644.571 |
| 2055 | 572.466   | 437.449 | 1.009.915 | 54.654.486 |
| 2056 | 554.652   | 423.503 | 978.154   | 55.632.640 |
| 2057 | 537.479   | 410.069 | 947.548   | 56.580.188 |
| 2058 | 520.921   | 397.127 | 918.049   | 57.498.236 |

|      |         |         |         |            |
|------|---------|---------|---------|------------|
| 2059 | 504.955 | 384.657 | 889.612 | 58.387.849 |
| 2060 | 489.556 | 372.640 | 862.196 | 59.250.045 |
| 2061 | 474.703 | 361.056 | 835.759 | 60.085.804 |
| 2062 | 460.373 | 349.890 | 810.263 | 60.896.066 |
| 2063 | 446.545 | 339.124 | 785.669 | 61.681.735 |
| 2064 | 433.201 | 328.741 | 761.942 | 62.443.678 |
| 2065 | 420.321 | 318.728 | 739.049 | 63.182.727 |
| 2066 | 407.888 | 309.068 | 716.956 | 63.899.683 |
| 2067 | 395.883 | 299.749 | 695.632 | 64.595.315 |
| 2068 | 384.291 | 290.756 | 675.047 | 65.270.362 |
| 2069 | 373.095 | 282.078 | 655.173 | 65.925.535 |
| 2070 | 362.280 | 273.701 | 635.981 | 66.561.516 |
| 2071 | 351.832 | 265.614 | 617.446 | 67.178.962 |
| 2072 | 341.737 | 257.806 | 599.543 | 67.778.504 |
| 2073 | 331.981 | 250.266 | 582.246 | 68.360.751 |
| 2074 | 322.551 | 242.983 | 565.534 | 68.926.285 |
| 2075 | 313.436 | 235.949 | 549.384 | 69.475.670 |
| 2076 | 304.623 | 229.152 | 533.775 | 70.009.445 |
| 2077 | 296.102 | 222.585 | 518.687 | 70.528.132 |
| 2078 | 287.861 | 216.238 | 504.099 | 71.032.230 |
| 2079 | 279.889 | 210.104 | 489.993 | 71.522.223 |
| 2080 | 272.178 | 204.174 | 476.352 | 71.998.575 |
| 2081 | 264.717 | 198.440 | 463.157 | 72.461.733 |
| 2082 | 257.498 | 192.896 | 450.394 | 72.912.126 |

## 5.2 Project Emissions

Some unplanned deforestation may happen in the project area despite the implemented REDD project activity. The level at which deforestation will actually be reduced in the project case depends on the effectiveness of the proposed activities, which cannot be measured *ex ante*. *Ex post* measurements of the project results will be important to determine actual emission reductions.

To allow *ex ante* projections to be made, a conservative assumption was made about the effectiveness of the proposed project activities in order to define the Effectiveness Index (*EI*). The estimated value of *EI* is used to multiply the baseline projections by the factor  $(1 - EI)$  and the result was considered to be the *ex ante* estimated emissions from unplanned deforestation in the project case. This is calculated as follows:

$$\Delta CUDdPat = \Delta CBSLPat \times (1 - EI)$$

Where:

|                  |                                                                                                                                            |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| $\Delta CUDdPat$ | Total ex ante actual carbon stock change due to unavoided unplanned deforestation at year <i>t</i> in the project area; tCO <sub>2</sub> e |
| $\Delta CBSLPat$ | Total baseline carbon stock change in the project area at year <i>t</i> ; tCO <sub>2</sub> e                                               |
| <i>EI</i>        | Ex ante estimated Effectiveness Index; %                                                                                                   |
| <i>t</i>         | 1, 2, 3 ... <i>T</i> , a year of the proposed project crediting period; dimensionless                                                      |

Based on the historical of deforestation happened in the area before Project start, the Effectiveness Index (*EI*) of project activities was conservatively assumed as 95% in the first five years of implementation and that this value will increase gradually with their efficiency over the years, considering an increase in patrolling land surveillance and the implementation of social activities in the local communities, until reaching the value of 99% in year 2040.

This range of values represents efficiency of the ARAGUAIA REDD+ BOCALON PROJECT activities in reducing deforestation within the project area. It was then applied to calculate the *ex-ante* estimate of net carbon stock change in the project area under the project scenario ( $\Delta CPSPat$ ), as follows.

$$\Delta CPSPat = (\Delta CPAdPat + \Delta CUDdPat) - \Delta CPAiPat$$

Where:

|                  |                                                                                                                                            |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| $\Delta CPSPat$  | Sum of ex ante estimated actual carbon stock changes in the project area at year <i>t</i> ; tCO <sub>2</sub> e                             |
| $\Delta CPAdPat$ | Total decrease in carbon stock due to all planned activities at year <i>t</i> in the project area; tCO <sub>2</sub> e                      |
| $\Delta CUDdPat$ | Total ex ante actual carbon stock change due to unavoided unplanned deforestation at year <i>t</i> in the project area; tCO <sub>2</sub> e |
| $\Delta CPAiPat$ | Total increase in carbon stock due to all planned activities at year <i>t</i> in the project area; tCO <sub>2</sub> e                      |

Forest fires have not been included in the baseline scenario, thus non-CO2 emissions from biomass burning shall not be included in the project scenario. Therefore, EBBPSPat equals to zero.

*Ex ante* estimated net carbon stock change in the project area under the project scenario is showed in Table 47 below.

**Table 47. Total ex ante estimated actual net carbon stock changes and emissions of non-CO2 gases in the Project Area**

| Project year | Total carbon stock decrease due to planned activities |                             | Total carbon stock increase due to planned activities |                             | Total carbon stock decrease due to unavioded unplanned deforestation |                             | Total carbon stock decrease change in the project case |                            | Total ex ante estimated actual non-CO2 emissions from forest fires in the project area |                     |
|--------------|-------------------------------------------------------|-----------------------------|-------------------------------------------------------|-----------------------------|----------------------------------------------------------------------|-----------------------------|--------------------------------------------------------|----------------------------|----------------------------------------------------------------------------------------|---------------------|
|              | $\Delta$ CPAdPat annual                               | $\Delta$ CPAdPat cumulative | $\Delta$ CPAiPat annual                               | $\Delta$ CPAiPat cumulative | $\Delta$ CUDdPat annual                                              | $\Delta$ CUDdPat cumulative | $\Delta$ CPSPat annual                                 | $\Delta$ CPSPat cumulative | EBBPSPat annual                                                                        | EBBPSPat cumulative |
|              | tCO <sub>2</sub> e                                    | tCO <sub>2</sub> e          | tCO <sub>2</sub> e                                    | tCO <sub>2</sub> e          | tCO <sub>2</sub> e                                                   | tCO <sub>2</sub> e          | tCO <sub>2</sub> e                                     | tCO <sub>2</sub> e         | tCO <sub>2</sub> e                                                                     | tCO <sub>2</sub> e  |
| 2023         | 0,00                                                  | 0,00                        | 0,00                                                  | 0,00                        | 2.000                                                                | 2.000                       | 2.000                                                  | 2.000                      | 0,00                                                                                   | 0,00                |
| 2024         | 0,00                                                  | 0,00                        | 0,00                                                  | 0,00                        | 2.061                                                                | 4.062                       | 2.061                                                  | 4.062                      | 0,00                                                                                   | 0,00                |
| 2025         | 0,00                                                  | 0,00                        | 0,00                                                  | 0,00                        | 2.121                                                                | 6.183                       | 2.121                                                  | 6.183                      | 0,00                                                                                   | 0,00                |
| 2026         | 0,00                                                  | 0,00                        | 0,00                                                  | 0,00                        | 2.179                                                                | 8.362                       | 2.179                                                  | 8.362                      | 0,00                                                                                   | 0,00                |
| 2027         | 0,00                                                  | 0,00                        | 0,00                                                  | 0,00                        | 2.235                                                                | 10.597                      | 2.235                                                  | 10.597                     | 0,00                                                                                   | 0,00                |
| 2028         | 0,00                                                  | 0,00                        | 0,00                                                  | 0,00                        | 1.832                                                                | 12.429                      | 1.832                                                  | 12.429                     | 0,00                                                                                   | 0,00                |
| 2029         | 0,00                                                  | 0,00                        | 0,00                                                  | 0,00                        | 1.875                                                                | 14.304                      | 1.875                                                  | 14.304                     | 0,00                                                                                   | 0,00                |
| 2030         | 0,00                                                  | 0,00                        | 0,00                                                  | 0,00                        | 1.916                                                                | 16.220                      | 1.916                                                  | 16.220                     | 0,00                                                                                   | 0,00                |
| 2031         | 0,00                                                  | 0,00                        | 0,00                                                  | 0,00                        | 1.957                                                                | 18.177                      | 1.957                                                  | 18.177                     | 0,00                                                                                   | 0,00                |
| 2032         | 0,00                                                  | 0,00                        | 0,00                                                  | 0,00                        | 1.497                                                                | 19.674                      | 1.497                                                  | 19.674                     | 0,00                                                                                   | 0,00                |
| 2033         | 0,00                                                  | 0,00                        | 0,00                                                  | 0,00                        | 1.452                                                                | 21.127                      | 1.452                                                  | 21.127                     | 0,00                                                                                   | 0,00                |
| 2034         | 0,00                                                  | 0,00                        | 0,00                                                  | 0,00                        | 1.408                                                                | 22.535                      | 1.408                                                  | 22.535                     | 0,00                                                                                   | 0,00                |
| 2035         | 0,00                                                  | 0,00                        | 0,00                                                  | 0,00                        | 1.366                                                                | 23.901                      | 1.366                                                  | 23.901                     | 0,00                                                                                   | 0,00                |
| 2036         | 0,00                                                  | 0,00                        | 0,00                                                  | 0,00                        | 884                                                                  | 24.785                      | 884                                                    | 24.785                     | 0,00                                                                                   | 0,00                |
| 2037         | 0,00                                                  | 0,00                        | 0,00                                                  | 0,00                        | 858                                                                  | 25.643                      | 858                                                    | 25.643                     | 0,00                                                                                   | 0,00                |



|      |      |      |      |      |     |        |     |        |      |      |
|------|------|------|------|------|-----|--------|-----|--------|------|------|
| 2038 | 0,00 | 0,00 | 0,00 | 0,00 | 832 | 26.475 | 832 | 26.475 | 0,00 | 0,00 |
| 2039 | 0,00 | 0,00 | 0,00 | 0,00 | 808 | 27.283 | 808 | 27.283 | 0,00 | 0,00 |
| 2040 | 0,00 | 0,00 | 0,00 | 0,00 | 392 | 27.675 | 392 | 27.675 | 0,00 | 0,00 |
| 2041 | 0,00 | 0,00 | 0,00 | 0,00 | 381 | 28.055 | 381 | 28.055 | 0,00 | 0,00 |
| 2042 | 0,00 | 0,00 | 0,00 | 0,00 | 370 | 28.425 | 370 | 28.425 | 0,00 | 0,00 |
| 2043 | 0,00 | 0,00 | 0,00 | 0,00 | 359 | 28.784 | 359 | 28.784 | 0,00 | 0,00 |
| 2044 | 0,00 | 0,00 | 0,00 | 0,00 | 349 | 29.133 | 349 | 29.133 | 0,00 | 0,00 |
| 2045 | 0,00 | 0,00 | 0,00 | 0,00 | 339 | 29.471 | 339 | 29.471 | 0,00 | 0,00 |
| 2046 | 0,00 | 0,00 | 0,00 | 0,00 | 329 | 29.801 | 329 | 29.801 | 0,00 | 0,00 |
| 2047 | 0,00 | 0,00 | 0,00 | 0,00 | 320 | 30.120 | 320 | 30.120 | 0,00 | 0,00 |
| 2048 | 0,00 | 0,00 | 0,00 | 0,00 | 311 | 30.431 | 311 | 30.431 | 0,00 | 0,00 |
| 2049 | 0,00 | 0,00 | 0,00 | 0,00 | 302 | 30.733 | 302 | 30.733 | 0,00 | 0,00 |
| 2050 | 0,00 | 0,00 | 0,00 | 0,00 | 294 | 31.027 | 294 | 31.027 | 0,00 | 0,00 |
| 2051 | 0,00 | 0,00 | 0,00 | 0,00 | 286 | 31.313 | 286 | 31.313 | 0,00 | 0,00 |
| 2052 | 0,00 | 0,00 | 0,00 | 0,00 | 278 | 31.590 | 278 | 31.590 | 0,00 | 0,00 |
| 2053 | 0,00 | 0,00 | 0,00 | 0,00 | 270 | 31.861 | 270 | 31.861 | 0,00 | 0,00 |
| 2054 | 0,00 | 0,00 | 0,00 | 0,00 | 263 | 32.123 | 263 | 32.123 | 0,00 | 0,00 |
| 2055 | 0,00 | 0,00 | 0,00 | 0,00 | 256 | 32.379 | 256 | 32.379 | 0,00 | 0,00 |
| 2056 | 0,00 | 0,00 | 0,00 | 0,00 | 249 | 32.627 | 249 | 32.627 | 0,00 | 0,00 |
| 2057 | 0,00 | 0,00 | 0,00 | 0,00 | 242 | 32.869 | 242 | 32.869 | 0,00 | 0,00 |
| 2058 | 0,00 | 0,00 | 0,00 | 0,00 | 235 | 33.105 | 235 | 33.105 | 0,00 | 0,00 |
| 2059 | 0,00 | 0,00 | 0,00 | 0,00 | 229 | 33.334 | 229 | 33.334 | 0,00 | 0,00 |
| 2060 | 0,00 | 0,00 | 0,00 | 0,00 | 223 | 33.557 | 223 | 33.557 | 0,00 | 0,00 |
| 2061 | 0,00 | 0,00 | 0,00 | 0,00 | 217 | 33.774 | 217 | 33.774 | 0,00 | 0,00 |
| 2062 | 0,00 | 0,00 | 0,00 | 0,00 | 211 | 33.986 | 211 | 33.986 | 0,00 | 0,00 |
| 2063 | 0,00 | 0,00 | 0,00 | 0,00 | 206 | 34.192 | 206 | 34.192 | 0,00 | 0,00 |
| 2064 | 0,00 | 0,00 | 0,00 | 0,00 | 201 | 34.392 | 201 | 34.392 | 0,00 | 0,00 |
| 2065 | 0,00 | 0,00 | 0,00 | 0,00 | 195 | 34.588 | 195 | 34.588 | 0,00 | 0,00 |

|      |      |      |      |      |     |        |     |        |      |      |
|------|------|------|------|------|-----|--------|-----|--------|------|------|
| 2066 | 0,00 | 0,00 | 0,00 | 0,00 | 190 | 34.778 | 190 | 34.778 | 0,00 | 0,00 |
| 2067 | 0,00 | 0,00 | 0,00 | 0,00 | 185 | 34.963 | 185 | 34.963 | 0,00 | 0,00 |
| 2068 | 0,00 | 0,00 | 0,00 | 0,00 | 181 | 35.144 | 181 | 35.144 | 0,00 | 0,00 |
| 2069 | 0,00 | 0,00 | 0,00 | 0,00 | 176 | 35.320 | 176 | 35.320 | 0,00 | 0,00 |
| 2070 | 0,00 | 0,00 | 0,00 | 0,00 | 172 | 35.492 | 172 | 35.492 | 0,00 | 0,00 |
| 2071 | 0,00 | 0,00 | 0,00 | 0,00 | 167 | 35.659 | 167 | 35.659 | 0,00 | 0,00 |
| 2072 | 0,00 | 0,00 | 0,00 | 0,00 | 163 | 35.823 | 163 | 35.823 | 0,00 | 0,00 |
| 2073 | 0,00 | 0,00 | 0,00 | 0,00 | 159 | 35.982 | 159 | 35.982 | 0,00 | 0,00 |
| 2074 | 0,00 | 0,00 | 0,00 | 0,00 | 155 | 36.137 | 155 | 36.137 | 0,00 | 0,00 |
| 2075 | 0,00 | 0,00 | 0,00 | 0,00 | 151 | 36.288 | 151 | 36.288 | 0,00 | 0,00 |
| 2076 | 0,00 | 0,00 | 0,00 | 0,00 | 148 | 36.436 | 148 | 36.436 | 0,00 | 0,00 |
| 2077 | 0,00 | 0,00 | 0,00 | 0,00 | 144 | 36.580 | 144 | 36.580 | 0,00 | 0,00 |
| 2078 | 0,00 | 0,00 | 0,00 | 0,00 | 140 | 36.720 | 140 | 36.720 | 0,00 | 0,00 |
| 2079 | 0,00 | 0,00 | 0,00 | 0,00 | 137 | 36.857 | 137 | 36.857 | 0,00 | 0,00 |
| 2080 | 0,00 | 0,00 | 0,00 | 0,00 | 134 | 36.991 | 134 | 36.991 | 0,00 | 0,00 |
| 2081 | 0,00 | 0,00 | 0,00 | 0,00 | 131 | 37.122 | 131 | 37.122 | 0,00 | 0,00 |
| 2082 | 0,00 | 0,00 | 0,00 | 0,00 | 127 | 37.249 | 127 | 37.249 | 0,00 | 0,00 |

### 5.3 Leakage Emissions

This step provides an ex-ante estimate of the possible decrease in carbon stock and increase in GHG emissions (other than carbon stock change) due to leakage. According to the applied methodology, two sources of leakage are considered: a) decrease in carbon stocks and increase in GHG emissions associated with leakage prevention measures; and b) decrease in carbon stocks and increase in GHG emissions associated with activity displacement leakage.

#### **Ex ante estimation of decrease in carbon stocks and increase in GHG emissions due to leakage prevention measures.**

Initially, it is expected that leakage prevention measures will be employed within the limits of Fazenda Tapirapé, conducting courses and training related to sustainable development and conservation and environmental awareness. Subsequently, outside the limits of the Project Area, through assistance to associations of small farmers in the environment. These initiatives will focus not only on training and guidance for farmers in the region but also on raising people's awareness of environmental issues and preserving the forest.

As already mentioned in this document, it is not expected to develop any activity that could lead to the reduction of carbon stocks or the increase of GHG emissions compared to the baseline scenario. If there are significant changes in carbon stock, these activities will be monitored, accounted for and reported. Activities that will cause deforestation within the project area in the baseline case could be displaced outside the project boundary due to the implementation of the AUD project activity. A greater decrease in carbon stocks within the leakage belt during the project scenario than those predicted *ex-ante* would indicate displacement of deforestation activities due to the project.

The *ex ante* activity displacement leakage was calculated based on the anticipated combined effectiveness of the proposed leakage prevention measures and project activities. This was done by multiplying the estimated baseline carbon stock changes for the project area by a “Displacement Leakage Factor” (DLF) representing the percent of deforestation expected to be displaced outside the project boundary, beginning with an index of 5% and decreasing it along the project life-time. The equation is presented below:

$$\Delta CADLK_t = \Delta CBSLP_{At} \times DLF$$

Where,

$\Delta CADLK_t$  Total decrease in carbon stocks due to displaced deforestation at year t; tCO<sub>2</sub>e

DLF Displacement leakage factor; %

Thus, a displacement factor of 5% was adopted for the first five years. Then the reduction of the leakage displacement factor is gradual, already considering the influence of the Project in this context. Thus, the leakage displacement factor tends to approach to zero during the 60 years of project implementation.

The actual calculated values for  $\Delta CADLK_t$ , annually and cumulatively, across the project crediting period are own in Table 48 below.

Table 48. Ex ante estimated total leakage

| Project year | Total ex ante GHG emissions from increased grazing activities |                   | Total ex ante increase in GHG emissions due to displaced forest fires |                   | Total ex ante decrease in carbon stocks due to displaced deforestation |                   | Carbon stock decrease due to leakage prevention measures |                    | Total net carbon stock change due to leakage within the leakage belt |                   | Total net increase in emissions due to leakage |                   |
|--------------|---------------------------------------------------------------|-------------------|-----------------------------------------------------------------------|-------------------|------------------------------------------------------------------------|-------------------|----------------------------------------------------------|--------------------|----------------------------------------------------------------------|-------------------|------------------------------------------------|-------------------|
|              | EgLKt annual                                                  | EgLK cumulative   | EADLKt annual                                                         | EADLK cumulative  | ΔCADLKt annual                                                         | ΔCADLK cumulative | ΔCLPMLKt annual                                          | ΔCLPMLK cumulative | ΔCLKt annual                                                         | ΔCLK cumulative   | ELKt annual                                    | ELK cumulative    |
|              | tCO <sub>2e</sub>                                             | tCO <sub>2e</sub> | tCO <sub>2e</sub>                                                     | tCO <sub>2e</sub> | tCO <sub>2e</sub>                                                      | tCO <sub>2e</sub> | tCO <sub>2e</sub>                                        | tCO <sub>2e</sub>  | tCO <sub>2e</sub>                                                    | tCO <sub>2e</sub> | tCO <sub>2e</sub>                              | tCO <sub>2e</sub> |
| 2023         | 0,00                                                          | 0,00              | 0,00                                                                  | 0,00              | 2.000                                                                  | 2.000             | 0,00                                                     | 0,00               | 2.000                                                                | 2.000             | 0,00                                           | 0,00              |
| 2024         | 0,00                                                          | 0,00              | 0,00                                                                  | 0,00              | 2.061                                                                  | 4.062             | 0,00                                                     | 0,00               | 2.061                                                                | 4.062             | 0,00                                           | 0,00              |
| 2025         | 0,00                                                          | 0,00              | 0,00                                                                  | 0,00              | 2.121                                                                  | 6.183             | 0,00                                                     | 0,00               | 2.121                                                                | 6.183             | 0,00                                           | 0,00              |
| 2026         | 0,00                                                          | 0,00              | 0,00                                                                  | 0,00              | 2.179                                                                  | 8.362             | 0,00                                                     | 0,00               | 2.179                                                                | 8.362             | 0,00                                           | 0,00              |
| 2027         | 0,00                                                          | 0,00              | 0,00                                                                  | 0,00              | 2.235                                                                  | 10.597            | 0,00                                                     | 0,00               | 2.235                                                                | 10.597            | 0,00                                           | 0,00              |
| 2028         | 0,00                                                          | 0,00              | 0,00                                                                  | 0,00              | 1.832                                                                  | 12.429            | 0,00                                                     | 0,00               | 1.832                                                                | 12.429            | 0,00                                           | 0,00              |
| 2029         | 0,00                                                          | 0,00              | 0,00                                                                  | 0,00              | 1.875                                                                  | 14.304            | 0,00                                                     | 0,00               | 1.875                                                                | 14.304            | 0,00                                           | 0,00              |
| 2030         | 0,00                                                          | 0,00              | 0,00                                                                  | 0,00              | 1.916                                                                  | 16.220            | 0,00                                                     | 0,00               | 1.916                                                                | 16.220            | 0,00                                           | 0,00              |
| 2031         | 0,00                                                          | 0,00              | 0,00                                                                  | 0,00              | 1.957                                                                  | 18.177            | 0,00                                                     | 0,00               | 1.957                                                                | 18.177            | 0,00                                           | 0,00              |
| 2032         | 0,00                                                          | 0,00              | 0,00                                                                  | 0,00              | 1.497                                                                  | 19.674            | 0,00                                                     | 0,00               | 1.497                                                                | 19.674            | 0,00                                           | 0,00              |
| 2033         | 0,00                                                          | 0,00              | 0,00                                                                  | 0,00              | 1.452                                                                  | 21.127            | 0,00                                                     | 0,00               | 1.452                                                                | 21.127            | 0,00                                           | 0,00              |
| 2034         | 0,00                                                          | 0,00              | 0,00                                                                  | 0,00              | 1.408                                                                  | 22.535            | 0,00                                                     | 0,00               | 1.408                                                                | 22.535            | 0,00                                           | 0,00              |
| 2035         | 0,00                                                          | 0,00              | 0,00                                                                  | 0,00              | 1.366                                                                  | 23.901            | 0,00                                                     | 0,00               | 1.366                                                                | 23.901            | 0,00                                           | 0,00              |
| 2036         | 0,00                                                          | 0,00              | 0,00                                                                  | 0,00              | 884                                                                    | 24.785            | 0,00                                                     | 0,00               | 884                                                                  | 24.785            | 0,00                                           | 0,00              |
| 2037         | 0,00                                                          | 0,00              | 0,00                                                                  | 0,00              | 858                                                                    | 25.643            | 0,00                                                     | 0,00               | 858                                                                  | 25.643            | 0,00                                           | 0,00              |
| 2038         | 0,00                                                          | 0,00              | 0,00                                                                  | 0,00              | 832                                                                    | 26.475            | 0,00                                                     | 0,00               | 832                                                                  | 26.475            | 0,00                                           | 0,00              |
| 2039         | 0,00                                                          | 0,00              | 0,00                                                                  | 0,00              | 808                                                                    | 27.283            | 0,00                                                     | 0,00               | 808                                                                  | 27.283            | 0,00                                           | 0,00              |
| 2040         | 0,00                                                          | 0,00              | 0,00                                                                  | 0,00              | 392                                                                    | 27.675            | 0,00                                                     | 0,00               | 392                                                                  | 27.675            | 0,00                                           | 0,00              |
| 2041         | 0,00                                                          | 0,00              | 0,00                                                                  | 0,00              | 381                                                                    | 28.055            | 0,00                                                     | 0,00               | 381                                                                  | 28.055            | 0,00                                           | 0,00              |
| 2042         | 0,00                                                          | 0,00              | 0,00                                                                  | 0,00              | 370                                                                    | 28.425            | 0,00                                                     | 0,00               | 370                                                                  | 28.425            | 0,00                                           | 0,00              |
| 2043         | 0,00                                                          | 0,00              | 0,00                                                                  | 0,00              | 359                                                                    | 28.784            | 0,00                                                     | 0,00               | 359                                                                  | 28.784            | 0,00                                           | 0,00              |

|      |      |      |      |      |     |        |      |      |     |        |      |      |
|------|------|------|------|------|-----|--------|------|------|-----|--------|------|------|
| 2044 | 0,00 | 0,00 | 0,00 | 0,00 | 349 | 29.133 | 0,00 | 0,00 | 349 | 29.133 | 0,00 | 0,00 |
| 2045 | 0,00 | 0,00 | 0,00 | 0,00 | 339 | 29.471 | 0,00 | 0,00 | 339 | 29.471 | 0,00 | 0,00 |
| 2046 | 0,00 | 0,00 | 0,00 | 0,00 | 329 | 29.801 | 0,00 | 0,00 | 329 | 29.801 | 0,00 | 0,00 |
| 2047 | 0,00 | 0,00 | 0,00 | 0,00 | 320 | 30.120 | 0,00 | 0,00 | 320 | 30.120 | 0,00 | 0,00 |
| 2048 | 0,00 | 0,00 | 0,00 | 0,00 | 311 | 30.431 | 0,00 | 0,00 | 311 | 30.431 | 0,00 | 0,00 |
| 2049 | 0,00 | 0,00 | 0,00 | 0,00 | 302 | 30.733 | 0,00 | 0,00 | 302 | 30.733 | 0,00 | 0,00 |
| 2050 | 0,00 | 0,00 | 0,00 | 0,00 | 294 | 31.027 | 0,00 | 0,00 | 294 | 31.027 | 0,00 | 0,00 |
| 2051 | 0,00 | 0,00 | 0,00 | 0,00 | 286 | 31.313 | 0,00 | 0,00 | 286 | 31.313 | 0,00 | 0,00 |
| 2052 | 0,00 | 0,00 | 0,00 | 0,00 | 278 | 31.590 | 0,00 | 0,00 | 278 | 31.590 | 0,00 | 0,00 |
| 2053 | 0,00 | 0,00 | 0,00 | 0,00 | 270 | 31.861 | 0,00 | 0,00 | 270 | 31.861 | 0,00 | 0,00 |
| 2054 | 0,00 | 0,00 | 0,00 | 0,00 | 263 | 32.123 | 0,00 | 0,00 | 263 | 32.123 | 0,00 | 0,00 |
| 2055 | 0,00 | 0,00 | 0,00 | 0,00 | 256 | 32.379 | 0,00 | 0,00 | 256 | 32.379 | 0,00 | 0,00 |
| 2056 | 0,00 | 0,00 | 0,00 | 0,00 | 249 | 32.627 | 0,00 | 0,00 | 249 | 32.627 | 0,00 | 0,00 |
| 2057 | 0,00 | 0,00 | 0,00 | 0,00 | 242 | 32.869 | 0,00 | 0,00 | 242 | 32.869 | 0,00 | 0,00 |
| 2058 | 0,00 | 0,00 | 0,00 | 0,00 | 235 | 33.105 | 0,00 | 0,00 | 235 | 33.105 | 0,00 | 0,00 |
| 2059 | 0,00 | 0,00 | 0,00 | 0,00 | 229 | 33.334 | 0,00 | 0,00 | 229 | 33.334 | 0,00 | 0,00 |
| 2060 | 0,00 | 0,00 | 0,00 | 0,00 | 223 | 33.557 | 0,00 | 0,00 | 223 | 33.557 | 0,00 | 0,00 |
| 2061 | 0,00 | 0,00 | 0,00 | 0,00 | 217 | 33.774 | 0,00 | 0,00 | 217 | 33.774 | 0,00 | 0,00 |
| 2062 | 0,00 | 0,00 | 0,00 | 0,00 | 211 | 33.986 | 0,00 | 0,00 | 211 | 33.986 | 0,00 | 0,00 |
| 2063 | 0,00 | 0,00 | 0,00 | 0,00 | 206 | 34.192 | 0,00 | 0,00 | 206 | 34.192 | 0,00 | 0,00 |
| 2064 | 0,00 | 0,00 | 0,00 | 0,00 | 201 | 34.392 | 0,00 | 0,00 | 201 | 34.392 | 0,00 | 0,00 |
| 2065 | 0,00 | 0,00 | 0,00 | 0,00 | 195 | 34.588 | 0,00 | 0,00 | 195 | 34.588 | 0,00 | 0,00 |
| 2066 | 0,00 | 0,00 | 0,00 | 0,00 | 190 | 34.778 | 0,00 | 0,00 | 190 | 34.778 | 0,00 | 0,00 |
| 2067 | 0,00 | 0,00 | 0,00 | 0,00 | 185 | 34.963 | 0,00 | 0,00 | 185 | 34.963 | 0,00 | 0,00 |
| 2068 | 0,00 | 0,00 | 0,00 | 0,00 | 181 | 35.144 | 0,00 | 0,00 | 181 | 35.144 | 0,00 | 0,00 |
| 2069 | 0,00 | 0,00 | 0,00 | 0,00 | 176 | 35.320 | 0,00 | 0,00 | 176 | 35.320 | 0,00 | 0,00 |
| 2070 | 0,00 | 0,00 | 0,00 | 0,00 | 172 | 35.492 | 0,00 | 0,00 | 172 | 35.492 | 0,00 | 0,00 |
| 2071 | 0,00 | 0,00 | 0,00 | 0,00 | 167 | 35.659 | 0,00 | 0,00 | 167 | 35.659 | 0,00 | 0,00 |

|      |      |      |      |      |     |        |      |      |     |        |      |      |
|------|------|------|------|------|-----|--------|------|------|-----|--------|------|------|
| 2072 | 0,00 | 0,00 | 0,00 | 0,00 | 163 | 35.823 | 0,00 | 0,00 | 163 | 35.823 | 0,00 | 0,00 |
| 2073 | 0,00 | 0,00 | 0,00 | 0,00 | 159 | 35.982 | 0,00 | 0,00 | 159 | 35.982 | 0,00 | 0,00 |
| 2074 | 0,00 | 0,00 | 0,00 | 0,00 | 155 | 36.137 | 0,00 | 0,00 | 155 | 36.137 | 0,00 | 0,00 |
| 2075 | 0,00 | 0,00 | 0,00 | 0,00 | 151 | 36.288 | 0,00 | 0,00 | 151 | 36.288 | 0,00 | 0,00 |
| 2076 | 0,00 | 0,00 | 0,00 | 0,00 | 148 | 36.436 | 0,00 | 0,00 | 148 | 36.436 | 0,00 | 0,00 |
| 2077 | 0,00 | 0,00 | 0,00 | 0,00 | 144 | 36.580 | 0,00 | 0,00 | 144 | 36.580 | 0,00 | 0,00 |
| 2078 | 0,00 | 0,00 | 0,00 | 0,00 | 140 | 36.720 | 0,00 | 0,00 | 140 | 36.720 | 0,00 | 0,00 |
| 2079 | 0,00 | 0,00 | 0,00 | 0,00 | 137 | 36.857 | 0,00 | 0,00 | 137 | 36.857 | 0,00 | 0,00 |
| 2080 | 0,00 | 0,00 | 0,00 | 0,00 | 134 | 36.991 | 0,00 | 0,00 | 134 | 36.991 | 0,00 | 0,00 |
| 2081 | 0,00 | 0,00 | 0,00 | 0,00 | 131 | 37.122 | 0,00 | 0,00 | 131 | 37.122 | 0,00 | 0,00 |
| 2082 | 0,00 | 0,00 | 0,00 | 0,00 | 127 | 37.249 | 0,00 | 0,00 | 127 | 37.249 | 0,00 | 0,00 |

## 5.4 Estimated GHG Emission Reductions and Carbon Dioxide Removals

The net anthropogenic GHG emission reduction of the proposed AUD project activity is calculated as follows:

$$\Delta REDDt = (\Delta CBSLPAt + EBBBSLPAt) - (\Delta CPSPAt + EBBPSPAt) - (\Delta CLKt + ELKt)$$

Where:

$\Delta REDDt$  Ex ante estimated net anthropogenic greenhouse gas emission reduction attributable to the AUD project activity at year t; tCO<sub>2</sub>e

$\Delta CBSLPAt$  Sum of baseline carbon stock changes in the project area at year t; tCO<sub>2</sub>e

$EBBBSLPAt$  Sum of baseline emissions from biomass burning in the project area at year t; tCO<sub>2</sub>e

$\Delta CPSPAt$  Sum of ex ante estimated actual carbon stock changes in the project area at year t; tCO<sub>2</sub>e

**Note:** If  $\Delta CPSPAt$  represents a net increase in carbon stocks, a negative sign before the absolute value of  $\Delta CPSPAt$  shall be used. If  $\Delta CPSPAt$  represents a net decrease, the positive sign shall be used.

$EBBPSPAt$  Sum of (ex ante estimated) actual emissions from biomass burning in the project area at year t; tCO<sub>2</sub>e

$\Delta CLKt$  Sum of ex ante estimated leakage net carbon stock changes at year t; tCO<sub>2</sub>e

**Note:** If the cumulative sum of  $\Delta CLKt$  within a fixed baseline period is  $> 0$ ,  $\Delta CLKt$  shall be set to zero.

|        |                                                                                |
|--------|--------------------------------------------------------------------------------|
| $ELKt$ | Sum of ex ante estimated leakage emissions at year $t$ ; tCO <sub>2</sub> e    |
| $t$    | 1, 2, 3 ... T, a year of the proposed project crediting period; dimensionless. |

The number of Verified Carbon Units (VCUs) to be generated through the proposed AUD project activity at year  $t$  is calculated as follows:

$$VCUt = \Delta REDDt - VBCt$$

$$VBCt = (\Delta CBSLPAt - \Delta CPSPAt) \times RFt$$

Where:

|                  |                                                                                                                                                 |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| $VCUt$           | Number of Verified Carbon Units that can be traded at time $t$ ; t CO <sub>2</sub> -e                                                           |
| $\Delta REDDt$   | Ex ante estimated net anthropogenic greenhouse gas emission reduction attributable to the AUD project activity at year $t$ ; tCO <sub>2</sub> e |
| $VBCt$           | Number of Buffer Credits deposited in the VCS Buffer at time $t$ ; t CO <sub>2</sub> -e                                                         |
| $\Delta CBSLPAt$ | Sum of baseline carbon stock changes in the project area at year $t$ ; tCO <sub>2</sub> e                                                       |
| $\Delta CPSPAt$  | Sum of ex ante estimated actual carbon stock changes in the project area at year $t$ ; tCO <sub>2</sub> -e ha <sup>-1</sup>                     |
| $RFt$            | Risk factor used to calculate VCS buffer credits; %                                                                                             |
| $t$              | 1, 2, 3 ... T, a year of the proposed project crediting period; dimensionless.                                                                  |

The  $RFt$  was estimated using the most recent version of the VCS-approved AFOLU Non-Permanence Risk Tool the resulting value of  $RFt$  was 20%.

The specific summary of GHG reductions and removals in the ARAGUAIA REDD+ BOCALON PROJECT is included in table 49. The latter table includes estimates of GHG emissions reduction ( $REDDt$ ), calculations of buffer and leakage, and the resulting calculation of tradable Verified Carbon Units ( $VCUt$ ).

### Non-Permanence Risk Analysis

In the development of this Project, the application of the AFOLU Non-Permanence Risk tool v4.2 (October 2023) was a critical step to ensure an accurate analysis of non-permanence risks and to determine the buffer contributions required for the project. This methodology, endorsed by VERRA, outlines specific requirements for assessing risks and determining the appropriate risk rating for projects under the REDD+ framework.

In line with the latest updates to the VCS standard for REDD+ projects, NovaAgri, the proponent of this project, committed to utilizing the most recent version of the tool, released in the 4<sup>th</sup> quarter of 2023. This updated version, revised by VERRA, incorporates more detailed criteria for risk assessment, drawing upon the scientific information provided by respected bodies such as the Intergovernmental Panel on Climate Change (IPCC). Specifically, it aligns with the contributions from Working Group I (the physical science basis) of the Sixth Assessment Report (AR6) and the Special Report on the Ocean and Cryosphere in a Changing Climate (SROCC, 2019).

NovaAgri's decision to employ this advanced tool reflects the company's commitment to driving real changes within the carbon market. By prioritizing integrity and transparency in REDD+ projects, NovaAgri aligns with the recently announced New Era of VERRA.

#### *Internal Risks*

The ARAGUAIA REDD+ BOCALON PROJECT is covered by an adaptive management plan, which includes a comprehensive monitoring strategy to ensure the project's effectiveness and adaptability over time. The management team is comprised of a team with extensive experience in carbon accounting, geospatial processing, and reporting under other greenhouse gas (GHG) programs, ensuring that the project adheres to the necessary standards of accuracy and reliability in its environmental impact assessments.

Financial stability is a basis of the project's feasibility, with secured budget to cover all expenses until the project reaches its breakeven point. This financial assurance is reinforced by corporate investments from a company whose plans are deeply integrated with real impact results and commitments through innovation and sustainability. In addition, the financial subsidize of the economic group to which NovaAgri belongs serves as a primary supporter of the project, providing a foundation for its long-term feasibility.

With a projected lifespan of 60 years, the project's duration far exceeds the minimum requirement of 40 years set by the latest Verified Carbon Standard (VCS), ensuring its long-term contribution to carbon sequestration efforts. This extended timeframe is safeguarded by a legal agreement between the project proponent and the landowner, who holds the forest areas involved in the project, committing to the continuation of management practices that support the project's goals.

Despite the potential for revenue loss when compared to the most profitable alternative land use activities, the project is driven by a genuine interest among the involved stakeholders to engage in preservation activities and social support initiatives.

Finally, it is known that financial return is subject to external factors, such as the volatility of carbon market prices. However, the strategic planning of the proponent company and its economic group has covered the investment costs, significantly reducing the risk of non-permanence due to financial feasibility.



### *External Risks*

This project has rigorously completed a due diligence process to identify any possible disputes related to ownership, land, resource access, and use rights. Ensuring clarity and legality in these matters, a binding legal agreement has been executed, granting the project the legal right to control and operate activities across the project area. This agreement encompasses all entities with verified claims of ownership, resource access, and use rights, establishing a solid legal foundation for project activities operation.

The project is fortunate to be located in a jurisdiction devoid of a history of local government intervention in land or resource use, and it has not been historically affected by disputes over land tenure or ownership. This stability is crucial for the project's long-term feasibility and integrity.

As mentioned before, the project area is safeguarded by a legally binding commitment to continue management practices that protect carbon stocks beyond the duration of the project crediting period. This ensures that the benefits of the project will persist long into the future, contributing to global efforts to mitigate climate change.

Local populations, including those living in the project area or within 20 kilometers of its boundary, have been consulted through visits and local meetings, as detailed in Section 2 of this Project Description. These consultations have been instrumental in aligning project activities with community needs and expectations, ensuring local support and participation.

In terms of political risks, Brazil displays a level of governance that, despite challenges, is active to the implementation of climate initiatives. The country is actively working to enhance its climate agenda, with a regulated market project currently under discussion in Congress, for instance. According to the World Bank's Worldwide Governance Indicators, Brazil has a median Governance score of -0.29. Moreover, the nation is a party to the Paris Agreement and has submitted a Nationally Determined Contribution (NDC) to the UNFCCC Secretariat in the past five years<sup>85</sup>, which includes commitments related to Agriculture, Forestry, and Other Land Use (AFOLU), such as plans for zero deforestation. The country also has a documented and active climate change plan that encompasses activities similar to those of this project, underscoring its commitment to environmental stewardship and sustainable development.

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<sup>85</sup> Worldbank Group – Climate Knowledge Portal. Brazil – <https://climateknowledgeportal.worldbank.org/country/brazil>

| Indicator                        | Country | Year | Number of Sources | Governance (-2.5 to +2.5) | Percentile Rank | Standard Error |
|----------------------------------|---------|------|-------------------|---------------------------|-----------------|----------------|
| Voice and Accountability         | Brazil  | 2022 | 13                | 0,21                      | 55,56           | 0,12           |
| Political Stability and Absenc.. | Brazil  | 2022 | 8                 | -0,33                     | 33,96           | 0,22           |
| Government Effectiveness         | Brazil  | 2022 | 8                 | -0,59                     | 30,66           | 0,24           |
| Regulatory Quality               | Brazil  | 2022 | 8                 | -0,22                     | 43,87           | 0,21           |
| Rule of Law                      | Brazil  | 2022 | 12                | -0,26                     | 43,40           | 0,16           |
| Control of Corruption            | Brazil  | 2022 | 11                | -0,57                     | 32,08           | 0,16           |

Source: The Worldwide Governance Indicators – Brazilian rank and scores<sup>86</sup>

### *Natural Risks*

In the analysis of climate studies especially regarding the average temperature raise due to the effects of global warming, the proponent conducted studies to highlight potential risks for the project area and its jurisdiction.

It is crucial to note that historical natural risk is assessed based on likelihood and significance, as proposed by the VCS tool. As an AFOLU project, it must evaluate, at a minimum, fire, pest and disease outbreaks, extreme weather events, and geological risks.

As a complement to this analysis, it is important to refer to Section 1.13 of this PD, which details the climatic and geomorphological conditions of the locality in question, forming the basis for understanding existing risks, as presented below:

### *Fire Risk (F)*

The project area is not exempt from this type of risk, as it is located in a transitional forest-savanna area, making it more vulnerable to natural and anthropogenic wildfires, especially in the savanna portions within the property boundaries. However, historical data on wildfires indicate that the vegetation area is nearly entirely preserved, demonstrating a low frequency of events and rapid regeneration. Although not insignificant, the fire risk exists and can be classified as minor, occurring at a frequency of approximately one event every ten years.

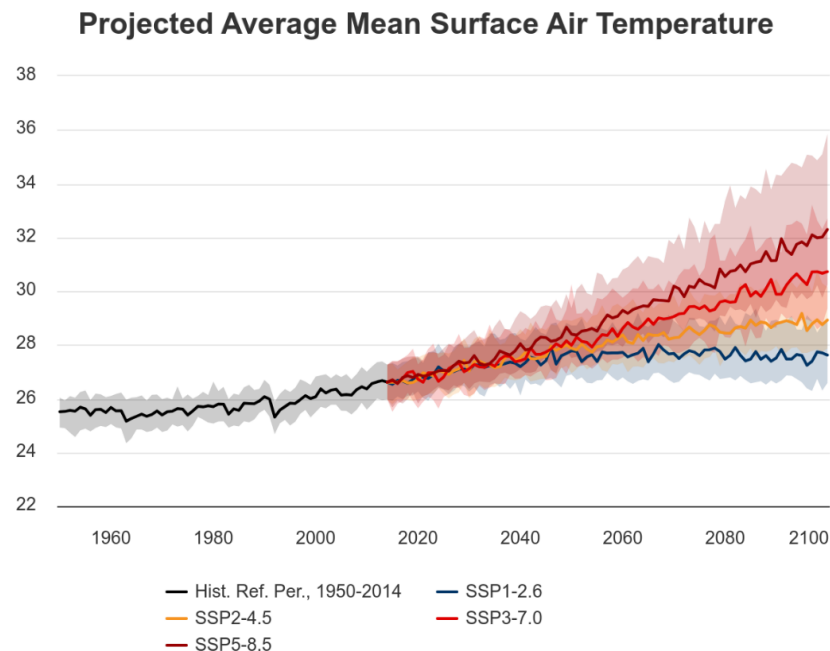
### *Pest and Disease Outbreaks (PD):*

It is understood that the risk is not applicable to the proposed project activities, which are essentially based on forest maintenance rather than agricultural activities, where pest risks are more prevalent. Even considering the conventional agricultural activities happening within the Project Area, pest and plant disease control is well-managed, and the soil itself is well-treated, (i.e. with straw cover between crop seasons). Such precautions are essential for mitigating potential small risks of pest exposure to surrounding forest species, which are generally more resilient and adapted to these natural adversities.

<sup>86</sup> The Worldwide Governance Indicators - <https://www.worldbank.org/en/publication/worldwide-governance-indicators/interactive-data-access>

### Extreme Weather (W):

Regarding the microclimate of the region, an analysis was conducted considering the state of Mato Grosso, comparing its geographical area with studies from the World Bank Group projecting climate change. The projections evaluations are based on the Shared Socio-economic Development Pathways (SSPs). According to the IPCC AR6 (2023)<sup>87</sup>, there are five SSPs (SSP1 to SSP5) designed to span a range of challenges to climate change mitigation and adaptation. For the State of Mato Grosso, the projections show a significant variation as climate scenarios progress in a growing and progressive movement. However, it can be observed that from the most optimistic scenario – with CO<sub>2</sub> emissions declining to net zero around 2050 and 2070, respectively, followed by varying levels of net negative CO<sub>2</sub> emissions – to the most pessimistic scenario – with CO<sub>2</sub> emissions roughly doubling from current levels by 2100 and 2050, respectively – the risks to the region's forests do not represent great alarm, considering that the maximum temperature that could be reached in 2100 is around 32 degrees Celsius as showed on the image below.



Graph Source: The World Bank Group – Climate Change Knowledge Portal<sup>88</sup>

<sup>87</sup> AR6 Synthesis Report: Climate Change 2023 - <https://www.ipcc.ch/report/sixth-assessment-report-cycle/>

<sup>88</sup> Climate Projections > Mean Projections > Brazil – Projected Average Mean Surface Air Temperature, Mato Grosso (ref. period: 1995-2014), Multi-Model Ensemble <https://climateknowledgeportal.worldbank.org/country/brazil/climate-data-projections>

According to the study by Sullivan et al (2020)<sup>89</sup>, the sensitivity of tropical forests to higher temperatures begins to have a greater effect in the short term when temperatures overcome above 32.2 degrees Celsius. In other words, even with significant changes in the worst-projected scenario, the tropical forests of Mato Grosso will likely remain resilient. The conclusion is that extreme climate events can be considered of lesser significance. It is also important to emphasize that the region does not experience alarming precipitation events, typhoons, or hurricanes due to Brazilian morphoclimatic conditions.

*Geological Risks (G) and Sea-Level Rise Risks (SLR)*

Considering the geological and geographical conditions presented in this PD (sections 1.12 and 1.13), on which the project area is defined and detailed, it is possible to assess that there are no risks related to relief amplitude or proximity to the sea. This excludes the chances of possible climate change impacts on natural disasters involving landslides, tsunamis, or any others related to the locality. Due to the distance from tectonic plate intersections, seismic events and similar occurrences can also be disregarded.

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<sup>89</sup> Sullivan et al., Science 368, 869–874 (2020) - <https://www.science.org/doi/epdf/10.1126/science.aaw7578>

Table 49. Ex ante estimated net anthropogenic GHG emission reductions (REDDt) and Verified Carbon Units (VCUt)

| Project year | Baseline carbon stock changes |                     | Baseline GHG emissions from biomass burning |                      | Ex ante project carbon stock changes |                     | Ex ante project GHG emissions from biomass burning |                      | Ex ante leakage carbon stock changes within the leakage belt |                    | Ex ante leakage GHG emissions |                    | Ex ante net anthropogenic GHG emission reductions |                    | Ex ante VCU's tradable |                    | Ex ante buffer credits |                    |
|--------------|-------------------------------|---------------------|---------------------------------------------|----------------------|--------------------------------------|---------------------|----------------------------------------------------|----------------------|--------------------------------------------------------------|--------------------|-------------------------------|--------------------|---------------------------------------------------|--------------------|------------------------|--------------------|------------------------|--------------------|
|              | ΔCBSLPat annual               | ΔCBSLPat cumulative | ΔEBBSLPat annual                            | ΔEBBSLPat cumulative | ΔCPSLPat annual                      | ΔCPSLPat cumulative | ΔEBBSLPat annual                                   | ΔEBBSLPat cumulative | ΔCLKt annual                                                 | ΔCLKt cumulative   | ELKt annual                   | ELKt cumulative    | ΔREDDt annual                                     | ΔREDDt cumulative  | ΔVCUt annual           | ΔVCUt cumulative   | ΔVBt annual            | ΔVBt cumulative    |
|              | tCO <sub>2</sub> e            | tCO <sub>2</sub> e  | tCO <sub>2</sub> e                          | tCO <sub>2</sub> e   | tCO <sub>2</sub> e                   | tCO <sub>2</sub> e  | tCO <sub>2</sub> e                                 | tCO <sub>2</sub> e   | tCO <sub>2</sub> e                                           | tCO <sub>2</sub> e | tCO <sub>2</sub> e            | tCO <sub>2</sub> e | tCO <sub>2</sub> e                                | tCO <sub>2</sub> e | tCO <sub>2</sub> e     | tCO <sub>2</sub> e | tCO <sub>2</sub> e     | tCO <sub>2</sub> e |
| 2023         | 40.008                        | 40.008              | 0.00                                        | 0.00                 | 2.000                                | 2.000               | 0                                                  | 0                    | 2.000                                                        | 2.000              | 0.00                          | 0.00               | 36.007                                            | 36.007             | 29.526                 | 29.526             | 6.481                  | 6.481              |
| 2024         | 41.230                        | 81.238              | 0.00                                        | 0.00                 | 2.061                                | 4.062               | 0                                                  | 0                    | 2.061                                                        | 4.062              | 0.00                          | 0.00               | 37.107                                            | 73.114             | 30.428                 | 59.954             | 6.679                  | 13.161             |
| 2025         | 42.419                        | 123.657             | 0.00                                        | 0.00                 | 2.121                                | 6.183               | 0                                                  | 0                    | 2.121                                                        | 6.183              | 0.00                          | 0.00               | 38.177                                            | 111.291            | 31.305                 | 91.259             | 6.872                  | 20.032             |
| 2026         | 43.576                        | 167.233             | 0.00                                        | 0.00                 | 2.179                                | 8.362               | 0                                                  | 0                    | 2.179                                                        | 8.362              | 0.00                          | 0.00               | 39.219                                            | 150.510            | 32.159                 | 123.418            | 7.059                  | 27.092             |
| 2027         | 44.703                        | 211.937             | 0.00                                        | 0.00                 | 2.235                                | 10.597              | 0                                                  | 0                    | 2.235                                                        | 10.597             | 0.00                          | 0.00               | 40.233                                            | 190.743            | 32.991                 | 156.409            | 7.242                  | 34.334             |
| 2028         | 45.801                        | 257.738             | 0.00                                        | 0.00                 | 1.832                                | 12.429              | 0                                                  | 0                    | 1.832                                                        | 12.429             | 0.00                          | 0.00               | 42.137                                            | 232.880            | 34.552                 | 190.962            | 7.585                  | 41.918             |
| 2029         | 46.870                        | 304.607             | 0.00                                        | 0.00                 | 1.875                                | 14.304              | 0                                                  | 0                    | 1.875                                                        | 14.304             | 0.00                          | 0.00               | 43.120                                            | 276.000            | 35.358                 | 226.320            | 7.762                  | 49.680             |
| 2030         | 47.911                        | 352.518             | 0.00                                        | 0.00                 | 1.916                                | 16.220              | 0                                                  | 0                    | 1.916                                                        | 16.220             | 0.00                          | 0.00               | 44.078                                            | 320.078            | 36.144                 | 262.464            | 7.934                  | 57.614             |
| 2031         | 48.925                        | 401.442             | 0.00                                        | 0.00                 | 1.957                                | 18.177              | 0                                                  | 0                    | 1.957                                                        | 18.177             | 0.00                          | 0.00               | 45.011                                            | 365.088            | 36.909                 | 299.372            | 8.102                  | 65.716             |
| 2032         | 49.912                        | 451.355             | 0.00                                        | 0.00                 | 1.497                                | 19.674              | 0                                                  | 0                    | 1.497                                                        | 19.674             | 0.00                          | 0.00               | 46.918                                            | 412.006            | 38.472                 | 337.845            | 8.445                  | 74.161             |
| 2033         | 48.404                        | 499.759             | 0.00                                        | 0.00                 | 1.452                                | 21.127              | 0                                                  | 0                    | 1.452                                                        | 21.127             | 0.00                          | 0.00               | 45.500                                            | 457.506            | 37.310                 | 375.155            | 8.190                  | 82.351             |
| 2034         | 46.948                        | 546.707             | 0.00                                        | 0.00                 | 1.408                                | 22.535              | 0                                                  | 0                    | 1.408                                                        | 22.535             | 0.00                          | 0.00               | 44.131                                            | 501.637            | 36.188                 | 411.343            | 7.944                  | 90.295             |
| 2035         | 45.543                        | 592.250             | 0.00                                        | 0.00                 | 1.366                                | 23.901              | 0                                                  | 0                    | 1.366                                                        | 23.901             | 0.00                          | 0.00               | 42.811                                            | 544.448            | 35.105                 | 446.447            | 7.706                  | 98.001             |
| 2036         | 44.186                        | 636.437             | 0.00                                        | 0.00                 | 884                                  | 24.785              | 0                                                  | 0                    | 884                                                          | 24.785             | 0.00                          | 0.00               | 42.419                                            | 586.867            | 34.784                 | 481.231            | 7.635                  | 105.636            |
| 2037         | 42.876                        | 679.313             | 0.00                                        | 0.00                 | 858                                  | 25.643              | 0                                                  | 0                    | 858                                                          | 25.643             | 0.00                          | 0.00               | 41.161                                            | 628.028            | 33.752                 | 514.983            | 7.409                  | 113.045            |
| 2038         | 41.611                        | 720.924             | 0.00                                        | 0.00                 | 832                                  | 26.475              | 0                                                  | 0                    | 832                                                          | 26.475             | 0.00                          | 0.00               | 39.947                                            | 667.975            | 32.756                 | 547.739            | 7.190                  | 120.235            |
| 2039         | 40.389                        | 761.314             | 0.00                                        | 0.00                 | 808                                  | 27.283              | 0                                                  | 0                    | 808                                                          | 27.283             | 0.00                          | 0.00               | 38.774                                            | 706.748            | 31.794                 | 579.534            | 6.979                  | 127.215            |
| 2040         | 39.209                        | 800.522             | 0.00                                        | 0.00                 | 392                                  | 27.675              | 0                                                  | 0                    | 392                                                          | 27.675             | 0.00                          | 0.00               | 38.424                                            | 745.173            | 31.508                 | 611.042            | 6.916                  | 134.131            |
| 2041         | 38.068                        | 838.590             | 0.00                                        | 0.00                 | 381                                  | 28.055              | 0                                                  | 0                    | 381                                                          | 28.055             | 0.00                          | 0.00               | 37.307                                            | 782.480            | 30.592                 | 641.633            | 6.715                  | 140.846            |
| 2042         | 36.966                        | 875.556             | 0.00                                        | 0.00                 | 370                                  | 28.425              | 0                                                  | 0                    | 370                                                          | 28.425             | 0.00                          | 0.00               | 36.227                                            | 818.706            | 29.706                 | 671.339            | 6.521                  | 147.367            |
| 2043         | 35.901                        | 911.457             | 0.00                                        | 0.00                 | 359                                  | 28.784              | 0                                                  | 0                    | 359                                                          | 28.784             | 0.00                          | 0.00               | 35.183                                            | 853.889            | 28.850                 | 700.189            | 6.333                  | 153.700            |
| 2044         | 34.872                        | 946.329             | 0.00                                        | 0.00                 | 349                                  | 29.133              | 0                                                  | 0                    | 349                                                          | 29.133             | 0.00                          | 0.00               | 34.174                                            | 888.064            | 28.023                 | 728.212            | 6.151                  | 159.851            |
| 2045         | 33.876                        | 980.206             | 0.00                                        | 0.00                 | 339                                  | 29.471              | 0                                                  | 0                    | 339                                                          | 29.471             | 0.00                          | 0.00               | 33.199                                            | 921.263            | 27.223                 | 755.435            | 5.976                  | 165.827            |
| 2046         | 32.914                        | 1.013.120           | 0.00                                        | 0.00                 | 329                                  | 29.801              | 0                                                  | 0                    | 329                                                          | 29.801             | 0.00                          | 0.00               | 32.256                                            | 953.519            | 26.450                 | 781.885            | 5.806                  | 171.633            |
| 2047         | 31.984                        | 1.045.104           | 0.00                                        | 0.00                 | 320                                  | 30.120              | 0                                                  | 0                    | 320                                                          | 30.120             | 0.00                          | 0.00               | 31.344                                            | 984.863            | 25.702                 | 807.588            | 5.642                  | 177.275            |
| 2048         | 31.084                        | 1.076.188           | 0.00                                        | 0.00                 | 311                                  | 30.431              | 0                                                  | 0                    | 311                                                          | 30.431             | 0.00                          | 0.00               | 30.463                                            | 1.015.326          | 24.979                 | 832.567            | 5.483                  | 182.759            |
| 2049         | 30.214                        | 1.106.403           | 0.00                                        | 0.00                 | 302                                  | 30.733              | 0                                                  | 0                    | 302                                                          | 30.733             | 0.00                          | 0.00               | 29.610                                            | 1.044.936          | 24.280                 | 856.847            | 5.330                  | 188.088            |
| 2050         | 29.372                        | 1.135.775           | 0.00                                        | 0.00                 | 294                                  | 31.027              | 0                                                  | 0                    | 294                                                          | 31.027             | 0.00                          | 0.00               | 28.785                                            | 1.073.721          | 23.604                 | 880.451            | 5.181                  | 193.270            |
| 2051         | 28.558                        | 1.164.333           | 0.00                                        | 0.00                 | 286                                  | 31.313              | 0                                                  | 0                    | 286                                                          | 31.313             | 0.00                          | 0.00               | 27.987                                            | 1.101.708          | 22.949                 | 903.400            | 5.038                  | 198.307            |
| 2052         | 27.770                        | 1.192.103           | 0.00                                        | 0.00                 | 278                                  | 31.590              | 0                                                  | 0                    | 278                                                          | 31.590             | 0.00                          | 0.00               | 27.215                                            | 1.128.922          | 22.316                 | 925.716            | 4.899                  | 203.206            |
| 2053         | 27.008                        | 1.219.111           | 0.00                                        | 0.00                 | 270                                  | 31.861              | 0                                                  | 0                    | 270                                                          | 31.861             | 0.00                          | 0.00               | 26.467                                            | 1.155.390          | 21.703                 | 947.419            | 4.764                  | 207.970            |
| 2054         | 26.269                        | 1.245.380           | 0.00                                        | 0.00                 | 263                                  | 32.123              | 0                                                  | 0                    | 263                                                          | 32.123             | 0.00                          | 0.00               | 25.744                                            | 1.181.134          | 21.110                 | 968.530            | 4.634                  | 212.604            |
| 2055         | 25.555                        | 1.270.935           | 0.00                                        | 0.00                 | 256                                  | 32.379              | 0                                                  | 0                    | 256                                                          | 32.379             | 0.00                          | 0.00               | 25.044                                            | 1.206.178          | 20.536                 | 989.066            | 4.508                  | 217.112            |
| 2056         | 24.863                        | 1.295.799           | 0.00                                        | 0.00                 | 249                                  | 32.627              | 0                                                  | 0                    | 249                                                          | 32.627             | 0.00                          | 0.00               | 24.366                                            | 1.230.544          | 19.980                 | 1.009.046          | 4.386                  | 221.498            |
| 2057         | 24.194                        | 1.319.992           | 0.00                                        | 0.00                 | 242                                  | 32.869              | 0                                                  | 0                    | 242                                                          | 32.869             | 0.00                          | 0.00               | 23.710                                            | 1.254.253          | 19.442                 | 1.028.488          | 4.268                  | 225.766            |
| 2058         | 23.545                        | 1.343.537           | 0.00                                        | 0.00                 | 235                                  | 33.105              | 0                                                  | 0                    | 235                                                          | 33.105             | 0.00                          | 0.00               | 23.074                                            | 1.277.328          | 18.921                 | 1.047.409          | 4.153                  | 229.919            |
| 2059         | 22.917                        | 1.368.454           | 0.00                                        | 0.00                 | 229                                  | 33.334              | 0                                                  | 0                    | 229                                                          | 33.334             | 0.00                          | 0.00               | 22.459                                            | 1.299.786          | 18.416                 | 1.065.825          | 4.043                  | 233.962            |
| 2060         | 22.309                        | 1.388.763           | 0.00                                        | 0.00                 | 223                                  | 33.557              | 0                                                  | 0                    | 223                                                          | 33.557             | 0.00                          | 0.00               | 21.863                                            | 1.321.649          | 17.927                 | 1.083.752          | 3.935                  | 237.897            |
| 2061         | 21.719                        | 1.410.482           | 0.00                                        | 0.00                 | 217                                  | 33.774              | 0                                                  | 0                    | 217                                                          | 33.774             | 0.00                          | 0.00               | 21.285                                            | 1.342.934          | 17.454                 | 1.101.206          | 3.831                  | 241.728            |
| 2062         | 21.148                        | 1.431.631           | 0.00                                        | 0.00                 | 211                                  | 33.986              | 0                                                  | 0                    | 211                                                          | 33.986             | 0.00                          | 0.00               | 20.725                                            | 1.363.659          | 16.995                 | 1.118.201          | 3.731                  | 245.459            |
| 2063         | 20.595                        | 1.452.226           | 0.00                                        | 0.00                 | 206                                  | 34.192              | 0                                                  | 0                    | 206                                                          | 34.192             | 0.00                          | 0.00               | 20.183                                            | 1.383.842          | 16.550                 | 1.134.751          | 3.633                  | 249.092            |
| 2064         | 20.059                        | 1.472.284           | 0.00                                        | 0.00                 | 201                                  | 34.392              | 0                                                  | 0                    | 201                                                          | 34.392             | 0.00                          | 0.00               | 19.658                                            | 1.403.500          | 16.119                 | 1.150.870          | 3.538                  | 252.630            |
| 2065         | 19.539                        | 1.491.823           | 0.00                                        | 0.00                 | 195                                  | 34.588              | 0                                                  | 0                    | 195                                                          | 34.588             | 0.00                          | 0.00               | 19.148                                            | 1.422.648          | 15.702                 | 1.166.571          | 3.447                  | 256.077            |
| 2066         | 19.035                        | 1.510.858           | 0.00                                        | 0.00                 | 190                                  | 34.778              | 0                                                  | 0                    | 190                                                          | 34.778             | 0.00                          | 0.00               | 18.654                                            | 1.441.302          | 15.297                 | 1.181.868          | 3.358                  | 259.434            |
| 2067         | 18.547                        | 1.529.405           | 0.00                                        | 0.00                 | 185                                  | 34.963              | 0                                                  | 0                    | 185                                                          | 34.963             | 0.00                          | 0.00               | 18.176                                            | 1.459.478          | 14.904                 | 1.196.772          | 3.272                  | 262.706            |
| 2068         | 18.073                        | 1.547.478           | 0.00                                        | 0.00                 | 181                                  | 35.144              | 0                                                  | 0                    | 181                                                          | 35.144             | 0.00                          | 0.00               | 17.712                                            | 1.477.190          | 14.524                 | 1.211.296          | 3.188                  | 265.894            |
| 2069         | 17.614                        | 1.565.092           | 0.00                                        | 0.00                 | 176                                  | 35.320              | 0                                                  | 0                    | 176                                                          | 35.320             | 0.00                          | 0.00               | 17.261                                            | 1.494.451          | 14.154                 | 1.225.450          | 3.107                  | 269.001            |
| 2070         | 17.168                        | 1.582.260           | 0.00                                        | 0.00                 | 172                                  | 35.492              | 0                                                  | 0                    | 172                                                          | 35.492             | 0.00                          | 0.00               | 16.825                                            | 1.511.276          | 13.796                 | 1.239.246          | 3.028                  | 272.030            |
| 2071         | 16.736                        | 1.598.997           | 0.00                                        | 0.00                 | 167                                  | 35.659              | 0                                                  | 0                    | 167                                                          | 35.659             | 0.00                          | 0.00               | 16.402                                            | 1.527.678          | 13.449                 | 1.252.696          | 2.952                  | 274.982            |
| 2072         | 16.317                        | 1.615.314           | 0.00                                        | 0.00                 | 163                                  | 35.823              | 0                                                  | 0                    | 163                                                          | 35.823             | 0.00                          | 0.00               | 15.991                                            | 1.543.668          | 13.112                 | 1.265.808          | 2.878                  | 277.860            |
| 2073         | 15.910                        | 1.631.224           | 0.00                                        | 0.00                 | 159                                  | 35.982              | 0                                                  | 0                    | 159                                                          | 35.982             | 0.00                          | 0.00               | 15.592                                            | 1.559.261          | 12.786                 | 1.278.594          | 2.807                  | 280.667            |
| 2074         | 15.516                        | 1.646.740           | 0.00                                        | 0.00                 | 155                                  | 36.137              | 0                                                  | 0                    | 155                                                          | 36.137             | 0.00                          | 0.00               | 15.205                                            | 1.574.466          | 12.468                 | 1.291.062          | 2.737                  | 283.404            |
| 2075         | 15.133                        | 1.661.872           | 0.00                                        | 0.00                 | 151                                  | 36.288              | 0                                                  | 0                    | 151                                                          | 36.288             | 0.00                          | 0.00               | 14.830                                            | 1.589.296          | 12.161                 | 1.303.223          | 2.669                  | 286.073            |
| 2076         | 14.761                        | 1.676.634           | 0.00                                        | 0.00                 | 148                                  | 36.436              | 0                                                  | 0                    | 148                                                          | 36.436             | 0.00                          | 0.00               | 14.466                                            | 1.603.762          | 11.862                 | 1.315.085          | 2.604                  | 288.677            |
| 2077         | 14.400                        | 1.691.034           | 0.00                                        | 0.00                 | 144                                  | 36.580              | 0                                                  | 0                    | 144                                                          | 36.580             | 0.00                          | 0.00               | 14.112                                            | 1.617.874          | 11.572                 | 1.326.657          | 2.540                  | 291.217            |
| 2078         | 14.050                        | 1.705.084           | 0.00                                        | 0.00                 | 140                                  | 36.720              | 0                                                  | 0                    | 140                                                          | 36.720             | 0.00                          | 0.00               | 13.769                                            | 1.631.643          | 11.291                 | 1.337.948          | 2.478                  | 293.696            |
| 2079         | 13.710                        | 1.718.794           | 0.00                                        | 0.00                 | 137                                  | 36.857              | 0                                                  | 0                    | 137                                                          | 36.857             | 0.00                          | 0.00               | 13.436                                            | 1.645.079          | 11.017                 | 1.348.965          | 2.418                  | 296.114            |
| 2080         | 13.380                        | 1.732.173           | 0.00                                        | 0.00                 | 134                                  | 36.991              | 0                                                  | 0                    | 134                                                          | 36.991             | 0.00                          | 0.00               | 13.112                                            | 1.658.191          | 10.752                 | 1.359.717          | 2.360                  | 298.474            |
| 2081         | 13.059                        | 1.745.232           | 0.00                                        | 0.00                 | 131                                  | 37.122              | 0                                                  | 0                    | 131                                                          | 37.122             | 0.00                          | 0.00               | 12.798                                            | 1.670.989          | 10.494                 | 1.370.211          | 2.                     |                    |

## 6 MONITORING

### 6.1 Data and Parameters Available at Validation

|                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Data / Parameter                                                                             | ABSLRR $t$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Data unit                                                                                    | ha                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Description                                                                                  | Annual area of deforestation in the reference region at year $t$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Source of data                                                                               | Calculated                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Value applied                                                                                | Tables Section 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Justification of choice of data or description of measurement methods and procedures applied | <p>The deforestation baseline is calculated based on a historical time-series of deforestation compiled within the HDAs that belong to the reference area, where regulatory and geomorphological characteristics (i.e. vegetation, soil, climate and slope) are similar.</p> <p>Data from PRODES produced based on satellite images with sufficient spatial resolution to identify deforestation size from 1ha were used to create an annual time series with at least 20 years of remaining forest area observations within the reference area.</p> <p>The logarithm of the observed values of the remaining forest area is the object of a linear regression that is then extrapolated into the future. The extrapolated values are then the object of an exponential function, which result in the predicted values of the remaining forest area.</p> |
| Purpose of Data                                                                              | Determination of baseline scenario                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Comments                                                                                     | Monitoring – each renewal of baseline validity period                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

|                  |                                                              |
|------------------|--------------------------------------------------------------|
| Data / Parameter | ABSLPA $t$                                                   |
| Data unit        | ha                                                           |
| Description      | Annual area of deforestation in the project area at year $t$ |

|                                                                                              |                                                                                                                                                 |
|----------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| Source of data                                                                               | <i>Calculated</i>                                                                                                                               |
| Value applied                                                                                | <i>Tables Section 3</i>                                                                                                                         |
| Justification of choice of data or description of measurement methods and procedures applied | <i>Forest cover change due to deforestation will be monitored through assessment of classified satellite imagery covering the project area.</i> |
| Purpose of Data                                                                              | <i>This parameter will be used to calculate baseline emissions and project emissions in the project scenario.</i>                               |
| Comments                                                                                     | <i>N/A</i>                                                                                                                                      |

|                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Data / Parameter                                                                             | <b>Cab icl</b>                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Data unit                                                                                    | tCO2e/ha                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Description                                                                                  | <i>Average carbon stock per hectare in aboveground biomass carbon pool of initial forest</i>                                                                                                                                                                                                                                                                                                                                     |
| Source of data                                                                               | <i>Estimated from FREL</i>                                                                                                                                                                                                                                                                                                                                                                                                       |
| Value applied                                                                                | <i>Tables Section 3</i>                                                                                                                                                                                                                                                                                                                                                                                                          |
| Justification of choice of data or description of measurement methods and procedures applied | <p><i>The Carbon stocks per hectare of initial forest existing in the project area were calculated based on official literature regarding specific characteristics of different biomes and vegetation aspects, all referenced on this PD.</i></p> <p><i>The calculation has been ran considering an area-weighted average carbon stocks of the post deforestation LU/LC classes existing within each zone are displayed.</i></p> |
| Purpose of Data                                                                              | <i>This parameter was used to calculate the baseline, project and leakage emissions from deforestation occurred in the baseline and project scenarios. Provides an average of the biomass stock per hectare in the above-ground biomass within the project reference region.</i>                                                                                                                                                 |
| Comments                                                                                     | <i>If new and more accurate biomass stock data become available, these can be used to estimate the net anthropogenic GHG emission reduction of the subsequent fixed baseline period.</i>                                                                                                                                                                                                                                         |

|                  |                |
|------------------|----------------|
| Data / Parameter | <b>Cbb icl</b> |
| Data unit        | tCO2e/ha       |

|                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-----------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b>                                                                                  | <i>Average carbon stock per hectare in belowground biomass carbon pool of initial forest</i>                                                                                                                                                                                                                                                                                                                                     |
| <b>Source of data</b>                                                                               | <i>Estimated from FREL</i>                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Value applied</b>                                                                                | <i>Tables Section 3</i>                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Justification of choice of data or description of measurement methods and procedures applied</b> | <p><i>The Carbon stocks per hectare of initial forest existing in the project area were calculated based on official literature regarding specific characteristics of different biomes and vegetation aspects, all referenced on this PD.</i></p> <p><i>The calculation has been ran considering an area-weighted average carbon stocks of the post deforestation LU/LC classes existing within each zone are displayed.</i></p> |
| <b>Purpose of Data</b>                                                                              | <i>This parameter was used to calculate the baseline, project and leakage emissions from deforestation occurred in the baseline and project scenarios. Provides an average of the biomass stock per hectare in the below-ground biomass within the project reference region.</i>                                                                                                                                                 |
| <b>Comments</b>                                                                                     | <i>If new and more accurate biomass stock data become available, these can be used to estimate the net anthropogenic GHG emission reduction of the subsequent fixed baseline period.</i>                                                                                                                                                                                                                                         |

|                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-----------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Data / Parameter</b>                                                                             | <b>EI</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Data unit</b>                                                                                    | %                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Description</b>                                                                                  | <i>Ex ante estimated effectiveness index</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Source of data</b>                                                                               | <i>IPCC 2016. Good practice guidance for land use, land-use change and forestry.</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Value applied</b>                                                                                | 95                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Justification of choice of data or description of measurement methods and procedures applied</b> | <p><i>The calculation of the effectiveness index was estimated based on the ARAGUAIA REDD+ BOCALON PROJECT's efficiency in reducing deforestation within the project area.</i></p> <p><i>Effectiveness Index (EI) of project activities was conservatively assumed as 95% in the first five years of implementation and that this value will increases gradually with their efficiency over the years, considering an increase in patrolling land surveillance and the implementation of social activities in the local communities, until reaching the value of 99% in year 2040.</i></p> |



|                 |                                                                                                                                                                                                                                                                                                                   |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Purpose of Data | <i>This parameter was used to calculate project emissions in the baseline scenario. Provides an ex ante estimation of the carbon stock changes due to unavoidable unplanned deforestation within the project area, based on the effectiveness of the proposed project activities to reduce the deforestation.</i> |
| Comments        | <i>Ex post monitoring of the project area shall be done to determine deforestation rate and the project emissions. This parameter will be updated at each renewal of fixed baseline period.</i>                                                                                                                   |

|                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                           |
|----------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Data / Parameter                                                                             | DLF                                                                                                                                                                                                                                                                                                                                                                                                       |
| Data unit                                                                                    | %                                                                                                                                                                                                                                                                                                                                                                                                         |
| Description                                                                                  | Displacement Leakage Factor                                                                                                                                                                                                                                                                                                                                                                               |
| Source of data                                                                               | IPCC 2016. Good practice guidance for land use, land-use change and forestry.                                                                                                                                                                                                                                                                                                                             |
| Value applied                                                                                | 5                                                                                                                                                                                                                                                                                                                                                                                                         |
| Justification of choice of data or description of measurement methods and procedures applied | <i>The DLF was estimated at 5% in the first five years, which is considered a conservative estimate based on the literature referenced, and then decreasing over the life of the project until reaching the value of 1% in the year 2040, due to the expected combined effectiveness of the proposed leakage prevention measures and project activities.</i>                                              |
| Purpose of Data                                                                              | <i>This parameter was used to calculate leakage emissions in the baseline scenario due to activity displacement leakage, providing an ex ante estimation of the decrease in carbon stocks and increase in GHG emissions. This value was calculated based on the percent of deforestation expected to be displaced outside the project boundary due to the implementation of the AUD project activity.</i> |
| Comments                                                                                     | <i>Ex post monitoring of the leakage belt shall be done to determine deforestation rate outside the project area and consequently, the leakage emissions and carbon stock decrease. This parameter will be updated at each renewal of fixed baseline period.</i>                                                                                                                                          |

|                  |                        |
|------------------|------------------------|
| Data / Parameter | $\Delta C_{abBSLRR} t$ |
|------------------|------------------------|

|                                                                                              |                                                                                                                                                     |
|----------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| Data unit                                                                                    | tCO <sub>2</sub> e                                                                                                                                  |
| Description                                                                                  | <i>Changes in the aboveground stock in reference region</i>                                                                                         |
| Source of data                                                                               | <i>Calculated</i>                                                                                                                                   |
| Value applied                                                                                | <i>Tables Section 5</i>                                                                                                                             |
| Justification of choice of data or description of measurement methods and procedures applied | <i>A carbon stock change factors for initial forest classes in aboveground carbon pool were applied to calculate baseline carbon stock changes.</i> |
| Purpose of Data                                                                              | <i>Calculation of baseline emissions</i>                                                                                                            |
| Comments                                                                                     | <i>Monitoring – each renewal of baseline validity period</i>                                                                                        |

|                                                                                              |                                                                                                                                                     |
|----------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| Data / Parameter                                                                             | $\Delta C_{bbBSLRR} \text{ t}$                                                                                                                      |
| Data unit                                                                                    | tCO <sub>2</sub> e                                                                                                                                  |
| Description                                                                                  | <i>Changes in the belowground stock in reference region</i>                                                                                         |
| Source of data                                                                               | <i>Calculated</i>                                                                                                                                   |
| Value applied                                                                                | <i>Tables Section 5</i>                                                                                                                             |
| Justification of choice of data or description of measurement methods and procedures applied | <i>A carbon stock change factors for initial forest classes in belowground carbon pool were applied to calculate baseline carbon stock changes.</i> |
| Purpose of Data                                                                              | <i>Calculation of baseline emissions</i>                                                                                                            |
| Comments                                                                                     | <i>Monitoring – each renewal of baseline validity period</i>                                                                                        |

|                  |                                                         |
|------------------|---------------------------------------------------------|
| Data / Parameter | $\Delta C_{abBSLPA} \text{ t}$                          |
| Data unit        | tCO <sub>2</sub> e                                      |
| Description      | <i>Changes in the aboveground stock in project area</i> |
| Source of data   | <i>Calculated</i>                                       |
| Value applied    | <i>Tables Section 5</i>                                 |

|                                                                                              |                                                                                                                                                     |
|----------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| Justification of choice of data or description of measurement methods and procedures applied | <i>A carbon stock change factors for initial forest classes in aboveground carbon pool were applied to calculate baseline carbon stock changes.</i> |
| Purpose of Data                                                                              | <i>Calculation of baseline emissions</i>                                                                                                            |
| Comments                                                                                     | <i>Monitoring – each renewal of baseline validity period</i>                                                                                        |

|                                                                                              |                                                                                                                                                     |
|----------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| Data / Parameter                                                                             | $\Delta C_{bbBSLPA}$ t                                                                                                                              |
| Data unit                                                                                    | tCO <sub>2</sub> e                                                                                                                                  |
| Description                                                                                  | <i>Changes in the belowground stock in project area</i>                                                                                             |
| Source of data                                                                               | <i>Calculated</i>                                                                                                                                   |
| Value applied                                                                                | <i>Tables Section 5</i>                                                                                                                             |
| Justification of choice of data or description of measurement methods and procedures applied | <i>A carbon stock change factors for initial forest classes in belowground carbon pool were applied to calculate baseline carbon stock changes.</i> |
| Purpose of Data                                                                              | <i>Calculation of baseline emissions</i>                                                                                                            |
| Comments                                                                                     | <i>Monitoring – each renewal of baseline validity period</i>                                                                                        |

|                                                                                              |                                                                                                                                                     |
|----------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| Data / Parameter                                                                             | $\Delta C_{abBSLLK}$ t                                                                                                                              |
| Data unit                                                                                    | tCO <sub>2</sub> e                                                                                                                                  |
| Description                                                                                  | <i>Changes in the aboveground stock in leakage belt</i>                                                                                             |
| Source of data                                                                               | <i>Calculated</i>                                                                                                                                   |
| Value applied                                                                                | <i>Tables Section 5</i>                                                                                                                             |
| Justification of choice of data or description of measurement methods and procedures applied | <i>A carbon stock change factors for initial forest classes in aboveground carbon pool were applied to calculate baseline carbon stock changes.</i> |
| Purpose of Data                                                                              | <i>Calculation of baseline emissions</i>                                                                                                            |
| Comments                                                                                     | <i>Monitoring – each renewal of baseline validity period</i>                                                                                        |

|                                                                                              |                                                                                                                                              |
|----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| Data / Parameter                                                                             | $\Delta C_{bbBSLLK\ t}$                                                                                                                      |
| Data unit                                                                                    | tCO <sub>2</sub> e                                                                                                                           |
| Description                                                                                  | Changes in the belowground stock in leakage belt                                                                                             |
| Source of data                                                                               | Calculated                                                                                                                                   |
| Value applied                                                                                | Tables Section 5                                                                                                                             |
| Justification of choice of data or description of measurement methods and procedures applied | A carbon stock change factors for initial forest classes in belowground carbon pool were applied to calculate baseline carbon stock changes. |
| Purpose of Data                                                                              | Calculation of baseline emissions                                                                                                            |
| Comments                                                                                     | Monitoring – each renewal of baseline validity period                                                                                        |

|                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|----------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Data / Parameter                                                                             | $\Delta C_{BSLPA\ t}$                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Data unit                                                                                    | tCO <sub>2</sub> e                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Description                                                                                  | Total baseline carbon stock change within project area at year t                                                                                                                                                                                                                                                                                                                                                                                                             |
| Source of data                                                                               | Calculated                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Value applied                                                                                | Tables Section 5                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Justification of choice of data or description of measurement methods and procedures applied | <p>The total baseline carbon stock change in the project area at year t is calculated as follows:</p> $C_{BSLPA\ t} = C_{abBSLPA\ t} + C_{bbBSLPA\ t}$ <p>Where,</p> <p><math>C_{BSLPA\ t}</math>: Total baseline carbon stock changes in the project area at year t; tCO<sub>2</sub>e</p> <p><math>C_{abBSLPA\ t}</math>: Total baseline carbon stock change for the above-ground biomass pool in the project area for initial forest class at year t; tCO<sub>2</sub>e</p> |

|                 |                                                                                                                                                       |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
|                 | <i>CbbBSLPA t: Total baseline carbon stock change for the below-ground biomass pool in the project area for initial forest class at year t; tCO2e</i> |
| Purpose of Data | <i>Calculation of baseline emissions</i>                                                                                                              |
| Comments        | <i>Monitoring – each renewal of baseline validity period</i>                                                                                          |

## 6.2 Data and Parameters Monitored

|                                                                 |                                                                                                                                                                                            |
|-----------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Data / Parameter                                                | ABSLRR t                                                                                                                                                                                   |
| Data unit                                                       | Ha                                                                                                                                                                                         |
| Description                                                     | <i>Annual area of deforestation in the reference region at year t</i>                                                                                                                      |
| Source of data                                                  | <i>Remote sensing and GIS</i>                                                                                                                                                              |
| Description of measurement methods and procedures to be applied | <i>Forest cover change due to deforestation is monitored through periodic assessment of classified satellite imagery covering the reference region</i>                                     |
| Frequency of monitoring/recording                               | <i>Annually</i>                                                                                                                                                                            |
| Value applied                                                   | <i>Tables Sections 5 and 7</i>                                                                                                                                                             |
| Monitoring equipment                                            | <i>Remote sensing and GIS</i>                                                                                                                                                              |
| QA/QC procedures to be applied                                  | <i>Best practices in remote sensing</i>                                                                                                                                                    |
| Purpose of data                                                 | <i>This parameter will be used to calculate baseline emissions in the project scenario. Provides the ex post value of the deforested area per forest class within the reference region</i> |
| Calculation method                                              | <i>Analysis of satellite images and maps</i>                                                                                                                                               |
| Comments                                                        | <i>N/A</i>                                                                                                                                                                                 |

|                  |          |
|------------------|----------|
| Data / Parameter | ABSLPA t |
| Data unit        | Ha       |

|                                                                 |                                                                                                                                                                                                               |
|-----------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Description                                                     | <i>Annual area of deforestation in the project area at year t</i>                                                                                                                                             |
| Source of data                                                  | <i>Remote sensing and GIS</i>                                                                                                                                                                                 |
| Description of measurement methods and procedures to be applied | <i>Forest cover change due to deforestation is monitored through periodic assessment of classified satellite imagery covering the project area.</i>                                                           |
| Frequency of monitoring/recording                               | <i>Annually</i>                                                                                                                                                                                               |
| Value applied                                                   | <i>Tables Sections 5 and 7</i>                                                                                                                                                                                |
| Monitoring equipment                                            | <i>Remote sensing and GIS</i>                                                                                                                                                                                 |
| QA/QC procedures to be applied                                  | <i>Best practices in remote sensing</i>                                                                                                                                                                       |
| Purpose of data                                                 | <i>This parameter will be used to calculate baseline emissions and project emissions in the project scenario. Provides the ex post value of the deforested area per forest class within the project area.</i> |
| Calculation method                                              | <i>Analysis of satellite images and maps</i>                                                                                                                                                                  |
| Comments                                                        | <i>N/A</i>                                                                                                                                                                                                    |

|                                                                 |                                                                                                                                                                                                                                                          |
|-----------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Data / Parameter                                                | <b>ABSLK t</b>                                                                                                                                                                                                                                           |
| Data unit                                                       | <i>Ha</i>                                                                                                                                                                                                                                                |
| Description                                                     | <i>Annual area of deforestation in the leakage belt at year t</i>                                                                                                                                                                                        |
| Source of data                                                  | <i>Remote sensing and GIS</i>                                                                                                                                                                                                                            |
| Description of measurement methods and procedures to be applied | <i>Deforestation in the leakage belt area will be considered activity displacement leakage. Activity data for the leakage belt area will be determined using the same methods applied to monitoring deforestation activity data in the project area.</i> |
| Frequency of monitoring/recording                               | <i>Annually</i>                                                                                                                                                                                                                                          |
| Value applied                                                   | <i>Tables Sections 5 and 7</i>                                                                                                                                                                                                                           |
| Monitoring equipment                                            | <i>Remote sensing and GIS</i>                                                                                                                                                                                                                            |

|                                |                                                                                                                                                                                                                                                                               |
|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| QA/QC procedures to be applied | <i>Best practices in remote sensing</i>                                                                                                                                                                                                                                       |
| Purpose of data                | <i>This parameter will be used to calculate leakage emissions in the project scenario. Provides the ex post value of the deforested area within the leakage belt.</i>                                                                                                         |
| Calculation method             | <i>Analysis of satellite images and maps</i>                                                                                                                                                                                                                                  |
| Comments                       | <i>Where strong evidence can be collected that deforestation in the leakage belt is attributable to deforestation agents that are not linked to the project area, the detected deforestation will not be attributed to the project activity, thus not considered leakage.</i> |

|                                                                 |                                                                                                                                                                                                                                                          |
|-----------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Data / Parameter                                                | $\Delta\text{CADLK}_t$                                                                                                                                                                                                                                   |
| Data unit                                                       | tCO <sub>2</sub> e                                                                                                                                                                                                                                       |
| Description                                                     | <i>Total decrease in carbon stocks due to displaced deforestation at year t</i>                                                                                                                                                                          |
| Source of data                                                  | <i>Remote sensing and GIS.</i>                                                                                                                                                                                                                           |
| Description of measurement methods and procedures to be applied | <i>Deforestation in the leakage belt area will be considered activity displacement leakage. Activity data for the leakage belt area will be determined using the same methods applied to monitoring deforestation activity data in the project area.</i> |
| Frequency of monitoring/recording                               | <i>Annually</i>                                                                                                                                                                                                                                          |
| Value applied                                                   | <i>Tables Sections 5 and 7</i>                                                                                                                                                                                                                           |
| Monitoring equipment                                            | <i>Remote sensing and GIS</i>                                                                                                                                                                                                                            |
| QA/QC procedures to be applied                                  | <i>Best practices in remote sensing</i>                                                                                                                                                                                                                  |
| Purpose of data                                                 | <i>This parameter will be used to calculate leakage emissions in the project scenario. Provides the ex post value of the decrease in carbon stocks due to displaced deforestation in the leakage belt.</i>                                               |

|                    |                                                                                                                                                                                                                                                                                |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Calculation method | <i>Emissions from deforestation are estimated by multiplying the detected area of forest loss by the average forest carbon stock per unit area</i>                                                                                                                             |
| Comments           | <i>Where evidence can be collected that deforestation in the leakage belt is attributable to deforestation agents that are not linked to the project area, the detected deforestation may not be attributed to the project activity and therefore, not considered leakage.</i> |

|                                                                 |                                                                                                                                                                                                                        |
|-----------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Data / Parameter                                                | $\Delta\text{CUDdPA}_t$                                                                                                                                                                                                |
| Data unit                                                       | tCO <sub>2</sub> e                                                                                                                                                                                                     |
| Description                                                     | <i>Total actual carbon stock change due to unavoided unplanned deforestation at year t in the project area</i>                                                                                                         |
| Source of data                                                  | <i>Remote sensing and GIS.</i>                                                                                                                                                                                         |
| Description of measurement methods and procedures to be applied | <i>Forest cover change due to unplanned deforestation will be monitored through assessment of classified satellite imagery covering the project area.</i>                                                              |
| Frequency of monitoring/recording                               | <i>Annually</i>                                                                                                                                                                                                        |
| Value applied                                                   | <i>Tables Sections 5 and 7</i>                                                                                                                                                                                         |
| Monitoring equipment                                            | <i>Remote sensing and GIS</i>                                                                                                                                                                                          |
| QA/QC procedures to be applied                                  | <i>Best practices in remote sensing</i>                                                                                                                                                                                |
| Purpose of data                                                 | <i>This parameter will be used to calculate project emissions in the project scenario. Provides the ex post value of the change in carbon stocks due to unavoided unplanned deforestation within the project area.</i> |
| Calculation method                                              | <i>Emissions from deforestation are estimated by multiplying the detected area of forest loss by the average forest carbon stock per unit area</i>                                                                     |
| Comments                                                        | <i>N/A</i>                                                                                                                                                                                                             |

|                  |                 |
|------------------|-----------------|
| Data / Parameter | RF <sub>t</sub> |
|------------------|-----------------|



|                                                                 |                                                                                                                                                                                                        |
|-----------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Data unit                                                       | %                                                                                                                                                                                                      |
| Description                                                     | <i>Risk factor used to calculate VCS buffer credits</i>                                                                                                                                                |
| Source of data                                                  | <ul style="list-style-type: none"> <li>- VCS Non-Permanence Risk Report (v3.1)</li> <li>- Remote sensing data and GIS</li> <li>- Supervisor report</li> <li>- Literature data</li> </ul>               |
| Description of measurement methods and procedures to be applied | <i>All sources of data from the VCS Non-Permanence Risk Report will be used to measure the various risk factors.</i>                                                                                   |
| Frequency of monitoring/recording                               | <i>Annually</i>                                                                                                                                                                                        |
| Value applied                                                   | <i>18</i>                                                                                                                                                                                              |
| Monitoring equipment                                            | <i>Remote sensing and GIS</i>                                                                                                                                                                          |
| QA/QC procedures to be applied                                  | <i>Best practices in remote sensing and GIS</i>                                                                                                                                                        |
| Purpose of data                                                 | <i>This parameter represents the non-permanence risk rating of the project, which was used to determine the number of buffer credits that shall be deposited into the AFOLU pooled buffer account.</i> |
| Calculation method                                              | <i>This parameter was calculated using the AFOLU Non-Permanence Risk Tool. All the risk factors described in the VCS Non-Permanence Risk Report were assessed.</i>                                     |
| Comments                                                        | <i>N/A</i>                                                                                                                                                                                             |

### 6.3 Monitoring Plan

| ASPECT                  | TASK(s)                                                                  | PROJECT ACTIVITY                                                           | TECHNICAL DESCRIPTION                                                                                                                                                                                                                                                                                                                                                                                                                                                         | RESPONSIBLE (R) AND PARTIES INVOLVED (PI) | DATA COLLECTION AND ARCHIVING PROCEDURES                                                                                                                                                                                                                                                                                         | QUALITY CONTROL AND ASSURANCE PROCEDURES                                                                                                                                                                          |
|-------------------------|--------------------------------------------------------------------------|----------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TACKLING CLIMATE CHANGE | Monitoring actual carbon stock changes and GHG<br><br>Monitoring leakage | Near real-time visualization of the Project Area by using satellite images | PlanetScope: Use of the biggest satellite constellation (about 150 satellites), which collect daily images from any spot on the planet: generating continuous imaging generated from PlanetScope orthorectified images with a 3-meter spatial resolution, accessible with RGB viewing functionality in XYZ format, including levels 15 (4.77 meters spatial resolution) and 16 (2.38 meters spatial resolution) with an 8-bit radiometric resolution.<br><br>Axelspace: - - - | R: NovaAgri<br>P.I.: Axelspace (partner)  | Visual access for all the area of interest, with daily images. The archive of current images will be archived in the internal data storage of NovaAgri - implemented on OneDrive (Microsoft) interface. The historical archive of daily PlanetScope images collected from 2017 onwards will also be available for visualization. | A image visualization routine will be implemented daily to check for any significant changes, minor alterations, or early signs of clearing, wildfires, or any other adverse conditions within the forested area. |

|                               |                                                               |                    |                                                                                                                                                                                                                                                                                                  |                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                               |
|-------------------------------|---------------------------------------------------------------|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TACKLING<br>CLIMATE<br>CHANGE | Monitoring<br>actual<br>carbon<br>stock<br>changes<br>and GHG | Dashboard -<br>Web | Implementation and provision<br>of a Web Dashboard integrated<br>with a robust and structured<br>database, allowing for the<br>retrieval, organization,<br>analysis, quantification, and<br>presentation of information<br>associated with the areas of<br>interest from NovaAgri's<br>database. | R: NovaAgri<br>P.I.: SCCON<br>(supplier) | <p>This is delivered through<br/>an interactive panel and<br/>the generation of<br/>analytical reports<br/>directly from the<br/>dashboard, reinforcing<br/>the governance and<br/>transparency for the<br/>NovaAgri's projects.</p> <p>The archive of reports<br/>will be archived in both<br/>dashboard storage<br/>system and the internal<br/>data storage of NovaAgri<br/>- implemented on<br/>OneDrive (Microsoft)<br/>interface.</p> | <p>The dashboard will<br/>be permanently<br/>active (24-7), making<br/>the consultation<br/>possible for anytime.<br/>The users will be<br/>configured<br/>accordingly to the<br/>function on the<br/>Project, making<br/>certain types of<br/>visualisation restrict<br/>for some people or<br/>parties.</p> |
|-------------------------------|---------------------------------------------------------------|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## TACKLING CLIMATE CHANGE

Monitoring actual carbon stock changes and GHG

Deforestation Alerts

Utilization of continuous generation of deforestation alerts throughout the project period. The method involves crossing the fire alert system by INPE with the daily Planet satellite images´ system in order to confirm the fire spots more precisely. Files provided by IPNPE contains information on forest fire and wildfire incidents at various intervals, ranging from annual to near real-time (10 minutes). These data are available in CSV and KML formats.

The alerts will be generated and made available in shapefile format and published on the Web Dashboard for reference. The alerts include before and after images of the occurrence. The alerts cover the following classes: - Clear-cutting; - Access; - Degradation; - Burn scar. A perpetual and unrestricted license for the generated alerts will be provided for use, distribution, and modification.

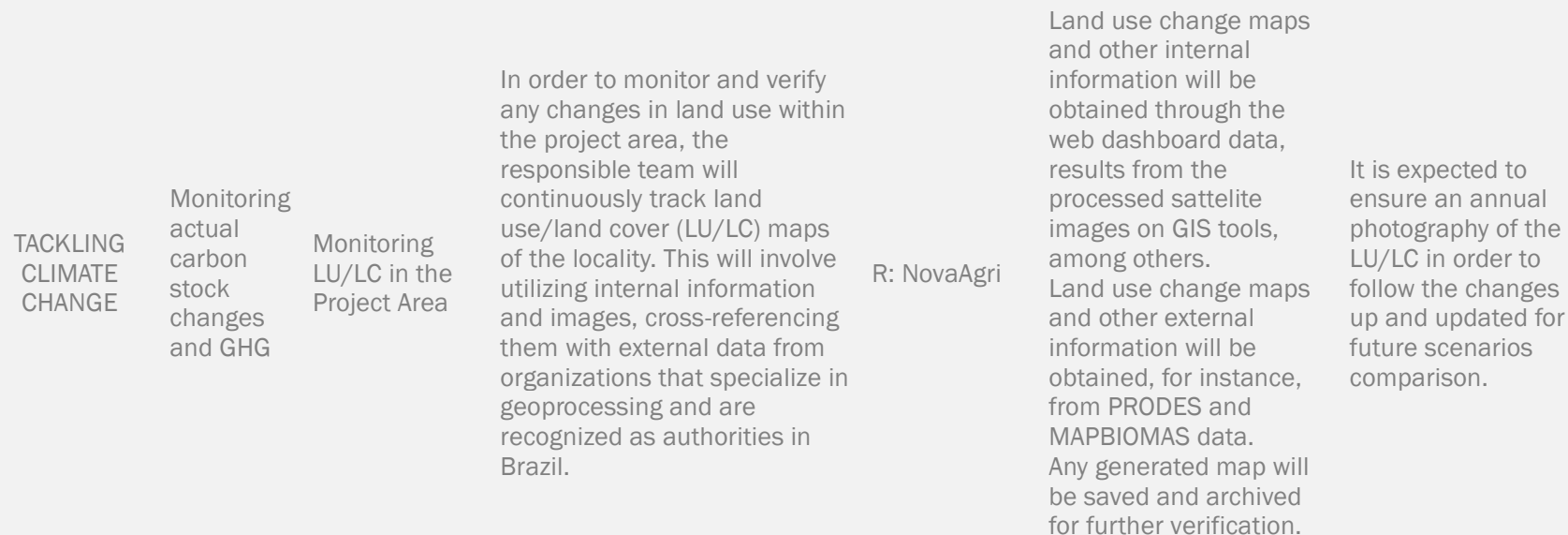
R: NovaAgri  
P.I.: SCCON  
(supplier)

This is delivered through an interactive panel and the generation of analytical reports directly from the dashboard, reinforcing the governance and transparency for the NovaAgri´s projects. As soon as the system alerts a signal, an automatic message is delivered to the team responsible for monitoring.

The archive of reports will be archived in both dashboard storage system and the internal data storage of NovaAgri - implemented on OneDrive (Microsoft) interface.

The alerts will be continually active during the longevity of the project and connected to the permanent web dashboard. The alert system is programmed to detect any signals as immediately as the responsible team runs the images processing, by using geospatial tools and transmitting through the dashboard web.

|                               |                                                               |                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                             |                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                  |
|-------------------------------|---------------------------------------------------------------|---------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TACKLING<br>CLIMATE<br>CHANGE | Monitoring<br>actual<br>carbon<br>stock<br>changes<br>and GHG | Training and<br>consciousness<br>activities | Periodic training sessions will be conducted to facilitate effective actions for the protection of forested areas. These sessions will cover practical issues such as firefighting and safety, as well as more general topics like the challenges of climate change and the biodiversity richness. To achieve this, actions such as visual arts, informative brochures, social media posts, and occasional lectures during in-person visits to stakeholders, will be employed. | R: NovaAgri<br>R: Landowner | All activities conducted during the defined crediting period will be documented in the monitoring reports. The documentation will include information such as the number of training sessions, photo records, post views, attendance lists, and other relevant details. | All records will adhere to the company's privacy policy to ensure the protection of third-party data. The mode of communication and any interactions with stakeholders will also have to comply with the Group's Code of Ethics. |
| PROTECTING<br>BIODIVERSITY    |                                                               |                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                             |                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                  |
| INVOLVING<br>COMMUNITIES      |                                                               |                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                             |                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                  |





|                                |       |                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|--------------------------------|-------|--------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| GOVERNANCE<br>AND<br>INTEGRITY | Extra | Land and<br>socioenvironm<br>ental<br>compliance | NovaAgri will apply its Sustainability Policy continuously during the course of the project in order to ensure the best integrity for the project area. This includes assessing potential issues such as regarding IBAMA/SEMA Embargoes, debts to IBAMA, Conservation Units, Indigenous and public lands. All the assessment will be done during the project longevity through the current systems used by the proponent. | R: NovaAgri<br>P.I.: SCCON<br>(supllier)<br>P.I.: VEGA<br>(supplier) | Application of updated shapefile layers downloaded from official government and interal data for the generation of overlay maps, in order to alert for any other area that intersects with the project area, especially areas corresponding to public properties or areas designated for preservation/demarcation of any kind. The results will be presented on the systems conventionally used by NovaAgri (LYRA) and also in the Web dashboard. Then, the maps can be managed and edited in GIS softwares and tools. Any data exported for any reasons will be archived in the internal data storage of NovaAgri - implemented on OneDrive (Microsoft) interface. | The compliance verification is firstly made at the moment of projects feasibility analysis and updated periodically when: 1) any new areas are included into the project area; 2) any changes on documents are communicated by the landowner or observed in the official registry websites (e.g. CAR website); (3) any areas exceptionally excluded from the area belowed by the landowner for any reasons; (4) everytime this risk analysis have to be ran by the credit/risks department. |
|--------------------------------|-------|--------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|



|                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>GOVERNANCE<br/>AND<br/>INTEGRITY</p> <p>Extra</p> <p>Legal binding<br/>agreement<br/>follow-up</p> | <p>The legal binding agreement signed between the project proponent and the project landowner will be periodically reviewed to ensure that all terms and clauses remain valid. It is expected that there will be no changes to the agreements made at the time of signing; however, any changes in legislation or force majeure issues may require the creation of contractual amendment.</p> <p>R: NovaAgri</p> <p>Any contractual amendment signed between the parties will be saved alongside the first version of the contract and make it available for the VVB verification.</p> <p>The changes on contract will only happen in cases of substantial needs´ occasions. The proponent´s legal department is responsible for any revision, modification and amendment of the contractual draft.</p> |
|-------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|                                         |                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                 |                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                          |
|-----------------------------------------|----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>GOVERNANCE<br/>AND<br/>INTEGRITY</p> | <p>Extra</p> <p>Avoiding double counting of Credits' sales and/or retirement</p> | <p>The term of the agreement also regards to not engaging in any other remunerated or non-remunerated activities in the same Project Area, generating any credits/assets/values/rights, incurring to double-counting and causing negative impacts on the project and parties' integrity and credibility, under penalty incurring fines. The VERRA registry itself is an essential tool for verification of any credits being registered and retired, thus any unauthorized action will be promptly accused on the website.</p> | <p>R: NovaAgri<br/>R: VERRA</p> | <p>The contract is already signed at the time of submitting this PD, so any issues related to the generation and retirement of credits are legally covered through a formal agreement (available for the VVB). The monitoring of records on the VERRA Registry website will also be conducted continuously and reported in the monitoring reports.</p> | <p>Continuous monitoring for checking any needs of contract's updates or actions against double-crediting identified. The proponent's legal team will promptly act in case of any divergence or illegitimate course.</p> |
|-----------------------------------------|----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|                                  |       |                                                                          |                                                                                                                                                                                                                                                                                                                                                                                       |             |                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|----------------------------------|-------|--------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DELIVERING<br>SOCIAL<br>BENEFITS | Extra | Returning<br>benefits for<br>communities<br>by allocating<br>investments | The company is committed to allocating 5% of its own returns – generated from the sale of Verified Carbon Units (VCUs) – to fund and implement impactful social initiatives within the communities surrounding our project area. The social actions may include, for instance, initiatives such as educational support, infrastructure development, access to natural resources etc.. | R: NovaAgri | Any evidence regarding the credits pricing, commercialization, total values received, and the distribution of these values among the parties will be documented to enable the verification of the amounts related to NovaAgri's social investments. Any additional funds invested directly in social responsibility will also be considered in the reports, including allocations made by other receiving party (the landowner). | The proponent will rely on its specialized departments, including financial/tax and legal, to ensure a fully compliant process with reference standards. To ensure that the entire investment process is also carried out in a proper manner, due diligence will be conducted on the partners involved - who will benefit from the allocated benefits, as this concern is also essential for the reputation of the proponent and its economic group. |
|----------------------------------|-------|--------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## 7 QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS

### 7.1 Data and Parameters Monitored

|                  |                                                                                                                                                                                                                                                                                              |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Data / Parameter | ABSLRR <sub>t</sub>                                                                                                                                                                                                                                                                          |
| Data unit        | Ha                                                                                                                                                                                                                                                                                           |
| Description      | Annual area of deforestation in the reference region at year t                                                                                                                                                                                                                               |
| Value applied:   | 23,676                                                                                                                                                                                                                                                                                       |
| Comments         | Forest cover change due to deforestation is monitored through periodic assessment of classified satellite imagery covering the reference region. Through analysis of satellite images and geoprocessed data from deforestation alerts, the applied value was found for the Reference Region. |

|                  |                                                                                                                                                                                                                                                                                          |
|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Data / Parameter | ABSLPA <sub>t</sub>                                                                                                                                                                                                                                                                      |
| Data unit        | Ha                                                                                                                                                                                                                                                                                       |
| Description      | Annual area of deforestation in the project area at year t                                                                                                                                                                                                                               |
| Value applied:   | 0                                                                                                                                                                                                                                                                                        |
| Comments         | Forest cover change due to deforestation is monitored through periodic assessment of classified satellite imagery covering the reference region. Through analysis of satellite images and geoprocessed data from deforestation alerts, the applied value was found for the Project Area. |

|                  |                    |
|------------------|--------------------|
| Data / Parameter | ABSLK <sub>t</sub> |
|------------------|--------------------|

|                |                                                                                                                                                                                                                                                                                          |
|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Data unit      | Ha                                                                                                                                                                                                                                                                                       |
| Description    | Annual area of deforestation in the leakage belt at year t                                                                                                                                                                                                                               |
| Value applied: | 1,216                                                                                                                                                                                                                                                                                    |
| Comments       | Forest cover change due to deforestation is monitored through periodic assessment of classified satellite imagery covering the reference region. Through analysis of satellite images and geoprocessed data from deforestation alerts, the applied value was found for the Leakage Belt. |

|                  |                                                                                                                                                                                                                                                |
|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Data / Parameter | $\Delta CADLK_t$                                                                                                                                                                                                                               |
| Data unit        | tCO <sub>2</sub> e                                                                                                                                                                                                                             |
| Description      | Total decrease in carbon stocks due to displaced deforestation at year t                                                                                                                                                                       |
| Value applied:   | 139,016                                                                                                                                                                                                                                        |
| Comments         | Through combination of satellite images and geoprocessed data from deforestation alerts, the $\Delta CADLK_t$ calculation was ran, considering the same biomass stock from the baseline, and the applied value was found for the Leakage Belt. |

|                  |                                                                                                                                                                                                                                                |
|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Data / Parameter | $\Delta CUDdPA_t$                                                                                                                                                                                                                              |
| Data unit        | tCO <sub>2</sub> e                                                                                                                                                                                                                             |
| Description      | Total actual carbon stock change due to unavoided unplanned deforestation at year t in the project area                                                                                                                                        |
| Value applied:   | 0                                                                                                                                                                                                                                              |
| Comments         | Through combination of satellite images and geoprocessed data from deforestation alerts, the $\Delta CADLK_t$ calculation was ran, considering the same biomass stock from the baseline, and the applied value was found for the Project Area. |

|                  |                                                                                                                                                                                                                                                                      |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Data / Parameter | RF <sub>t</sub>                                                                                                                                                                                                                                                      |
| Data unit        | %                                                                                                                                                                                                                                                                    |
| Description      | Risk factor used to calculate VCS buffer credits                                                                                                                                                                                                                     |
| Value applied:   | 18                                                                                                                                                                                                                                                                   |
| Comments         | This parameter was calculated using the AFOLU Non-Permanence Risk Tool. All the risk factors described in the VCS Non-Permanence Risk Report were assessed. There was no change during this monitoring period, which is still part of the first baseline assessment. |

## 7.2 Baseline Emissions

Find below the baseline emissions and/or carbon stock changes for the monitoring period in accordance with the applied methodology. The calculations in the emission reduction and removal were presented in the Sections 3 (Application of Methodology) and 5 (Quantification Of Estimated Ghg Emission Reductions And Removals).

| (RR) baseline Projection |                             |                               |                        |                        |
|--------------------------|-----------------------------|-------------------------------|------------------------|------------------------|
| Monitoring Period        | Baseline deforestation (ha) | Cumulative deforestation (ha) | Forest cover area (ha) | Deforestation rate (%) |
| 2023-2024                | 32,030                      | 32,030                        | 2,353,538              | 1.34%                  |

| (RR) Monitoring             |                               |                        |                        |
|-----------------------------|-------------------------------|------------------------|------------------------|
| Baseline deforestation (ha) | Cumulative deforestation (ha) | Forest cover area (ha) | Deforestation rate (%) |
| 23,676                      | 23,676                        | 2,361,892              | 0.99%                  |

|                                                                    |  |
|--------------------------------------------------------------------|--|
| (RR) Baseline net carbon stock change of the reference region area |  |
| PDD baseline                                                       |  |
| tCO2e                                                              |  |
| 3,662,574                                                          |  |

|                                                                         |           |
|-------------------------------------------------------------------------|-----------|
| (RR) Total ex-post net carbon stock change of the reference region area |           |
| Monitoring                                                              |           |
| tCO2e                                                                   | tCO2e     |
| 2,707,294                                                               | 2,707,294 |

| (PA) baseline Projection |                             |                               |                        |                        |
|--------------------------|-----------------------------|-------------------------------|------------------------|------------------------|
| Monitoring Period        | Baseline deforestation (ha) | Cumulative deforestation (ha) | Forest cover area (ha) | Deforestation rate (%) |
| 2023-2024                | 350                         | 350                           | 13,576                 | 2.51%                  |

| (PA) Monitoring             |                               |                        |                        |
|-----------------------------|-------------------------------|------------------------|------------------------|
| Baseline deforestation (ha) | Cumulative deforestation (ha) | Forest cover area (ha) | Deforestation rate (%) |
| 0.00                        | 0.00                          | 13,926                 | 0.00%                  |

|                                                             |  |
|-------------------------------------------------------------|--|
| (PA) Total net carbon stock change of the leakage belt area |  |
| PDD baseline                                                |  |
| tCO2e                                                       |  |
| 40,008                                                      |  |

| (PA) Total ex-post net carbon stock change of the project area |       |
|----------------------------------------------------------------|-------|
| Monitoring                                                     |       |
| tCO2e                                                          | tCO2e |
| 0.00                                                           | 0.00  |

| (LL) baseline Projection |                             |                               |                        |                        |
|--------------------------|-----------------------------|-------------------------------|------------------------|------------------------|
| Monitoring Period        | Baseline deforestation (ha) | Cumulative deforestation (ha) | Forest cover area (ha) | Deforestation rate (%) |
| 2023-2024                | 16,059                      | 16,059                        | 511,180                | 3.05%                  |

| (LL) Monitoring             |                               |                        |                        |
|-----------------------------|-------------------------------|------------------------|------------------------|
| Baseline deforestation (ha) | Cumulative deforestation (ha) | Forest cover area (ha) | Deforestation rate (%) |
| 1,216                       | 1,216                         | 526,023                | 0.23%                  |

| (LL) Total net carbon stock change of the leakage belt area |
|-------------------------------------------------------------|
| PDD baseline                                                |
| tCO2e<br>1,836,324                                          |

| (LL) Total ex-post net carbon stock change of the Leakage Belt |
|----------------------------------------------------------------|
| Monitoring                                                     |
| tCO2e<br>139,016                                               |
| tCO2e<br>139,016                                               |



### 7.3 Project Emissions

Find below the project emission and/or carbon stock changes for the monitoring period in accordance with the applied methodology. The calculations in the emission reduction and removal were presented in the Sections 3 (Application of Methodology) and 5 (Quantification Of Estimated Ghg Emission Reductions And Removals).

| Project year | Total carbon stock decrease due to planned activities |                        | Total carbon stock increase due to planned activities |                        | Total carbon stock decrease due to unavoids unplanned deforestation |                        | Total carbon stock decrease change in the project case |                       | Total ex ante estimated actual non-CO <sub>2</sub> emissions from forest fires in the project area |                   |
|--------------|-------------------------------------------------------|------------------------|-------------------------------------------------------|------------------------|---------------------------------------------------------------------|------------------------|--------------------------------------------------------|-----------------------|----------------------------------------------------------------------------------------------------|-------------------|
|              | $\Delta\text{CPAdPat}$                                | $\Delta\text{CPAdPat}$ | $\Delta\text{CPAiPat}$                                | $\Delta\text{CPAiPat}$ | $\Delta\text{CUDdPat}$                                              | $\Delta\text{CUDdPat}$ | $\Delta\text{CPSPat}$                                  | $\Delta\text{CPSPat}$ | $\text{EBBPSPat}$                                                                                  | $\text{EBBPSPat}$ |
|              | annual                                                | cumulative             | annual                                                | cumulative             | annual                                                              | cumulative             | annual                                                 | cumulative            | annual                                                                                             | cumulative        |
|              | tCO <sub>2e</sub>                                     | tCO <sub>2e</sub>      | tCO <sub>2e</sub>                                     | tCO <sub>2e</sub>      | tCO <sub>2e</sub>                                                   | tCO <sub>2e</sub>      | tCO <sub>2e</sub>                                      | tCO <sub>2e</sub>     | tCO <sub>2e</sub>                                                                                  | tCO <sub>2e</sub> |
| 2023         | 0,00                                                  | 0,00                   | 0,00                                                  | 0,00                   | 0,00                                                                | 0,00                   | 0,00                                                   | 0,00                  | 0,00                                                                                               | 0,00              |

### 7.4 Leakage Emissions

Find below the quantified leakage emissions for the monitoring period in accordance with the applied methodology. The calculations in the emission reduction and removal were presented in the Sections 3 (Application of Methodology) and 5 (Quantification Of Estimated Ghg Emission Reductions And Removals).

| Project year | Total ex post GHG emissions from increased grazing activities |                   | Total ex post increase in GHG emissions due to displaced forest fires |                   | Total ex post decrease in carbon stocks due to displaced deforestation |                      | Carbon stock decrease due to leakage prevention measures |                       | Total net carbon stock change due to leakage within the leakage belt |                    | Total net increase in emissions due to leakage |                   |
|--------------|---------------------------------------------------------------|-------------------|-----------------------------------------------------------------------|-------------------|------------------------------------------------------------------------|----------------------|----------------------------------------------------------|-----------------------|----------------------------------------------------------------------|--------------------|------------------------------------------------|-------------------|
|              | $\text{EgLKt}$                                                | $\text{EgLK}$     | $\text{EADLKt}$                                                       | $\text{EADLK}$    | $\Delta\text{CADLKt}$                                                  | $\Delta\text{CADLK}$ | $\Delta\text{CLPMLKt}$                                   | $\Delta\text{CLPMLK}$ | $\Delta\text{CLKt}$                                                  | $\Delta\text{CLK}$ | $\text{ELKt}$                                  | $\text{ELK}$      |
|              | annual                                                        | cumulative        | annual                                                                | cumulative        | annual                                                                 | cumulative           | annual                                                   | cumulative            | annual                                                               | cumulative         | annual                                         | cumulative        |
|              | tCO <sub>2e</sub>                                             | tCO <sub>2e</sub> | tCO <sub>2e</sub>                                                     | tCO <sub>2e</sub> | tCO <sub>2e</sub>                                                      | tCO <sub>2e</sub>    | tCO <sub>2e</sub>                                        | tCO <sub>2e</sub>     | tCO <sub>2e</sub>                                                    | tCO <sub>2e</sub>  | tCO <sub>2e</sub>                              | tCO <sub>2e</sub> |
| 2023         | 0,00                                                          | 0,00              | 0,00                                                                  | 0,00              | 0,00                                                                   | 0,00                 | 0,00                                                     | 0,00                  | 0,00                                                                 | 0,00               | 0,00                                           | 0,00              |

As a result, the leakage emissions equalled 'zero' because the monitoring demonstrated that the deforestation in the leakage belt showed values below to those projected in the baseline.

## 7.5 GHG Emission Reductions and Carbon Dioxide Removals

Summary of GHG Emission Reductions and Removals

| Baseline carbon stock changes |                    |                    | Ex post project carbon stock changes |                    |                    | Ex post leakage carbon stock changes |                    |                    | Ex post net anthropogenic GHG carbon stock changes |                    |                    |
|-------------------------------|--------------------|--------------------|--------------------------------------|--------------------|--------------------|--------------------------------------|--------------------|--------------------|----------------------------------------------------|--------------------|--------------------|
| Annual                        | Cumulative         |                    | Annual                               | Cumulative         |                    | Annual                               | Cumulative         |                    | Annual                                             | Cumulative         |                    |
|                               | tCO <sub>2</sub> e | tCO <sub>2</sub> e |                                      | tCO <sub>2</sub> e | tCO <sub>2</sub> e |                                      | tCO <sub>2</sub> e | tCO <sub>2</sub> e |                                                    | tCO <sub>2</sub> e | tCO <sub>2</sub> e |
|                               | 40,008             | 40,008             |                                      | 0.00               | 0.00               |                                      | 0.00               | 0.00               |                                                    | 40,008             | 40,008             |

18% Risk Factor parameter

| Ex post VCUs tradable |                    |                    | Ex post buffer credit |                    |                    |
|-----------------------|--------------------|--------------------|-----------------------|--------------------|--------------------|
| Annual                | Cumulative         |                    | Annual                | Cumulative         |                    |
|                       | tCO <sub>2</sub> e | tCO <sub>2</sub> e |                       | tCO <sub>2</sub> e | tCO <sub>2</sub> e |
|                       | 32,807             | 32,807             |                       | 7,201              | 7,201              |

Requested information about permanence risk assessment:

|                                                                                                       |                                                                                   |
|-------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| State the non-permanence risk rating (%)                                                              | 18                                                                                |
| Has the non-permanence risk report been attached as either an appendix or a separate document?        | <input checked="" type="checkbox"/> Yes <input type="checkbox"/> No               |
| For ARR and IFM projects with harvesting, state, in tCO <sub>2</sub> e the Long-term Average (LTA).   | N/A                                                                               |
| Has the LTA been updated based on monitored data, if applicable?                                      | <input type="checkbox"/> Yes <input checked="" type="checkbox"/> No               |
| If a loss occurred (including a loss event or reversal), state the amount of tCO <sub>2</sub> e lost: | There was no loss stated for the project area during the first monitoring period. |

# APPENDIX 1: COMMERCIALLY SENSITIVE INFORMATION

*Use the table below to describe the commercially sensitive information included in the project description to be excluded in the public version.*

| Section      | Information        | Justification                                                                                                                                                                                                                             |
|--------------|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <u>3.5.2</u> | Investments values | in order to protect sensible commercial or strategic information, the values presented above do not represent the actual numbers – all the items were shown in a range basis just to demonstrate the magnitude order of those investments |

## APPENDIX 2: OWNERSHIP AND ENVIRONMENTAL COMPLIANCE DOCUMENTS

*Document 1: MATRÍCULA DE POSSE PRIVADA (Land Registry of private ownership)*

*Find below sections (partial) of the official registry documents mentioned in the Section 1.8 (Ownership)*

### **1º REGISTRO GERAL DE IMÓVEIS, TÍTULOS E DOCUMENTOS**

**COMARCA DE PORTO ALEGRE DO NORTE - ESTADO DE MATO GROSSO**

Av. Betumarco, 35-B - Centro - Porto Alegre do Norte - MT - CEP: 78.655-000 - Fone:(66) 3569-2058

██████████ - Oficial Registrador

### **LIVRO 02 - REGISTRO GERAL**

**MATRÍCULA: 2.032 LIVRO: 02 FICHA: 01**

(Protocolo nº 4278, em 28/11/2011). **IMÓVEL: FAZENDA TAPIRAPÉ - GLEBA "B"**, situada na zona rural do Município e Comarca de **Porto Alegre do Norte**, Estado de Mato Grosso, com **área** de **4.764,0651ha** (quatro mil e setecentos e sessenta e quatro hectares, seis ares e cinquenta e um centiares) e perímetro georreferenciado de 35.809,61 metros, conforme **Certificação do INCRA nº ██████████ 31**, de 15 de junho de 2011, de seguinte **descrição**: "Inicia-se a descrição deste perímetro no vértice A0P-M-1154, de coordenadas N 8.802.474,644m e E 494.791,626m, situado na margem esquerda do Rio Tapirapé, com a Fazenda Tapirapé – matrícula nº 172, código INCRA: 901.180.000.078-0; deste, segue confrontando com a referida fazenda, com azimuth de 268°45'25" e distância de 3.770,47 metros, até o vértice A0P-M-1153, de coordenadas N 8.802.392,849m e E 491.022,044m, situado no limite da Fazenda Tapirapé – matrícula nº 172 com a ██████████, código INCRA: não consta; deste, segue confrontando com a ██████████, com azimuth de 1°10'13" e distância de 8.126,01 metros, até o vértice A0P-V-1003, de coordenadas N 8.810.517,168m e E 491.188,009m, situado no limite da Fazenda ██████████ com a Fazenda Tapirapé – matrícula nº 170 Gleba "A", código INCRA: ██████████;

# 1º REGISTRO GERAL DE IMÓVEIS, TÍTULOS E DOCUMENTOS

## COMARCA DE PORTO ALEGRE DO NORTE - ESTADO DE MATO GROSSO

Av. Betumarco, 35-B - Centro - Porto Alegre do Norte - MT - CEP: 78.655-000 - Fone:(66) 3569-2058

- Oficial Registrador

### LIVRO 02 - REGISTRO GERAL

MATRÍCULA: 2.036 LIVRO: 02 FICHA: 01

(Protocolo nº 4278, em 28/11/2011). **IMÓVEL:** FAZENDA TAPIRAPÉ, situada na zona rural do Município e Comarca de Porto Alegre do Norte, Estado de Mato Grosso, com área de 1.078,8164ha (mil e setenta e oito hectares, oitenta e um ares e sessenta e quatro centiares) e perímetro georreferenciado de 19.039,94 metros, conforme **Certificação do INCRA nº 131106000033-31**, de 15 de junho de 2011, de seguinte **descrição:** "Inicia-se a descrição desse perímetro no vértice AOP-M-1153, de coordenadas N=8.802.392,849m e E=491.022,044m cravado na confrontação [REDACTED], deste, segue confrontando com FAZENDA TAPIRAPÉ ([REDACTED]), Código INCRA nº [REDACTED], com o azimute de 088°45'25" e distância de

| Matricula | Ficha | 1º OFÍCIO - REGISTRO DE IMÓVEIS<br>COMARCA DE VILA RICA - ESTADO DE MATO GROSSO<br>Caixa Postal 51 - Cep 78.645-000. Fone (66) 3554-2669<br>[REDACTED] - Oficial |
|-----------|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5.073     | 1     | LIVRO 02 - REGISTRO GERAL                                                                                                                                        |

**IMÓVEL:** Um **LOTE RURAL**, situado no município de Santa Terezinha, Comarca de Vila Rica, Estado de Mato Grosso, denominado "**FAZENDA TAPIRAPÉ**", com a área de **2.110,1585 ha (Dois mil e cento e dez hectares, quinze ares e oitenta e cinco centiares)**, com os seguintes limites e confrontações: Perímetro: 22.687,32 metros. "Inicia-se a descrição desse perímetro no vértice AOP-M-1112(SAT09), de coordenadas N=8.829.930,424m e E=488.343,824m cravado na confrontação da RODOVIA ESTADUAL MT-432, deste, segue confrontando com a faixa de domínio da mesma, no sentido da cidade de Santa Terezinha, com o azimute de 088°41'52" e distância de 2.318,55m, indo até o vértice AOP-M-1150, de coordenadas N=8.829.983,121m e E=490.661,776m; deste, segue

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|-----------|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5.074     | 1     | LIVRO 02 - REGISTRO GERAL                                                                                                                                        |

**IMÓVEL:** Um **LOTE RURAL**, situado no município de Santa Terezinha, Comarca de Vila Rica, Estado de Mato Grosso, denominado "**FAZENDA TAPIRAPÉ**", com a área de **6.383,3917 ha (Seis mil e trezentos e oitenta e três hectares, trinta e nove ares e dezessete centiares)**, com os seguintes limites e confrontações: Perímetro: 32.243,71 metros. "Inicia-se a descrição desse perímetro no vértice AOP-M-0322, de coordenadas N=8.830.234,400m e E=497.721,521m cravado na confrontação da RODOVIA ESTADUAL MT-432, deste, segue confrontando com [REDACTED], Código INCRA [REDACTED] MATRÍCULAS: 9.849 / 14.475 / 14.476 RGI DE SÃO FÉLIX DO ARAGUAIA, com o azimute de 179°32'10" e distância de 9.214,43m, indo até o vértice AOP-V-0006, de coordenadas N=8.821.020,273m e E=497.796,107m;



| Matricula | Ficha | 1º OFÍCIO - REGISTRO DE IMÓVEIS<br>COMARCA DE VILA RICA - ESTADO DE MATO GROSSO<br>Caixa Postal 51 - Cep 78.645-000. Fone (66) 3554-2669<br>[REDACTED] - Oficial |
|-----------|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5.075     | 1     | LIVRO 02 - REGISTRO GERAL                                                                                                                                        |

**IMÓVEL:** Um **LOTE RURAL**, situado no município de Santa Terezinha, Comarca de Vila Rica, Estado de Mato Grosso, denominado "**FAZENDA TAPIRAPÉ**", com a área de **6.500,7657 ha (Seis mil e quinhentos hectares, setenta e seis ares e cinquenta e sete centiares)**, com os seguintes limites e confrontações: Perímetro: 33.750,94 metros. "Inicia-se a descrição desse perímetro no vértice AOP-V-0006, de coordenadas N=8.821.020,273m e E=497.796,107m cravado na confrontação da FAZENDA TAPIRAPÉ, ([REDACTED]), Código INCRA nº 901.180.000.078-0 MATRÍCULA Nº. 148 RGI DE VILA RICA; deste, segue confrontando com [REDACTED] (Proprietário: [REDACTED]), Código INCRA nº. [REDACTED] MATRÍCULAS: 9.849 / 14.475 / 14.476 [REDACTED], com o azimuth de 179º32'10" e distância de 1.036,98m,

| Matricula | Ficha | 1º OFÍCIO - REGISTRO DE IMÓVEIS<br>COMARCA DE VILA RICA - ESTADO DE MATO GROSSO<br>Caixa Postal 51 - Cep 78.645-000. Fone (66) 3554-2669<br>[REDACTED] - Oficial |
|-----------|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5.076     | 1     | LIVRO 02 - REGISTRO GERAL                                                                                                                                        |

**IMÓVEL:** Um **LOTE RURAL**, GLEBA A, situado no município de Santa Terezinha, Comarca de Vila Rica, Estado de Mato Grosso, denominado "**FAZENDA TAPIRAPÉ**", com a área de **1.058,4773 ha (Um mil e cinquenta e oito hectares, quarenta e sete ares e setenta e três centiares)**, com os seguintes limites e confrontações: Perímetro: 18.823,17 metros. "Inicia-se a descrição deste perímetro no vértice AOP-M-1091, de coordenadas N 8.811.600,288m e E 491.210,135m, situado no limite ca [REDACTED] com a Fazenda Tapirapé - matrícula n.º 171 e com a Fazenda Tapirapé - matrícula n.º 169, código INCRA: [REDACTED] deste, segue confrontando com a Fazenda Tapirapé - matrícula n.º 169, com azimuth de 88º26'24" e distância de 7.552,67 metros, até o vértice AOP-V-0019, de coordenadas N 8.811.805,908m e E 498.760,005m, situado no limite da Fazenda Tapirapé -

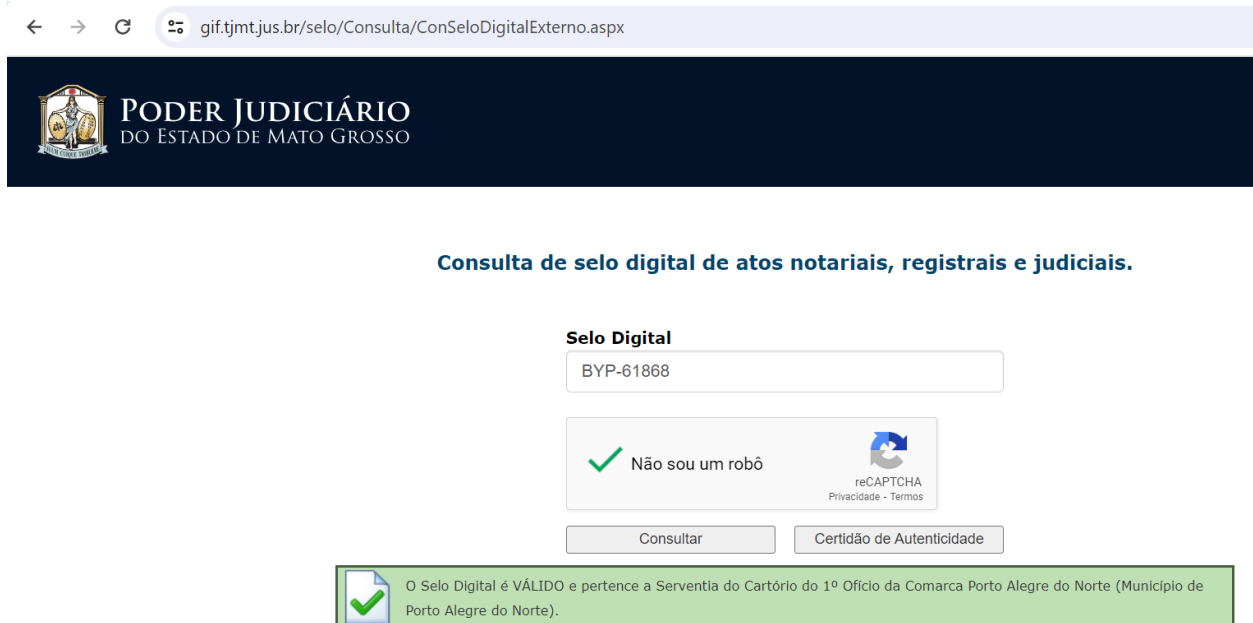
| Matricula | Ficha | 1º OFÍCIO - REGISTRO DE IMÓVEIS<br>COMARCA DE VILA RICA - ESTADO DE MATO GROSSO<br>Caixa Postal 51 - Cep 78.645-000. Fone (66) 3554-2669<br>[REDACTED] - Oficial |
|-----------|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5.077     | 1     | LIVRO 02 - REGISTRO GERAL                                                                                                                                        |

**IMÓVEL:** Um **LOTE RURAL**, situado no município de Santa Terezinha, Comarca de Vila Rica, Estado de Mato Grosso, denominado "**FAZENDA TAPIRAPÉ**", com a área de **2.204,2556 ha (Dois mil e duzentos e quatro hectares, vinte e cinco ares e cinquenta e seis centiares)**, com os seguintes limites e confrontações: Perímetro: 23.367,71 metros. "Inicia-se a descrição desse perímetro no vértice AOP-M-1102(M50), de coordenadas N=8.820.923,822m e E=488.550,209m cravado na confrontação [REDACTED], (HOMOLOGADA PELO DECRETO SEM NÚMERO DE 09/09/1993); deste, segue confrontando com FAZENDA TAPIRAPÉ, (Proprietário: [REDACTED]), Código INCRA [REDACTED] MATRÍCULA Nº. 66 RGI DE VILA RICA, com o azimuth de 089º26'27" e distância de 2.320,41m, indo até o vértice AOP-M-1151, de coordenadas N=8.820.946,464m e

*Document 2: SELO DE AUTENTICIDADE DOS ATOS NOTARIAIS, REGISTRAIS E JUDICIAIS<sup>90</sup> (Seal of authenticity of the notarial, registration and legal documents in Mato Grosso)*



*QR CODE – Mato Grosso State Registry Official Website (for codes consulting)*



The screenshot shows a web browser window with the URL [gif.tjmt.jus.br/selo/Consulta/ConSeloDigitalExterno.aspx](http://gif.tjmt.jus.br/selo/Consulta/ConSeloDigitalExterno.aspx). The header features the logo of the Poder Judiciário do Estado de Mato Grosso. The main content area is titled "Consulta de selo digital de atos notariais, registrais e judiciais." Below this, there is a section for "Selo Digital" with a text input field containing "BYP-61868". A green checkmark and the text "Não sou um robô" are displayed next to a reCAPTCHA logo. Below the input field are two buttons: "Consultar" and "Certidão de Autenticidade". At the bottom, a green banner with a checkmark icon contains the text: "O Selo Digital é VÁLIDO e pertence a Serventia do Cartório do 1º Ofício da Comarca Porto Alegre do Norte (Município de Porto Alegre do Norte)."

*Image – [Registry 2.032] OFFICIAL WEBSITE consulting print screen on 25/03/2024*

<sup>90</sup> <https://gif.tjmt.jus.br/selo/Consulta/ConSeloDigitalExterno.aspx>

← → ↻ gif.tjmt.jus.br/selo/Consulta/ConSeloDigitalExterno.aspx



PODER JUDICIÁRIO  
DO ESTADO DE MATO GROSSO

### Consulta de selo digital de atos notariais, registrais e judiciais.

#### Selo Digital

BYP-61867

✓ Não sou um robô



reCAPTCHA  
Privacidade - Termos

Consultar

Certidão de Autenticidade



O Selo Digital é VÁLIDO e pertence a Serventia do Cartório do 1º Ofício da Comarca Porto Alegre do Norte (Município de Porto Alegre do Norte).

Image – [Registry 2.036] OFFICIAL WEBSITE consulting print screen on 25/03/2024

← → ↻ gif.tjmt.jus.br/selo/Consulta/ConSeloDigitalExterno.aspx



PODER JUDICIÁRIO  
DO ESTADO DE MATO GROSSO

### Consulta de selo digital de atos notariais, registrais e judiciais.

#### Selo Digital

BYF-77169

✓ Não sou um robô



reCAPTCHA  
Privacidade - Termos

Consultar

Certidão de Autenticidade



O Selo Digital é VÁLIDO e pertence a Serventia do Cartório do 1º Ofício da Comarca Vila Rica (Município de Vila Rica).

Image – [Registry 5.073] OFFICIAL WEBSITE consulting print screen on 25/03/2024



← → ↻ gif.tjmt.jus.br/selo/Consulta/ConSeloDigitalExterno.aspx

 **PODER JUDICIÁRIO**  
DO ESTADO DE MATO GROSSO

**Consulta de selo digital de atos notariais, registrais e judiciais.**

**Selo Digital**

BYF-77174

✓ Não sou um robô  reCAPTCHA  
Privacidade - Termos

Consultar Certidão de Autenticidade

 O Selo Digital é VÁLIDO e pertence a Serventia do Cartório do 1º Ofício da Comarca Vila Rica (Município de Vila Rica).

Image – [Registry 5.074] OFFICIAL WEBSITE consulting print screen on 25/03/2024

← → ↻ gif.tjmt.jus.br/selo/Consulta/ConSeloDigitalExterno.aspx

 **PODER JUDICIÁRIO**  
DO ESTADO DE MATO GROSSO

**Consulta de selo digital de atos notariais, registrais e judiciais.**

**Selo Digital**

BYF-77173

✓ Não sou um robô  reCAPTCHA  
Privacidade - Termos

Consultar Certidão de Autenticidade

 O Selo Digital é VÁLIDO e pertence a Serventia do Cartório do 1º Ofício da Comarca Vila Rica (Município de Vila Rica).

Image – [Registry 5.075] OFFICIAL WEBSITE consulting print screen on 25/03/2024

← → ↻ gif.tjmt.jus.br/selo/Consulta/ConSeloDigitalExterno.aspx

 **PODER JUDICIÁRIO**  
DO ESTADO DE MATO GROSSO

**Consulta de selo digital de atos notariais, registrais e judiciais.**

**Selo Digital**

BYF-77167

✓ Não sou um robô  reCAPTCHA  
Privacidade - Termos

Consultar Certidão de Autenticidade

 O Selo Digital é VÁLIDO e pertence a Serventia do Cartório do 1º Ofício da Comarca Vila Rica (Município de Vila Rica).

Image – [Registry 5.076] OFFICIAL WEBSITE consulting print screen on 25/03/2024

← → ↻ gif.tjmt.jus.br/selo/Consulta/ConSeloDigitalExterno.aspx

 **PODER JUDICIÁRIO**  
DO ESTADO DE MATO GROSSO

**Consulta de selo digital de atos notariais, registrais e judiciais.**

**Selo Digital**

BYF-77168

✓ Não sou um robô  reCAPTCHA  
Privacidade - Termos

Consultar Certidão de Autenticidade

 O Selo Digital é VÁLIDO e pertence a Serventia do Cartório do 1º Ofício da Comarca Vila Rica (Município de Vila Rica).

Image – [Registry 5.077] OFFICIAL WEBSITE consulting print screen on 25/03/2024

**Document 3: CAR (Rural Environmental Registry)**

- Active (most recent check: 25/03/2024)
- Analysed by the government institution.

- Regulated in environmental compliance.

#### Demonstrativo da Situação das Informações Declaradas no CAR



Situação do Cadastro:

Ativo

Registro de Inscrição no CAR:

MT-5107776-D0F70258A57348CF817E9838DBA02012

Condição Externa:

Analisado, em regularização ambiental (Lei nº 12.651/2012)

#### Dados do Imóvel Rural

Área do Imóvel Rural: 24.100,02 ha

Data da Inscrição: 30/10/2015

Módulos fiscais: 301,25

Data da Última Retificação: 14/10/2022

Município / UF: Santa Terezinha (MT)

Coordenadas

Geográficas do Lat: 10°42'15,65" S

Centróide: Long: 51°03'34,87" O